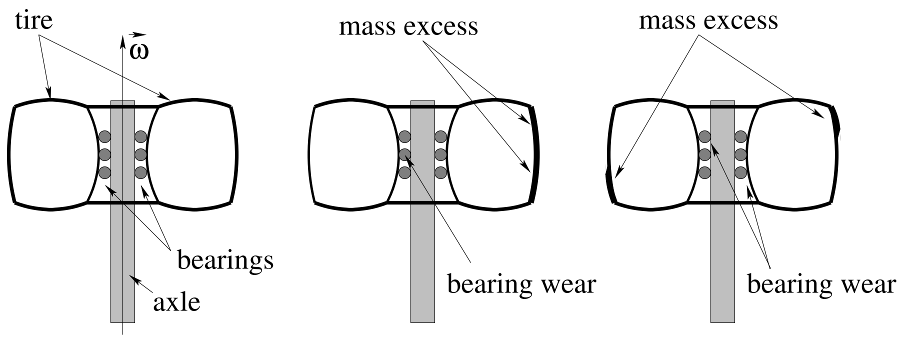


Figure 90: Three tires viewed in cross section. The first one is perfectly symmetric and balanced. The second one has a static imbalance – one side is literally more massive than the other. It will stress the bearings as it rotates as the bearings have to exert a differential centripetal force on the more massive side. The third is dynamically imbalanced – it has a non-planar mass assymetry and the bearings will have to exert a constantly precessing *torque* on the tire. Both of the latter situations will make the drive train noisy, the car more difficult and dangerous to drive, and will wear your bearings and tires out much faster.

plane of rotation but not across the axis of rotation. One side has thicker tread than the other (and would tend to rotate down if the tire were elevated and allowed to spin freely). When a statically imbalanced tire rotates while driving, the center of mass of the tire moves in a circle around the axle and the bearings on the *opposite* side from the increased mass are anomalously compressed in order to provide the required centripetal force. The car bearings will wear faster than they should, and the car will have an annoying vibration and make a wubba-wubba noise as you drive (the latter can occur for many reasons but this is one of them).

The third is a tire that is *dynamically* imbalanced. The surplus masses shown are balanced well enough from left to right – neither side would roll down as in static imbalance, but the mass distribution does *not* have mirror symmetry across the axis of rotation *nor* does it have mirror symmetry across the plane of rotation through the pivot. Like the unbalanced mass sweeping out a cone, the bearings have to exert a *dynamically changing torque* on the tires as they rotate because their angular momentum is *not parallel to the axis of rotation*. At the instant shown, the bearings are stressed in *two* places (to exert a net torque on the hub *out of the page* if the direction of $\vec{\Omega}$ is up as drawn).

The solution to the problem of tire imbalance is simple – rotate your tires (to maintain even wear) and have your tires regularly *balanced* by adding compensatory weights on the "light" side of a static imbalance and to restore relative cylindrical symmetry for dynamical imbalance, the way adding a second mass did to our single mass sweeping out a cone.

I should point out that there are *other* ways to balance a rotating rigid object, and that *every* rigid object, no matter how its mass is distributed, has at least certain axes through the center of mass that can "diagonalize" the moment of inertia with respect to those axes. These are the axes that you can spin the object about and it will rotate freely without any application of an outside force or torque. Sadly, though, this is beyond the scope of this course[121]

[121] Yes, I know, I know – this was a *joke!* I know that you aren't, in fact, terribly sad about this...;-)

At this point you should understand how easy it is to evaluate the angular momentum of symmetric rotating rigid objects (given their moment of inertia) using $L = I\Omega$, and how very difficult it can be to manage the angular momentum in the cases where the mass distribution is asymmetric and unbalanced. In the latter case we will usually find that the angular momentum vector will *precess* around the axis of rotation, necessitating the application of a *continuous torque* to maintain the (somewhat "unnatural") motion.

There is one other very important context where precession occurs, and that is when a *symmetric* rigid body is rapidly rotating and has a large angular momentum, and a torque is applied to it in just the right way. This is one of the most important problems we will learn to solve this week, one essential for *everybody* to know, future physicians, physicists, mathematicians, engineers: The Precession of a Top (or other symmetric rotator).

6.7: Precession of a Top

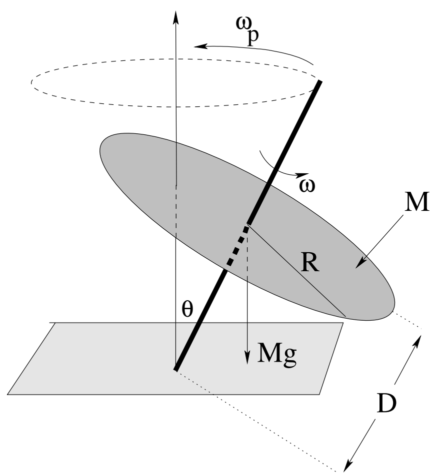


Figure 91:

Nowhere is the vector character of torque and angular momentum more clearly demonstrated than in the phenomenon of ***precession*** of a top, a spinning proton, or the Earth itself. This is also an *important* problem to clearly and quantitatively understand even in this introductory course because it is ***the basis of Magnetic Resonance Imaging (MRI)*** in medicine, the basis of understanding quantum phenomena ranging from ***spin resonance*** to *resonant emission from two-level atoms* for physicists, and the basis for ***gyroscopes*** to the engineer. Evvybody need to know it, in other words, no fair hiding behind the sputtered "but I don't need to know this crap" weasel-squirm all too often uttered by frustrated students.

Look, I'll bribe you. This is one of those topics/problems that I *guarantee* will be on ***at least one quiz, hour exam, or the final*** this semester. That is, mastering it will be worth ***at least*** ten points of your total credit, and in most years it is more likely to be thirty or even fifty points. It's that important! If you master it now, then understanding the precession of a proton around a magnetic field will be a breeze next semester. Otherwise, you risk the *additional* 10-50 points it might be worth next semester.

If you are a sensible person, then, you will recognize that I've just made it worth your while to invest the time required to *completely understand* precession in terms of vector torque, and to be able to ***derive*** the angular frequency of precession, ω_p – which is our goal. I'll show you three ways to do it, don't worry, and any of the three will be acceptable (although I prefer that you use the second or third as the first is a bit *too* simple, it gets the right answer but doesn't give you a good feel for what happens if the forces that produce the torque change in time).

First, though, let's understand the phenonenon. If figure **??** above, a simple top is shown spinning at some angular velocity Ω (not to be confused with ω_p, the precession frequency). We will idealize this particular top as a massive disk with mass M, radius R, spinning around a rigid massless spindle that is resting on the ground, tipped at an angle θ with respect to the vertical.

We begin by noting that this top is symmetric, so we can easily compute the magnitude and direction of its ***angular momentum***:

$$L = |\vec{L}| = I\Omega = \frac{1}{2}MR^2\Omega \tag{6.64}$$

Using the "grasp the axis" version of the right hand rule, we see that in the example portrayed this angular momentum points *in the direction from the pivot at the point of contact between the spindle and the ground and the center of mass of the disk* along the axis of rotation.

Second, we note that there is a ***net torque*** exerted on the top relative to this spindle-ground contact point pivot due to gravity. There is no torque, of course, from the normal force that holds up the top, and we are assuming the spindle and ground are frictionless. This torque is a *vector*:

$$\vec{\tau} = \vec{D} \times (-Mg\hat{z}) \tag{6.65}$$

or

$$\tau = MgD\sin(\theta) \tag{6.66}$$

into the page as drawn above, where z is vertical.

Third we note that the torque will change the angular momentum by displacing it *into the page* in the direction of the torque. But as it does so, the plane containing the $\vec{D}$ and $Mg\hat{z}$ will be rotated a tiny bit around the z (precession) axis, and the torque will *also* shift its direction to *remain* perpendicular to this plane. It will shift $\vec{L}$ a bit more, which shifts $\vec{\tau}$ a bit more and so on. In the end $\vec{L}$ will ***precess around*** z, with $\vec{\tau}$ precessing as well, always $\pi/2$ ahead .

Since $\vec{\tau} \perp \vec{L}$, the magnitude of $\vec{L}$ does not change, only the direction. $\vec{L}$ sweeps out a cone, *exactly like the cone swept out in the unbalanced rotation problems above* and we can use similar considerations to relate the magnitude of the torque (known above) to the change of angular momentum per period. This is the simplest (and least accurate) way to find the precession frequency. Let's start by giving this a try:

Example 6.7.1: Finding ω_p From $\Delta L/\Delta t$ (Average)

We already derived above that

$$\tau_{\rm avg} = \tau = MgD\sin(\theta) \tag{6.67}$$

is the *magnitude* of the torque at all points in the precession cycle, and hence is also the average of the magnitude of the torque over a precession cycle (as opposed to the average of the vector torque over a cycle, which is obviously zero).

We now compute the average torque algebraically from $\Delta L/\Delta t$ (average) and set it equal to the computed torque above. That is:

$$\Delta L_{\rm cycle} = 2\pi L_\perp = 2\pi L\sin(\theta) \tag{6.68}$$

and:

$$\Delta t_{\rm cycle} = T = \frac{2\pi}{\omega_p} \tag{6.69}$$

where ω_p is the (desired) precession frequency, or:

$$\tau_{\rm avg} = MgD\sin(\theta) = \frac{\Delta L_{\rm cycle}}{\Delta t_{\rm cycle}} = \omega_p L\sin(\theta) \tag{6.70}$$

We now solve the the precession frequency ω_p:

$$\omega_p = \frac{MgD}{L} = \frac{MgD}{I\Omega} = \frac{2gD}{R^2\Omega} \tag{6.71}$$

Note well that the precession frequencey is *independent of* θ – this is extremely important next semester when you study the precession of spinning charged particles around an applied magnetic field, the basis of ***Magnetic Resonance Imaging*** (MRI).

Example 6.7.2: Finding ω_p from ΔL and Δt Separately

Simple and accurate as it is, the previous derivation has a few “issues” and hence it is not my favorite one; averaging in this way doesn’t give you the most insight into what’s going on and doesn’t help you get a good feel for the calculus of it all. This *matters* in MRI, where the magnetic field varies in time, although it is a bit more difficult to make gravity vary in a similar way so it doesn’t matter so much *this* semester.

Nevertheless, to get you off on the right foot, so to speak, for E&M[122], let’s do a second derivation that is in between the very crude averaging up above and a rigorous application of calculus to the problem that we can’t even understand properly until we reach the week where we cover oscillation.

Let’s repeat the general idea of the previous derivation, only instead of setting τ equal to the *average* change in $\vec{L}$ per unit time, we will set it equal to the *instantaneous* change in $\vec{L}$ per unit time, writing everything in terms of very small (but finite) intervals and then taking the appropriate limits.

[122] Electricity and Magnetism

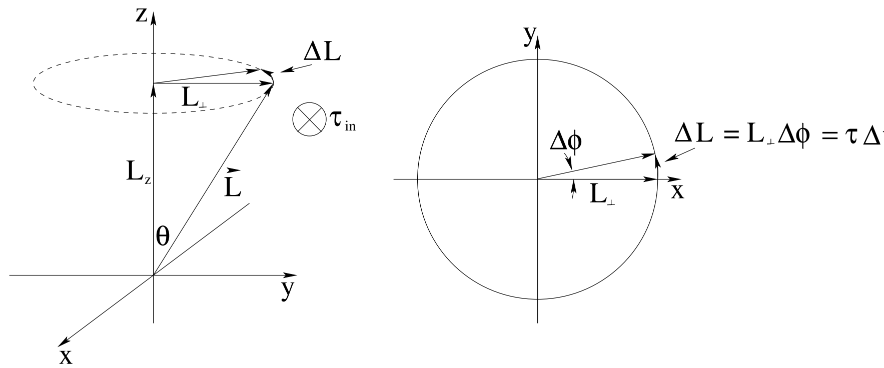


Figure 92: The cone swept out by the precession of the angular momentum vector, $\vec{L}$, as well as an "overhead view" of the trajectory of $\vec{L}_\perp$, the component of $\vec{L}$ perpendicular to the z-axis.

To do this, we need to *picture* how $\vec{L}$ changes in time. Note well that at any given instant, $\vec{\tau}$ is perpendicular to the plane containing $-\hat{y}$ (the direction of the gravitational force) and $\vec{L}$. This direction is thus always perpendicular to *both* $\vec{L}$ and $-Mg\hat{z}$ and ***cannot change the magnitude of $\vec{L}$ or its z component L_z***. Over a very short time Δt, the change in $\vec{L}$ is thus $\Delta\vec{L}$ in the plane perpendicular to $\hat{z}$. As the angular momentum changes *direction only* into the $\vec{\tau}$ direction, the $\vec{\tau}$ direction also changes to remain perpendicular. This should remind you have our long-ago discussion of the kinematics of circular motion – $\vec{L}$ sweeps out a *cone* around the z-axis where L_z and the magnitude of L remain constant.

$L_\perp$ sweeps out a circle as we saw in the previous example, and we can visualize both the cone swept out by $\vec{L}$ and the change over a short time Δt, $\Delta\vec{L}$, in figure 92. We can see from the figure that:

$$\Delta L = L_\perp \Delta\phi = \tau\Delta t \tag{6.72}$$

where $\Delta\phi$ is the angle the angular momentum vector precesses through in time Δt. We substitute $L_\perp = L\sin(\theta)$ and $\tau = MgD\sin(\theta)$ as before, and get:

$$L\sin(\theta)\ \Delta\phi = MgD\sin(\theta)\ \Delta t \tag{6.73}$$

Finally we solve for:

$$\omega_p = \frac{d\phi}{dt} = \lim_{\Delta t\to 0}\frac{\Delta\phi}{\Delta t} = \frac{MgD}{L} = \frac{2gD}{R^2\Omega} \tag{6.74}$$

as before. Note well that because this time we could take the limit as $\Delta t \to 0$, we get an expression that is good at *any* instant in time, even if the top is in a rocket ship (an accelerating frame) and $g \to g'$ is varying in time!

This still isn't the most elegant approach. The *best* approach, although it *does* use some real calculus, is to just write down the equation of motion for the system as differential equations and solve them for *both* $\vec{L}(t)$ *and* for ω_p.

Example 6.7.3: Finding ω_p from Calculus

Way back at the beginning of this section we wrote down Newton's Second Law for the rotation of the gyroscope *directly*:

$$\vec{\boldsymbol{\tau}} = \vec{\boldsymbol{D}} \times (-Mg\hat{\boldsymbol{z}}) \tag{6.75}$$

Because $-Mg\hat{\boldsymbol{z}}$ points only in the negative z-direction and

$$\vec{\boldsymbol{D}} = D\frac{\vec{\boldsymbol{L}}}{L} \tag{6.76}$$

because $\vec{\boldsymbol{L}}$ is parallel to $\vec{\boldsymbol{D}}$, for a general $\vec{\boldsymbol{L}} = L_x\hat{\boldsymbol{x}} + L_y\hat{\boldsymbol{y}}$ we get precisely *two terms* out of the cross product:

$$\frac{dL_x}{dt} = \tau_x = MgD\sin(\theta) = \frac{MgDL_y}{L} \tag{6.77}$$

$$\frac{dL_y}{dt} = \tau_y = -MgD\sin(\theta) = -\frac{MgDL_x}{L} \tag{6.78}$$

or

$$\frac{dL_x}{dt} = \frac{MgD}{L}L_y \tag{6.79}$$

$$\frac{dL_y}{dt} = \frac{MgD}{L}L_x \tag{6.80}$$

If we differentiate the first expression and substitute the second into the first (and vice versa) we transform this pair of coupled *first* order differential equations for L_x and L_y into the following pair of *second* order differential equations:

$$\frac{d^2L_x}{dt^2} = \left(\frac{MgD}{L}\right)^2 L_x = \omega_p^2 L_x \tag{6.81}$$

$$\frac{d^2L_y}{dt^2} = \left(\frac{MgD}{L}\right)^2 L_y = \omega_p^2 L_y \tag{6.82}$$

These (either one) we will learn to recognize as the differential equation of motion for the *simple harmonic oscillator*. A particular solution of interest (that satisfies the first order equations above) might be:

$$L_x(t) = L_\perp \cos(\omega_p t) \tag{6.83}$$

$$L_y(t) = L_\perp \sin(\omega_p t) \tag{6.84}$$

$$L_z(t) = L_{z0} \tag{6.85}$$

where $L_\perp$ is the magnitude of the component of $\vec{\boldsymbol{L}}$ which is perpendicular to $\hat{z}$. This is then the *exact solution* to the equation of motion for $\vec{\boldsymbol{L}}(t)$ that describes the actual cone being swept out, with no hand-waving or limit taking required.

The only catch to this approach is, of course, that we don't know how to solve the equation of motion yet, and the very *phrase* "second order differential equation" strikes terror into our hearts ***in spite of the fact that we've been solving one after another since week 1 in this class! All*** of the equations of motion we have solved from Newton's

Second Law have been second order ones, after all – it is just that the ones for constant acceleration were directly *integrable* where this set is not, at least not easily.

It is pretty easy to solve for all of that, but we will postpone the actual solution until later. In the meantime, remember, you must know how to reproduce *one of the three derivations above* for ω_p, the angular precession frequency, for *at least one quiz, test, or hour exam*. Not to mention a homework problem, below. Be sure that you master this because precession is *important*.

One last suggestion before we move on to treat angular collisions. Most students have a lot of experience with pushes and pulls, and so far it has been pretty easy to come up with everyday experiences of force, energy, one dimensional torque and rolling, circular motion, and all that. It's not so easy to come up with everyday experiences involving vector torque and precession. Yes, may of you played with tops when you were kids, yes, nearly everybody *rode bicycles* and balancing and steering a bike involves torque, but you haven't really *felt* it knowing what was going on.

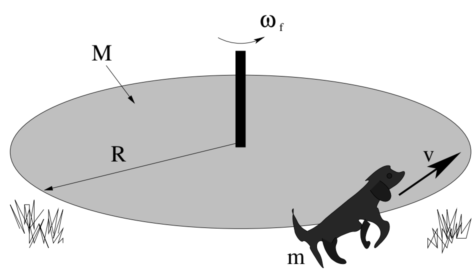
The only device likely to help you to personally experience torque "with your own two hands" is the bicycle wheel with handles and the string that was demonstrated in class. I ***urge you to take a turn spinning this wheel*** and trying to turn it by means of its handles while it is spinning, to spin it and suspend it by the rope attached to one handle and watch it precess. Get to where you can predict the direction of precession given the direction of the spin and the handle the rope is attached to, get so you have *experience* of pulling it (say) in and out and feeling the handles deflect up and down or vice versa. Only thus can you ***feel*** the nasty old cross product in both the torque and the angular momentum, and only thus can you come to understand torque with your gut as well as your head.

Homework for Week 6

Problem 1.

Physics Concepts: Make this week's physics concepts summary as you work all of the problems in this week's assignment. Be sure to cross-reference each concept in the summary to the problem(s) they were key to, and include concepts from previous weeks as necessary. Do the work carefully enough that you can (after it has been handed in and graded) punch it and add it to a three ring binder for review and study come finals!

Problem 2.



This problem will help you learn required concepts such as:

- Angular Momentum Conservation
- Inelastic Collisions

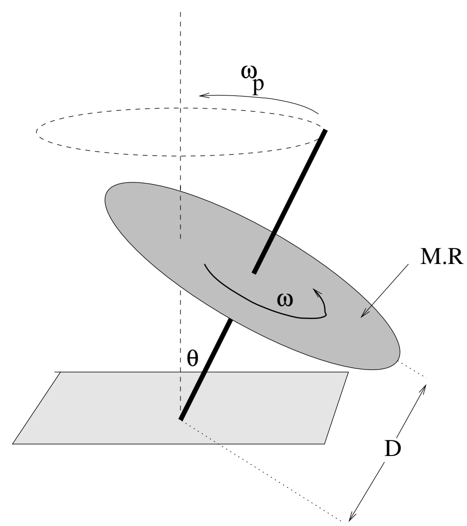
so please review them before you begin.

Satchmo, a dog with mass m runs and jumps onto the edge of a merry-go-round (that is initially at rest) and then sits down there to take a ride as it spins. The merry-go-round can be thought of as a disk of radius R and mass M, and has approximately frictionless bearings in its axle. At the time of this angular "collision" Satchmo is travelling at speed v perpendicular to the radius of the merry-go-round and you can neglect Satchmo's moment of inertia about an axis through his OWN center of mass compared to that of Satchmo travelling around the merry-go-round axis (because R of the merry-go-round is much larger than Satchmo's size, so we can treat him like "a particle").

a) What is the angular velocity ω_f of the merry-go-round (and Satchmo) right after the collision?

b) How much of Satchmo's initial kinetic energy was lost, compared to the final kinetic energy of Satchmo plus the merry-go-round, in the collision? (Believe me, no matter what he's lost it isn't enough – he's a border collie and he has plenty more!)

Problem 3.



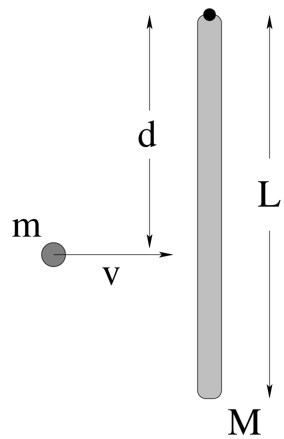
This problem will help you learn required concepts such as:

- Vector Torque
- Vector Angular Momentum
- Geometry of Precession

so please review them before you begin.

A top is made of a disk of radius R and mass M with a very thin, light nail ($r \ll R$ and $m \ll M$) for a spindle so that the disk is a distance D from the tip. The top is spun with a large angular velocity ω. When the top is spinning at a small angle θ with the vertical (as shown) what is the angular frequency ω_p of the top's precession? Note that this is a ***required problem*** that *will be on at least one test* in one form or another, so be sure that you have mastered it when you are done with the homework!

Problem 4.



This problem will help you learn required concepts such as:

- Angular Momentum Conservation
- Momentum Conservation
- Inelastic Collisions
- Impulse

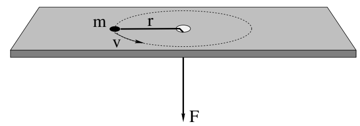
so please review them before you begin. You may also find it useful to read: Wikipedia: http://www.wikipedia.org/wiki/Center of Percussion.

A rod of mass M and length L rests on a frictionless table and is pivoted on a frictionless nail at one end as shown. A blob of putty of mass m approaches with velocity v from the left and strikes the rod a distance d from the end as shown, sticking to the rod.

- Find the angular velocity ω of the system about the nail after the collision.
- Is the linear momentum of the rod/blob system conserved in this collision for a general value of d? If not, why not?
- Is there a value of d for which it *is* conserved? If there were such a value, it would be called the *center of percussion* for the rod for this sort of collision.

All answers should be in terms of M, m, L, v and d as needed. Note well that you should clearly indicate what physical principles you are using to solve this problem at the *beginning* of the work.

Problem 5.



This problem will help you learn required concepts such as:

- Torque Due to Radial Forces
- Angular Momentum Conservation
- Centripetal Acceleration
- Work and Kinetic Energy

so please review them before you begin.

A particle of mass m is tied to a string that passes through a hole in a frictionless table and held. The mass is given a push so that it moves in a circle of radius r at speed v.

a) What is the torque exerted on the particle by the string?

b) What is the magnitude of the angular momentum L of the particle in the direction of the axis of rotation (as a function of m, r and v)?

c) Show that the magnitude of the force (the tension in the string) that must be exerted to keep the particle moving in a circle is:

$$F = \frac{L^2}{mr^3}$$

d) Show that the kinetic energy of the particle in terms of its angular momentum is:

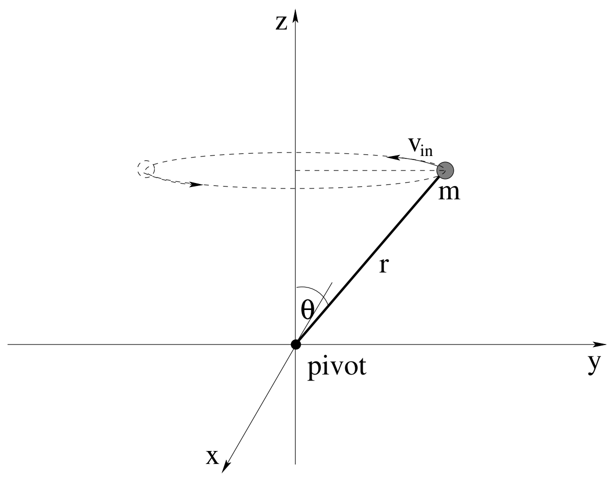
$$K = \frac{L^2}{2mr^2}$$

Now, suppose that the radius of the orbit and initial speed are r_i and v_i, respectively. From under the table, the string is *slowly* pulled down (so that the puck is always moving in an approximately circular trajectory and the tension in the string remains radial) to where the particle is moving in a circle of radius r_2.

e) Find its velocity v_2 using angular momentum conservation. This should be very easy.

f) Compute the work done by the force from part c) above and identify the answer *as* the work-kinetic energy theorem.

Problem 6.



This problem will help you learn required concepts such as:

- Vector Torque
- Non-equivalence of $\vec{L}$ and $I\vec{\omega}$.

so please review them before you begin.

In the figure above, a mass m is being spun in a circular path at a constant speed v around the z-axis on the end of a massless rigid rod of length r pivoted at the orgin (that itself is being pushed by forces not shown). All answers should be given in terms of m, r, θ and v. Ignore gravity and friction.

a) Find the angular momentum vector of this system *relative to the pivot* at the instant shown and draw it in on a copy of the figure.

b) Find the angular velocity vector of the mass m at the instant shown and draw *it* in on the figure above. Is $\vec{L} = I_z\vec{\omega}$, where $I_z = mr^2\sin^2(\theta)$ is the moment of inertia around the z axis? What *component* of the angular momentum is equal to $I_z\omega$?

c) What is the magnitude and direction of the torque exerted by the rod on the mass (relative to the pivot shown) in order to keep it moving in this circle at constant speed? (Hint: Consider the *precession* of the angular momentum around the z-axis where the precession frequency is ω, and follow the method presented in the textbook above.)

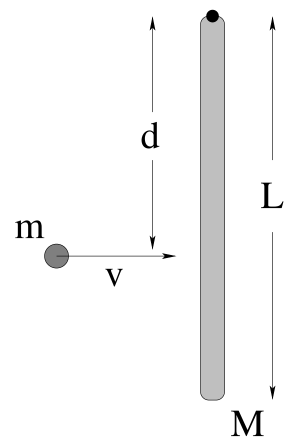
d) What is the *direction* of the force exerted by the rod in order to create this torque? (Hint: Shift gears and think about the actual trajectory of the particle. What must the direction of the force be?)

e) Set $\tau = rF_\perp$ or $\tau = r_\perp F$ and determine the magnitude of this torque. Note that it is equal in magnitude to your answer to c) above and has just the right direction!

Optional Problems

The following problems are **not required or to be handed in**, but are provided to give you some extra things to work on or test yourself with.

Optional Problem 7.



A rod of mass M and length L is hanging vertically from a frictionless pivot (where gravity is "down"). A blob of putty of mass m approaches with velocity v from the left and strikes the rod a distance d from its center of mass as shown, sticking to the rod.

a) Find the angular velocity ω_f of the system about the pivot (at the top of the rod) after the collision.

b) Find the distance $x_{\rm cm}$ from the pivot of the center of mass of the rod-putty system immediately after the collision.

c) Find the velocity of the center of mass $v_{\rm cm}$ of the system (immediately) after the collision.

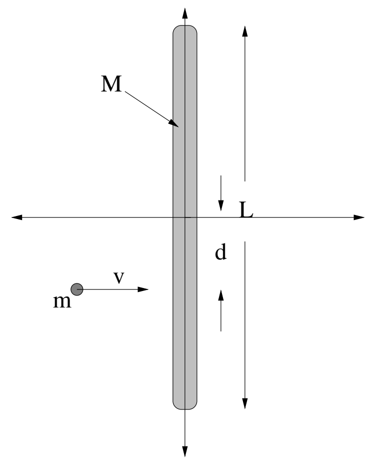
d) Find the kinetic energy of the rod and putty immediately after the collision. Was kinetic energy conserved in this collision? If not, how much energy was lost to heat?

e) After the collision, the rod swings up to a maximum angle $\theta_{\rm max}$ and then comes momentarily to rest. Find $\theta_{\rm max}$.

f) In general (for most values of d is linear momentum conserved in this collision? At the risk of giving away the answer, why not? What exerts an external force on the system during the collision when momentum is not conserved?

All answers should be in terms of M, m, L, v, g and d as needed.

Optional Problem 8.



This problem will help you learn required concepts such as:

- Angular Momentum Conservation
- Momentum Conservation
- Inelastic Collisions
- Impulse

so please review them before you begin.

A rod of mass M and length L rests on a frictionless table. A blob of putty of mass m approaches with velocity v from the left and strikes the rod a distance d from the end as shown, sticking to the rod.

- Find the angular velocity ω of the system about the center of mass *of the system* after the collision. Note that the rod and putty will not be rotating about the center of mass of the rod!
- Is the linear momentum of the rod/blob system conserved in this collision for a general value of d? If not, why not?
- Is kinetic energy conserved in this collision? If not, how much is lost? Where does the energy go?

All answers should be in terms of M, m, L, v and d as needed. Note well that you should clearly indicate what physical principles you are using to solve this problem at the *beginning* of the work.

Week 7: Statics

Statics Summary

- A rigid object is in ***static equilibrium*** when ***both*** the ***vector torque and the vector force*** acting on it are ***zero***. That is:

  > ***If $\vec{F}_{\rm tot} = 0$ and $\vec{\tau}_{\rm tot} = 0$, then an object initially at rest will remain at rest and neither accelerate nor rotate.***

  This rule applies to particles with intrinsic spin as well as “rigid objects”, but this week we will primarily concern ourselves with rigid objects as static *force* equilibrium for particles was previously discussed.

  Note well that the torque and force in the previous problem are both ***vectors***. In many problems they will effectively be one dimensional, but in some they will ***not*** and you must establish e.g. the torque equilibrium condition for several different directions.

- A common question that arises in statics is the ***tipping problem***. For an object placed on a slope or pivoted in some way such that *gravity opposed by normal forces* provides one of the sources of torque that tends to keep the object *stable*, while some variable force provides a torque that tends to *tip the object over* the pivot, one uses the condition of *marginal static equilibrium* to determine, e.g. the lowest value of the variable force that will tip the object over.

- A force ***couple*** is defined to be a pair of forces that are equal and opposite but that ***do not necessarily or generally act along the same line upon an object***. The point of this definition is that it is easily to see that force couples ***exert no net force*** on an object but they ***will exert a net torque on the object*** as long as they do *not* act along the same line. Furthermore:

- The vector torque exerted on a rigid object by a force couple is the ***same for all choices of pivot!*** This (and the frequency with which they occur in problems) is the basis for the definition.

As you can see, this is a short week, just perfect to share with the midterm hour exam.

7.1: Conditions for Static Equilibrium

We already know ***well*** (I hope) from our work in the first few weeks of the course that an object at rest remains at rest *unless acted on by a net external force!* After all, this is just Newton's First Law! If a particle is located at some position in any inertial reference frame, and isn't moving, it won't *start* to move unless we push on it with some force produced by a law of nature.

Newton's Second Law, of course, applies only to *particles* – infinitesimal chunklets of mass in extended objects or elementary particles that really appear to have no finite extent. However, in week 4 we showed that it also applies to *systems* of particles, with the replacement of the position of the particle by the position of the center of mass of the system and the force with the total *external* force acting on the entire system (internal forces cancelled), and to *extended objects* made up of many of those infinitesimal chunklets. We could then extend Newton's First Law to apply as well to amorphous systems such as clouds of gas or structured systems such as "rigid objects" as long as we considered being "at rest" a statement concerning the motion of their *center of mass*. Thus a "baseball", made up of a truly staggering number of elementary microscopic particles, becomes a "particle" in its own right located at its center of mass.

We also learned that the *force* equilibrium of particles acted on by conservative force occurred at the points where the potential energy was maximum or minimum or neutral (flat), where we named maxima "unstable equilibrium points", minima "stable equilibrium points" and flat regions "neutral equilibria"[123].

However, in weeks 5 and 6 we learned enough to now be able to see that force equilibrium *alone* is ***not sufficient*** to cause an extended object or collection of particles to be in equilibrium. We can easily arrange situations where *two* forces act on an object in opposite directions (so there is no net force) but along lines such that together they exert a nonzero *torque* on the object and hence cause it to angularly accelerate and gain kinetic energy without bound, hardly a condition one would call "equilibrium".

The good news is that Newton's Second Law for Rotation is sufficient to imply Newton's First Law for Rotation:

> ***If, in an inertial reference frame, a rigid object is initially at rotational rest (not rotating), it will remain at rotational rest unless acted upon by a net external torque.***

That is, $\vec{\tau} = I\vec{\alpha} = 0$ implies $\vec{\Omega} = 0$ and constant[124]. We will call the condition where $\vec{\tau} = 0$ and a rigid object is not rotating ***torque equilibrium***.

We can make a baseball (initially with its center of mass at rest and not rotating) spin without exerting a net force on it that makes its center of mass move – it can be in force

[123] Recall that neutral equilibria were generally closer to being unstable than stable, as any nonzero velocity, no matter how small, would cause a particle to move continuously across the neutral region, making no particular point *stable* to say the least. That same particle would *oscillate*, due to the restoring force that traps it between the *turning points* of motion that we previously learned about and at least remain in the "vicinity" of a true stable equilibrium point for small enough velocities/kinetic energies.

[124] Whether or not I is a scalar or a tensor form...

equilibrium but not torque equilibrium. Similarly, we can throw a baseball without imparting any rotational spin – it can be in torque equilibrium but not force equilibrium. If we want the baseball (or any rigid object) to be in a *true* static equilibrium, one where it is *neither* translating *nor* rotating in the future if it is at rest and not rotating initially, we need both the conditions for force equilibrium and torque equilibrium to be true.

Therefore we now *define* the conditions for the ***static equilibrium of a rigid body*** to be:

> ***A rigid object is in static equilibrium when both the vector torque and the vector force acting on it are zero.***

That is:

> ***If $\vec{F}_{\rm tot} = 0$ and $\vec{\tau}_{\rm tot} = 0$, then an object initially at translational and rotational rest will remain at rest and neither accelerate nor rotate.***

That's it. Really, pretty simple.

Needless to say, the *idea* of stable versus unstable and neutral equilibrium still holds for torques as well as forces. We will consider an equilibrium to be stable only if ***both*** the force ***and*** the torque are "restoring" – and push or twist the system *back* to the equilibrium if we make small linear or angular displacements away from it.

7.2: Static Equilibrium Problems

After working so long and hard on solving actual *dynamical* problems involving force and torque, static equilibrium problems sound like they would be pretty trivial. After all, *nothing happens!* It seems as though solving for what happens when *nothing* happens would be easy.

Not so. To put it in perspective, let's consider *why* we might want to solve a problem in static equilibrium. Suppose we want to build a house. A properly built house is one that won't just *fall down*, either all by itself or the first time the wolf huffs and puffs at your door. It seems as though building a house that is *stable* enough not to fall down when you move around in it, or load it with furniture in different ways, or the first time a category 2 hurricane roars by overhead and whacks it with 160 kilometer per hour winds is a worthy design goal. You might even want it to survive earthquakes!

If you think that building a stable house is *easy*, I commend trying to build a house out of cards[125] . You will soon learn that balancing force loads, taking advantage of friction (or other "fastening" forces", avoiding unbalanced torques is all actually remarkably tricky, for all that we can learn to do it without solving actual equations. Engineers who want to build *serious* structures such as bridges, skyscrapers, radio towers, cars, airplanes, and so on spend a *lot* of time learning statics (and a certain amount of dynamics, because no

[125] Wikipedia: http://www.wikipedia.org/wiki/House of cards. Yes, even this has a wikipedia entry. Pretty cool, actually!

structure in a dynamical world filled with Big Bad Wolves is truly "static") because it is *very expensive* when buildings, bridges, and so on fall down, when their structural integrity fails.

Stability is just as important for physicians to understand. The human body is not the world's most stable structure, as it turns out. If you have ever played with Barbie or G.I. Joe dolls, then you understand that getting them to stand up on their own takes a bit of doing – just a bit of bend at the waist, just a bit too much weight at the side, or the feet not adjusted just so, and they fall right over. Actual humans stabilize their erect stance by *constantly adjusting the force balance* of their feet, shifting weight without thinking to the heel or to the toes, from the left foot to the right foot as they move their arms or bend at the waist or lift something. Even healthy, coordinated adults who are paying attention nevertheless sometimes *lose their balance* because their motions exceed the fairly narrow tolerance for stability in some stance or another.

This ability to remain stable standing up or walking rapidly disappears as one's various sensory feedback mechanisms are impaired, and many, many health conditions impair them. Drugs or alcohol, neuropathy, disorders of the vestibular system, pain and weakness due to arthritis or aging. Many injuries (especially in the elderly) occur because people just plain *fall over*.

Then there is the fact that nearly all of our physical activity involves the adjustment of static balance between muscles (providing tension) and bones (providing compression), with our joints becoming stress-points that have to provide enormous forces, painlessly, on demand. In the end, physicians have to have a *very good conceptual understanding* of static equilibrium in order to help their patients acheive it and maintain it in the many aspects of the "mechanical" operation of the human body where it is essential.

For this reason, no matter who you are taking this course, you need to learn to solve real statics problems, ones that can help you later understand and work with statics as your life and career demand. This may be nothing more than helping your kid build a stable tree house, but y'know, you don't want to help them build a tree house and have that house *fall out of the tree* with your kid in it, nor do you want to deny them the joy of hanging out in their own tree house up in the trees!

As has been our habit from the beginning of the course, we start by considering the simplest problems in static equilibrium and then move on to more difficult ones. The simplest problems cannot, alas, be truly *one dimensional* because if the forces involved are truly one dimensional (and act to the right or left along a single line) there *is* no possible torque and force equilibrium suffices for both.

The simplest problem involving both force *and* torque is therefore at least three dimensional – two dimensional as far as the forces are concerned and one dimensional as far as torque and rotation is concerned. In other words, it will involve force balance in some plane of (possible) rotation and torque balance perpendicular to this plane alone a (possible) axis of rotation.

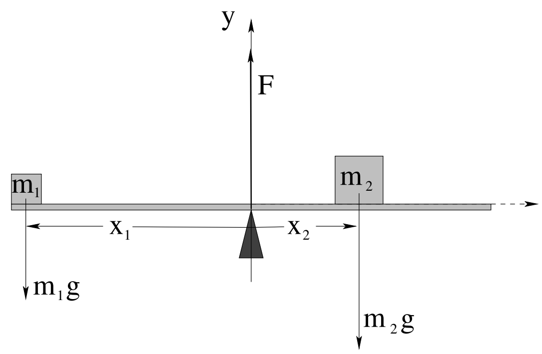


Figure 93: You are given m_1, x_1, and x_2 and are asked to find m_2 and F such that the see-saw is in *static equilibrium*.

Example 7.2.1: Balancing a See-Saw

One typical problem in statics is balancing weights on a see-saw type arrangement – a uniform plank supported by a fulcrum in the middle. This particular problem is really only one dimensional as far as force is concerned, as there is no force acting in the x-direction or z-direction. It is one dimensional as far as torque is concerned, with rotation around any pivot one might select either into or out of the paper.

Static equilibrium requires force balance (one equation) and torque balance (one equation) and therefore we can solve for pretty much any two variables (unknowns) visible in figure 93 above. Let's imagine that in *this* particular problem, the mass m_1 and the distances x_1 and x_2 are given, and we need to find m_2 and F.

We have two choices to make – where we select the pivot and which direction (in or out of the page) we are going to define to be "positive". A perfectly reasonable (but not unique or necessarily "best") choice is to select the pivot at the fulcrum of the see-saw where the unknown force F is exerted, and to select the $+z$-axis as positive rotation (out of the page as drawn).

We then write:

$$\sum F_y = F - m_1 g - m_2 g = 0 \tag{7.1}$$

$$\sum \tau_z = x_1 m_1 g - x_2 m_2 g = 0 \tag{7.2}$$

This is almost embarrassingly simple to solve. From the second equation:

$$m_2 = \frac{m_1 g x_1}{g x_2} = \left(\frac{x_1}{x_2}\right) m_1 \tag{7.3}$$

It is worth noting that this is precisely the mass that moves the *center of mass* of the system so that it is square over the fulcrum/pivot.

From the first equation and the solution for m_2:

$$F = m_1 g + m_2 g = m_1 g \left(1 + \left(\frac{x_1}{x_2}\right)\right) = m_1 g \left(\frac{x_1 + x_2}{x_2}\right) \tag{7.4}$$

That's all there is to it! Obviously, we could have been given m_1 and m_2 and x_1 and been asked to find x_2 and F, etc, just as easily.

Example 7.2.2: Two Saw Horses

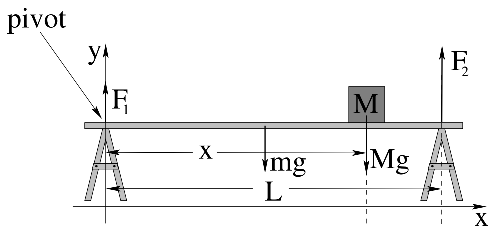


Figure 94: Two saw horses separated by a distance L support a plank of mass m symmetrically placed across them as shown. A block of mass M is placed on the plank a distance x from the saw horse on the left.

In figure 94, two saw horses separated by a distance L support a symmetrically placed plank. The rigid plank has mass m and supports a block of mass M placed a distance x from the left-hand saw horse. Find F_1 and F_2, the upward (normal) force exerted by each saw horse in order for this system to be in static equilibribum.

First let us pick something to put into equilibrium. The saw horses look pretty stable. The mass M does need to be in equilibrium, but that is pretty trivial – the plank exerts a normal force on M equal to its weight. The only tricky thing is the plank itself, which could and would rotate or collapse if F_1 and F_2 don't correctly balance the load created by the weight of the plank plus the weight of the mass M.

Again there are no forces in x or z, so we simply ignore those directions. In the y direction:

$$F_1 + F_2 - mg - Mg = 0 \tag{7.5}$$

or "the two saw horses must support the total weight of the plank plus the block", $F_1 + F_2 = (m + M)g$. This is not unreasonable or even unexpected, but it doesn't tell us how this weight is distributed between the two saw horses.

Once again we much choose a pivot. Four possible points – the point on the left-hand saw horse where F_1 is applied, the point at $L/2$ that is the center of mass of the plank (and half-way in between the two saw horses), the point under mass M where its gravitational force acts, and the point on the right-hand saw horse where F_2 is applied. Any of these will eliminate the torque due to one of the forces and presumably will simplify the problem relative to more arbitrary points. We select the left-hand point as shown – why not?

Then:

$$\tau_z = F_2 L - mgL/2 - Mgx = 0 \tag{7.6}$$

states that the torque around this pivot must be zero. We can easily solve for F_2:

$$F_2 = \frac{mgL/2 + Mgx}{L} = \frac{mg}{2} + Mg\frac{x}{L} \tag{7.7}$$

Finally, we can solve for F_1:

$$F_1 = (m + M)g - F_2 = \frac{mg}{2} + Mg\frac{L - x}{L} \tag{7.8}$$

Does this make sense? Sure. The two saw horses *share* the weight of the symmetrically placed plank, obviously. The saw horse *closest* to the block M supports most of its weight, in a completely symmetric way. In the picture above, with $x > L/2$, that is saw horse 2, but if $x < L/2$, it would have been saw horse 1. In the middle, where $x = L/2$, the two saw horses symmetrically share the weight of the block as well!

This picture and solution are worth studying until all of this makes sense. Carrying things like sofas and tables (with the load shared between a person on either end) is a frequent experience, and from the solution to this problem you can see that if the load is not *symmetrical*, the person closest to the center of gravity will carry the largest load.

Let's do a slightly more difficult one, one involving equilibrium in *two* force directions (and one torque direction). This will allow us to solve for *three* unknown quantities.

Example 7.2.3: Hanging a Tavern Sign

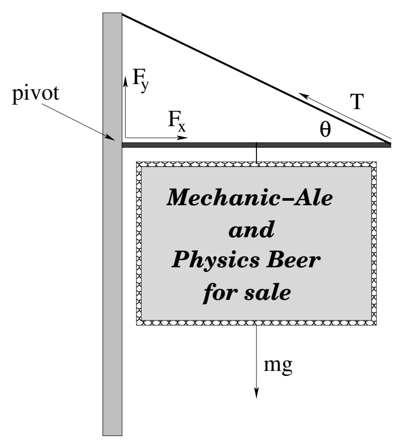


Figure 95: A sign with mass m is hung from a massless rigid pole of length L attached to a post and suspended by means of a wire at an angle θ relative to the horizontal.

Suppose that one day you get tired being a hardworking professional and decide to give it all up and open your own tavern/brewpub. Naturally, you site it in a lovely brick building close to campus (perhaps in Brightleaf Square). To attract passers-by you need a really good *sign* – the old fashioned sort made out of solid oak that hangs from a pole, one that (with the pole) masses $m = 50$ kg.

However, you *really* don't want the sign to either punch through the brick wall or break the suspension wire you are going to use to support the end farthest from the wall and you don't trust your architect because he seems way too interested in your future wares, so you decide to work out for yourself just what the forces are that the wall and wire have to support, as a function of the angle θ between the support pole and the support wire.

The physical arrangement you expect to end up with is shown in figure 95. You wish to find F_x, F_y and T, given m and θ.

By now the idea should be sinking in. Static equilibrium requires $\sum \vec{F} = 0$ and $\sum \vec{\tau} = 0$. There are no forces in the z direction so we ignore it. There is *only* torque in the $+z$ direction. In this problem there is a clearly-best pivot to choose – one at the point of contact with the wall, where the two forces F_x and F_y are exerted. If we choose *this* as our pivot, these forces will not contribute to the net torque!

Thus:

$$\begin{aligned} F_x - T\cos(\theta) &= 0 & (7.9)\\ F_y + T\sin(\theta) - mg &= 0 & (7.10)\\ T\sin(\theta)L - mgL/2 &= 0 & (7.11)\end{aligned}$$

The last equation involves only T, so we solve it for:

$$T = \frac{mg}{2\sin(\theta)} \tag{7.12}$$

We can substitute this into the first equation and solve for F_x:

$$F_x = \frac{1}{2} mg\cot(\theta) \tag{7.13}$$

Ditto for the second equation:

$$F_y = mg - \frac{1}{2}mg = \frac{1}{2}mg \tag{7.14}$$

There are several features of interest in this solution. One is that the wire and the wall support each must support half of the weight of the sign. However, in order to accomplish this, the tension in the wire will be strictly greater than half the weight!

Consider $\theta = 30^\circ$. Then $T = mg$ (the entire weight of the sign) and $F_x = \frac{\sqrt{3}}{2}mg$. The magnitude of the force exerted by the wall on the pole equals the tension.

Consider $\theta = 10^\circ$. Now $T \approx 2.9mg$ (which still must equal the magnitude of the force exerted by the wall. Why?). The smaller the angle, the larger the tension (and force exerted by/on the wall). Make the angle *too* small, and your pole will punch right through the brick wall!

7.2.1: Equilibrium with a Vector Torque

So far we've only treated problems where all of the forces and moment arms live in a *single plane* (if not in a single direction). What if the moment arms themselves live in a plane? What if the forces exert torques in different directions?

Nothing changes a whole lot, actually. One simply has to set each *component* of the force and torque to zero *separately*. If anything, it may give us more equations to work with, and hence the ability to deal with more unknowns, at the cost of – naturally – some algebraic complexity.

Static equilibrium problems involving multiple torque directions are actually rather common. Every time you sit in a chair, every time food is placed on a table, the legs increase the forces they supply to the seat to maintain force *and* torque equilibrium. In fact, every time *any* two-dimensional sheet of mass, such as a floor, a roof, a tray, a table is suspended horizontally, one must solve a problem in vector torque to keep it from rotating around *any* of the axes in the plane.

We don't need to solve the most algebraically complex problems in the Universe in order to learn how to balance both multiple force components and multiple torque components, but we do need to do one or two to get the idea, because nearly everybody who is taking this course needs to be able to actually work with static equilibrium in multiple dimensions. Physicists need to be able to understand it both to teach it and to prime themselves for the full-blown theory of angular momentum in a more advanced course. Engineers, well, we don't want those roofs to tip *over*, those bridges to fall *down*. Physicians and veterinarians – balancing human or animal bodies so that they don't tip over one way *or another* seems like a good idea. Things like canes, four point walkers, special support shoes all are tools you may one day use to help patients retain the precious ability to navigate the world in an otherwise precarious vertical static equilibrium.

Example 7.2.4: Building a Deck

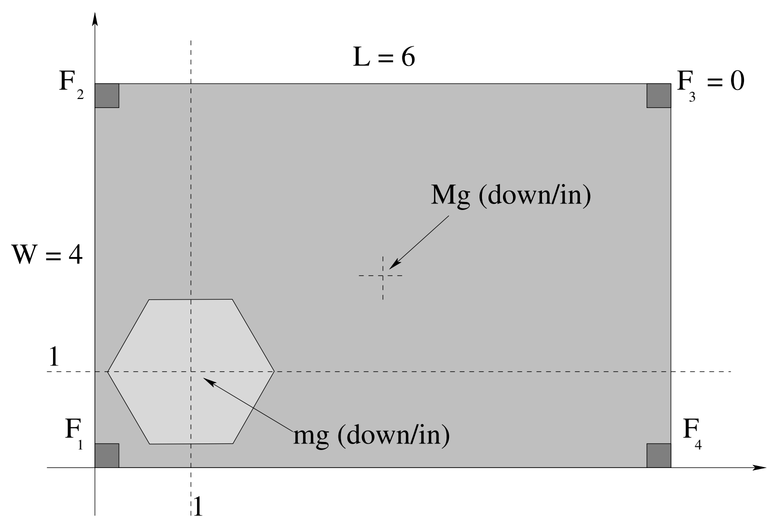


Figure 96:

In figure 96 a very simple deck layout is shown. The deck is 4 meters wide and 6 meters long. It is supported by four load-bearing posts, one at each corner. You would like to put a hot tub on the deck, one that has a loaded mass of $m = 2000$ kg, so that its center of mass is 1 meter in and 1 meter up from the lower left corner (as drawn) next to the house. The deck itself is a uniform concrete slab of mass $M = 4000$ kg with its center of mass is at (3,2) from the lower left corner.

You would like to know if putting the hot tub on the deck will exceed the safe load capacity of the nearest corner support. It seems to you that as it is loaded with the hot tub, it will actually reduce the load on F_3. So find F_1, F_2, and F_4 when the deck is loaded in this way, assuming a perfectly rigid plane deck and $F_3 = 0$.

First of all, we need to choose a pivot, and I've chosen a fairly obvious one – the lower left corner, where the x axis runs through F_1 and F_4 and the y axis runs through F_1 and F_2.

Second, we need to note that $\sum F_x$ and $\sum F_y$ can be ignored – there are no lateral forces at all at work here. Gravity pulls the masses down, the corner beams push the deck itself up. We can solve the normal force and force transfer in our heads – supporting the hot tub, the deck experiences a force equal to the *weight* of the hot tub right below the center of mass of the hot tub.

This gives us only *one* force equation:

$$F_1 + F_2 + F_4 - mg - Mg = 0 \tag{7.15}$$

That is, yes, the three pillars we've selected must support the total weight of the hot tub and deck together, since the F_3 pillar refuses to help out.

It gives us *two* torque equations, as hopefully it is obvious that $\tau_z = 0$! To make this nice and algebraic, we will set $h_x = 1$, $h_y = 1$ as the position of the hot tub, and use $L/2$ and $W/2$ as the position of the center of mass of the deck.

$$\sum \tau_x = WF_2 - \frac{W}{2}Mg - h_y mg = 0 \tag{7.16}$$

$$\sum \tau_y = h_x mg + \frac{L}{2}Mg - LF_4 = 0 \tag{7.17}$$

Note that if F_3 was acting, it would contribute to both of these torques and to the force above, and there would be an infinite number of possible solutions. As it is, though, solving this is pretty easy. Solve the last two equations for F_2 and F_4 respectively, then substitute the result into F_1. Only at the end substitute numbers in and see roughly what F_1 might be. Bear in mind that 1000 kg is a "metric ton" and weighs roughly 2200 pounds. So the deck and hot tub together, in this not-too-realistic problem, weigh over 6 tons!

Oops, we forgot the people in the hot tub and the barbecue grill at the far end and the furniture and the dog and the dancing. Better make the corner posts twice as strong as they need to be. Or even four times.

Once you see how this one goes, you should be ready to tackle the homework problem involving three legs, a tabletop, and a weight – same problem, really, but more complicated numbers.

7.3: Tipping

Another important application of the ideas of static equilibrium is to ***tipping problems***. A tipping problem is one where one uses the ideas of static equilibrium to identify the particular angle or force combination that will *marginally* cause some object to tip over. Sometimes this is presented in the context of objects on an inclined plane, held in place by static friction, and a tipping problem can be combined with a ***slipping problem***: determining if a block placed on an incline that is gradually raised tips first or slips first.

The idea of tipping is simple enough. An object placed on a flat surface is typically stable as long as the center of gravity is vertically ***inside the edges*** that are in contact with the surface, so that the torque created by the gravitational force around this limiting pivot is opposed by the torque exerted by the (variable) normal force.

That's all there is to it! Look at the center of gravity, look at the corner or edge intuition tells you the object will "tip over", done.

Example 7.3.1: Tipping Versus Slipping

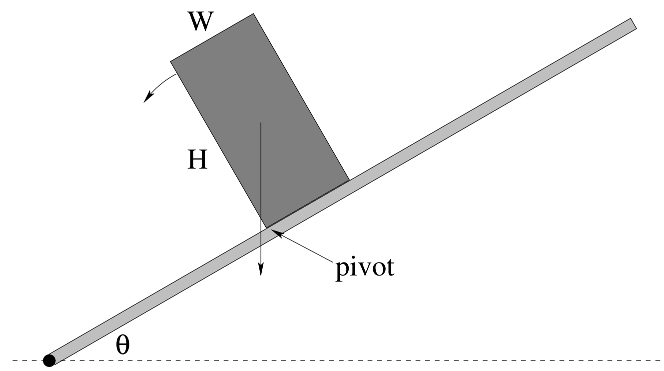


Figure 97: A rectangular block either tips first or slips (slides down the incline) first as the incline is gradually increased. Which one happens first? The figure is show with the block just past the ***tipping angle***.

In figure 97 a rectangular block of height H and width W is sitting on a rough plank that is gradually being raised at one end (so the angle it makes with the horizontal, θ, is slowly increasing).

At some angle we know that the block will start to slide. This will occur because the normal force is decreasing with the angle (and hence, so is the maximum force static friction can exert) and at the same time, the component of the weight of the object that points *down* the incline is increasing. Eventually the latter will exceed the former and the block will slide.

However, at some angle the block will *also* tip *over*. We know that this will happen because the normal force can only prevent the block from rotating *clockwise* (as drawn) around the pivot consisting of the lower left corner of the block. Unless the block has a magnetic lower surface, or a lower surface covered with velcro or glue, the plank cannot *attract* the lower surface of the block and prevent it from rotating *counterclockwise*.

As long as the net torque due to *gravity* (about this lower left pivot point) is *into* the page, the plank itself can exert a countertorque out of the page sufficient to keep the block from rotating down through the plank. If the torque due to gravity is *out of the page* – as it is in the figure 97 above, when the center of gravity moves over and to the *vertical left* of the pivot corner – the normal force exerted by the plank cannot oppose the counterclockwise torque of gravity and the block will fall over.

The ***tipping point***, or ***tipping angle*** is thus the angle where the ***center of gravity is directly over the pivot*** that the object will “tip” around as it falls over.

A very common sort of problem here is to determine whether some given block or shape will tip first or slip first. This is easy to find. First let’s find the slipping angle θ_s. Let “down” mean “down the incline”. Then:

$$\sum F_{\text{down}} = mg\sin(\theta) - F_s = 0 \tag{7.18}$$

$$\sum F_{\perp} = N - mg\cos(\theta) = 0 \tag{7.19}$$

From the latter, as usual:

$$N = mg\cos(\theta) \tag{7.20}$$

and $F_s \leq F_s^{\max} = \mu_s N$.

When

$$mg\sin(\theta_s) = F_s^{\max} = \mu_s \cos(\theta_s) \tag{7.21}$$

the force of gravity down the incline precisely balances the force of static friction. We can solve for the angle where this occurs:

$$\theta_s = \tan^{-1}(\mu_s) \tag{7.22}$$

Now let's determine the angle where it tips over. As noted, this is where the torque to to gravity around the pivot that the object will tip over changes sign from in the page (as drawn, stable) to out of the page (unstable, tipping over). This happens when the center of mass passes ***directly over the pivot***.

From inspection of the figure (which is drawn *very close* to the tipping point) it should be clear that the tipping angle θ_t is given by:

$$\theta_t = \tan^{-1}\left(\frac{W}{H}\right) \tag{7.23}$$

So, which one wins? The ***smaller*** of the two, θ_s or θ_t, of course – that's the one that happens *first* as the plank is raised. Indeed, since both are inverse tangents, the smaller of:

$$\mu_s, W/H \tag{7.24}$$

determines whether the system slips first or tips first, no need to actually *evaluate* any tangents or inverse tangents!

Example 7.3.2: Tipping While Pushing

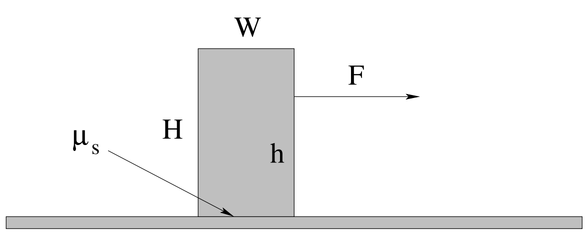


Figure 98: A uniform rectangular block with dimensions W by H (which has its center of mass at $W/2$, $H/2$) is pushed at a height h by a force F. The block sits on a horizontal smooth table with coefficient of static friction μ_s.

A uniform block of mass M being pushed by a horizontal outside force (say, a finger) a height h above the flat, smooth surface it is resting on (say, a table) as portrayed in figure 98. If it is pushed down low (small h) the block slides. If pushed up high (large h) it tips. Find a condition for the height h at which it tips and slips at the same time.

The solution here is much the same as the solution to the previous problem. We *independently* determine the condition for slipping, as that is rather easy, and then using the maximum force that can be applied without it quite slipping, find the height h at which the block barely starts to tip over.

"Tipping over" in this case means that all of the normal force will be concentrated right at the pivot corner (the one the block will rotate around as it tips over) because the rest of the bottom surface is *barely* starting to leave the ground. All of the force of static friction is similarly concentrated at this one point. This is convenient to us, since neither one will therefore contribute to the torque around this point!

Conceptually, then, we seek the point h where, pushing with the maximum non-slipping force, the torque due to gravity *alone* is exactly equal to the torque exerted by the force F. This seems simple enough.

To find the force we need only to examine the usual force balance equations:

$$\begin{aligned} F - F_s &= 0 \quad (7.25) \\ N - Mg &= 0 \quad (7.26) \end{aligned}$$

and hence $N = Mg$ (as usual) and:

$$F_{\rm max} = F_s^{\rm max} = \mu_s N = \mu_s Mg \quad (7.27)$$

(also as usual). Hopefully by now you had this completely solved in your head before I even wrote it all down neatly and were saying to yourself "$F_{\rm max}$, yeah, sure, that's $\mu_s Mg$, let's get on with it..."

So we shall. Consider the torque around the bottom right hand corner of the block (which is clearly and intuitively the "tipping pivot" around which the block will rotate as it falls over when the torque due to F is large enough to overcome the torque of gravity). Let us choose the positive direction for torque to be out of the page. It should then be quite obvious that when the block is *barely* tipping over, so that we can ignore any torque due to N and F_s:

$$\frac{WMg}{2} - h_{\rm crit} F_{\rm max} = \frac{WMg}{2} - h_{\rm crit}\mu_s Mg = 0 \quad (7.28)$$

or (solving for $h_{\rm crit}$, the critical height where it *barely* tips over even as it starts to slip):

$$h_{\rm crit} = \frac{W}{2\mu_s} \quad (7.29)$$

Now, does this make sense? If $\mu_s \to 0$ (a frictionless surface) we will never tip it *before* it starts to slide, although we might well push hard enough to tip it over in spite of it sliding. We note that in this limit, $h_{\rm crit} \to \infty$, which makes sense. On the other hand for finite μ_s if we let W become very small then $h_{\rm crit}$ similarly becomes very small, because the block is now very *thin* and is indeed rather precariously balanced.

The last bit of "sense" we need to worry about is $h_{\rm crit}$ compared to H. If $h_{\rm crit}$ is *larger* than H, this basically means that we *can't* tip the block over before it slips, for any reasonable μ_s. This limit will always be realized for $W \gg H$. Suppose, for example, $\mu_s = 1$ (the upper limit of "normal" values of the coefficient of static friction that doesn't describe actual adhesion). Then $h_{\rm crit} = W/2$, and if $H < W/2$ there is no way to push in the block to make it tip before it slips. If μ_s is more reasonable, say $\mu_s = 0.5$, then only pushing at the very top of a block that is $W \times W$ in dimension marginally causes the block to tip. We can thus easily determine blocks "can" be tipped by a horizontal force and which ones cannot, just by knowing μ_s and looking at the blocks!

7.4: Force Couples

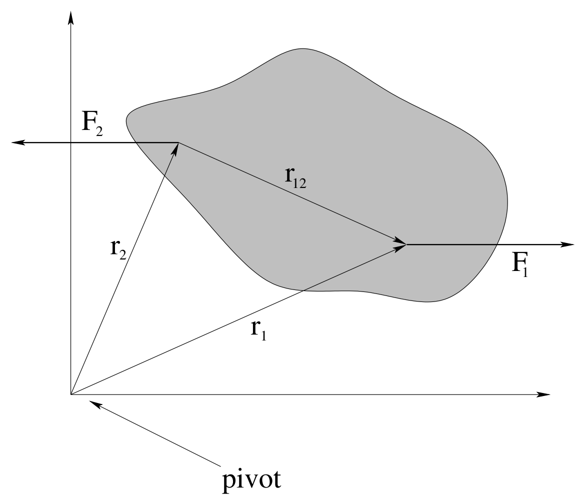


Figure 99: A ***Force Couple*** is a pair of equal and opposite forces that may or may not act along the line between the points where they are applied to a rigid object. Force couples exert a torque that is ***indenpedent of the pivot*** on an object and (of course) do not accelerate the center of mass of the object.

Two equal forces that act in opposite directions but not necessarily along the same line are called a ***force couple***. Force couples are important both in torque and rotation problems and in static equilibrium problems. One doesn't have to be able to name them, of course – we know everything we need to be able to handle the physics of such a pair already without a name.

One important property of force couples does stand out as being worth deriving and learning on its own, though – hence this section. Consider the total torque exerted by a force couple in the coordinate frame portrayed in figure **??**:

$$\vec{\boldsymbol{\tau}} = \vec{\boldsymbol{r}}_1 \times \vec{\boldsymbol{F}}_1 + \vec{\boldsymbol{r}}_2 \times \vec{\boldsymbol{F}}_2 \tag{7.30}$$

By hypothesis, $\vec{\boldsymbol{F}}_2 = -\vec{\boldsymbol{F}}_1$, so:

$$\vec{\boldsymbol{\tau}} = \vec{\boldsymbol{r}}_1 \times \vec{\boldsymbol{F}}_1 - \vec{\boldsymbol{r}}_2 \times \vec{\boldsymbol{F}}_1 = (\vec{\boldsymbol{r}}_1 - \vec{\boldsymbol{r}}_2) \times \vec{\boldsymbol{F}}_1 = \vec{\boldsymbol{r}}_{12} \times \vec{\boldsymbol{F}}_1 \tag{7.31}$$

This torque ***no longer depends on the coordinate frame!*** It depends only on the *difference* between $\vec{r}_1$ and $\vec{r}_2$, which is independent of coordinate system.

Note that we already used this property of couples when proving the law of conservation of angular momentum – it implies that internal Newton's Third Law forces can exert no torque on a system independent of intertial reference frame. Here it has a slightly different implication – it means that ***if the net torque produced by a force couple is zero in one frame, it is zero in all frames!*** The idea of static equilibrium itself is independent of frame!

It also means that equilibrium implies that ***the vector sum of all forces form force couples in each coordinate direction that are equal and opposite and that ultimately pass through the center of mass of the system***. This is a conceptually useful way to think about some tipping or slipping or static equilibrium problems.

Example 7.4.1: Rolling the Cylinder Over a Step

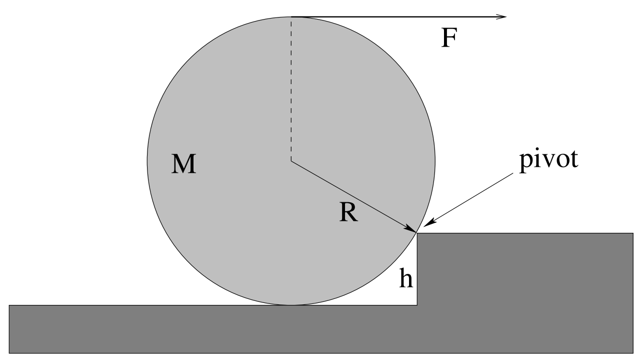


Figure 100:

One classic example of static equilibrium and force couples is that of a ball or cylinder being rolled up over a step. The way the problem is typically phrased is:

a) Find the minimum force F that must be applied (as shown in figure 100) to cause the cylinder to *barely* lift up off of the bottom step and rotate up around the corner of the next one, assuming that the cylinder does not slip on the corner of the next step.

b) Find the force exerted by that corner at this marginal condition.

The simplest way to solve this is to recognize the point of the term “barely”. When the force F is zero, gravity exerts a torque around the pivot *out* of the page, but the normal force of the tread of the lower stair exerts a countertorque precisely sufficient to keep the cylinder from rolling down into the stair itself. It also supports the weight of the cylinder. As F is increased, it exerts a torque around the pivot that is *into* the page, also opposing the gravitational torque, and the normal force *decreases* as less is needed to prevent rotation down into the step. At the same time, the pivot exerts a force that has to *both* oppose F (so the cylinder doesn’t translate to the right) *and* support more and more of the weight of the cylinder as the normal force supports less.

At some particular point, the force exerted by the step *up* will precisely equal the weight of the step down. The force exerted by the step to the *left* will exactly equal the force F to the right. These forces will (vector) sum to zero and will incidentally exert no net torque either, as a pair of opposing couples.

That is enough that we could almost *guess* the answer (at least, if we drew some very nice pictures). However, we should work the problem algebraically to make sure that we all understand it. Let us assume that $F = F_m$, the desired minimum force where $N \to 0$. Then (with out of the page positive):

$$\sum \tau = mg\sqrt{R^2 - (R-h)^2} - F_m(2R-h) = 0 \tag{7.32}$$

where I have used the $r_\perp F$ form of the torque in both cases, and used the pythagorean theorem and/or inspection of the figure to determine $r_\perp$ for each of the two forces. No torque due to N is present, so F_m in this case is indeed the minimum force F at the marginal point where rotation *just* starts to happen:

$$F_m = \frac{mg\sqrt{R^2 - (R-h)^2}}{2R-h} \tag{7.33}$$

Next summing the forces in the x and y direction and solving for F_x and F_y exerted by the pivot corner itself we get:

$$F_x = -F_m = -\frac{mg\sqrt{R^2 - (R-h)^2}}{2R-h} \tag{7.34}$$

$$F_y = mg \tag{7.35}$$

Obviously, these forces form a perfect couple such that the torques vanish.

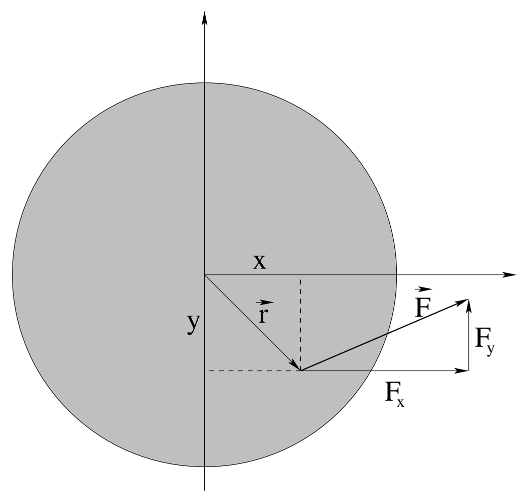
That's all there is to it! There are probably other questions one could ask, or other ways to ask the main question, but the idea is simple – look for the marginal static condition where rotation, or tipping, occur. Set it up algebraically, and then solve!

Homework for Week 7

Problem 1.

Physics Concepts: Make this week's physics concepts summary as you work all of the problems in this week's assignment. Be sure to cross-reference each concept in the summary to the problem(s) they were key to, and include concepts from previous weeks as necessary. Do the work carefully enough that you can (after it has been handed in and graded) punch it and add it to a three ring binder for review and study come finals!

Problem 2.



In the figure above, a force

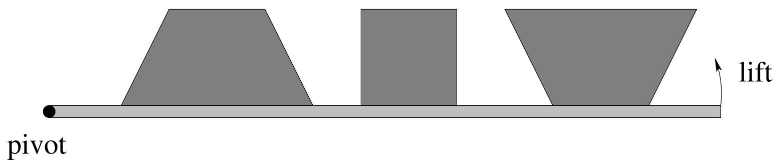
$$\vec{F} = 2\hat{x} + 1\hat{y}$$

Newtons is applied to a disk at the point

$$\vec{r} = 2\hat{x} - 2\hat{y}$$

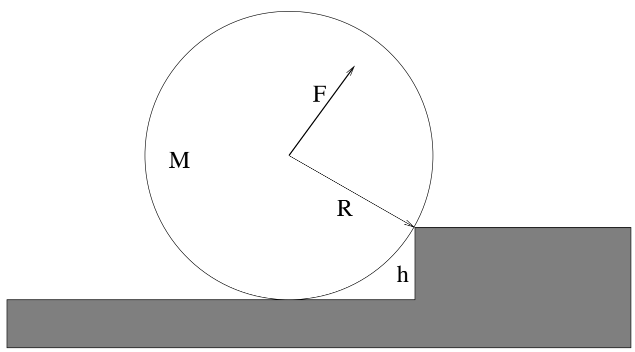
as shown. (That is, $F_x = 2$ N, $F_y = 1$ N, $x = 2$ m, $y = -2$ m). Find the *total torque* about a pivot *at the origin*. Don't forget that torque is a ***vector***, so specify its direction as well as its magnitude (or give the answer as a cartesian vector)! Show your work!

Problem 3.



In the figure above, three shapes (with uniform mass distribution and thickness) are drawn sitting on a plane that can be tipped up gradually. Assuming that static friction is great enough that all of these shapes will tip over before they slide, rank them in the order they will tip over as the angle of the board they are sitting on is increased.

Problem 4.



This problem will help you learn required concepts such as:

- Static Equilibrium
- Torque (about selected pivots)
- Geometry of Right Triangles

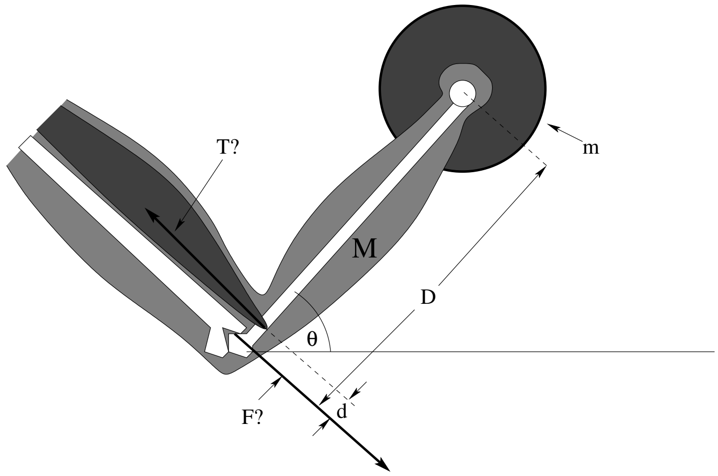
so please review them before you begin.

A cylinder of mass M and radius R sits against a step of height $h = R/2$ as shown above. A force $\vec{F}$ is applied at right angles to the line connecting the corner of the step and the center of the cylinder. All answers should be in terms of M, R, g.

a) Find the minimum value of $|\vec{F}|$ that will roll the cylinder over the step if the cylinder does not slide on the corner.

b) What is the force exerted by the corner (magnitude and direction) when that force $\vec{F}$ is being exerted on the center?

Problem 5.



This problem will help you learn required concepts such as:

- Static Equilibrium
- Force and Torque

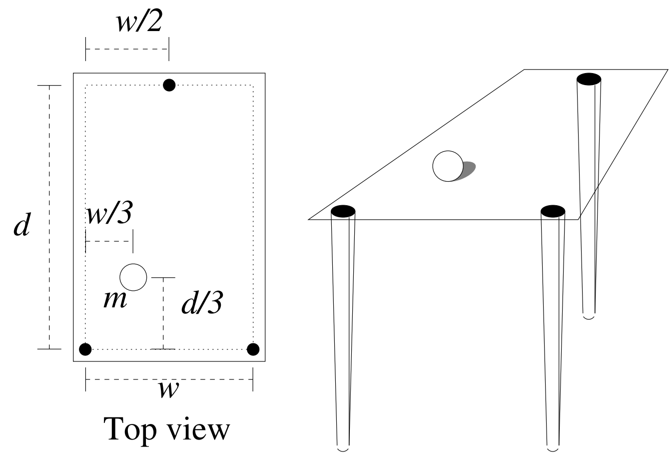
so please review them before you begin.

An exercising human person holds their arm of mass M and a barbell of mass m at rest at an angle θ with respect to the horizontal in an isometric curl as shown. Their bicep muscle that supports the suspended weight is connected at right angles to the bone a short distance d up from the elbow joint. The bone that supports the weight has length D.

a) Find the tension T in the muscle, assuming for the moment that the center of mass of the forearm is in the middle at $D/2$. Note that it is *much larger* than the weight of the arm and barbell combined, assuming a reasonable ratio of $D/d \approx 25$ or thereabouts.

b) Find the force $\vec{F}$ (magnitude *and* direction) exerted on the supporting bone by the elbow joint in the geometry shown. Again, note that it is much larger than "just" the weight being supported.

Problem 6.



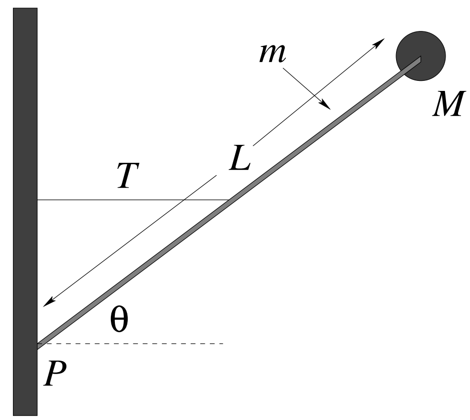
This problem will help you learn required concepts such as:

- Force Balance
- Torque Balance
- Static Equilibrium

so please review them before you begin.

The figure below shows a mass m placed on a table consisting of three narrow cylindrical legs at the positions shown with a light (presume massless) sheet of Plexiglas placed on top. What is the vertical force exerted by the Plexiglas on each leg when the mass is in the position shown?

Problem 7.



This problem will help you learn required concepts such as:

- Force Balance
- Torque Balance
- Static Equilibrium

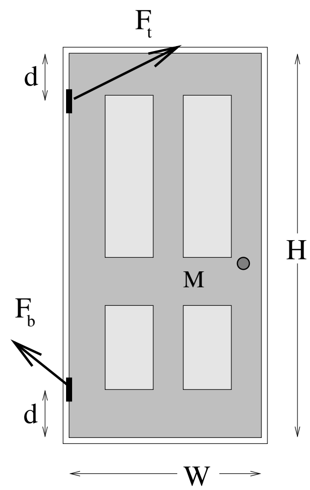
so please review them before you begin.

A small round mass M sits on the end of a rod of length L and mass m that is attached to a wall with a hinge at point P. The rod is kept from falling by a thin (massless) string attached horizontally between the midpoint of the rod and the wall. The rod makes an angle θ with the ground. Find:

a) the tension T in the string;

b) the force $\vec{F}$ exerted by the hinge on the rod.

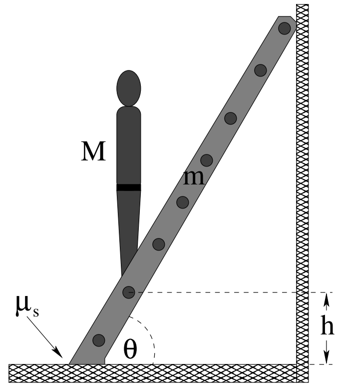
Problem 8.



A door of mass M that has height H and width W is hung from two hinges located a distance d from the top and bottom, respectively. ***Assuming that the vertical weight of the door is equally distributed between the two hinges***, find the total force (magnitude and direction) exerted by each hinge.

Neglect the mass of the doorknob and assume that the center of mass of the door is at $W/2, H/2$. The force directions drawn for you are **NOT** likely to be correct or even close.

Problem 9.



This problem will help you learn required concepts such as:

- Torque Balance
- Force Balance
- Static Equilibrium
- Static Friction

so please review them before you begin.

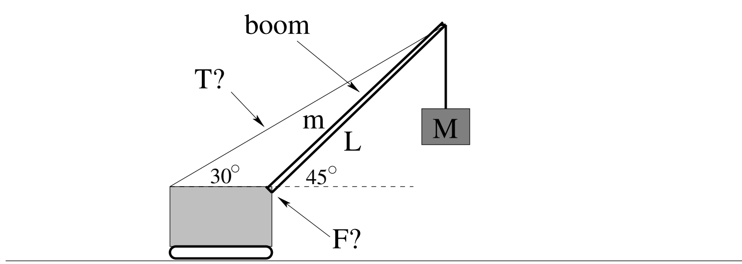
In the figure above, a ladder of mass m and length L is leaning against a wall at an angle θ. A person of mass M begins to climb the ladder. The ladder sits on the ground with a coefficient of static friction μ_s between the ground and the ladder. The wall is frictionless – it exerts only a normal force on the ladder.

If the person climbs the ladder, find the height h where the ladder slips.

Optional Problems

The following problems are **not required or to be handed in**, but are provided to give you some extra things to work on or test yourself with *after* mastering the required problems and concepts above and to prepare for quizzes and exams.

Optional Problem 10.



A crane with a boom (the long support between the body and the load) of mass m and length L holds a mass M suspended as shown. Assume that the center of mass of the boom is at $L/2$. Note that the wire with the tension T is **fixed** to the top of the boom, not run over a pulley to the mass M.

a) Find the tension in the wire.

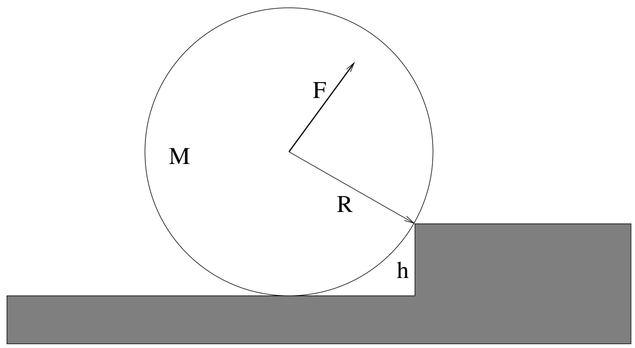
b) Find the force exerted on the boom by the crane body.

Note:

$$\sin(30^\circ) = \cos(60^\circ) = \frac{1}{2}$$
$$\cos(30^\circ) = \sin(60^\circ) = \frac{\sqrt{3}}{2}$$
$$\sin(45^\circ) = \cos(45^\circ) = \frac{\sqrt{2}}{2}$$

*

Optional Problem 11.



A cylinder of mass M and radius R sits against a step of height $h = R/2$ as shown above. A force $\vec{F}$ is applied parallel to the ground as shown. All answers should be in terms of M, R, g.

a) Find the minimum value of $|\vec{F}|$ that will roll the cylinder over the step if the cylinder does not slide on the corner.

b) What is the force exerted by the corner (magnitude and direction) when that force $\vec{F}$ is being exerted on the center?

III: Applications of Mechanics

Week 8: Fluids

Fluids Summary

- **Fluids** are states of matter characterized by a lack of long range order. They are characterized by their **density** ρ and their **compressibility**. Liquids such as water are (typically) relatively incompressible; gases can be significantly compressed. Fluids have other characteristics, for example viscosity (how strongly a fluid communicates *shear* stress from a boundary into its bulk volume) and adhesion (how "wet" water is, that is, how strongly it binds to material that is in contact with it at a confining surface).

- **Pressure** is the force per unit area exerted by a fluid on its surroundings:

$$P = F/A \tag{8.1}$$

  Its SI units are *pascals* where 1 pascal = 1 newton/meter squared. Pressure is also measured in "atmospheres" (the pressure of air at or near sea level) where 1 atmosphere $\approx 10^5$ pascals. The pressure in an incompressible fluid varies with depth according to:

$$P = P_0 + \rho g D \tag{8.2}$$

  where P_0 is the pressure at the top and D is the depth.

- **Pascal's Principle** Pressure applied to a fluid is transmitted undiminished to all points of the fluid.

- **Archimedes' Principle** The buoyant force on an object

$$F_b = \rho g V_{\rm disp} \tag{8.3}$$

  where frequency $V_{\rm disp}$ is the volume of fluid displaced by an object.

- **Conservation of Flow** We will study only steady/laminar flow in the absence of turbulence and viscosity.

$$I = A_1 v_1 = A_2 v_2 \tag{8.4}$$

  where I is the ***flow***, the volume per unit time that passes a given point in e.g. a pipe.

- For a circular smooth round pipe of length L and radius r carrying a fluid in ***laminar flow*** with ***dynamical viscosity***[126] μ, the flow is related to the pressure difference across the pipe by the ***resistance*** R:

$$\Delta P = IR \tag{8.5}$$

  It is worth noting that this is the fluid-flow version of ***Ohm's Law***, which you will learn next semester if you continue. We will generally omit the modifier "dynamical" from the term viscosity in this course, although there is actually another, equivalent measure of viscosity called the ***kinematic viscosity***, $\nu = \mu/\rho$. The primary difference is the units – μ has the SI units of pascal-seconds where ν has units of meters square per second.

- The resistance R is given by the follow formula:

$$R = \frac{8L\mu}{\pi r^4} \tag{8.6}$$

  and the flow equation above becomes ***Poiseuille's Law***[127] :

$$I = \frac{\Delta P}{R} = \frac{\pi r^4\,\Delta P}{8L\mu} \tag{8.7}$$

  The key facts from this series of definitions are that flow increases linearly with pressure (so to achieve a given e.g. perfusion in a system of capillaries one requires a sufficient pressure difference across them), increases with the ***fourth power*** of the radius of the pipe (which is why narrowing blood vessels become so dangerous past a certain point) and decreases with the length (longer blood vessels have a greater resistance).

- If we ***neglect resistance*** (an idealization roughly equivalent to neglecting friction) and consider the flow of fluid in a closed pipe that can e.g. go up and down, the ***work-mechanical energy theorem*** per unit volume of the fluid can be written as **Bernoulli's Equation**:

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant} \tag{8.8}$$

- **Venturi Effect** At constant height, the pressure in a fluid *decreases* as the velocity of the fluid *increases* (the work done by the pressure difference is what speeds up the fluid!. This is responsible for e.g. the lift of an airplane wing and the force that makes a spinning baseball or golf ball curve.

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[126] Wikipedia: http://www.wikipedia.org/wiki/viscosity. We will defer any actual statement of how viscosity is related to forces until we cover shear stress in a couple of weeks. It's just too much for now. Oh, and sorry about the symbol. Yes, we already have used μ for e.g. static and kinetic friction. Alas, we will use μ for still more things later. Even with both greek and roman characters to draw on, there just aren't enough characters to cover all of the quantities we want to algebraically work with, so you have to get used to their reuse ***in different contexts*** that hopefully make them easy enough to keep straight. I decided that it is better to use the accepted symbol in this textbook rather than make one up myself or steal a character from, say, Urdu or a rune from Ancient Norse.

[127] Wikipedia: http://www.wikipedia.org/wiki/Hagen-Poiseuille equation. The derivation of this result isn't *horribly* difficult or hard to understand, but it is long and beyond the scope of this course. Physics and math majors are encouraged to give it a peek though, if only to learn where it comes from.

- **Torricelli's Rule:** If a fluid is flowing through a very small hole (for example at the bottom of a large tank) then the velocity of the fluid at the large end can be neglected in Bernoulli's Equation. In that case the exit speed is the same as the speed of a mass dropped the same distance:

$$v = \sqrt{2gH} \tag{8.9}$$

where H is the depth of the hole relative to the top surface of the fluid in the tank.

8.1: General Fluid Properties

Fluids are the generic name given to two states of matter, liquids and gases[128] characterized by a lack of long range order and a high degree of mobility at the molecular scale. Let us begin by visualizing fluids microscopically, since we like to build our understanding of matter from the ground up.

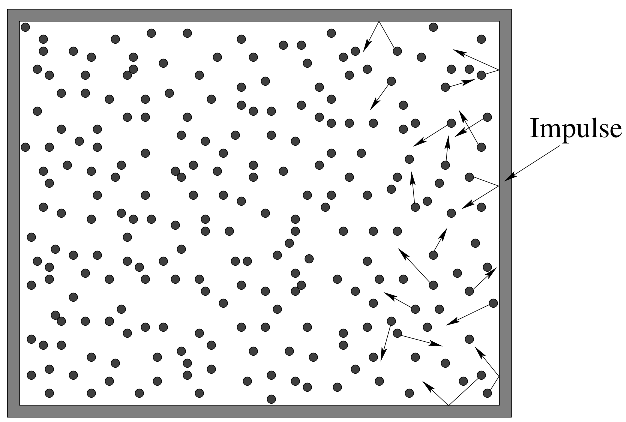

Figure 101: A large number of atoms or molecules are confined within in a "box", where they bounce around off of each other and the walls. They exert a *force* on the walls equal and opposite the the *force* the walls exert on them as the collisions more or less elastically reverse the particles' momenta perpendicular to the walls.

In figure 101 we see a highly idealized picture of what we might see looking into a tiny box full of gas. Many particles all of mass m are *constantly moving* in random, constantly changing directions (as the particles collide with each other and the walls) with an average kinetic energy related to the temperature of the fluid. Some of the particles (which might be atoms such as helium or neon or molecules such as H_2 or O_2) happen to be close to the walls of the container and moving in the right direction to bounce (elastically) off of those walls.

When they do, their *momentum* perpendicular to those walls is *reversed*. Since *many, many* of these collisions occur each second, there is a nearly continuous momentum transfer between the walls and the gas and the gas and the walls. This transfer, per unit time,

[128] We will not concern ourselves with "plasma" as a possible fourth state of matter in this class, viewing it as just an "ionized gas" although a very dense plasma might well be more like a liquid. Only physics majors and perhaps a few engineers are likely to study plasmas, and you have plenty of time to figure them out after you have learned some electromagnetic theory.

becomes the *average force* exerted by the walls on the gas and the gas on the walls (see the problem in Week 4 with beads bouncing off of a pan).

Eventually, we will transform this simple picture into the ***Kinetic Theory of Gases*** and use it to derive the venerable ***Ideal Gas Law*** (physicist style)[129] :

$$PV = Nk_bT \tag{8.10}$$

but for now we will ignore the role of temperature and focus more on understanding the physical characteristics of the fluid such as its ***density***, the idea of ***pressure itself*** and the ***force*** exerted by fluids on themselves (internally) and on anything the fluid presses upon along the lines of the particles above and the walls of the box.

8.1.1: Pressure

As noted above, the walls of the container exert an average force on the fluid molecules that confine them by reversing their perpendicular momenta in collisions. The total momentum transfer is proportional to the number of molecules that collide per unit time, and this in turn is (all things being equal) clearly proportional to the area of the walls. Twice the surface area – when confining the same number of molecules over each part – has to exert twice the force as twice the number of collisions occur per unit of time, each transferring (on average) the same impulse. It thus makes sense, when considering fluids, to describe the forces that confine and act on the fluids in terms of ***pressure***, defined to be the ***force per unit area*** with which a fluid pushes on a confining wall or the confining wall pushes on the fluid:

$$P = \frac{F}{A} \tag{8.11}$$

Pressure gets its own SI units, which clearly must be Newtons per square meter. We give these units their own name, ***Pascals***:

$$1 \text{ Pascal} = \frac{\text{Newton}}{\text{meter}^2} \tag{8.12}$$

A Pascal is a tiny unit of pressure – a Newton isn't very big, recall (one kilogram weighs roughly ten Newtons or 2.2 pounds) so a Pascal is the weight of a quarter pound spread out over a square meter. Writing out "pascal" is a bit cumbersome and you'll see it sometimes abbreviated ***Pa*** (with the usual power-of-ten modifications, kPa, MPa, GPa, mPa and so on).

A more convenient measure of pressure in our everyday world is a form of the unit called a ***bar***:

$$1 \text{ bar} = 10^5 \text{ Pa} = 100 \text{ kPa} \tag{8.13}$$

As it happens, the average air pressure at sea level is *very nearly* 1 bar, and varies by at most a few percent on either side of this. For that reason, air pressure in the modern world is generally reported on the scale of ***millibars***, for example you might see air pressure given as 959 mbar (characteristic of the low pressure in a major storm such as a hurricane), 1023 mbar (on a fine, sunny day).

[129]Wikipedia: http://www.wikipedia.org/wiki/Ideal Gas Law.

The mbar is probably the "best" of these units for describing everyday air pressure (and its temporal and local and height variation without the need for a decimal or power of ten), with Pascals being an equally good and useful general purpose arbitrary precision unit

The symbol ***atm*** stands for ***one standard atmosphere***. The connection between atmospheres, bars, and pascals is:

$$1 \text{ standard atmosphere} = 101.325 \text{ kPa} = 1013.25 \text{ mbar} \tag{8.14}$$

Note that *real* air pressure at sea level is most unlikely to be this exact value, and although this pressure is often referred to in textbooks and encylopedias as "the average air pressure at sea level" this is not, in fact, the case. The extra significant digits therefore refer only to a fairly arbitrary value (in pascals) historically related to the original definition of a standard atmosphere in terms of "millimeters of mercury" or *torr*:

$$1 \text{ standard atmosphere} = 760.00 \text{ mmHg} = 760.00 \text{ torr} \tag{8.15}$$

that is of no practical or immediate use. All of this is discussed in some detail in the section on barometers below.

In this class we will use the simple rule 1 bar $\approx$ 1 atm to avoid having to divide out the extra digits, just as we approximated $g \approx 10$ when it is really closer to 9.8. This rule is more than adequate for nearly all purposes and makes pressure arithmetic something you can often do with fingers and toes or the back of an envelope, with around a 1% error if somebody actually gave a pressure in atmospheres with lots of significant digits instead of the superior pascal or bar SI units.

Note well: in the field of medicine ***blood pressures*** are given in mm of mercury (or torr) by long standing tradition (largely because for at least a century blood pressure was measured with a mercury-based sphygmomanometer). This is discussed further in the section below on the human heart and circulatory system. These can be converted into atmospheres by dividing by 760, remembering that one is measuring the *difference* between these pressures and the standard atmosphere (so the actual blood pressure is always *greater* than one atmosphere).

Pressure isn't only exerted at the *boundaries* of fluids. Pressure also describes the internal transmission of forces *within* a fluid. For example, we will soon ask ourselves "Why don't fluid molecules all fall to the ground under the influence of gravity and stay there?" The answer is that (at a sufficient temperature) the *internal pressure* of the fluid suffices to *support* the fluid above upon the back (so to speak) of the fluid below, all the way down to the ground, which of course has to support the weight of the entire column of fluid. Just as "tension" exists in a stretched string at all points along the string from end to end, so the pressure within a fluid is well-defined at all points from one side of a volume of the fluid to the other, although in neither case will the tension or pressure in general be *constant*.

8.1.2: Density

As we have done from almost the beginning, let us note that even a very tiny volume of fluid has many, many atoms or molecules in it, at least under ordinary circumstances in our everyday lives. True, we can work to create a *vacuum* – a volume that has relatively *few* molecules in it per unit volume, but it is almost impossible to make that number zero – even the hard vacuum of outer space has on average one molecule per cubic meter or thereabouts[130] . We live at the bottom of a gravity well that confines our *atmosphere* – the air that we breathe – so that it forms a relatively thick soup that we move through and breathe with order of *Avogadro's Number* (6×10^{23}) molecules per *liter* – hundreds of billions of billions per cubic centimeter.

At this point we cannot possibly track the motion and interactions of all of the individual molecules, so we *coarse grain* and *average*. The coarse graining means that we once again consider volumes that are large relative to the sizes of the atoms but small relative to our macroscopic length scale of meters – cubic millimeters or cubic microns, for example – that are large enough to contain many, many molecules (and hence a well defined *average* number of molecules) but small enough to treat like a differential volume for the purposes of using calculus to add things up.

We could just count molecules in these tiny volumes, but the properties of *oxygen* molecules and *helium* molecules might well be very different, so the molecular count alone may not be the most useful quantity. Since we are interested in how forces might act on these small volumes, we need to know their mass, and thus we define the *density* of a fluid to be:

$$\rho = \frac{dm}{dV}, \tag{8.16}$$

the ***mass per unit volume*** we are all familiar with from our discussions of the center of mass of continuous objects and moments of inertia of rigid objects.

Although the definition itself is the same, the density of a fluid behaves in a manner that is similar, but not quite identical in its properties, to the density of a solid. The density of a fluid usually varies *smoothly* from one location to another, because an excess of density in one place will spread out as the molecules travel and collide to smooth out, on average. The particles in some fluids (or almost any fluid at certain temperatures) are "sticky", or ***strongly interacting***, and hence the fluid coheres together in clumps where the particles are mostly touching, forming a *liquid*. In other fluids (or all fluids at higher temperatures) the molecules move so fast that they do *not* interact much and spend most of their time relatively far apart, forming a *gas*.

A gas spreads itself out to fill any volume it is placed in, subject only to forces that confine it such as the walls of containers or gravity. It assumes the shape of containers, and forms a (usually nearly spherical) layer of atmosphere around planets or stars when confined by gravity. Liquids also spread themselves out to some extent to fill containers they are place in or volumes they are confined to by a mix of surface forces and gravity, but they also have the property of *surface tension* that can permit a liquid to exert a force of confinement on *itself*. Hence water fills a glass, but water also forms nearly spherical droplets when falling freely as surface tension causes the droplet to minimize its surface area relative to its volume, forming a sphere.

[130]Wikipedia: http://www.wikipedia.org/wiki/Vacuum. Vacuum is, of course, "nothing", and if you take the time to read this Wikipedia article on it you will realize that even *nothing* can be pretty amazing. In man-made vacuums, there are nearly always as many as hundreds of molecules per cubic centimeter.

Surface chemistry or surface adhesion can also exert forces on fluids and initiate things like *capillary flow* of e.g. water up into very fine tubes, drawn there by the surface interaction of the hydrophilic walls of the tube with the water. Similarly, hydrophobic materials can actually repel water and cause water to bead up instead of spreading out to wet the surface. We will largely ignore these phenomena in this course, but they are very interesting and are actually *useful* to physicians as they use pipettes to collect fluid samples that draw themselves up into sample tubes as if by magic. It's not magic, it's just physics.

8.1.3: Compressibility

A major difference between fluids and solids, and liquids and gases within the fluids, is the *compressibility* of these materials. Compressibility describes how a material responds to changes in *pressure*. Intuitively, we expect that if we change the volume of the container (making it smaller, for example, by pushing a piston into a confining cylinder) while holding the amount of material inside the volume constant we will change the pressure; a smaller volume makes for a larger pressure. Although we are not quite prepared to derive and fully justify it, it seems at least *reasonable* that this can be expressed as a simple *linear relationship*:

$$\Delta P = -B\frac{\Delta V}{V} \tag{8.17}$$

Pressure up, volume down and vice versa, where the amount it goes up or down is related, not unreasonably, to the total volume that was present in the first place. The constant of proportionality B is called the ***bulk modulus*** of the material, and it is very much like (and closely related to) the *spring constant* in Hooke's Law for springs.

Note well that we haven't really specified *yet* whether the "material" is solid, liquid or gas. All three of them have densities, all three of them have bulk moduli. Where they differ is in the *qualitative* properties of their compressibility.

Solids are typically *relatively* incompressible (large B), although there are certainly exceptions. The have long range order – all of the molecules are packed and tightly bonded together in structures and there is usually very little free volume. Atoms themselves violently oppose being "squeezed together" because of the ***Pauli exclusion principle*** that forbids electrons from having the same set of quantum numbers as well as straight up ***Coulomb repulsion*** that you will learn about next semester.

Liquids are also relatively incompressible (large B). They differ from solids in that they lack long range order. All of the molecules are constantly moving around and any small "structures" that appear due to local interaction are short-lived. The molecules of a liquid are close enough together that there is often significant physical and chemical interaction, giving rise to surface tension and wetting properties – especially in water, which is (as one sack of water speaking to another) an amazing fluid!

Gases are in contrast quite *compressible* (small B). One can usually squeeze gases smoothly into smaller and smaller volumes, until they reach the point where the molecules are basically all touching and the gas converts to a liquid! Gases per se (especially hot gases) usually remain "weakly interacting" right up to where they become a liquid, although the correct (non-ideal) equation of state for a real gas often displays features that are the results of moderate interaction, depending on the pressure and temperature.

Water[131] is, as noted, a remarkable liquid. H_2O is a polar molecules with a permanent dipole moment, so water molecules are very strongly interacting, both with each other and with other materials. It organizes itself quickly into a state of relative order that is ***very incompressible***. The bulk modulus of water is 2.2×10^9 Pa, which means that even deep in the ocean where pressures can be measured in the tens of millions of Pascals (or hundreds of atmospheres) the density of water only varies by a few percent from that on the surface. Its density varies much more rapidly with *temperature* than with pressure[132]. We will idealize water by considering it to be *perfectly incompressible* in this course, which is close enough to true for nearly any mundane application of hydraulics that you are most unlikely to ever observe an exception that matters.

8.1.4: Viscosity and fluid flow

Fluids, whether liquid or gas, have some internal "stickiness" that resists the relative motion of one part of the fluid compared to another, a kind of internal "friction" that tries to equilibrate an entire body of fluid to move together. They also interact with the walls of any container in which they are confined. The ***viscosity*** of a fluid (symbol μ) is a measure of this internal friction or stickiness. Thin fluids have a low viscosity and flow easily with minimum resistance; thick sticky fluids have a high viscosity and resist flow.

Fluid, when flowing through (say) a cylindrical pipe tends to organize itself in one of two very different ways – a state of *laminar flow* where the fluid at the very edge of the flowing volume is at rest where it is in contact with the pipe and the speed concentrically and symmetrically increases to a maximum in the center of the pipe, and *turbulent flow* where the fluid tumbles and rolls and forms eddies as it flows through the pipe. Turbulence and flow and viscosity are properties that will be discussed in more detail below.

8.1.5: Properties Summary

To summarize, fluids have the following properties that you should conceptually and intuitively understand and be able to use in working fluid problems:

- They usually assume the shape of any vessel they are placed in (exceptions are associated with confinement due to gravity and surface effects such as surface tension and how well the fluid adheres to the surface in question).
- They are characterized by a mass per unit volume *density* ρ.
- They exert a *pressure* P (force per unit area) on themselves and any surfaces they are in contact with.

[131] Wikipedia: http://www.wikipedia.org/wiki/Properties of Water. As I said, water is amazing. This article is well worth reading just for fun.

[132] A fact that impacts my beer-making activities quite significantly, as the specific gravity of hot wort fresh off of the boil is quite different from the specific gravity of the same wort cooled to room temperature. The specific gravity of the wort is related to the sugar content, which is ultimately related to the alcohol content of the fermented beer. Just in case this interests you...

- The pressure can vary according to the dynamic and static properties of the fluid.
- The fluid has a measure of its "stickiness" and resistance to flow called *viscosity*. Viscosity is the internal friction of a fluid, more or less. We will treat fluids as being "ideal" and ignore viscosity in this course.
- Fluids are *compressible* – when the pressure in a fluid is increased, its volume descreases according to the relation:

$$\Delta P = -B\frac{\Delta V}{V} \tag{8.18}$$

  where B is called the ***bulk modulus*** of the fluid (the equivalent of a spring constant). An alternative formulation of this equation uses the ***compressibility*** β:

$$\beta = -\frac{1}{V}\left(\frac{dV}{dP}\right) = \frac{1}{B}$$

- Fluids where β is a very small number (so large changes in pressure create only tiny changes in fractional volume) are called ***incompressible*** and their density is roughly constant. Water is an example of an incompressible fluid, as are most liquids. Gases have a higher compressibility and pressure changes can produce a large change in occupied volume (and hence density).
- Below a critical speed, the dynamic flow of a moving fluid tends to be laminar, where every bit of fluid moves parallel to its neighbors in response to pressure differentials and around obstacles. Above that speed it becomes turbulent flow. Turbulent flow is quite difficult to treat mathematically and is hence beyond the scope of this introductory course – we will restrict our attention to ideal fluids either static or in laminar flow.

We will now use these general properties and definitions, plus our existing knowledge of physics, to deduce a number of important properties of and laws pertaining to *static fluids*, fluids that are in static equilibrium.

Static Fluids

8.1.6: Pressure and Confinement of Static Fluids

In figure 102 we see a box of a fluid that is confined within the box by the rigid walls of the box. We will imagine that this particular box is in "free space" far from any gravitational attractor and is therefore at rest with no external forces acting on it. We know from our intuition based on things like cups of coffee that no matter how this fluid is initially stirred up and moving *within* the container, after a very long time the fluid will damp down any initial motion by interacting with the walls of the container and arrive at *static equilibrium*[133].

[133]This state will also entail *thermodynamic equilibrium* with the box (which must be at a uniform temperature) and hence the fluid in this particular non-accelerating box has a uniform density.

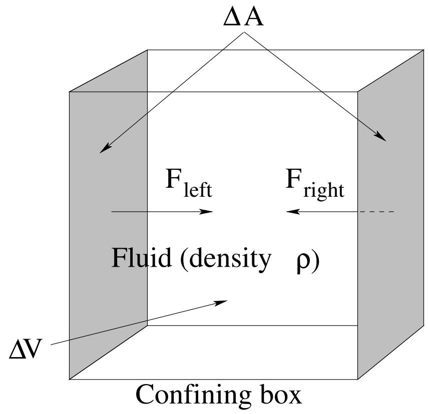

Figure 102: A fluid in static equilibrium confined to a sealed rectilinear box in zero gravity.

A fluid in static equilibrium has the property that every single tiny chunk of volume in the fluid has to *independently* be in *force equilibrium* – the total force acting on the differential volume chunk must be *zero*. In addition the net torques acting on all of these differential subvolumes must be zero, and the fluid must be at rest, neither translating nor rotating. Fluid rotation is more complex than the rotation of a static object because a fluid can be *internally* rotating even if all of the fluid in the outermost layer is in contact with a contain and is *stationary*. It can also be *turbulent* – there can be lots of internal eddies and swirls of motion, including some that can exist at very small length scales and persist for fair amounts of time. We will idealize all of this – when we discuss static properties of fluids we will assume that all of this sort of internal motion has disappeared.

We can now make a few very simple observations about the forces exerted by the walls of the container on the fluid within. First of all the mass of the fluid in the box above is clearly:

$$\Delta M = \rho \Delta V \tag{8.19}$$

where ΔV is the volume of the box. Since it is at rest and remains at rest, the net external force exerted on it (only) by the the box must be zero (see Week 4). We drew a *symmetric* box to make it easy to see that the magnitudes of the forces exerted by opposing walls are equal $F_{\rm left} = F_{\rm right}$ (for example). Similarly the forces exerted by the top and bottom surfaces, and the front and back surfaces, must cancel.

The average velocity of the molecules in the box must be zero, but the molecules themselves will generally not be at rest at any nonzero temperature. They will be in a state of constant motion where they *bounce elastically* off of the walls of the box, both giving and receiving an impulse (change in momentum) from the walls as they do. The walls of any box large enough to contain many molecules thus exerts a *nearly continuous* nonzero force that confines any fluid not at zero temperature[134].

From this physical picture we can also deduce an important *scaling property* of the force exerted by the walls. We have deliberately omitted giving any actual dimensions to our box in figure 102. Suppose (as shown) the cross-sectional area of the left and right walls are ΔA originally. Consider now what we expect if we *double the size* of the box

[134]Or at a temperature low enough for the fluid to *freeze* and becomes a *solid*

and at the same time add enough additional fluid for the fluid density to remain the same, making the side walls have the area $2\Delta A$. With twice the area (and twice the volume and twice as much fluid), we have twice as many molecular collisions per unit time on the doubled wall areas (with the same average impulse per collision). The average force exerted by the doubled wall areas therefore *also doubles*.

From this simple argument we can conclude that the average force exerted by any wall is *proportional to the area of the wall*. This force is therefore most naturally expressible in terms of *pressure*, for example:

$$F_{\rm left} = P_{\rm left}\Delta A = P_{\rm right}\Delta A = F_{\rm right} \tag{8.20}$$

which implies that the *pressure* at the left and right confining walls is the same:

$$P_{\rm left} = P_{\rm right} = P \tag{8.21}$$

, and that this pressure describes the force exerted by the fluid on the walls and vice versa. Again, the exact same thing is true for the other four sides.

There is nothing special about our particular choice of left and right. If we had originally drawn a *cubic* box (as indeed we did) we can easily see that the pressure P on *all* the faces of the cube must be the same and indeed (as we shall see more explicitly below) the pressure everywhere in the fluid must be the same!

That's quite a lot of mileage to get out of symmetry and the definition of static equilibrium, but there is one more important piece to get. This last bit involves forces exerted by the wall *parallel* to its surface. On average, there cannot be any! To see why, suppose that one surface, say the left one, exerted a force tangent to the surface itself on the fluid in contact with that surface. An important property of fluids is that *one part of a fluid can move independent of another* so the fluid in at least *some* layer with a finite thickness near the wall would therefore experience a *net* force and would *accelerate*. But this violates our assumption of static equilibrium, so a fluid in *static equilibrium* exerts no tangential force on the walls of a confining container and vice versa.

We therefore conclude that the *direction* of the force exerted by a confining surface with an area ΔA on the fluid that is in contact with it is:

$$\vec{\boldsymbol{F}} = P\ \Delta A\hat{\boldsymbol{n}} \tag{8.22}$$

where $\hat{\boldsymbol{n}}$ is an inward-directed unit vector *perpendicular to* (normal to) the surface. This final rule permits us to handle the force exerted on fluids confined to *irregular* amoebic blob shaped containers, or balloons, or bags, or – well, us, by our skins and vascular system.

Note well that this says nothing about the tangential force exerted by fluids in *relative motion* to the walls of the confining container. We already know that a fluid moving across a solid surface will exert a *drag force*, and later this week we'll attempt to at least approximately quantify this.

Next, let's consider what happens when we bring this box of fluid[135] *down to Earth* and consider what happens to the pressure in a box in *near-Earth gravity*.

[135]It's just a box of rain. I don't know who put it there...

8.1.7: Pressure and Confinement of Static Fluids in Gravity

The principle change brought about by setting our box of fluid down on the ground in a gravitational field (or equivalently, accelerating the box of fluid uniformly in some direction to develop a *pseudo*-gravitational field in the non-inertial frame of the box) is that an additional external force comes into play: The weight of the fluid. A static fluid, confined in some way in a gravitational field, must ***support the weight of its many component parts*** internally, and of course the box itself must support the weight of the entire mass ΔM of the fluid.

As hopefully you can see if you carefully read the previous section. The only force available to provide the necessary internal support or confinement force is the ***variation of pressure within the fluid***. We would like to know how the pressure *varies* as we move up or down in a static fluid so that it supports its own weight.

There are countless reasons that this knowledge is valuable. It is this pressure variation that hurts your ears if you dive deep into the water or collapses submarines if they dive too far. It is this pressure variation that causes your ears to pop as you ride in a car up the side of a mountain or your blood to boil into the vacuum of space if you ride in a rocket all the way out of the atmosphere without a special suit or vehicle that provides a personally pressurized environment. It is this pressure variation that will one day very likely cause you to have varicose veins and edema in your lower extremities from standing on your feet all day – and can help treat/reverse both if you stand in 1.5 meter deep water instead of air. The pressure variation drives water out of the pipes in your home when you open the tap, helps lift a balloon filled with helium, floats a boat but fails to float a rock.

We need to *understand* all of this, whether our eventual goal is to become a physicist, a physician, an engineer, or just a scientifically literate human being. Let's get to it.

Here's the general idea. If we consider a tiny (eventually differentially small) chunk of fluid in force equilibrium, gravity will pull it down and the only thing that can push it up is a *pressure difference* between the top and the bottom of the chunk. By requiring that the force exerted by the pressure difference balance the weight, we will learn how the pressure varies with increasing depth.

For incompressible fluids, this is really all there is to it – it takes only a few lines to derive a lovely formula for the increase in pressure as a function of depth in an incompressible *liquid*.

For gases there is, alas, a small complication. Compressible fluids have densities that *increase* as the pressure increases. This means that boxes of the same size also have *more mass* in them as one descends. More mass means that the pressure difference has to increase, faster, which makes the density/mass greater still, and one discovers (in the end) that the pressure varies *exponentially* with depth. Hence the air pressure drops relatively *quickly* as one goes up from the Earth's surface to very *close* to zero at a height of ten miles, but the atmosphere itself extends for a very long way into space, never quite dropping to "zero" even when one is twenty or a hundred miles high.

As it happens, the calculus for the two kinds of fluids is the *same* up to a given (very important) common point, and then differs, becoming very simple indeed for incompressible fluids and a bit more difficult for compressible ones. Simple solutions suffice to help us build our conceptual understanding; we will therefore treat incompressible fluids first and ***everybody*** is responsible for understanding them. Physics majors, math majors, engineers, and people who love a good bit of calculus now and then should probably continue on and learn how to integrate the simple model provided for compressible fluids.

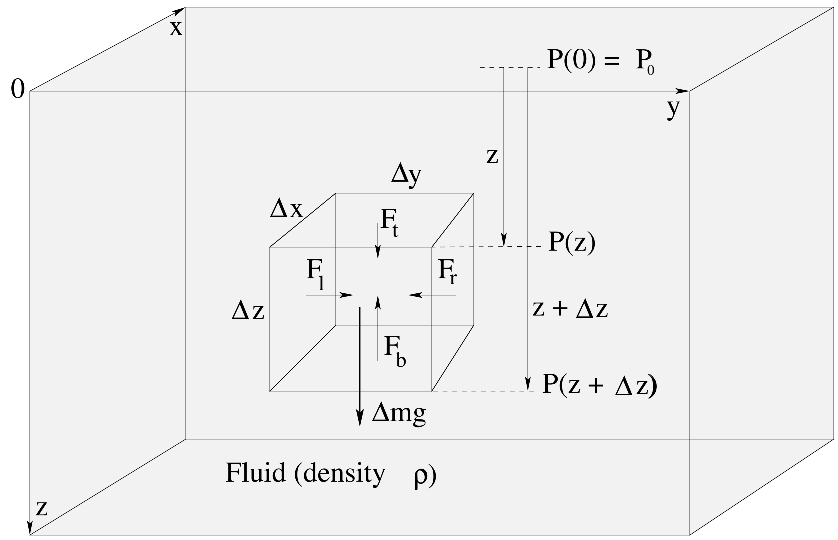

Figure 103: A fluid in static equilibrium confined to a sealed rectilinear box in a near-Earth gravitational field $\vec{g}$. Note well the small chunk of fluid with dimensions $\Delta x, \Delta y, \Delta z$ in the middle of the fluid. Also note that the coordinate system selected has z increasing from the top of the box *down*, so that z can be thought of as the *depth* of the fluid.

In figure 103 a (portion of) a fluid confined to a box is illustrated. The box could be a completely sealed one with rigid walls on all sides, or it could be something like a cup or bucket that is open on the top but where the fluid is still confined there by e.g. atmospheric pressure.

Let us consider a small (eventually infinitesimal) chunk of fluid somewhere in the *middle* of the container. As shown, it has physical dimensions $\Delta x, \Delta y, \Delta z$; its upper surface is a distance z below the origin (where z increases down and hence can represent "depth") and its lower surface is at depth $z + \Delta z$. The areas of the top and bottom surfaces of this small chunk are e.g. $\Delta A_{tb} = \Delta x \Delta y$, the areas of the sides are $\Delta x \Delta z$ and $\Delta y \Delta z$ respectively, and the volume of this small chunk is $\Delta V = \Delta x \Delta y \Delta z$.

This small chunk is itself in static equilibrium – therefore the forces between any pair of its horizontal sides (in the x or y direction) must cancel. As before (for the box in space) $F_l = F_r$ in magnitude (and opposite in their y-direction) and similarly for the force on the front and back faces in the x-direction, which will always be true if the pressure does not vary *horizontally* with variations in x or y. In the z-direction, however, force equilibrium requires that:

$$F_t + \Delta mg - F_b = 0 \tag{8.23}$$

(where recall, down is positive).

The only possible *source* of F_t and F_b are the ***pressure in the fluid itself*** which will ***vary with the depth*** z: $F_t = P(z)\Delta A_{tb}$ and $F_b = P(z + \Delta z)\Delta A_{tb}$. Also, the mass of fluid in the (small) box is $\Delta m = \rho \Delta V$ (using our ritual incantation "the mass of the chunks is...").

We can thus write:

$$P(z)\Delta x\Delta y + \rho(\Delta x\Delta y\Delta z)g - P(z+\Delta z)\Delta x\Delta y = 0 \tag{8.24}$$

or (dividing by $\Delta x\Delta y\Delta z$ and rearranging):

$$\frac{\Delta P}{\Delta z} = \frac{P(z+\Delta z) - P(z)}{\Delta z} = \rho g \tag{8.25}$$

Finally, we take the limit $\Delta z \to 0$ and identify the ***definition of the derivative*** to get:

$$\frac{dP}{dz} = \rho g \tag{8.26}$$

Identical arguments but *without* any horizontal external force followed by $\Delta x \to 0$ and $\Delta y \to 0$ lead to:

$$\frac{dP}{dx} = \frac{dP}{dy} = 0 \tag{8.27}$$

as well – P does not vary with x or y as already noted[136].

In order to find $P(z)$ from this differential expression (which applies, recall, to *any* confined fluid in static equilibrium in a gravitational field) we have to *integrate* it. This integral is very simple if the fluid is incompressible because in that case ***ρ is a constant***. The integral isn't *that* difficult if ρ is *not* a constant as implied by the equation we wrote above for the bulk compressibility. We will therefore first do incompressible fluids, then compressible ones.

8.1.8: Variation of Pressure in Incompressible Fluids

In the case of incompressible fluids, ρ is a constant and does not vary with pressure and/or depth. Therefore we can easily multiple $dP/dz = \rho g$ above by dz on both sides and integrate to find P:

$$\begin{aligned} dP &= \rho g\ dz \\ \int dP &= \int \rho g\ dz \\ P(z) &= \rho g z + P_0 \end{aligned} \tag{8.28}$$

where P_0 is the *constant of integration* for both integrals, and practically speaking is the pressure in the fluid at zero depth (wherever that might be in the coordinate system chosen).

Example 8.1.1: Barometers

Mercury barometers were originally invented by Evangelista Torricelli[137] a natural philosopher who acted as Galileo's secretary for the last three months of Galileo's life under house

[136] Physics and math majors and other students of multivariate calculus will recognize that I should probably be using partial derivatives here and establishing that $\vec{\nabla} P = \rho\vec{g}$, where in free space we should instead have had $\vec{\nabla} P = 0 \to P$ constant.

[137] Wikipedia: http://www.wikipedia.org/wiki/Evangelista_Torricelli. ,

arrest. The invention was inspired by Torricelli's attempt to solve an important engineering problem. The pump makers of the Grand Duke of Tuscany had built powerful pumps intended to raise water twelve or more meters, but discovered that no matter how powerful the pump, water stubbornly refused to rise more than ten meters into a pipe evacuated at the top.

Torricelli demonstrated that a shorter glass tube filled with mercury, when inverted into a dish of mercury, would fall back into a column with a height of roughly 0.76 meters with a vacuum on top, and soon thereafter discovered that the height of the column fluctuated with the pressure of the outside air pressing down on the mercury in the dish, correctly concluding that water would behave exactly the same way[138]. Torricelli made a number of other important 17th century discoveries, correctly describing the causes of wind and discovering "Torricelli's Law" (an aspect of the Bernoulli Equation we will note below).

In honor of Torricelli, a unit of pressure was named after him. The ***torr*** is the pressure required to push the mercury in Torricelli's barometer up one millimeter. Because mercury barometers were at one time nearly ubiquitous as the most precise way to measure the pressure of the air, a specific height of the mercury column was the original definition of the standard atmosphere. For better or worse, Torricelli's original observation defined one standard atmosphere to be *exactly* "760 millimeters of mercury" (which is a lot to write or say) or as we would now say, "760 torr"[139].

Mercury barometers are now more or less banned, certainly from the workplace, because mercury is a potentially toxic heavy metal. In actual fact, *liquid* mercury is not biologically active and hence is not particularly toxic. Mercury vapor *is* toxic, but the amount of mercury vapor emitted by the exposed surface of a mercury barometer at room temperature is well below the levels considered to be a risk to human health by OSHA unless the barometer is kept in a small, hot, poorly ventilated room with someone who works there over years. This isn't all that common a situation, but with all toxic metals we are probably better safe than sorry[140].

At this point mercury barometers are rapidly disappearing everywhere but from the hands of collectors. Their manufacture is banned in the U.S., Canada, Europe, and many other nations. We had a lovely one (probably more than one, but I recall one) in the Duke Physics Department up until sometime in the 90's[141], but it was – sanely enough – removed and retired during a renovation that also cleaned up most if not all of the asbestos in the building. Ah, my toxic youth...

[138]You, too, get to solve Torricelli's problem as one of your homework problems, but armed with a lot better understanding.

[139]Not to be outdone, one standard atmosphere (or atmospheric pressures in weather reporting) in the U.S. is often given as 29.92 barleycorn-derived *inches* of mercury instead of millimeters. Sigh.

[140]The single biggest risk associated with uncontained liquid mercury (in a barometer or otherwise) is that you can easily spill it, and once spilled it is fairly likely to sooner or later make its way into either *mercury vapor* or ***methyl mercury***, both of which are biologically active and highly toxic. Liquid mercury itself you could drink a glass of and it would pretty much pass straight through you with minimal absorption and little to no damage *if* you – um – "collected" it carefully and disposed of it properly on the other side.

[141]I used to work in the small, cramped space with poor circulation where it was located from time to time but never very long at a time and besides, the room was *cold*. But if I seem "mad as a hatter" – mercury nitrate was used in the making of hats and the vapor used to poison the hatters vapor used to poison hat makers – it probably isn't from this...

Still, at one time they were *extremely* common – most ships had one, many households had one, businesses and government agencies had them – knowing the pressure of the air is an important factor in weather prediction. Let's see how they work(ed).

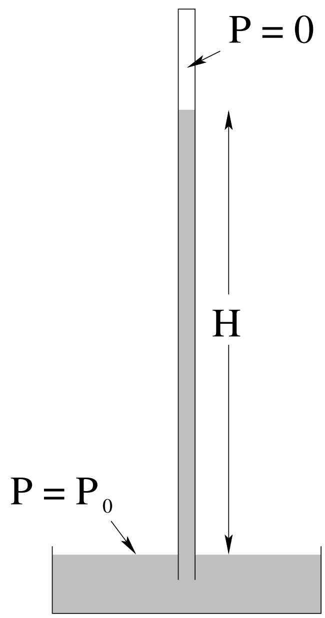

Figure 104: A simple fluid barometer consists of a tube with a vacuum at the top filled with fluid supported by the air pressure outside.

A simple mercury barometer is shown in figure 104. It consists of a tube that is completely filled with mercury. Mercury has a specific gravity of 13.534 at a typical room temperature, hence a density of 13534 kg/m^3). The filled tube is then inverted into a small reservoir of mercury (although other designs of the bottom are possible, some with smaller exposed surface area of the mercury). The mercury falls (pulled down by gravity) out of the tube, leaving behind a ***vacuum*** at the top.

We can easily compute the expected height of the mercury column if P_0 is the pressure on the exposed surface of the mercury in the reservoir. In that case

$$P = P_0 + \rho g z \tag{8.29}$$

as usual for an incompressible fluid. Applying this formula to both the top and the bottom,

$$P(0) = P_0 \tag{8.30}$$

and

$$P(H) = P_0 - \rho g H = 0 \tag{8.31}$$

(recall that the upper surface is *above* the lower one, $z = -H$). From this last equation:

$$P_0 = \rho g H \tag{8.32}$$

and one can easily convert the measured height H of mercury above the top surface of mercury in the reservoir into P_0, the air pressure on the top of the reservoir.

At one standard atmosphere, we can easily determine what a mercury barometer at room temperature will read (the height H of its column of mercury above the level of mercury in the reservoir):

$$P_0 = 13534 \frac{\mathrm{kg}}{\mathrm{m}^3} \times 9.80665 \frac{\mathrm{m}}{\mathrm{sec}^2} \times H = 101325\mathrm{Pa} \tag{8.33}$$

Note well, we have used the precise SI value of g in this expression, and the density of mercury at "room temperature" around 20°C or 293°K. Dividing, we find the expected height of mercury in a barometer at room temperature and one standard atmosphere is $H = 0.763$ meters or 763 torr

Note that this is not exactly the 760 torr we expect to read for a standard atmosphere. This is because for high precision work one cannot just use *any old* temperature (because mercury has a significant thermal expansion coefficient and was then and continues to be used today in mercury thermometers as a consequence). The unit definition is based on using the density of mercury at 0°C or 273.16°K, which has a specific gravity (according to NIST, the National Institute of Standards in the US) of 13.595. Then the precise connection between SI units and torr follows from:

$$P_0 = 13595 \frac{\mathrm{kg}}{\mathrm{m}^3} \times 9.80665 \frac{\mathrm{m}}{\mathrm{sec}^2} \times H = 101325\mathrm{Pa} \tag{8.34}$$

Dividing we find the value of H expected at one standard atmosphere:

$$H_{\mathrm{atm}} = 0.76000 = 760.00 \text{ millimeters} \tag{8.35}$$

Note well the precision, indicative of the fact that the SI units for a standard atmosphere *follow* from their definition in torr, not the other way around.

Curiously, this value is invariably given in both textbooks and even the wikipedia article on atmospheric pressure as the *average* atmospheric pressure at sea level, which it almost certainly is not – a spatiotemporal averaging of sea level pressure would have been utterly impossible during Torricelli's time (and would be difficult today!) and if it was done, could not possibly have worked out to be *exactly* 760.00 millimeters of mercury at 273.16°K.

Example 8.1.2: Variation of Oceanic Pressure with Depth

The pressure on the surface of the ocean is, approximately, by definition, one atmosphere. Water is a highly incompressible fluid with $\rho_w = 1000$ kilograms per cubic meter[142]. $g \approx 10$ meters/second^2. Thus:

$$P(z) = P_0 + \rho_w g z = \left(10^5 + 10^4 z\right) \text{ Pa} \tag{8.36}$$

or

$$P(z) = (1.0 + 0.1z) \text{ bar} = (1000 + 100z) \text{ mbar} \tag{8.37}$$

Every ten meters of depth (either way) increases water pressure by (approximately) one atmosphere!

Wow, *that* was easy. This is a *very important rule of thumb* and is actually fairly easy to remember! How about compressible fluids?

[142]Good number to remember. In fact, *great* number to remember.

8.1.9: Variation of Pressure in Compressible Fluids

Compressible fluids, as noted, have a density which *varies* with pressure. Recall our equation for the compressibility:

$$\Delta P = -B\frac{\Delta V}{V} \tag{8.38}$$

If one increases the pressure, one therefore *decreases* occupied volume of any given chunk of mass, and hence increases the density. However, to predict precisely how the density will depend on pressure requires more than just this – it requires a *model* relating pressure, volume and mass.

Just such a model for a compressible gas is provided (for example) by the ***Ideal Gas Law***[143] :

$$PV = Nk_bT = nRT \tag{8.39}$$

where N is the number of molecules in the volume V, k_b is Boltzmann's constant[144] n is the number of moles of gas in the volume V, R is the ideal gas constant[145] and T is the temperature in degrees Kelvin (or Absolute)[146] . If we assume constant temperature, and convert N to the mass of the gas by multiplying by the molar mass and dividing by Avogadro's Number[147] 6×10^{23}.

(***Aside:*** If you've never taken chemistry a lot of this is going to sound like Martian to you. Sorry about that. As always, consider visting the e.g. Wikipedia pages linked above to learn enough about these topics to get by for the moment, or just keep reading as the *details* of all of this won't turn out to be very important...)

When we do this, we get the following formula for the density of an ideal gas:

$$\rho = \frac{M}{RT}P \tag{8.40}$$

where M is the molar mass[148] , the number of kilograms of the gas per mole. Note well that this result is *idealized* – that's why they call it the *Ideal* Gas Law! – and that no real gases are "ideal" for all pressures and temperatures because sooner or later they all become *liquids* or *solids* due to molecular interactions. However, the gases that make up "air" are all reasonably ideal at temperatures in the ballpark of room temperature, and in any event it is worth seeing how the pressure of an ideal gas varies with z to get an *idea* of how air pressure will vary with height. Nature will probably be somewhat different than this prediction, but we ought to be able to make a *qualitatively* accurate model that is also moderately *quantitatively* predictive as well.

As mentioned above, the formula for the derivative of pressure with z is unchanged for compressible or incompressible fluids. If we take $dP/dz = \rho g$ and multiply both sides by

[143] Wikipedia: http://www.wikipedia.org/wiki/Ideal Gas Law.
[144] Wikipedia: http://www.wikipedia.org/wiki/Boltzmann's Constant. ,
[145] Wikipedia: http://www.wikipedia.org/wiki/Gas Constant. ,
[146] Wikipedia: http://www.wikipedia.org/wiki/Temperature.
[147] Wikipedia: http://www.wikipedia.org/wiki/Avogadro's Number.
[148] Wikipedia: http://www.wikipedia.org/wiki/Molar Mass.

dz as before and integrate, now we get (assuming a fixed temperature T):

$$\begin{aligned}
dP &= \rho g\, dz \;=\; \frac{Mg}{RT} P\, dz \\
\frac{dP}{P} &= \frac{Mg}{RT} dz \\
\int \frac{dP}{P} &= \frac{Mg}{RT} \int dz \\
\ln(P) &= \frac{Mg}{RT} z + C
\end{aligned}$$

We now do the usual[149] – exponentiate both sides, turn the exponential of the sum into the product of exponentials, turn the exponential of a constant of integration into a constant of integration, and match the units:

$$P(z) = P_0 e^{\left(\frac{Mg}{RT}\right) z} \tag{8.41}$$

where P_0 is the pressure at zero *depth*, because (recall!) z is measured positive *down* in our expression for dP/dz.

Example 8.1.3: Variation of Atmospheric Pressure with Height

Using z to describe depth is moderately inconvenient, so let us define the ***height h above sea level*** to be $-z$. In that case P_0 is (how about that!) *1 Atmosphere*. The molar mass of dry air is $M = 0.029$ kilograms per mole. $R = 8.31$ Joules/(mole-K°). Hence a bit of multiplication at $T = 300°$:

$$\frac{Mg}{RT} = \frac{0.029 \times 10}{8.31 \times 300} = 1.12 \times 10^{-4} \text{ meters}^{-1} \tag{8.42}$$

or:

$$P(h) = 10^5 \exp(-0.00012\; h) \text{ Pa} = 1000 \exp(-0.00012\; h) \text{ mbar} \tag{8.43}$$

Note well that the temperature of air is *not* constant as one ascends – it drops by a fairly significant amount, even on the absolute scale (and higher still, it *rises* by an even *greater* amount before dropping again as one moves through the layers of the atmosphere. Since the pressure is found from an integral, this in turn means that the exponential behavior itself is rather inexact, but still it isn't a *terrible* predictor of the variation of pressure with height. This equation predicts that air pressure should drop to $1/e$ of its sea-level value of 1000 mbar at a height of around 8000 meters, the height of the so-called ***death zone***[150] . We can compare the actual (average) pressure at 8000 meters, 356 mbar, to $1000 \times e^{-1} = 368$ mbar. We get remarkably good agreement!

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[149] Which should be familiar to you both from solving the linear drag problem in Week 2 *and* from the online Math Review.

[150] Wikipedia: http://www.wikipedia.org/wiki/Effects of high altitude on humans. This is the height where air pressure drops to where humans are at extreme risk of dying if they climb without supplemental oxygen support – beyond this height hypoxia reduces one's ability to make important and life-critical decisions during the very last, most stressful, part of the climb. Mount Everest (for example) can only be climbed with oxygen masks and some of the greatest disasters that have occurred climbing it and other peaks are associated with a lack of or failure of supplemental oxygen.

This agreement rapidly breaks down, however, and meteorologists actually use a patchwork of formulae (both algebraic and exponential) to give better agreement to the actual variation of air pressure with height as one moves up and down through the various named layers of the atmosphere with the pressure, temperature and even molecular composition of "air" varying all the way. This simple model explains a *lot* of the variation, but its assumptions are not really correct.

8.2: Pascal's Principle and Hydraulics

We note that (from the above) the *general form* of P of a fluid confined to a sealed container has the most general form:

$$P(z) = P_0 + \int_0^z \rho g dz \tag{8.44}$$

where P_0 is the constant of integration or value of the pressure at the reference ***depth*** $z = 0$. This has an important consequence that forms the basis of *hydraulics*.

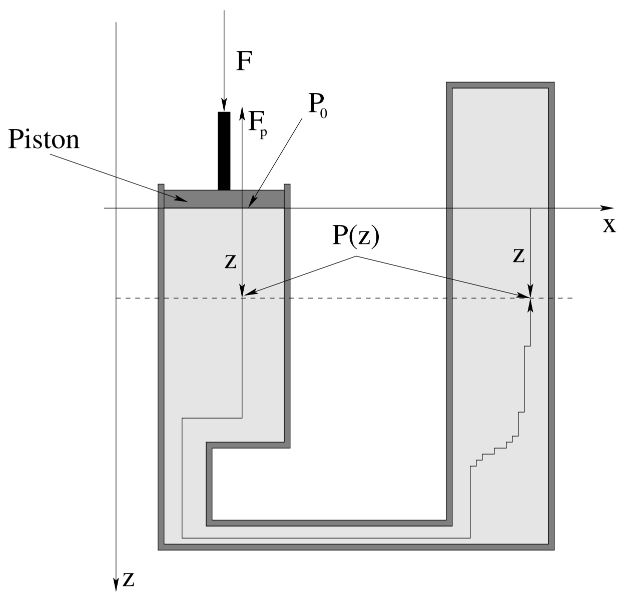

Figure 105: A single piston seated tightly in a frictionless cylinder of cross-sectional area A is used to compress water in a sealed container. Water is incompressible and does not significantly change its volume at $P = 1$ bar (and a constant room temperature) for pressure changes on the order of 0.1-100 bar.

Suppose, then, that we have an ***incompressible fluid*** e.g. water confined within a sealed container by e.g. a *piston* that can be pushed or pulled on to *increase or decrease* the *confinement pressure* on the surface of the piston. Such an arrangement is portrayed in figure 105.

We can push down (or pull back) on the piston with any total downward force F that we like that leaves the system in equilibrium. Since the piston itself is in static equilibrium, the force we push with must be opposed by the pressure in the fluid, which exerts an equal and opposite upwards force:

$$F = F_p = P_0 A \tag{8.45}$$

where A is the cross sectional area of the piston and where we've put the cylinder face at $z = 0$, which we are obviously free to do. For all practical purposes this means that we can make P_0 "anything we like" within the range of pressures that are unlikely to make water at room temperature change it's state or volume do other bad things, say $P = (0.1, 100)$ bar.

The pressure at a depth z in the container is then (from our previous work):

$$P(z) = P_0 + \rho g z \tag{8.46}$$

where $\rho = \rho_w$ if the cylinder is indeed filled with water, but the cylinder could equally well be filled with hydraulic fluid (basically oil, which assists in lubricating the piston and ensuring that it remains "frictionless' while assisting the seal), alcohol, mercury, or any other incompressible liquid.

We recall that the pressure changes *only* when we change our depth. Moving laterally does not change the pressure, because e.g. $dP/dx = dP/dy = 0$. We can always find a path consisting of vertical and lateral displacements from $z = 0$ to any other point in the container – two such points at the same depth z are shown in 105, along with a (deliberately ziggy-zaggy[151]) vertical/horizontal path connecting them. Clearly these two points ***must have the same pressure*** $\boldsymbol{P(z)}$***!***

Now consider the following. Suppose we start with pressure P_0 (so that the pressure at these two points is $P(z)$, but then change F to make the pressure P_0' and the pressure at the two points $P'(z)$. Then:

$$\begin{aligned} P(z) &= P_0 + \rho g z \\ P'(z) &= P_0' + \rho g z \\ \Delta P(z) &= P'(z) - P(z) = P_0' - P_0 = \Delta P_0 \end{aligned} \tag{8.47}$$

That is, the pressure change at depth z does not depend on z *at any point in the fluid!* It depends *only on the change in the pressure exerted by the piston!*

This result is known as ***Pascal's Principle*** and it holds (more or less) for *any* compressible fluid, not just incompressible ones, but in the case of compressible fluids the piston will move up or down or in or out and the density of the fluid will change and hence the treatment of the integral will be too complicated to cope with. Pascal's Principle is more commonly given in *English words* as:

> ***Any change in the pressure exerted at a given point on a confined fluid is transmitted, undiminished, throughout the fluid.***

Pascal's principle is the basis of ***hydraulics***. Hydraulics are a kind of fluid-based simple machine that can be used to greatly amplify an applied force. To understand it, consider the following figure:

[151]Because we can make the zigs and zags *differentially small*, at which point this piecewise horizontal-vertical line becomes an *arbitrary curve* that remain in the fluid. Multivariate calculus can be used to formulate all of these results more prettily, but the *reasoning* behind them is completely contained in the picture and this text explanation.

Example 8.2.1: A Hydraulic Lift

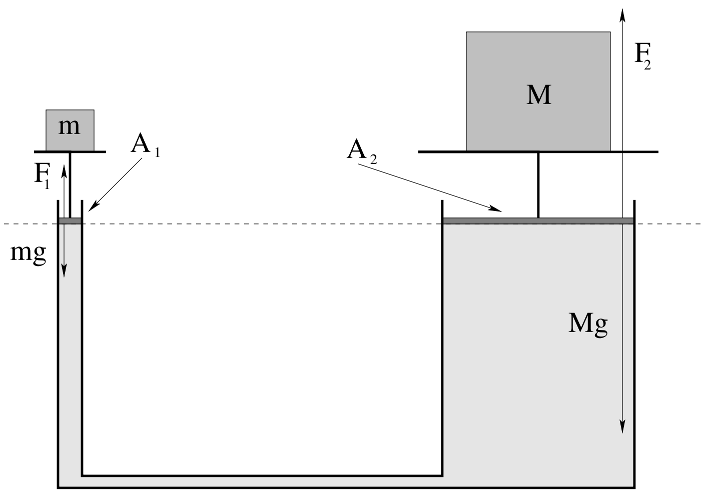

Figure 106: A simple schematic for a hydraulic lift of the sort used in auto shops to lift your car.

Figure 106 illustrates the way we can multiply forces using Pascal's Principle. Two pistons seal off a pair of cylinders connected by a closed tube that contains an incompressible fluid. The two pistons are deliberately given the same height (which might as well be $z = 0$, then, in the figure, although we could easily deal with the variation of pressure associated with them being at different heights since we know $P(z) = P_0 + \rho g z$. The two pistons have cross sectional areas A_1 and A_2 respectively, and support a small mass m on the left and large mass M on the right in static equilibrium.

For them to be in equilibrium, clearly:

$$F_1 - mg = 0 \quad (8.48)$$

$$F_2 - Mg = 0 \quad (8.49)$$

We also/therefore have:

$$F_1 = P_0 A_1 = mg \quad (8.50)$$

$$F_2 = P_0 A_2 = Mg \quad (8.51)$$

Thus

$$\frac{F_1}{A_1} = P_0 = \frac{F_2}{A_2} \quad (8.52)$$

or (substituting and cancelling g):

$$M = \frac{A_2}{A_1} m \quad (8.53)$$

A small mass on a small-area piston can easily balance a *much larger mass on an equally larger area piston!*

Just like a lever, we can balance or lift a large weight with a small one. Also just as was the case with a lever, *there ain't no such thing as a free lunch!* If we try to *lift* (say) a car with a hydraulic lift, we have to move the same volume $\Delta V = A\Delta z$ from under the small piston (as it descends) to under the large one (as it ascends). If the small one goes down a distance z_1 and the large one goes up a distance z_2, then:

$$\frac{z_1}{z_2} = \frac{A_2}{A_1} \tag{8.54}$$

The *work* done by the two cylinders thus *precisely balances*:

$$W_2 = F_2 z_2 = F_1 \frac{A_2}{A_1} z_2 = F_1 \frac{A_2}{A_1} z_1 \frac{A_1}{A_2} = F_1 z_1 = W_1 \tag{8.55}$$

The hydraulic arrangement thus transforms pushing a small force through a large distance into a large force moved through a small distance so that the work done ***on*** piston 1 matches the work done ***by*** piston 2. No energy is created or destroyed (although in the real world a bit will be lost to heat as things move around) and all is well, quite literally, with the Universe.

This example is pretty simple, but it should suffice to guide you through doing a work-energy conservation problem where (for example) the mass m goes down a distance d (losing gravitational energy) and the mass M goes up a distance D (gaining gravitational energy *while the fluid itself also is net moved up above its former level!* Don't forget that last, tricky bit if you ever have a problem like that!

8.3: Fluid Displacement and Buoyancy

First, a story. Archimedes[152] was, quite possibly, the smartest person who has ever lived (so far). His day job was being the "court magician" in the island kingdom of Syracuse in the third century BCE, some 2200 years ago; in his free time he did things like invent primitive integration, accurately compute pi, invent amazing machines of war and peace, determine the key principles of both statics and fluid statics (including the one we are about to study and the principles of the lever – "Give me put a place to stand and I can move the world!" is a famous Archimedes quote, implying that a sufficiently long lever would allow the small forces humans can exert to move even something as large as the Earth, although yeah, there are a few problems with that that go beyond just a place to stand[153]).

The king (Hieron II) of Syracuse had a problem. He had given a goldsmith a mass of pure gold to make him a ***votive crown***, but when the crown came back he had the niggling suspicion that the goldsmith had substituted cheap silver for some of the gold and kept the gold. It was keeping him awake at nights, because if somebody can steal from the king and get away with it (and word gets out) it can only encourage a loss of respect and rebellion.

So he called in his court magician (Archimedes) and gave him the task of determining whether or not the crown had been made by adulterated gold – or else. And oh, yeah –

[152] Wikipedia: http://www.wikipedia.org/wiki/Archimedes. A very, very interesting person. I strongly recommend that my students read this short article on this person who came *within a hair* of inventing physics and calculus and starting the Enlightenment some 1900 years before Newton. Scary supergenius polymath guy. Would have won multiple Nobel prizes, a Macarthur "Genius" grant, and so on if alive today. Arguably the smartest person who has ever lived – so far.

[153] The "sound bite" is hardly a modern invention, after all. Humans have always loved a good, pithy statement of insight, even if it isn't actually even approximately true...

you can't melt down the crown and cast it back into a regular shape whose dimensions can be directly compared to the same shape of gold, permitting a direct comparison of their *densities* (the density of pure gold is not equal to the density of gold with an admixture of silver). And don't forget the "or else".

Archimedes puzzled over this for some days, and decided to take a bath and cool off his overheated brain. In those days, baths were large public affairs – you *went* to the baths as opposed to having one in your home – where a filled tub was provided, sometimes with attendants happy to help you wash. As the possibly apocryphal story has it, Archimedes lowered himself into the overfull tub and as he did so, water sloshed out as he *displaced its volume* with his own volume. In an intuitive, instantaneous flash of insight – a "light bulb moment" – he realized that *displacement of a liquid by an irregular shaped solid* can be used to measure its volume, and that such a measurement of displaced volume would allow the king's problem to be solved.

Archimedes then leaped out of the tub and ran naked through the streets of Syracuse (which we can only imagine provided its inhabitants with as much amusement then as it would provide now) yelling "Eureka!", which in Greek means "I have found it!" The test (two possible versions of which are supplied below, one more probable than the other but less instructive for our own purposes) was performed, and revealed that the goldsmith was indeed dishonest and had stolen some of the king's gold. Bad move, goldsmith! We will draw a tasteful veil over the probable painful and messy fate of the goldsmith.

Archimedes transformed his serendipitous discovery of static fluid displacement into an elaborate physical principle that explained *buoyancy*, the tendency of fluids to support all or part of the weight of objects immersed in them.

The fate of Archimedes himself is worth a moment more of our time. In roughly 212 BCE, the Romans invaded Syracuse in the Second Punic War after a two year siege. As legend has it, as the city fell and armed soldiers raced through the streets "subduing" the population as only soldiers can, Archimedes was in his court chambers working on a problem in the geometry of circles, which he had drawn out in the sand boxes that then served as a "chalkboard". A Roman soldier demanded that he leave his work and come meet with the conquering general, Marcus Claudius Marcellus. Archimedes declined, replying with his last words "Do not disturb my circles" and the soldier killed him. Bad move, soldier – Archimedes himself was a major part of the loot of the city and Marcellus had ordered that he was not to be harmed. The fate of the soldier that killed him is unknown, but it wasn't really a very good idea to anger a conquering general by destroying an object or person of enormous value, and I doubt that it was very good.

Anyway, let's see the modern version of Archimedes' discovery and see as well how Archimedes very probably used it to test the crown.

8.3.1: Archimedes' Principle

If you are astute, you will note that figure 107 is ***exactly like figure 103*** above, except that the internal chunk of *fluid* has been replace by *some other material*. The point is that this replacemend does not matter – ***the net force exerted on the cube by the fluid is the***

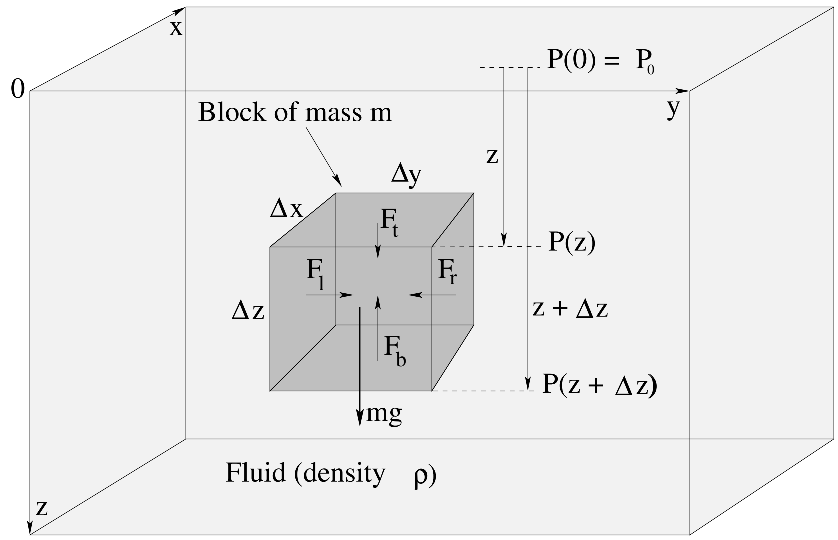

Figure 107: A solid chunk of “stuff” of mass m and the dimensions shown is immersed in a fluid of density ρ at a depth z. The vertical pressure difference in the fluid (that arises as the fluid itself becomes static static) exerts a vertical force on the cube.

same!

Hopefully, that is obvious. The net upward force exerted by the fluid is called the ***buoyant force*** F_b and is equal to:

$$\begin{aligned} F_b &= P(z+\Delta z)\Delta x \Delta y - P(z)\Delta x \Delta y \\ &= \left((P_0 + \rho g(z+\Delta z)) - (P_0 + \rho g z)\right)\Delta x \Delta y \\ &= \rho g \Delta z \Delta x \Delta y \\ &= \rho g \Delta V \end{aligned} \tag{8.56}$$

where ΔV is the volume of the small block.

The buoyant force is thus the ***weight of the fluid displaced*** by this single tiny block. This is all we need to show that the same thing is true for an *arbitrary* immersed shape of object.

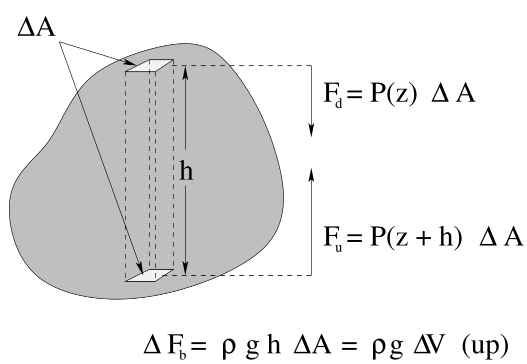

Figure 108: An arbitrary chunk of stuff is immersed in a fluid and we consider a vertical cross section with horizontal ends of area ΔA and height h through the chunk.

In figure 108, an arbitrary blob-shape is immersed in a fluid (not shown) of density ρ.

Imagine that we've taken a french-fry cutter and cuts the whole blob into nice rectangular segments, one of which (of length h and cross-sectional area ΔA) is shown. We can trim or average the end caps so that they are all perfectly horizontal by making all of the rectangles arbitrarily small (in fact, differentially small in a moment). In that case the *vertical* force exerted by the fluid on just the two lightly shaded surfaces shown would be:

$$F_d = P(z)\Delta A \tag{8.57}$$
$$F_u = P(z+h)\Delta A \tag{8.58}$$

where we assume the upper surface is at depth z (this won't matter, as we'll see in a moment). Since $P(z+h) = P(z) + \rho g h$, we can find the *net upward buoyant force* exerted on this little cross-section by subtracting the first from the second:

$$\Delta F_b = F_u - F_d = \rho g \, h\Delta A = \rho g \, \Delta V \tag{8.59}$$

where the *volume* of this piece of the entire blob is $\Delta V = h \, \Delta A$.

We can now let $\Delta A \to dA$, so that $\Delta V - > dV$, and write

$$F_b = \int dF_b = \int_{V \text{ of blob}} \rho g \, dV = \rho g V = m_f g \tag{8.60}$$

where $m_f = \rho V$ is the mass of the fluid displaced, so that $m_f g$ is its weight.

That is:

> ***The total buoyant force on the immersed object is the weight of the fluid displaced by the object.***

This is really an adequate *proof* of this statement, although if we were really going to be picky we'd use the fact that P doesn't vary in x or y to show that the net force in the x or y direction is zero independent of the shape of the blob, using our differential french-fry cutter mentally in the x direction and then noting that the blob is *arbitrary* in shape and we could have just as easily labelled or oriented the blob with this direction called y so it must be true in *any* direction perpendicular to $\vec{g}$.

This statement – in the English or algebraic statement as you prefer – is known as ***Archimedes' Principle***, although Archimedes could hardly have formulated it quite the way we did algebraically above as he died before he could *quite* finish inventing the calculus and physics.

This principle is enormously important and ubiquitous. Buoyancy is why boats float, but rocks don't. It is why childrens' helium-filled balloons do odd things in accelerating cars. It exerts a subtle force on everything submerged in the air, in water, in beer, in liquid mercury, as long as the fluid itself is either in a gravitational field (and hence has a pressure gradient) or is in an accelerating container with its own "pseudogravity" (and hence has a pressure gradient).

Let's see how Archimedes could have used this principle to test the crown two ways. The first way is very simple and *conceptually* instructive; the second way is more practical to us as it illustrates the way we generally do algebra associated with buoyancy problems.

Example 8.3.1: Testing the Crown I

The tools Archimedes probably had available to him were balance-type scales, as these tools for comparatively measuring the weight were well-known in antiquity. He certainly had vessels he could fill with water. He had thread or string, he had the crown itself, and he had access to pure gold from the king's treasury (at least for the duration of the test.

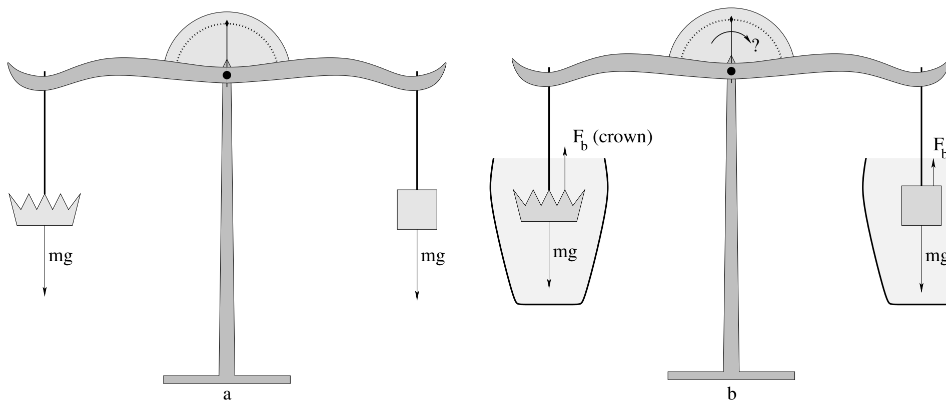

Figure 109: In a), the crown is balanced against an equal weight/mass of pure gold *in air*. In b) the crown *and* the gold are symmetrically submerged in containers of still water.

Archimedes very likely used his balance to first select and trim a piece of gold so it had *exactly the same weight as the crown* as illustrated in figure 109a. Then all he had to do was submerge the crown and the gold symmetrically in two urns filled with water, taking care that they are both fully underwater.

Pure gold is *more dense* than gold adulterated with silver (the most likely metal the goldsmith would have used; although a few others such as copper might have also been available and/or used they are also less dense than gold). This means that any given mass/weight (in air, with its negligible buoyant force) of adulterated gold would have a ***greater volume*** than an equal mass/weight (in air) of pure gold.

If the crown were made of pure gold, then, the buoyant forces acting on the gold and the crown would be equal. The weights of the gold and crown are equal. Therefore the submerged crown and submerged gold would be supported in static equilibrium by the *same force* on the ropes, and the balance would indicate "equal" (the indicator needle straight up). The goldsmith lives, the king is happy, Archimedes lives, everybody is happy.

If the crown is made of less-dense gold *alloy*, then its volume will be greater than that of pure gold. The buoyant force acting on it when submerged will therefore *also* be greater, so the tension in the string supporting it needed to keep it in static equilibrium will be smaller.

But this smaller tension then would fail to balance the *torque* exerted on the balance arms by the string attached to the gold, and the whole balance would rotate to the right, with the more dense gold sinking relative to the less dense crown. The balance needle would *not* read "equal". In the story, it didn't read equal. So sad – for the goldsmith.

Example 8.3.2: Testing the Crown II

Of course nowadays we don't do things with balance-type scales so often. More often than not we would use a *spring balance* to weigh something from a string. The good thing about a spring balance is that you can *directly read off the weight* instead of having to delicately balance some force or weight with masses in a counterbalance pan. Using such a balance (or any other accurate scale) we can measure and record the *density of pure gold* once and for all.

Let us imagine that we have done so, and discovered that:

$$\rho_{\rm Au} = 19300 \text{ kilograms/meter}^3 \tag{8.61}$$

For grins, please note that $\rho_{\rm Ag} = 10490 \text{ kilograms/meter}^3$. This is a bit over half the density of gold, so that adulterating the gold of the crown with 10% silver would have decreased its density by around 5%. If the mass of the crown was (say) a kilogram, the goldsmith would have stolen 100 grams – almost four ounces – of pure gold at the cost of 100 grams of silver. Even if he stole twice that, the 9% increase in volume would have been nearly impossible to directly observe in a worked piece. At that point the *color* of the gold would have been off, though. This could be remedied by adding *copper* ($\rho_{\rm Cu} = 8940 \text{ kilograms/meter}^3$) along with the silver. Gold-Silver-Copper all three alloy together, with silver whitening and yellowing the natural color of pure gold and copper reddening it, but with the two *balanced* one can create an alloy that is perhaps 10% *each* copper and silver that has almost exactly the same color as pure gold. This would *harden* and *strengthen* the gold of the crown, but you'd have to damage the crown to discover this.

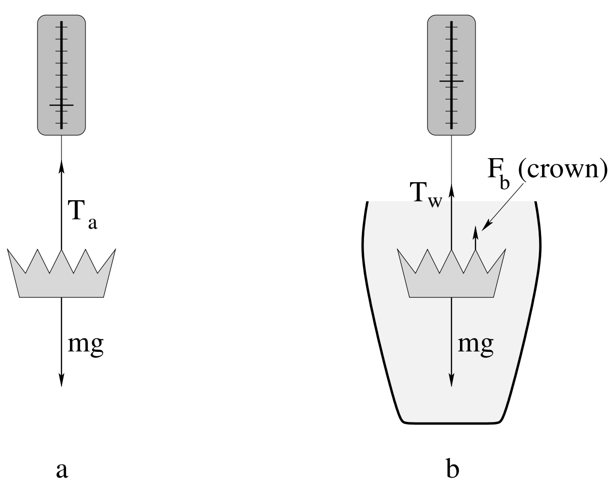

Figure 110: We now measure the effective weight of the *one* crown both in air (very close to its true weight) and in water, where the measured weight is *reduced* by the buoyant force.

Instead we hang the crown (of mass m) as before, but this time from a spring balance, both in air and in the water, recording both weights *as measured by the balance* (which measures, recall, the tension in the supporting string). This is illustrated in figure 110, where we note that the measured weights in a) and b) are T_a, the weight in air, and T_w, the measured weight while immersed in water.

Let's work this out. a) is simple. In static equilibrium:

$$\begin{aligned} T_a - mg &= 0 \\ T_a &= mg \\ T_a &= \rho_{\text{crown}} V g \end{aligned} \tag{8.62}$$

so the scale in a) just measures the almost-true weight of the crown (off by the buoyant force exerted by the *air* which, because the density of air is very small at $\rho_{\text{air}} \approx 1.2$ kilograms/meter^3, which represents around a 0.1% error in the measured weight of objects roughly the density of water, and an even smaller error for denser stuff like gold).

In b):

$$\begin{aligned} T_w + F_b - mg &= 0 \\ T_w &= mg - F_b \\ &= \rho_{\text{crown}} V g - \rho_{\text{water}} V g \\ &= (\rho_{\text{crown}} - \rho_{\text{water}}) V g \end{aligned} \tag{8.63}$$

We know (we *measured*) the values of T_a and T_w, but we don't know V or ρ_{crown}. We have two equations and two unknowns, and we would like most of all to solve for ρ_{crown}. To do so, we *divide these two equations by one another* to eliminate the V:

$$\frac{T_w}{T_a} = \frac{\rho_{\text{crown}} - \rho_{\text{water}}}{\rho_{\text{crown}}} \tag{8.64}$$

Whoa! g went away too! This means that from here on we don't even care what g is – we could make these weight measurements on the moon or on mars and we'd still get the *relative* density of the crown (compared to the density of water) right!

A bit of algebra-fu:

$$\rho_{\text{water}} = \rho_{\text{crown}}(1 - \frac{T_w}{T_a}) = \rho_{\text{crown}} \frac{T_a - T_w}{T_a} \tag{8.65}$$

or finally:

$$\rho_{\text{crown}} = \rho_{\text{water}} \frac{T_a}{T_a - T_w} \tag{8.66}$$

We are now prepared to be precise. Suppose that the color of the crown is very good. We perform the measurements above (using a scale accurate to *better* than a hundredth of a Newton or we might end up condemning our goldsmith due to a *measurement error!*) and find that $T_a = 10.00$ Newtons, $T_w = 9.45$ Newtons. Then

$$\rho_{\text{crown}} = 1000 \frac{10.00}{10.00 - 9.45} = 18182 \text{ kilograms/meter}^3 \tag{8.67}$$

We subtract, $19300 - 18182 = 1118$; divide, $1118/19300 \times 100 = 6$%. Our crown's material is around *six percent less dense than gold* which means that our clever goldsmith has adulterated the gold by removing some 12% of the gold (give or take a percent) and replaced it with some mixture of silver and copper. Baaaaad goldsmith, bad.

If the goldsmith were smart, of course, he could have beaten Archimedes (and us). What he needed to do is adulterate the gold with a mixture of metals that have *exactly the* *same density as gold!* Not so easy to do, but tungsten's density, $\rho_{\rm W} = 19300$ (to three digits) almost exactly matches that of gold. Alas, it has the highest melting point of all metals at 3684°K, is enormously hard, and might or might not alloy with gold or change the color of the gold if alloyed. It is also pretty expensive in its own right. Platinum, Plutonium, Iridium, and Osmium are all even denser then gold, but three of these are very expensive (even more expensive than gold!) and one is very explosive, a transuranic compound used to make nuclear bombs, enormously expensive *and* illegal to manufacture or own (and rather toxic as well). Not so easy, matching the density via adulteration and making a profit out of it...

Enough of all of this *fluid statics*. Time to return to some *dynamics*.

8.4: Fluid Flow

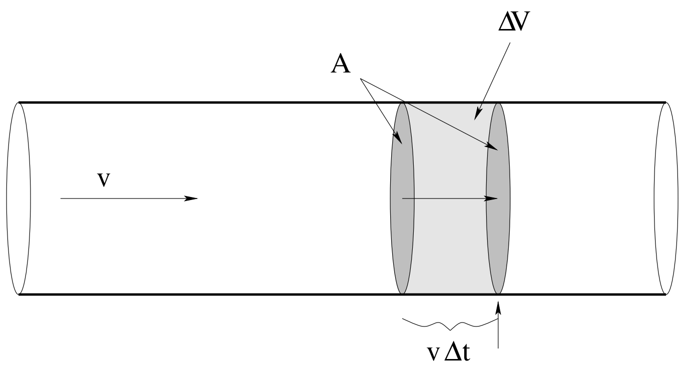

Figure 111: Fluid in uniform flow is transported down a pipe with a constant cross-section at a constant speed v. From this we can easily compute the *flow*, the volume per unit time that passes (through a surface that cuts the pipe at) a point on the pipe.

In figure 111 we see fluid flowing from left to right in a circular pipe. The pipe is assumed to be "frictionless" for the time being – to exert no drag force on the fluid flowing within – and hence all of the fluid is moving *uniformly* (at the same speed v with no relative *internal* motion) in a state of ***dynamic equilibrium***.

We are interested in understanding the ***flow*** or ***current*** of water carried by the pipe, which we will define to be the ***volume per unit time*** that passes any given point in the pipe. Note well that we could instead talk about the *mass per unit time* that passes a point, but this is just the volume per unit time times the density and hence for fluids with a more or less *uniform* density the two are the same within a constant.

For this reason we will restrict our discussion in the following to incompressible fluids, with constant ρ. This means that the concepts we develop will work gangbusters well for understanding water flowing in pipes, beer flowing from kegs, blood flowing in veins, and even rivers flowing slowly in not-too-rocky river beds but not so well to describe the dynamical evolution of weather patterns or the movement of oceanic currents. The *ideas* will still be extensible, but future climatologists or oceanographers will have to work a bit harder to understand the correct theory when dealing the compressibility.

We expect a "big pipe" (one with a large cross-sectional area) to carry more fluid per unit time, all other things being equal, than a "small pipe". To understand the relationship between area, speed and flow we turn our attention to figure 111. In a time Δt, all of the water within a distance $v\Delta t$ to the left of the second shaded surface (which is strictly imaginary – there is nothing actually in that pipe at that point but fluid) will pass *through* this surface and hence past the point indicated by the arrow underneath. The volume of this fluid is just the area of the surface times the height of the cylinder of water:

$$\Delta V = Av\Delta t \tag{8.68}$$

If we divide out the Δt, we get:

$$I = \frac{\Delta V}{\Delta t} = Av \tag{8.69}$$

This, then is the ***flow***, or ***volumetric current*** of fluid in the pipe.

This is an *extremely important relation*, but the picture and derivation itself is arguably even *more* important, as this is the first time – but ***not the last time*** – you have seen it, and it will be a crucial part of understanding things like ***flux*** and ***electric current*** in the second semester of this course. Physics and math majors will want to consider what happens when they take the quantity v and make it a *vector field* $\vec{v}$ that might *not* be flowing uniformly in the pipe, which might *not* have a uniform shape or cross section, and thence think still more generally to fluids flowing in arbitrary streamlined patterns. Future physicians, however, can draw a graceful curtain across these meditations for the moment, although they too will benefit next semester if they at least *try* to think about them now.

8.4.1: Conservation of Flow

Fluid does not, of course, only flow in smooth pipes with a single cross-sectional area. Sometimes it flows from large pipes into smaller ones or vice versa. We will now proceed to derive an important aspect of that flow for incompressible fluids and/or steady state flows of compressible ones.

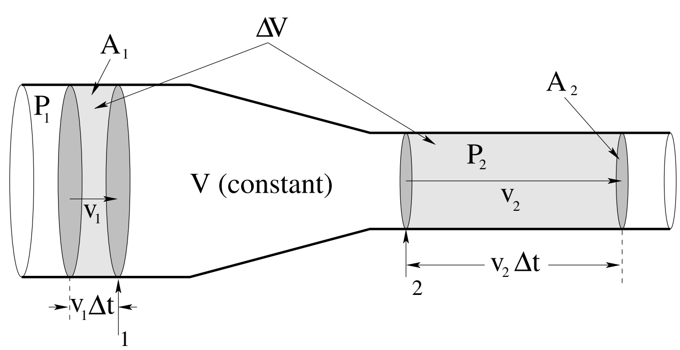

Figure 112: Water flows from a wider pipe with a "larger" cros-sectional area A_1 into a narrower pipe with a smaller cross-sectional area A_2. The speed of the fluid in the wider pipe is v_1, in the narrower one it is v_2. The pressure in the wider pipe is P_1, in the narrower one it is P_2.

Figure 112 shows a fluid as it flows from just such a wider pipe down a gently sloping neck into a narrower one. As before, we will ignore drag forces and assume that the flow is as uniform as possible as it narrows, while remaining completely uniform in the wider pipe and smaller pipe on either side of the neck. The pressure, speed of the (presumed incompressible) fluid, and cross sectional area for either pipe are P_1, v_1, and A_1 in the wider one and P_2, v_2, and A_2 in the narrower one.

Pay careful attention to the following reasoning. In a time Δt then – as before – a volume of fluid $\Delta V = A_1 v_1 \Delta t$ passes through the surface/past the point 1 marked with an arrow in the figure. In the volume between this surface and the next grey surface at the point 2 marked with an arrow ***no fluid can build up*** so actual quantity of mass in this volume must be a *constant*.

This is very important. The argument is simple. If more fluid flowed into this volume through the first surface than escaped through the second one, then fluid would be *building up* in the volume. This would increase the density. But the fluid's density *cannot* change – it is (by hypothesis) incompressible. Nor can *more* fluid escape through the second surface than enters through the first one.

Note well that this assertion implies that the fluid itself ***cannot be created or destroyed***, it can only flow *into* the volume through one surface and *out* through another, and because it is incompressible and uniform and the walls of the vessel are impermeable (don't leak) the quantity of fluid inside the surface cannot change in any other way.

This is a kind of *conservation law* which, for a continuous fluid or similar medium, is called a *continuity equation*. In particular, we are postulating the law of conservation of matter, implying a continuous flow of matter from one place to another! Strictly speaking, continuity alone would permit fluid to build up in between the surfaces (as this can be managed without creating or destroying the mass of the fluid) but we've trumped that by insisting that the fluid be incompressible.

This means that however much fluid *enters* on the left *must exit on the right* in the time Δt; the shaded volumes on the left and right in the figure above must be *equal*. If we write this out algebraically:

$$\begin{aligned} \Delta V &= A_1 v_1 \,\Delta t \;=\; A_2 v_2 \,\Delta t \\ I \;=\; \frac{\Delta V}{\Delta t} &= A_1 v_1 \;=\; A_2 v_2 \end{aligned} \tag{8.70}$$

Thus the current or flow through the two surfaces marked 1 and 2 must be the *same*:

$$A_1 v_1 = A_2 v_2 \tag{8.71}$$

Obviously, this argument would continue to work if it necked down (or up) further into a pipe with cross sectional area A_3, where it had speed v_3 and pressure P_3, and so on. The flow of water in the pipe must be *uniform*, $I = Av$ must be a *constant independent of where you are in the pipe!*

There are two more meaty results to extract from this picture before we move on, that combine into one "named" phenomenon. The first is that conservation of flow implies that the ***fluid speeds up*** when it flows from a wide tube and into a narrow one or vice versa, it slows down when it enters a wider tube from a narrow one. This means that every little chunk of mass in the fluid on the right is moving *faster* than it is on the left. The fluid has ***accelerated!***

Well, by now you should very well appreciate that *if* the fluid accelerates *then* there must be a net external force that acts on it. The only catch is, where is that force? What exerts it?

The force is exerted by the ***pressure difference*** $\boldsymbol{\Delta P}$ between P_1 and P_2. The force exerted by pressure at the walls of the container points only perpendicular to the pipe at that point; the fluid is moving parallel to the surface of the pipe and hence this "normal" confining force does no work and cannot speed up the fluid.

In a bit we will work out *quantitatively* how much the fluid speeds up, but even now we can see that since $A_1 > A_2$, it must be true that $v_2 > v_1$, and hence $P_1 > P_2$. This is a general result, which we state in words:

The pressure decreases in the direction that fluid velocity increases.

This might well be stated (in other books or in a paper you are reading) the other way: When a fluid slows *down*, the pressure in it *increases*. Either way the result is the same.

This result is responsible for many observable phenomena, notably the mechanism of the lift that supports a frisbee or airplane wing or the ***Magnus effect***Wikipedia: http://www.wikipedia.org that causes a spinning thrown baseball ball to curve.

Unfortunately, treating these phenomena *quantitatively* is beyond, and I do mean *way* beyond, the scope of this course. To correctly deal with lift for compressible or incompressible fluids one must work with and solve either the ***Euler equations***[154] , which are coupled partial differential equations that express Newton's Laws for fluids dynamically moving themselves in terms of the local density, the local pressure, and the local fluid velocity, or the ***Navier-Stokes Equations***[155] , ditto but *including the effects of viscosity* (neglected by Euler). Engineering students (especially those interested in aerospace engineering and real fluid dynamics) and math and physics majors are encouraged to take a peek at these articles, but not too long a peek lest you decide that perhaps majoring in advanced basket weaving really *was* the right choice after all. They are really, really difficult; on the high end "supergenius" difficult[156].

This isn't surprising – the equations have to be able to describe every possible dynamical state of a fluid as it flows in every possible environment – laminar flow, rotational flow, turbulence, drag, around/over smooth shapes, horribly not smooth shapes, and everything in between. At that, they don't account for things like temperature and the mixing of fluids and fluid chemistry – reality is more complex still. That's why we are stopping with the simple rules above – fluid flow is conserved (safe enough) and pressure decreases as fluid velocity increases, all things being equal.

All things are, of course, *not* always equal. In particular, one thing that can easily vary in the case of fluid flowing in pipes is *the height of the pipes*. The increase in velocity caused by a pressure differential can be interpreted or predicted by the Work-Kinetic Energy theorem, but if the fluid is moving *up or down hill* then we may discover that *gravity* is doing work as well!

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[154]Wikipedia: http://www.wikipedia.org/wiki/Euler Equations (fluid dynamics).

[155]Wikipedia: http://www.wikipedia.org/wiki/Navier-Stokes equations.

[156]To give you an idea of *how* difficult they are, note that there is a $1,000,000 prize just for showing that solutions to the 3 dimensional Navier-Stokes equations generally *exist* and/or are *not singular*.

In this case we should really use the Work-*Mechanical* Energy theorem to determine how pressure changes can move fluids. This is actually pretty easy to do, so let's do it.

8.4.2: Work-Mechanical Energy in Fluids: Bernoulli's Equation

Daniel Bernoulli was a third generation member of the famous Bernoulli family[157] who worked on (among many other things) fluid dynamics, along with his good friend and contemporary, Leonhard Euler. In 1738 he published a long work wherein he developed the idea of energy conservation to fluid motion. We'll try to manage it in a page or so.

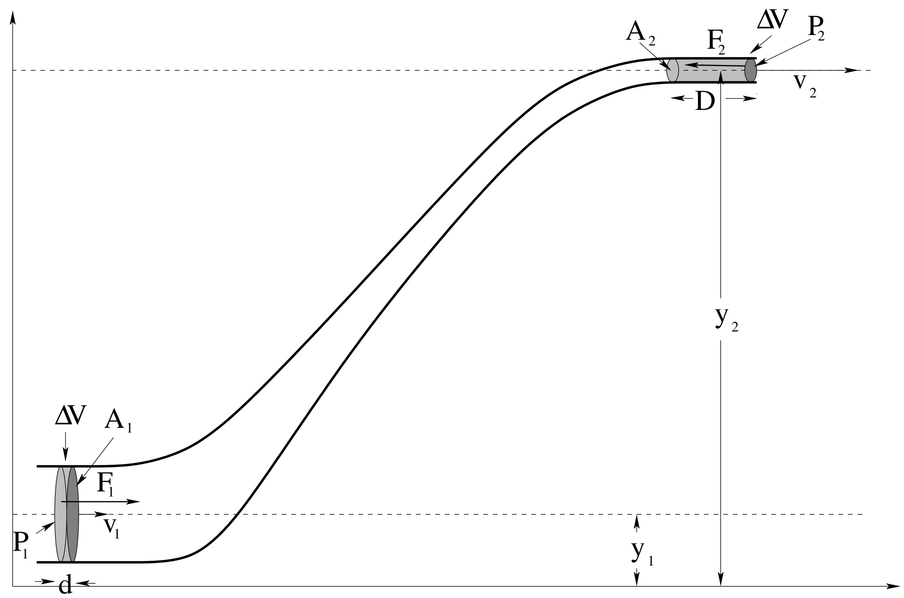

Figure 113: A circular cross-sectional necked pipe is arranged so that the pipe *changes height* between the larger and smaller sections. We will assume that both pipe segments are narrow compared to the height change, so that we don't have to account for a potential energy difference (per unit volume) between water flowing at the top of a pipe compared to the bottom, but for ease of viewing we do not draw the *picture* that way.

In figure 113 we see the same pipe we used to discuss conservation of flow, only now it is bent uphill so the 1 and 2 sections of the pipe are at heights y_1 and y_2 respectively. This really is the only change, otherwise we will recapitulate the same reasoning. The fluid is incompressible and the pipe itself does not leak, so fluid cannot build up between the bottom and the top. As the fluid on the bottom moves to the left a distance d (which might be $v_1\Delta t$ but we don't insist on it as rates will not be important in our result) exactly the same amount fluid must move to the left a distance D up at the top so that fluid is conserved.

The total *mechanical* consequence of this movement is thus the disappearance of a chunk of fluid mass:

$$\Delta m = \rho\Delta V = \rho A_1 d = \rho A_2 D \tag{8.72}$$

[157]Wikipedia: http://www.wikipedia.org/wiki/Bernoulli family. The Bernoullis were in on many of major mathematical and physical discoveries of the eighteenth and nineteenth century. Calculus, number theory, polynomial algebra, probability and statistics, fluid dynamics – if a theorem, distribution, principle has the name "Bernoulli" on it, it's gotta be good...

that is moving at speed v_1 and at height y_1 at the bottom and it's appearance moving at speed v_2 and at height y_2 at the top. Clearly ***both*** the kinetic energy ***and*** the potential energy of this chunk of mass have changed.

What caused this change in mechanical energy? Well, it can only be *work*. What does the work? The walls of the (frictionless, drag free) pipe can do no work as the only force it exerts is perpendicular to the wall and hence to $\vec{v}$ in the fluid. The only thing left is the ***pressure*** that acts on the entire block of water between the first surface (lightly shaded) drawn at both the top and the bottom as it moves forward to become the second surface (darkly shaded) drawn at the top and the bottom, effecting this net transfer of mass Δm.

The force F_1 exerted to the *right* on this block of fluid at the bottom is just $F_1 = P_1 A_1$; the force F_2 exerted to the *left* on this block of fluid at the top is similarly $F_2 = P_2 A_2$. The work done by the pressure acting over a distance d at the bottom is $W_1 = P_1 A_1 d$, at the top it is $W_2 = -P_2 A_2 D$. The total work is equal to the total change in mechanical energy of the chunk Δm:

$$\begin{aligned}
W_{\rm tot} &= \Delta E_{\rm mech} \\
W_1 + W_2 &= E_{\rm mech}(final) - E_{\rm mech}(initial) \\
P_1 A_1 d - P_2 A_2 D &= (\frac{1}{2}\Delta m v_2^2 + \Delta m g y_2) - (\frac{1}{2}\Delta m v_1^2 + \Delta m g y_1) \\
(P_1 - P_2)\Delta V &= (\frac{1}{2}\rho \Delta V v_2^2 + \rho \Delta V g y_2) - (\frac{1}{2}\rho \Delta V v_1^2 + \rho \Delta V g y_1) \\
(P_1 - P_2) &= (\frac{1}{2}\rho v_2^2 + \rho g y_2) - (\frac{1}{2}\rho v_1^2 + \rho y_1) \\
P_1 + \frac{1}{2}\rho v_1^2 + \rho y_1 &= P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2 \;=\; \text{a constant (units of pressure)} \qquad (8.73)
\end{aligned}$$

There, that wasn't so difficult, was it? This lovely result is known as ***Bernoulli's Principle*** (or the Bernoulli fluid equation). It contains pretty much *everything* we've done so far except conservation of flow (which is a distinct result, for all that we used it in the derivation) and Archimedes' Principle.

For example, if $v_1 = v_2 = 0$, it describes a static fluid:

$$P_2 - P_1 = -\rho g (y_2 - y_1) \qquad (8.74)$$

and if we change variables to make z (depth) $-y$ (negative height) we get the familiar:

$$\Delta P = \rho g \Delta z \qquad (8.75)$$

for a static incompressible fluid. It also not only tells us that pressure drops where fluid velocity increases, it tells us *how much* the pressure drops when it increases, allowing for things like the fluid flowing up or downhill at the same time! Very powerful formula, and all it is is the Work-Mechanical Energy theorem (per unit volume, as we divided out ΔV in the derivation, note well) applied to the fluid!

Example 8.4.1: Emptying the Iced Tea

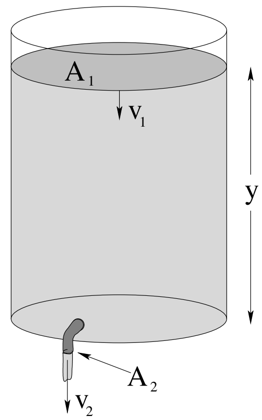

Figure 114:

In figure 114 above, a cooler full of iced tea is displayed. A tap in the bottom is opened, and the iced tea is running out. The cross-sectional area of the top surface of the tea (open to the atmosphere) is A_1. The cross-sectional area of the tap is $A_2 \ll A_1$, and it is also open to the atmosphere. The depth of iced tea (above the tap at the bottom) is y. The density of iced tea is basically identical to that of water, ρ_w. Ignore viscosity and resistance and drag.

What is the speed v_2 of the iced tea as it exits the tap at the bottom? How rapidly is the top of the iced tea descending at the top v_1? What is the rate of flow (volume per unit time) of the iced tea when the height is y?

This problem is fairly typical of Bernoulli's equation problems. The two concepts we will need are:

a) Conservation of flow: $A_1 v_1 = A_2 v_2 = I$

b) Bernoulli's formula: $P + \rho g y + \frac{1}{2}\rho v^2 = \mathrm{constant}$

First, let's write Bernoulli's formula for the top of the fluid and the fluid where it exits the tap. We'll choose $y = 0$ at the height of the tap.

$$P_0 + \rho g y + \frac{1}{2}\rho v_1^2 = P_0 + \rho g(0) + \frac{1}{2}\rho v_2^2 \tag{8.76}$$

We have two unknowns, v_1 and v_2. Let's eliminate v_1 in favor of v_2 using the flow equation and substitute it into Bernoulli.

$$v_1 = \frac{A_2}{A_1} v_2 \tag{8.77}$$

so (rearranging):

$$\rho g y = \frac{1}{2}\rho v_2^2 \left\{ 1 - \left(\frac{A_2}{A_1}\right)^2 \right\} \tag{8.78}$$

At this point, since $A_1 \gg A_2$ we will often want to approximate:

$$\left(\frac{A_2}{A_1}\right)^2 \approx 0 \tag{8.79}$$

and solve for

$$v_2 \approx \sqrt{2gy} \tag{8.80}$$

but it isn't *that* difficult to leave the factor in.

This latter result (equation 8.80) is known as *Torricelli's Law*. Torricelli is known to us as the inventor of the barometer, as a moderately famous mathematician, and as the final secretary of Galileo Galilei in the months before his death in 1642 – he completed the last of Galileo's dialogues under Galileo's personal direction and later saw to its publication. It basically states that a fluid exits any (sufficiently large) container through a (sufficiently small) hole at the same speed a rock would have if dropped from the height from the top of the fluid to the hole. This was a profound observation of gravitational energy conservation in a context quite different from Galileo's original observations of the universality of gravitation.

Given this result, it is now trivial to obtain v_1 from the relation above (which should be quite accurate even if you assumed the ratio to be zero initially) and compute the rate of flow from e.g. $I = A_2 v_2$. An interesting exercise in calculus is to estimate the time required for the vat of iced tea to empty through the lower tap if it starts at initial height y_0. It requires realizing that $\frac{dy}{dt} = -v_1$ and transforming the result above into a differential equation. Give it a try!

Example 8.4.2: Flow Between Two Tanks

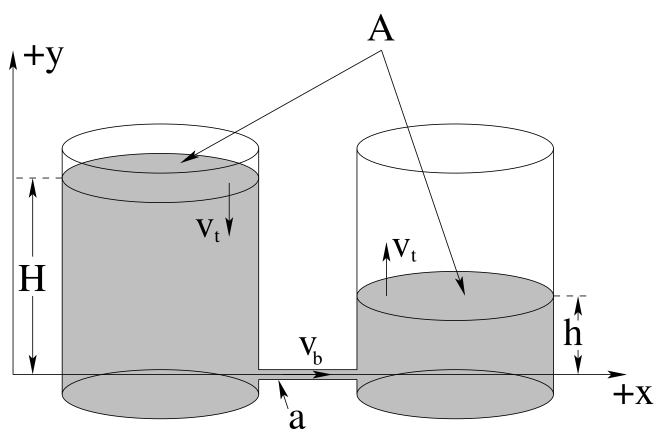

Figure 115:

In figure 115 two water tanks are filled to different heights. The two tanks are connected at the bottom by a narrow pipe through which water (density ρ_w) flows *without resistance* (see the next section to understand what one might do to include resistance in this pipe, but for the moment this will be considered too difficult to include in an introductory course). Both tanks are open to ordinary air pressure P_0 (one atmosphere) at the top. The cross sectional area of both tanks is A and the cross sectional area of the pipe is $a \ll A$.

Once again we would like to know things like: What is the speed v_b with which water flows through the small pipe from the tank on the left to the tank on the right? How fast does the water level of the tank on the left/right fall or rise? How long would it take for the two levels to become equal, starting from heights H and h as shown?

As before, we will need to use:

a) Conservation of flow: $A_1 v_1 = A_2 v_2 = I$

b) Bernoulli's formula: $P + \rho g y + \frac{1}{2}\rho v^2 = \mathrm{constant}$

This problem is actually more difficult than the previous problem. If one naively tries to express Bernoulli for the top of the tank on the left and the top of the tank on the right, one gets:

$$P_0 + \rho g H + \frac{1}{2}\rho v_t^2 = P_0 + \rho g h + \frac{1}{2}\rho v_t^2 \tag{8.81}$$

Note that we've equated the speeds *and* pressures on both sides because the tanks have equal cross-sectional areas so that they have to be the same. But this makes no sense! $\rho g H \neq \rho g h$!

The problem is this. The pressure is not constant across the pipe on the bottom. If you think about it, the pressure at the bottom of a nearly static fluid column on the left (just outside the mouth of the pipe) has to be approximately $P_0 + \rho g H$ (we can and will do a bit better than this, but this is what we expect when $a \ll A$). The pressure just outside of the mouth of the pipe in the fluid column on the right must be $P_0 + \rho g h$ from the same argument. *Physically*, it is this pressure difference that forces the fluid through the pipe, speeding it up as it enters on the left. The pressure in the pipe has to *drop* as the fluid speeds up entering the pipe from the bottom of the tank on the left (the Venturi effect).

This suggests that we write Bernoulli's formula for the top of the left hand tank and a point just inside the pipe at the bottom of the left hand tank:

$$P_0 + \rho g H + \frac{1}{2}\rho v_t^2 = P_p + \rho g(0) + \frac{1}{2}\rho v_b^2 \tag{8.82}$$

This equation has three unknowns: v_t, v_b, and P_p, the pressure just inside the pipe. As before, we can easily eliminate e.g. v_t in favor of v_b:

$$v_t = \frac{a}{A} v_b \tag{8.83}$$

so (rearranging):

$$P_0 - P_p + \rho g H = \frac{1}{2}\rho v_b^2 \left\{ 1 - \left(\frac{a}{A}\right)^2 \right\} \tag{8.84}$$

Just for fun, this time we won't approximate and throw away the a/A term, although in most cases we could and we'd never be able to detect the perhaps 1% or even less difference.

The problem now is: What is P_p? That we get on the other side, but not the way you might expect. Note that the pressure must be *constant* all the way across the pipe as neither y nor v can change. The pressure in the pipe must therefore match the pressure at the bottom of the other tank. But that pressure is just $P_0 + \rho g h$! Note well that this is completely consistent with what we did for the iced tea – the pressure at the outflow had to match the air pressure in the room.

Substituting, we get:

$$\rho g H - \rho g h = \frac{1}{2}\rho v_b^2 \left\{ 1 - \left(\frac{a}{A}\right)^2 \right\} \tag{8.85}$$

or

$$v_b = \sqrt{\left\{ \frac{2g(H-h)}{1 - \left(\frac{a}{A}\right)^2} \right\}} \tag{8.86}$$

which ***makes sense!*** What pushes the fluid through the pipe, speeding it up along the way? The difference in pressure between the ends. But the pressure inside the pipe itself has to match the pressure at the outflow because it has *already accelerated to this speed* across the tiny distance between a point on the bottom of the left tank "outside" of the pipe and a point just inside the pipe.

From v_b one can as before find v_t, and from v_t one can do some calculus to at the very least write an integral expression that can yield the time required for the two heights to come to equilibrium, which will happen when $H - \int v_t(t)\ dt = h + \int v_t(t)\ dt$. That is, the rate of change of the *difference* of the two heights is twice the velocity of either side.

Note well: This example is also highly relevant to the way a ***siphon*** works, whether the siphon empties into air or into fluid in a catch vessel. In particular, the pressure drops at the intake of the siphon to match the height-adjusted pressure at the siphon outflow, which could be in air or fluid in the catch vessel. The main difference is that the siphon tube is *not horizontal*, so according to Bernoulli the pressure has to increase or decrease as the fluid goes up and down in the siphon tube at constant velocity.

Questions: Is there a maximum height a siphon can function? Is the maximum height measured/determined by the intake side or the outflow side?

8.4.3: Fluid Viscosity and Resistance

In the discussion above, we have consistently ignored viscosity and drag, which behave like "friction", exerting a force *parallel* to the confining walls of a pipe in the direction opposite to the relative motion of fluid and pipe. We will now present a very simple (indeed, oversimplified!) treatment of viscosity, but one that (like our similarly oversimplified treatment of static and kinetic friction) is sufficient to give us a good conceptual understanding of what viscosity is and that will work well quantitatively – until it doesn't.

In figure 116 above a two horizontal plates with cross-sectional areas A have a layer of fluid with thickness d trapped between them. The plates are assumed to be very large, with a comparatively thin layer of fluid in between, so that we can neglect what happens near their edges. The bottom plate is fixed in our (inertial) frame of reference and the upper plate is moving to the right, pushed by a *constant* force F so that it maintains a ***constant speed*** v against the drag force exerted on it by the viscous fluid.

Many fluids, especially "wet" liquids like water (a polar molecule) strongly interact with the solid wall and basically stick to it at very short range – this is known as the "no-slip

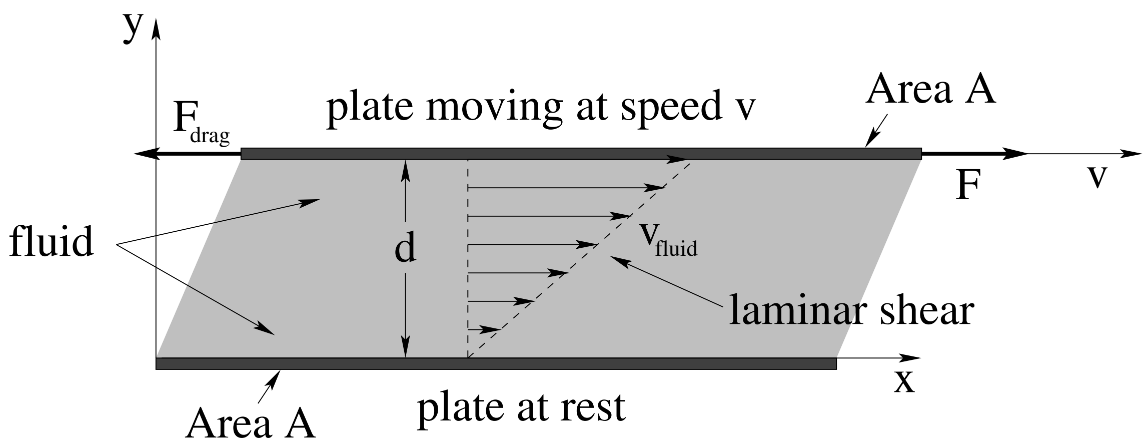

Figure 116: Dynamic viscosity is defined by the scaling of the force F required to keep a plate of cross-sectional area A moving at constant speed v on top of a layer of fluid of thickness d. This causes the fluid to ***shear***. Shear stress is explained below and in more detail in the chapter on oscillations.

condition". This means that a thin layer (a few molecules thick) of the fluid in "direct contact" with the upper or lower plates will generally not be moving *relative* to the plate it is touching. That is, in the figure the fluid is at rest where it touches the bottom plate, and is moving at speed v (in the x-direction) where it touches the top plate. In between the speed of the fluid must vary from 0 at the bottom to v at the top.

When the speed v is not too large (and various other parametric stars are in alignment), many fluids can be treated as ***Newtonian fluids***: layers of fluid such that each layer is going a bit faster than the layer below it and a bit slower than the one above it all the way from the bottom fixed layer all the way up to the top layer stuck to the moving plate. As discussed in week 2, this is called ***laminar flow***, and results in this simple case in the *linear* velocity profile illustrated in 116 above.

Consider two adjacent layers in the fluid with an imaginary surface separating them. The lower layer is moving more slowly than the upper one and hence there is "dynamic friction" between the two layers across the surface. This friction pulls *forward* (in the direction of the movement of the top plate) on the lower layer and *backward* on the upper layer. For each differentially thin layer in between the plates, then, there is a force pulling it forward from the layer above it and a force pulling it backwards from the layer below it, and in dynamic equilibrium (where the layers are all moving at their own constant speeds) these forces have to be *equal and opposite*.

We see that the total force pulling each layer *forward* has to *strictly increase* as one moves from the bottom plate/layer to the top plate/layer. To keep the top plate moving at a constant speed, it has to be pushed *forward* by an external force equal to the ***total (shear) force accumulated across all the layers*** that makes up the drag force acting opposite to its velocity relative to the bottom plate.

For a *Newtonian fluid*[158] it is empirically observed that the (shear) force required to

[158]Wikipedia: http://www.wikipedia.org/wiki/Newtonian Fluid. More advanced students will want to note that the formula presented here is just a part of a much more general partial differential equation for the *shear*

keep the top plate moving at speed v in the x direction relative to the fixed bottom plate in figure 116 is:

$$F = \mu A \frac{v}{d} \tag{8.87}$$

This formula incidentally ***defines*** the constant of proportionality μ, characteristic of the fluid, called the ***dynamic viscosity***. It is the moral equivalent of the "coefficient of kinetic friction" between fluid layers. However, if we divide the force F by the area A in contact we get the ***shear stress*** F/A and that this has SI *units of pressure*, pascals, but is ***not*** a pressure. The shear force F is directed *parallel* to the surface area A and not perpendicular to it. The units of v/d are inverse time. Consequently, the SI units of μ are pascal-seconds, unlike the coefficients of friction which are dimensionless.

This expression gives the *total* force transmitted from the bottom plate to the top plate through all of the layers of the fluid in between. Since the geometry of the figure is very simple, we expect the force to be distributed more or less *linearly* across the many layers in between. We express this assumption by looking at two layers of fluid, one below and one above, as drawn in figure 117.

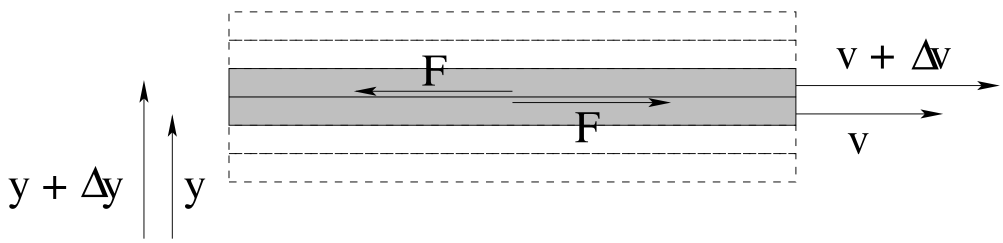

Figure 117: Two layers of fluid moving at slightly different velocities exert an equal and opposite force on each other at the boundary in between.

In order to get the empirical result above, we expect that the magnitude of the force between the layers is given by:

$$F = \mu A \frac{\Delta v}{\Delta y} \tag{8.88}$$

Note well that this is not the *net* force on a layer! The lower (shaded) layer in the figure above acts to slow the (shaded) layer just above, dragging it backwards, as the upper layer acts to drage the lower forward. But the layer *above* the upper layer *also* acts on the upper layer with the exact same magnitude of force, dragging it forward. The total *vector* force on a layer is thus *zero*, as must be the case for the layers in the fluid to have constant speed in dynamic equilibrium!

We would like to be able to use this expression for fluid layers that are not simple plane sheets between two plates, where we can't guess that the velocity profile of the fluid is a simple linear variation from one boundary surface to another. In that case, we expect that – subject to various conditions, such as the areas in contact being at least *locally* flat – the magnitude of the force between neighboring layers is proportional to the (partial) *derivative* of the fluid velocity in the direction of flow with respect to the direction perpendicular to the

stress tensor, and that the math required to do fluid dynamics correctly is not at all as simple as it might appear from these idealized cases.

flow:

$$F = \mu A \frac{\partial v}{\partial y} \tag{8.89}$$

This expression isn't quite correct – it is a single part of a more general tensor form – but it will do for the next section, where we give a simple derivation of the velocity profile and drag resistance of a circular pipe.

8.4.4: Resistance of a Circular Pipe: Poiseuille's Equation

The case of greatest interest to many of the students taking this course is not, actually, the drag force between parallel plates separated by a viscous fluid. It is the case of a viscous fluid in laminar flow in a ***pipe with a circular cross-section***. This is an excellent model for water flowing in ordinary pipes as well as blood flowing in arteries and veins in living organisms. It is thus useful to engineers and life science students alike, and of course physics or math majors should become as familiar as possible with *all* simple systems of fluid dynamics in preparation to tackling (eventually) the fiendish difficulty of the Navier-Stokes equation.

Note Well!

This section is split at the line below (as are several others in this textbook) into two parts. In the first, we explicitly step through at least one (comparatively simple, believe it or not) derivation of the Hagen-Poiseuille equation[159]. After the *second* separator line we will start *with* the Poiseuille equation as a given result and observe its general scaling and how to apply it without worrying about exactly where it comes from.

Physics and math majors (and perhaps some engineering students interested in fluid engineering, aerodynamics, and so on) *should* at least read through and study the actual derivation, and *all* students are welcome to take a look, but *all* students (even engineers and physics majors) can skip all of the content between the two separator lines and start reading again with the conceptual/application part that follows and still do perfectly well on the actual problems.

As was the case in our discussion of shear stress and viscosity above, we expect a viscous fluid moving at a reasonably low speed in a circular pipe to stick to the walls, creating a boundary layer of fluid that does not move. We will therefore make this "no-slip" condition a boundary condition of our problem. Unlike the shear stress example above, however, there is no possibility of moving two surfaces to maintain a laminar velocity profile in between, as there is only *one* surface bounding the fluid flowing in a pipe, at rest!

Let's assume that the fluid is moving slowly enough that laminar flow is established inside the pipe in steady state. The fluid touching the walls is not moving from the no-slip

[159]For the record, Poiseuille is correctly pronounced "Pwa-zoo-ee" – pwazooee. Not "poise-you-ill" or any of the myriad of other ways this might be interpreted by English-speaking students. And no, I didn't get it right either and had to use the Internet to look it up...

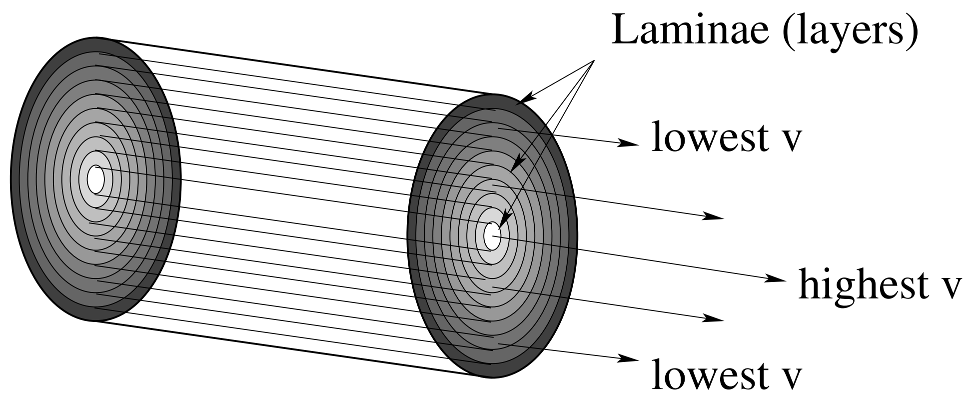

Figure 118: Cross-section of laminar flow in a pipe, where the darker shades correspond to slower-moving fluid in concentric layers (laminae) from the wall of the pipe (slowest speed) to the center (highest speed).

condition, so there must be an *annular velocity profile* where each annular layer is moving a bit faster as one moves in from the edges from the pipe to the center. Figure 118 can help you visualize this sort of laminar flow where the highest speed of the fluid is found in the center of the pipe, the greatest possible distance from the walls.

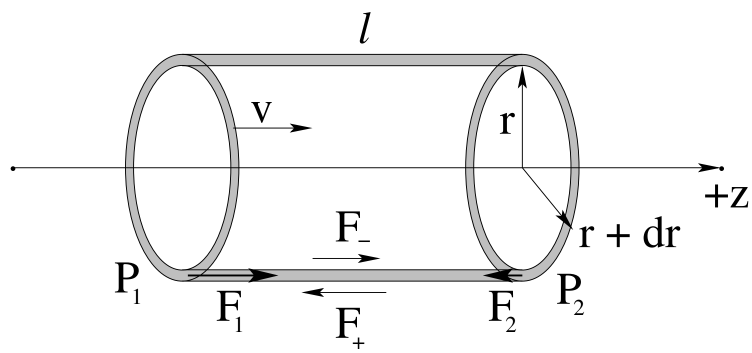

Figure 119: A differentially thin annular cylindrical shell of the fluid in the pipe. The pressure difference across the ends of the shell exerts a net force to the right. The differential viscous force between the inside and out surfaces of the shell exerts a net drag force on the shell. These must balance, maintaining a constant speed for the laminar shell v.

Here we cannot assume that the velocity profile is linear with the distance from the pipe walls, and in fact it turns out *not* to be linear. We have to apply the force rule we deduced for shear forces between layers in the previous section to try to obtain a differential equation representing Newton's Second Law for an *arbitrary* thin annular layer of fluid at radius r and with thickness dr. This is just *one* of the many annular cylindrical shell's of fluid portrayed above and is illustrated in figure 119.

In order for this laminar shell to move through the pipe in the direction indicated with a contant speed v (which so far is an unknown function of radius r) the forces on it must balance. There are two distinct sources of force that push on the shell. One is there must be a *pressure difference* between the pressure P_1 at the left side of the shell and P_2 at the right side of the shell, with $P_1 > P_2$, pushing the shell to the right. The other is the drag force due to viscosity acting on both the inner (-) and outer (+) surfaces of the layer, at the radii r and $r + dr$ respectively. We expect the force on the *inner* layer to be *in* the direction of fluid flow as the fluid is moving faster than the fluid in the shell there, and the force on the outer layer to be *against* the direction of fluid flow as the fluid outside of the shell is moving more slowly than the fluid in the shell.

We will use the differential form of the formula we deduced in the previous section to describe the drag forces between the surfaces of the shell and the surrounding fluid. The area of the inside of the annular cylinder in contact with the faster moving fluid towards the center is $2\pi r\ell$. We therefore expect the drag force due to viscosity acting on this layer to be:

$$F_{-} = -\mu(2\pi r\ell)\left.\frac{dv}{dr}\right|_{r} \tag{8.90}$$

Note that with the minus sign this force points towards the *right* – in the direction of flow – as expected because dv/dr is *negative*, with the highest speed at $r = 0$ in the middle.

The layer just *outside* of the illustrated laminar shell also exerts a drag force on our annular cylinder. The fluid there is moving more slowly, so it *slows it down*. We can use exactly the same relation to determine this force, but the area in contact is slightly differentially larger and the derivative has to be evaluated at a slightly (differentially) larger value of r:

$$F_{+} = \mu(2\pi(r + dr)\ell)\left.\frac{dv}{dr}\right|_{r+dr} \tag{8.91}$$

Again, dv/dr is negative, so this (z-directed) force points to the left as expected.

Finally, there are the two forces $F_1 = P_1 dA$ and $F_2 = P_2 dA$ acting on the ends of the shell due to the pressure. The cross-sectional area of the cylindrical shell is $dA = 2\pi r dr$ (its circumference times its thickness) so:

$$F_{\Delta P} = F_1 - F_2 = (P_1 - P_2)dA = \Delta P(2\pi r dr) \tag{8.92}$$

Assembling the entire expression, force balance is thus:

$$\Delta P(2\pi r dr) - \mu(2\pi r\ell)\left.\frac{dv}{dr}\right|_{r} + \mu(2\pi(r + dr)\ell)\left.\frac{dv}{dr}\right|_{r+dr} = 0 \tag{8.93}$$

So far, this has been pretty straightforward. Now things get a bit tricky. We use a Taylor series expansion for the derivative in the last term so we can evaluate all the derivatives in question at r only:

$$\left.\frac{dv}{dr}\right|_{r+dr} = \left.\frac{dv}{dr}\right|_{r} + \left.\frac{d^2v}{dr^2}\right|_{r} dr \tag{8.94}$$

or:

$$\Delta P(2\pi r dr) - \mu(2\pi r\ell)\frac{dv}{dr} + \mu(2\pi(r + dr)\ell)\left\{\frac{dv}{dr} + \frac{d^2v}{dr^2}dr\right\} = 0 \tag{8.95}$$

where all derivatives are evaluated at r so we no longer need to indicate the limits explicitly.

Two terms cancel, and we end up with:

$$\Delta P(2\pi r dr) + \mu(2\pi\ell)\frac{dv}{dr}dr + \mu(2\pi r\ell)\frac{d^2v}{dr^2}dr + \mu(2\pi\ell)\frac{d^2v}{dr^2}(dr)^2 = 0 \tag{8.96}$$

We then divide out the common factors of $2\pi r dr$ and throw out the last term as it vanishes like dr. Finally we rearrange a bit to get:

$$-\frac{1}{\mu}\frac{\Delta P}{\ell} = \left\{\frac{1}{r}\frac{dv}{dr} + \frac{d^2v}{dr^2}\right\} = \frac{1}{r}\frac{d}{dr}\left(r\frac{dv}{dr}\right) \tag{8.97}$$

The right hand side is (as it turns out) the radial part of the Laplacian in cylindrical coordinates, so this is a form of the Poisson (inhomogeneous Laplace) equation. Integrating this is beyond the scope of this course but is straightforward and taught in more advanced math and physics classes. We will simply jump to the result. If we integrate this differential equation subject to the boundary conditions $v(R) = 0$ for a pipe of radius R, we get:

$$v(r) = \frac{1}{4\mu}\frac{\Delta P}{\ell}\left(R^2 - r^2\right) \tag{8.98}$$

We can write this one last way:

$$v(r) = (1/4\mu)R^2\frac{\Delta P}{\ell}\left\{1 - \left(\frac{r}{R}\right)^2\right\} = v_{\rm max}\left\{1 - \left(\frac{r}{R}\right)^2\right\} \tag{8.99}$$

where $v_{\rm max}$ is the maximum speed of the fluid in the center of the pipe.

This is a *quadratic* profile to the velocity cross-section, peaked in the center, quite different from the linear profile we expected from symmetry if nothing else for two flat plates.

We are finally in position to get Poiseuille's Equation and determine the ***resistance*** to flow of a circular pipe carrying a fluid as a function of its parameters. Note well that the volume of fluid being carried from left to right per unit time – the *total volumetric current* dI or the *volumetric flow* of the annular shell – is just this speed times the cross-sectional area:

$$dI = \frac{dV}{dt} = vdA = (1/4\mu)R^2\frac{\Delta P}{\ell}\left\{1 - \left(\frac{r}{R}\right)^2\right\}2\pi r dr \tag{8.100}$$

We can easily integrate this from 0 to R to find the total current:

$$I = \int dI = \frac{\pi R^2}{2\mu}\frac{\Delta P}{\ell}\int_0^R\left(r - \frac{r^3}{R^2}\right)dr = \frac{\pi R^2}{2\mu}\frac{\Delta P}{\ell}\frac{R^2}{4} = \frac{\pi R^4}{8\mu}\frac{\Delta P}{\ell} \tag{8.101}$$

This equation relates the volumetric current in a circular pipe to the pressure difference one must maintain across the pipe in order to overcome the drag force between the walls of the pipe and the moving fluid. This concludes this section.

In the previous, possibly omitted, section we saw that – after a great deal of work – we could show that for a *circular pipe of radius r and length ℓ, carrying a fluid with dynamic viscosity μ in laminar flow*:

$$\Delta P = I\frac{8\mu\ell}{\pi r^4} \tag{8.102}$$

where I is the volumetric current, ΔP is the pressure difference across the pipe. Note well that we switched variables from R being the radius of the pipe to r being the radius in order to not confuse it with the pipe's resistance below. This is an example of a more general relation that holds for *any* pipe with a uniform, given, cross-section:

$$\Delta P = IR \tag{8.103}$$

where R is called the ***resistance*** of the pipe. Obviously,

$$R = \frac{8\mu\ell}{\pi r^4} \tag{8.104}$$

for a normal circular pipe (or vein, or artery, or hydraulic fluid line, or...).

This expression is the analog, in fluid mechanics, of *Ohm's Law* in electrical circuits. The pressure difference across the pipe is directly proportional to the rate of fluid flow. Double the pressure difference and you get twice as much fluid out of a given pipe (all things being equal, especially the maintenance of laminar flow). However, for a *fixed* pressure difference, you can also get more or less water out of a pipe by varying its parameters and hence its resistance. If you double the length of a pipe, this distributes the pressure difference over twice as much pipe so you expect to get half of the flow. If you increase the viscosity – the "stickiness" of the fluid – you expect to get less fluid through the pipe for a given pressure difference. Finally, if you reduce the cross-sectional area of a pipe, you expect to get less fluid through it for a given length, viscosity, and pressure difference.

The only real *surprise* in this is that the resistance goes down like the area of the pipe *squared* (like the fourth power of r). This is really quite unexpected, and arises because the velocity of the fluid in the pipe is not uniform over the cross-sectional area. When you double r you do indeed double the area (and so expect on that basis alone a doubling of the volume), but you also double the maximum velocity of the fluid in the pipe, which results in a *second* doubling!

Equation 8.102 is known as ***Poiseuille's Law*** and is a key relation for physicians, plumbers, physicists, and engineers to know because it describes both *flow of water in pipes* and the *flow of blood in blood vessels* whereever the flow is slow enough that it is laminar and not turbulent (which is actually "mostly", so that the expression is *useful*).

8.4.5: Turbulence

Although turbulence has been *observed* by humans from prehistoric times to the present, the systematic observation and description of turbulence begins with none other than Leonardo da Vinci. Da Vinci observed and sketched the turbulent flow that results when a stream of water falls into a pool, and noted that in the chaos where the two met there were eddies[160] of all sizes, from large ones down to tiny microscopic ones. He noted that large objects responded only to the large rotations, but that tiny objects were swept along and rotated by eddies of all sizes, eddies within eddies within eddies. Da Vinci named this phenomena "turbolenza" – hence our modern term for it.

We have already seen (in week/chapter 2) that there is a dramatic shift in the drag force exerted on an object by a turbulent fluid compared to that of a fluid in laminar flow.

[160] An "eddy" is also known as a "whirlpool" or "vortex" – a volume of water that can be almost any size that is *rotating* around a central line of some finite length that may not be straight.

In equation 8.87 above, we obtained a justification of sorts for a linear dependence of drag on relative fluid velocity as long as the velocity profile in the fluid near the surface was approximately linear. Of course for fluid flowing in a circular pipe this profile was *not* linear, but if we look more carefully at the form of the force between layers adjacent layers, as long as dy is perpendicular to dv_x:

$$dF = \mu A \frac{dv_x}{dy} dy = \mu A dv_x \tag{8.105}$$

You will not be responsible for the formulas or numbers in this section, but you should be conceptually aware of the *phenomenon* of the "onset of turbulence" in fluid flow. If we return to our original picture of laminar flow between two plates above, consider a small chunk of fluid somewhere in the middle. We recall that there is a (small) shear force across this constant-velocity chunk, which means that there is a drag force to the right at the top and to the left at the bottom. These forces form a ***force couple*** (two equal and opposite forces that do not act along the same line) which exerts a ***net torque on the fluid block***.

Here's the pretty problem. If the torque is *small* (whatever that word might mean relative to the parameters such as density and viscosity that describe the fluid) then the fluid will ***deform continually*** as described by the laminar viscosity equations above, without actually rotating. The fluid will, in other words, shear instead of rotate in response to the torque. However, as one increases the shear (and hence the velocity gradient and peak velocity) one can get to where the torque across some small cube of fluid causes it to rotate faster than it can shear! Suddenly small *vortices* of fluid appear throughout the laminae! These tiny whirling tubes of fluid have axes (generally) perpendicular to the direction of flow and *add a chaotic, constantly changing structure to the fluid*.

Once tiny vortices start to appear and reach a certain size, they rapidly grow with the velocity gradient and cause a change in the *character* of the drag force coupling across the fluid. There is a dimensionless scale factor called the ***Reynolds Number*** $\boldsymbol{Re}$[161] that has a certain characteristic value (different for different pipe or plate geometries) at the point where the vortices start to grow so that the fluid flow becomes ***turbulent*** instead of ***laminar***.

The Reynolds number for a circular pipe is:

$$Re = \frac{\rho v D}{\mu} = \frac{\rho v\ 2r}{\mu} \tag{8.106}$$

where $D = 2r$ is the ***hydraulic diameter***[162] , which in the case of a circular pipe is the actual diameter. If we multiply this by one in a particular form:

$$Re = \frac{\rho v D}{\mu}\frac{Dv}{Dv} = \frac{\rho v^2 D^2}{\mu D v} \tag{8.107}$$

$\rho v^2 D^2$ has units of momentum per unit time (work it out). It is proportional to the *inertial force* acting on a differential slice of the fluid. $\mu D v$ *also* has units of force (recall that the units of μ are N-sec/m^2). This is proportional to the "viscous force" propagated through the

[161] Wikipedia: http://www.wikipedia.org/wiki/Reynolds Number.
[162] Wikipedia: http://www.wikipedia.org/wiki/hydraulic diameter.

fluid between layers. The Reynolds number can be thought of as a ratio between the force pushing the fluid forward and the shear force holding the fluid together against rotation.

The one thing the Reynolds number does for *us* is that it serves as a ***marker for the transition to turbulent flow***. For $Re < 2300$ flow in a circular pipe is *laminar* and all of the relations above hold. Turbulent flow occurs for $Re > 4000$. In between is the region known as the ***onset of turbulence***, where the ***resistance of the pipe depends on flow in a very nonlinear fashion***, and among other things *dramatically increases* with the Reynolds number to eventually be proportional to v. As we will see in a moment (in an example) this means that the partial occlusion of blood vessels can have a profound effect on the human circulatory system – basically, instead of $\Delta P = IR$ one has $\Delta P = I^2 R'$ so doubling flow at constant resistance (which may itsef change form) requires four times the pressure difference in the turbulent regime!

At this point you should understand fluid statics and dynamics quite well, armed both with equations such as the Bernoulli Equation that describe idealized fluid dynamics and statics as well as with conceptual (but possibly quantitative) ideas such Pascal's Principle or Archimedes' principle as the relationship between pressure differences, flow, and the geometric factors that contribute to resistance. Let's put some of this nascent understanding to the test by looking over and analyzing the human circulatory system.

8.5: The Human Circulatory System

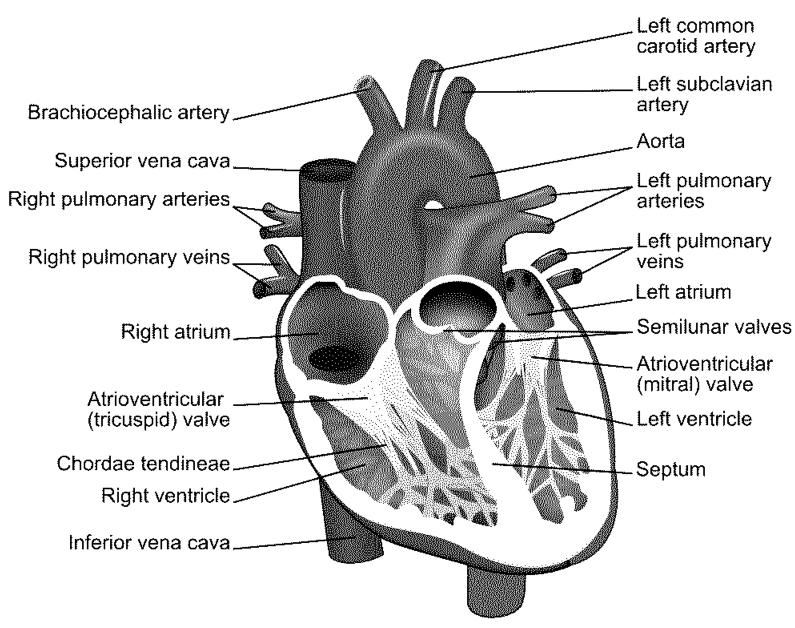

Figure 120: A simple cut-away diagram of the human heart.

For once, this is a chapter that math majors, physics majors, and engineers may, if they wish, skip, although personally I think that any intellectually curious person would want to learn all sorts of things that *sooner or later will impact on their own health and life*. To put that rather bluntly, kids, sure ***now*** you're all young and healthy and everything, but in thirty or forty more years (if you survive) you *won't be*, and understanding the things taught in this chapter will be extremely useful to you then, if not now as you choose a lifestyle and diet that might get you *through* to then in reasonably good cardiovascular shape!

Here is a list of True Facts about the human cardiovascular system, in no particular order, that you should now be able to understand at least qualitatively and conceptually if not quantitatively.

- The heart, illustrated in the schematic in figure 120 above[163] is the "pump" that drives blood through your blood vessels.
- The blood vessels are differentiated into three distinct types:
  - ***Arteries***, which lead strictly ***away from the heart*** and which contain a muscular layer that elastically dilates and contracts the arteries in a synchronous way to help carry the surging waves of blood. This acts as a "shock absorber" and hence reduces the peak systolic blood pressure (see below). As people age, this muscular tissue becomes less elastic – "hardening of the arteries" – as collagen repair mechanisms degrade or plaque (see below) is deposited and systolic blood pressure often increases as a result.

    Arteries split up the farther one is from the heart, eventually becoming ***arterioles***, the very small arteries that actually split off into capillaries.
  - ***Capillaries***, which are a dense network of very fine vessels (often only a single cell thick) that ***deliver oxygenated blood throughout all living tissue*** so that the oxygen can disassociate from the carrying hemoglobin molecules and diffuse into the surrounding cells in systemic circulation, or ***permit the oxygenation of blood*** in pulmonary circulation.
  - ***Veins***, which lead strictly ***back to the heart*** from the capillaries. Veins also have a muscle layer that expand or contract to aid in thermoregulation and regulation of blood pressure as one lies down or stands up. Veins also provide "capacitance" to the circulatory system and store the body's "spare" blood; 60% of the body's total blood supply is usually in the veins at any one time. Most of the veins, especially long vertical veins, are equipped with one-way ***venous valves*** every 4-9 cm that prevent backflow and pooling in the lower body during e.g. diastoli (see below).

    Blood from the capillaries is collected first in ***venules*** (the return-side equivalent of arterioles) and then into veins proper.
- There are two distinct circulatory systems in humans (and in the rest of the mammals and birds):
  - ***Systemic circulation***, where oxygenated blood enters the heart via pulmonary veins *from* the lungs and is pumped at high pressure *into* systemic arteries that deliver it through the capillaries and (deoxygenated) back via systemic veins to the heart.

[163]Wikipedia: http://www.wikipedia.org/wiki/Human heart. The diagram itself is borrowed from the wikipedia creative commons, and of course you can learn a lot more of the anatomy and function of the heart and circulation by reading the wikipedia articles on the heart and following links.

  - ***Pulmonary circulation***, where deoxgenated blood that has returned *from* the system circulation is pumped *into* pulmonary arteries that deliver it to the lungs, where it is oxygenated and returned to the heart by means of pulmonary veins. These two distinct circulations ***do not mix*** and together, ***form a closed double circulation loop***.

- The heart is the pump that serves ***both systemic and pulmonary circulation***. Blood enters into the ***atria*** and is expelled into the two circulatory system from the ***ventricles***. Systemic circulation enters from the pulmonary veins into the ***left atrium***, is pumped into the ***left ventricle*** through the one-way ***mitral valve***, which then pumps the blood at high pressure into the systemic arteries via the ***aorta*** through the one-way ***aortic valve***. It is eventually returned by the systemic veins (the ***superior and inferior vena cava***) to the ***right atrium***, pumped into the ***right ventricle*** through the one-way ***tricuspid valve***, and then pumped at high pressure into the ***pulmonary artery*** for delivery to the lungs.

  The human heart (as well as the hearts of birds and mammals in general) is thus ***four-chambered*** – two atria and two ventricles. The total resistance of the systemic circulation is generally larger than that of the pulmonary circulation and hence systemic arterial blood pressure must be higher than pulmonary arterial blood pressure in order to maintain the same flow. The left ventricle (primary systemic pump) is thus typically composed of thicker and stronger muscle tissue than the right ventricle. Certain reptiles also have four-chambered hearts, but their pulmonary and systemic circulations are not completely distinct and it is thought that their hearts became four-chambered by a different evolutionary pathway.

- ***Blood pressure*** is generally measured and reported in terms of two numbers:

  - ***Systolic*** blood pressure. This is the ***peak/maximum arterial pressure*** in the wave pulse generated that drives ***systemic circulation***. It is measured in the (brachial artery of the) arm, where it is supposed to be a reasonably accurate reflection of ***peak aortic pressure*** just outside of the heart, where, sadly, it cannot easily be *directly* measured without resorting to invasive methods that are, in fact, used e.g. during surgery.
  - ***Diastolic*** blood pressure. This is the ***trough/minimum arterial pressure*** in the wave pulse of systemic circulation.

  Blood pressure has historically been measured in ***millimeters of mercury***, in part because until fairly recently a sphygnomometer built using an integrated mercury barometer was by far the most accurate way to measure blood pressure, and it still extremely widely used in situations where high precision is required. Recall that 760 mmHg (torr) is 1 atm or 101325 Pa.

"Normal" Systolic systemic blood pressure can fairly accurately be estimated on the basis of the distance between the heart and the feet; a distance on the order of 1.5 meters leads to a pressure difference of around 0.15 atm or 120 mmHg.

Blood is driven through the relatively high resistance of the capillaries by the ***difference*** in arterial pressure and venous pressure. The venous system is entirely a ***low pressure***

return; its peak pressure is typically order of 0.008 bar (6 mmHg). This can be understood and predicted by the mean distance between valves in the venous system – the pressure difference between one valve and another (say) 8 cm higher is approximately $\rho_b g \times 0.08 \approx= 0.008$ bar. However, this pressure is not really static – it varies with the delayed pressure wave that causes blood to surge its way up, down, or sideways through the veins on its way back to the atria of the heart.

This difference in pressure means one very important thing. If you puncture or sever a vein, blood runs out relatively slowly and is fairly easily staunched, as it is driven by a pressure only a small bit higher than atmospheric pressure. Think of being able to easily plug a small leak in a glass of water (where the fluid height is likely very close to 8-10 cm) by putting your finger over the hole. When blood is drawn, a vein in e.g. your arm is typically tapped, and afterwards the hole almost immediately seals well enough with a simple clot sufficient to hold in the venous pressure.

If you puncture an *artery*, on the other hand, especially a large artery that still has close to the full systolic/diastolic pressure within it, blood ***spurts*** out of it driven by considerable pressure, the pressure one might see at the bottom of a *barrel* or a back yard *swimming pool*. Not so easy to stop it with light pressure from a finger, or seal up with a clot! It is proportionately more difficult to stop arterial bleeding and one can lose considerable blood volume in a very short time, leading to a fatal hypotension. Of course if one severs a *large* artery *or* vein (so that clotting has no chance to work) this is a very bad thing, but in general always worse for arteries than for veins, all other things being equal.

An exception of sorts is found in the jugular vein (returning from the brain). It ***has no valves*** because it is, after all, ***downhill*** from the brain back to the heart! As a consequence of this a human who is *inverted* (suspended by their feet, standing on their head) experiences a variety of circulatory problems in their head. Blood pools in the head, neck and brain until blood pressure there (increased by distension of the blood vessels) is enough to lift the blood back up to the heart *without* the help of valves, increasing venous return pressure in the brain itself by a factor of 2 to 4. This higher pressure is transmitted back throughout the brain's vasculature, and, if sustained, can easily cause aneurisms or ruptures of blood vessels (see below)) and death from blood clots or stroke. It can also cause permanent blindness as circulation through the eyes is impaired[164]

Note, however, that blood is pushed through the systemic and pulmonary circulation in *waves* of peak pressure that actually propagate faster than the flow and that the elastic aorta acts like a balloon that is constantly refilled during the left ventricular constraction and

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[164]

"You are old, Father William," the young man said,
"And your hair has become very white;
And yet you incessantly stand on your
Do you think, at your age, it is right?"

"In my youth," Father William replied to his son,
"I feared it might injure the brain;
But now that I'm perfectly sure I have
none, Why, I do it again and again."

Not really that good an idea...

"buffers" the arterial pressure – in fact, it behaves a great deal like a ***capacitor in an RC circuit*** behaves if it is being driven by a series of voltage pulses! Even this model isn't quite adequate. The systolic pulse propagates down through the (elastic) arteries (which dilate and contract to accomodate and maintain it) at a speed ***faster*** than the actual fluid velocity of the blood in the arteries. This effectively rapidly transports a new ejection volume of blood out to the arterioles to replace the blood that made it through the capillaries in the previous heartbeat and maintain a positive pressure difference *across* the capillary system even during diastole so that blood continues to move forward around the circulation loop.

Unfortunately we won't learn the math to be able to better understand this (and be able to at least qualitatively plot the expected time variation of flow) until next semester, so in the meantime we have to be satisfied with this heuristic explanation based on a dynamical "equilibrium" picture of flow plus a bit of intuition as to how an elastic "balloon" can store pressure pulses and smooth out their delivery.

There is a systolic peak in the blood pressure delivered to the pulmonary system as well, but it is ***much lower*** than the systolic systemic pressure, because (after all) the lungs are at pretty much the same height as the heart. The pressure difference needs only to be high enough to maintain the same average flow as that in the systemic circulation through the much lower resistance of the capillary network in the lungs. Typical healthy normal resting pulmonary systole pressures are around 15 mmHg; pressures higher than 25 mmHg are likely to be associated with ***pulmonary edema***, the pooling of fluid in the interstitial spaces of the lungs. This is a bad thing.

The return pressure in the pulmonary veins (to the right atrium, recall) is 2-15 mmHg, roughly consistent with central (systemic) venous pressure at the left atrium but with a larger variation.

The actual rhythm of the heartbeat is as follows. Bearing in mind that it is a continuous cycle, "first" blood from the last beat, which has accumulated in the left and right atrium during the heart's resting phase between beats, is expelled through the mitral and tricuspid valves from the left and right atrium respectively. The explusion is accomplished by the contraction of the atrial wall muscles, and the backflow of blood into the venous system is (mostly) prevented by the valves of the veins although there can be some small regurgitation. This expulsion which is nearly simultaneous with mitral barely leading tricuspid pre-compresses the blood in the ventricles and is the first "lub" one hears in the "lub-*dub*" of the heartbeat observed with a stethoscope.

Immediately following this a (nearly) simultanous contraction of the ventricular wall muscles causes the expulsion of the blood from the left and right ventricles through the semilunar valves into the aorta and pulmonary arteries, respectively. As noted above, the left-ventricular wall is thicker and stronger and produces a substantially higher peak systolic pressure in the aorta and systemic circulation than the right ventricle produces in the pulmonary arteries. The elastic aorta distends (reducing the peak systemic pressure as it stores the energy produced by the heartbeak) and partially sustains the higher pressure across the capillaries throughout the resting phase. This is the "dub" in the "lub-*dub*" of the heartbeat.

This isn't anywhere near *all* of the important physics in the circulatory system, but it should be enough for you to be able to understand the basic plumbing well enough to learn more later[165]. Let's do a few very important examples of how things go *wrong* in the circulatory system and how fluid physics helps you understand and detect them!

Example 8.5.1: Atherosclerotic Plaque Partially Occludes a Blood Vessel

Humans are not yet evolved to live 70 or more years. Mean life expectancy as little as a hundred years ago was in the mid-50's, *if* you only average the people that survived to age 15 – otherwise it was in the 30's! The average age when a woman bore her first child throughout most of the period we have been considered "human" has been perhaps 14 or 15, and a woman in her thirties was often a grandmother. Because evolution works best if parents don't hang around *too* long to compete with their own offspring, we are very likely *evolved to die* somewhere around the age of 50 or 60, three to four generations (old style) after our own birth. Humans begin to really experience the effects of aging – failing vision, incipient cardiovascular disease, metabolic slowing, greying hair, wearing out teeth, cancer, diminished collagen production, arthritis, around age 45 (give or take a few years), and it once it starts it just gets worse. Old age *physically* sucks, I can say authoritatively as I type this peering through reading glasses with my mildly arthritic fingers over my gradually expanding belly at age ~~56~~,~~59~~,61.

One of the many ways it sucks is that the 40's and 50's is where people usually show the first signs of *cardiovascular disease*, in particular ***atherosclerosis***[166] – granular deposits of fatty material called ***plaques*** that attach to the walls of e.g. arteries and gradually thicken over time, generally associated with high blood cholesterol and lipidemia. The risk factors for atherosclerosis form a list as long as your arm and its fundamental causes are not well understood, although they are currently believed to form as an inflammatory response to surplus low density lipoproteins (one kind of cholesterol) in the blood.

Our purpose, however, is not to think about causes and cures but instead what *fluid physics* has to say about the disorder, its diagnosis and effects. In figure 121 two arteries are illustrated. Artery a) is "clean", has a radius of r_1, and (from the Poiseuille Equation above) has a very *low resistance* to any given flow of blood. Because R_a over the length L is low, there is very little pressure drop between P_+ and P_- on the two sides of any given stretch of length L. The velocity profile of the fluid is also more or less uniform in the artery, slowing a bit near the walls but generally moving smoothly throughout the entire cross-section.

Artery b) has a significant deposit of atherosclerotic plaques that have coated the walls and reduced the effective radius of the vessel to $\sim r_2$ over an extended length L. The vessel is perhaps 90% occluded – only 10% of its normal cross-sectional area is available to carry fluid.

[165]...And understand why we spent time learning to solve first order differential equations this semester so that we can understand RC circuits in detail next semester so we can think *back* and further understand the remarks about how the aorta acts like a capacitor this semester.

[166]Wikipedia: http://www.wikipedia.org/wiki/Atherosclerosis. As always, there is far, far more to say about this subject than I can cover here, all of it interesting and capable of helping you to select a lifestyle that prolongs a high quality of life.

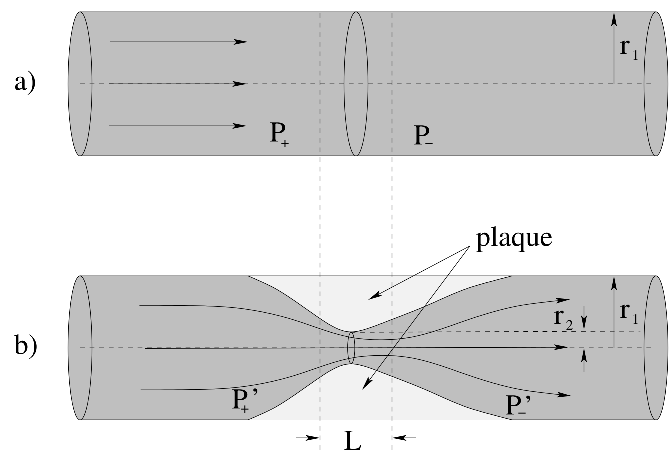


Figure 121: Two "identical" blood vessels with circular cross-sections, one that is clean (of radius r_1) and one that is perhaps 90% occluded by plaque that leaves an aperture of radius $r_2 < r_1$ in a region of some length L.

We can now easily understand several things about this situation. First, if the total *flow* in artery b) is still being maintained at close to the levels of the flow in artery a) (so that tissue being oxygenated by blood delivered by this artery is not being critically starved for oxygen yet) the ***fluid velocity in the narrowed region is ten times higher than normal!*** Since the Reynolds number for blood flowing in primary arteries is normally around 1000 to 2000, increasing v by a factor of 10 increases the Reynolds number by a factor of 10, causing the flow to become *turbulent* in the obstruction. This tendency is even more pronounced than this figure suggests – I've drawn a nice symmetric occlusion, but the atheroma (lesion) is more likely to grow predominantly on one side and irregular lesions are more likely to disturb laminar flow even for smaller Reynolds numbers.

This turbulence provides the basis for one method of possible detection and diagnosis – you can *hear* the turbulence (with luck) through the stethoscope during a physical exam. Physicians get a lot of practice listening for turbulence since turbulence produced by *artificially* restricting blood flow in the brachial artery by means of a constricting cuff is basically what one listens for when taking a patient's blood pressure. It really shouldn't be there, especially during diastole, the rest of the time.

Next, consider what the vessel's resistance across the lesion of length L should do. Recall that $R \propto 1/A^2$. That means that the resistance is at least ***100 times larger*** than the resistance of the healthy artery over the same distance. In truth, it is almost certainly much greater than this, because as noted, one has converted to turbulent flow and our expression for the resistance assumed *laminar* flow. The pressure difference required to maintain the same flow scales up by the ***square*** of the flow itself! Ouch!

A hundredfold to thousandfold increase in the resistance of the segment means that *either* the fluid flow itself will be reduced, assuming a constant upstream pressure, *or* the pressure upstream will increase – substantially – to maintain adequate flow and perfusion. In practice a certain amount of both can occur – the stiffening of the artery due to the lesion and an increased resting heart rate[167] can raise systolic blood pressure, which

[167] Among many other things. High blood pressure is extremely multifactorial.

tends to maintain flow, but as narrowing proceeds it cannot raise it *enough* to compensate, especially not without causing far greater damage somewhere else.

At some point, the tissue downstream from the occluded artery begins to suffer from lack of oxygen, especially during times of metabolic stress. If the tissue in question is in a leg or an arm, weakness and pain may result, not good, but arguably recoverable. If the tissue in questions is ***the heart itself*** or ***the lungs*** or ***the brain*** this is ***very bad indeed***. The failure to deliver sufficient oxygen to the heart over the time required to cause actual death of heart muscle tissue is what is commonly known as a heart attack. The same failure in an artery that supplies the brain is called a stroke. The heart and brain have very limited ability to regrow damaged tissue after either of these events. Occlusion and hardening of the pulmonary arteries can lead to pulmonary hypertension, which in turn (as already noted) can lead to pulmonary edema and a variety of associated problems.

Example 8.5.2: Aneurisms

An aneurism is basically the *opposite* of an atherosclerotic lesion. A portion of the walls of an artery or, less commonly, a vein *thins* and begins to ***dilate*** or stretch in response to the pulsing of the systolic wave. Once the artery has “permanently” stretched along some short length to a larger radius than the normal artery on either side, a nasty feedback mechanism ensues. Since the cross-sectional area of the dilated area is ***larger***, fluid flow there ***slows*** from conservation of flow. At the same time, the pressure in the dilated region must *increase* according to Bernoulli’s equation – the pressure increase is responsible for slowing the fluid as it enters the aneurism and re-accelerating it back to the normal flow rate on the far side.

The higher pressure, of course, then makes the already weakened arterial wall stretch more, which dilates the aneurism more, which slows the blood more which increases the pressure, until some sort of limit is reached: extra pressure from surrounding tissue serves to reinforce the artery and keeps it from continuing to grow or the aneurism *ruptures*, spilling blood into surrounding (low pressure) tissue with every heartbeat. While there aren’t a lot of places a ruptured aneurism is *good*, in the brain it is very bad magic, causing the same sort of damage as a stroke as the increased pressure in the tissue surrounding the rupture becomes so high that normal capillary flow through the tissue is compromised. Aortic aneurisms are also far from unknown and, because of the high blood flow directly from the heart, can cause almost instant death as one bleeds, under substantial pressure, internally.

Example 8.5.3: The Giraffe

“The Giraffe” isn’t really an example *problem*, it is more like a nifty/cool True Fact but I haven’t bothered to make up a nifty cool true fact header for the book (at least not yet). Full grown adult giraffes are animals (you may recall, or not[168]) that stand roughly 5 meters high.

[168] Wikipedia: http://www.wikipedia.org/wiki/Giraffe. And Wikipedia stands ready to educate you further, if you have never seen an actual Giraffe in a zoo and want to know a bit about them

Because of their height, giraffes have a uniquely evolved circulatory system[169] . In order to drive blood from its feet up to its brain, especially in times of stress when it is e.g. running, it's heart has to be able to maintain a pressure difference of close to *half an atmosphere of pressure* (using the rule that 10 meters of water column equals one atmosphere of pressure difference and assuming that blood and water have roughly the same density). A giraffe heart is correspondingly huge: roughly 60 cm long and has a mass of around 10 kg in order to accomplish this.

When a giraffe is erect, its cerebral blood pressure is "normal" (for a giraffe), but when it bends to drink, its head goes down to the ground. This rapidly increases the blood pressure being delivered by its heart to the brain by 50 kPa or so. It has evolved a complicated set of pressure controls in its neck to reduce this pressure so that it doesn't have a brain aneurism every time it gets thirsty!

Giraffes, like humans and most other large animals, have a second problem. The heart doesn't maintain a *steady* pressure differential in and of itself; it expels blood in *beats*. In between contractions that momentarily increase the pressure in the arterial (delivery) system to a ***systolic peak*** that drives blood over into the venous (return) system through capillaries that either oxygenate the blood in the pulmonary system or give up the oxygen to living tissue in the rest of the body, the arterial pressure decreases to a ***diastolic minimum***.

Even in relatively short (compared to a giraffe!) adult humans, the blood pressure differential between our nose and our toes is around 0.16 bar, which not-too-coincidentally (as noted above) is equivalent to the 120 torr (mmHg) that constitutes a fairly "normal" systolic blood pressure. The normal diastolic pressure of 70 torr (0.09 bar) is insufficient to keep blood in the venous system from "falling back" out of the brain and pooling in and distending the large veins of the lower limbs.

To help prevent that, long (especially vertical) veins have ***one-way valves*** that are spaced roughly every 4 to 8 cm along the vein. During systoli, the valves open and blood pulses forward. During diastoli, however, the valves *close* and *distribute the weight of the blood in the return system* to $\sim$6 cm segments of the veins while preventing backflow. The pressure differential across a valve and supported by the smooth muscle that gives tone to the vein walls is then just the pressure accumulated across 6 cm (around 5 torr).

Humans get ***varicose veins***[170] when these valves *fail* (because of gradual loss of tone in the veins with age, which causes the vein to swell to where the valve flaps don't properly meet, or other factors). When a valve fails, the next-lower valve has to support twice the pressure difference (say 10 torr) which in turn swells that vein close to the valve (which can cause it to fail as well) passing three times this differential pressure down to the next valve and so on. Note well that there are two aspects of this extra pressure to consider – one is the increase in pressure differential across the valve, but perhaps the greater one is the increase in pressure differential between the inside of the vein and the outside tissue. The latter causes the vein to *dilate* (swell, increase its radius) as the tissue stretches until its tension can supply the pressure needed to confine the blood column.

[169] Wikipedia: http://www.wikipedia.org/wiki/Giraffe#Circulatory_system.
[170] Wikipedia: http://www.wikipedia.org/wiki/Varicose Veins.

Opposing this positively fed-back tendency to dilate, which compromises valves, which in turn increases the dilation to eventual destruction are things like muscle use (contracting surrounding muscles exerts extra pressure on the outside of veins and hence decreases the pressure differential and stress on the venous tissue), general muscle and skin tone (the skin and surrounding tissue helps maintain a pressure outside of the veins that is already higher than ambient air pressure, and keeping one's blood pressure under control, as the diastolic pressure sets the scale for the venous pressure during diastoli and if it is high then the minimum pressure differential across the vein walls will be correspondingly high. Elevating one's feet can also help, exercise helps, wearing support stockings that act like a second skin and increase the pressure outside of the veins can help.

Carrying the extra pressure below compromised valves nearly all of the time, the veins gradually dilate until they are many times their normal diameter, and significant amounts of blood pool in them – these "ropy", twisted, fat, deformed veins that not infrequently visibly pop up out of the skin in which they are embedded are the varicose veins.

Naturally, giraffes have this problem in spades. The pressure in their lower extremities, even allowing for their system of valves, is great enough to rupture their capillaries. To keep this from happening, the skin on the legs of a giraffe is among the thickest and strongest found in nature – it functions like an astronaut's "g-suit" or a permanent pair of support stockings, maintaining a high baseline pressure in the *tissue* of the legs outside of the veins and capillaries, and thereby reducing the pressure differential.

Another cool fact about giraffes – as noted above, they pretty much live with "high blood pressure" – their normal pressure of 280/180 torr (mmHg) is 2-3 times that of humans (because their height is 2-3 times that of humans) in order to keep their brain perfused with blood. This pressure has to further elevate when they e.g. run away from predators or are stressed. Older adult giraffes have a tendency to ***die of a heart attack*** if they run for too long a time, so zoos have to take care to avoid stressing them if they wish to capture them!

Finally, giraffes splay their legs when they drink so that they reduced the pressure differential the peristalsis of their gullet has to maintain to pump water up and over the hump down into their stomach. Even this doesn't completely exhaust the interesting list of giraffe facts associated with their fluid systems. Future physicians would be well advised to take a closer peek at these very large mammals (as well as at elephants, who have many of the same problems but very different evolutionary adaptations to accommodate them) in order to gain insight into the complex fluid dynamics to be found in the human body.

Homework for Week 8

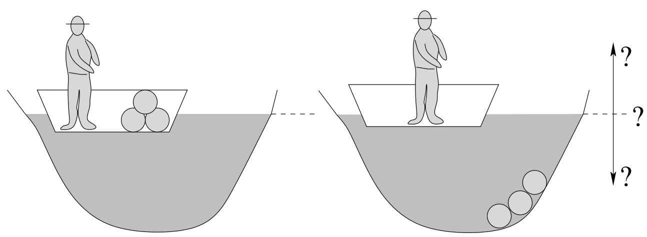
Problem 1.

Physics Concepts: Make this week's physics concepts summary as you work all of the problems in this week's assignment. Be sure to cross-reference each concept in the summary to the problem(s) they were key to, and include concepts from previous weeks as necessary. Do the work carefully enough that you can (after it has been handed in and graded) punch it and add it to a three ring binder for review and study come finals!

Problem 2.

A small boy is riding in a minivan with the windows closed, holding a helium balloon. The van goes around a corner to the left. Does the balloon swing to the left, still pull straight up, or swing to the right as the van swings around the corner?

Problem 3.



A person stands in a boat floating on a pond and containing several large, round, rocks. He throws the rocks out of the boat so that they sink to the bottom of the pond. The water level of the pond will:

a) Rise a bit.

b) Fall a bit.

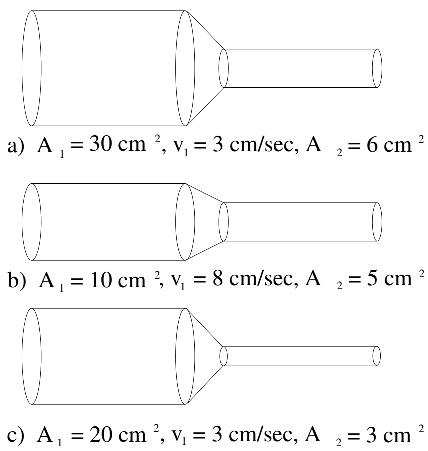
c) Remain unchanged.

d) Can't tell from the information given (it depends on, for example, the shape of the boat, the mass of the person, whether the pond is located on the Earth or on Mars...).

Problem 4.

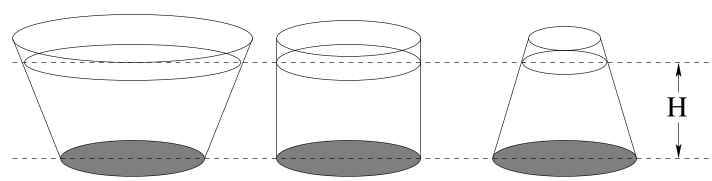
People with vascular disease or varicose veins (a disorder where the veins in one's lower extremeties become swollen and distended with fluid) are often told to walk in water 1-1.5 meters deep. Explain why.

Problem 5.



In the figure above three different pipes are shown, with cross-sectional areas and flow speeds as shown. Rank the three diagrams a, b, and c in the order of the speed of the outgoing flow.

Problem 6.



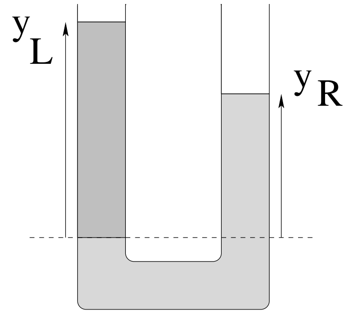
In the figure above three flasks are drawn that have the same (shaded) cross sectional area of the bottom. The depth of the water in all three flasks is H, and so the pressure at the bottom in all three cases is the same. Explain how the force exerted **by the fluid** on the circular bottom can be the same for all three flasks when all three flasks contain different weights of water.

Problem 7.

This problem will help you learn required concepts such as:

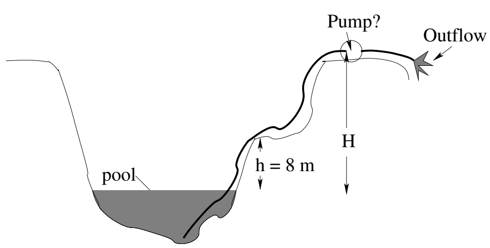
- Pascal's Principle
- Static Equilibrium

so please review them before you begin.



A vertical U-tube open to the air at the top is filled with oil (density ρ_o) on one side and water (density ρ_w) on the other, where $\rho_o < \rho_w$. Find y_L, the height of the column on the left, in terms of the densities, g, and y_R as needed. Clearly label the oil and the water in the diagram below and show all reasoning including the basic principle(s) upon which your answer is based.

Problem 8.



This problem will help you learn required concepts such as:

- Static Pressure
- Barometers

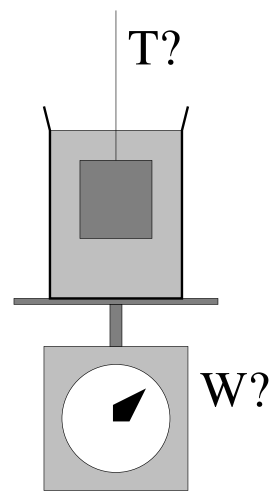
so please review them before you begin.

A pump is a machine that can maintain a pressure differential between its two sides. A particular pump that can maintain a pressure differential of as much as 10 atmospheres of pressure between the low pressure side and the high pressure side is being used on a construction site.

a) Your construction boss has just called you into her office to either explain why they aren't getting any water out of the pump on top of the $H = 25$ meter high cliff shown above. Examine the schematic above and show (algebraically) why it cannot possibly deliver water that high. Your explanation should include an invocation of the appropriate physical law(s) and an explicit calculation of the highest distance the a pump *could* lift water in this arrangement. Why is the notion that the pump "sucks water up" misleading? What really moves the water up?

b) If you answered a), you get to keep your job. If you answer b), you might even get a raise (or at least, get full credit on this problem)! Tell your boss where this single pump should be located to move water up to the top and show (draw a picture of) how it should be hooked up.

Problem 9.



This problem will help you learn required concepts such as:

- Archimedes Principle
- Weight

so please review them before you begin.

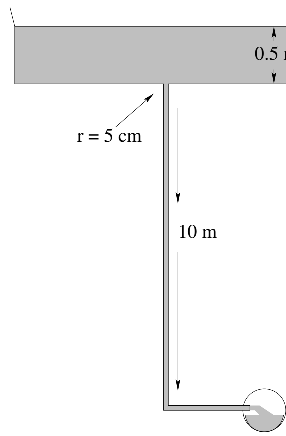
A block of density ρ and volume V is suspended by a thin thread and is immersed completely in a jar of oil (density $\rho_o < \rho$) that is resting on a scale as shown. The total mass of the oil and jar (alone) is M.

a) What is the buoyant force exerted by the oil on the block?

b) What is the tension T in the thread?

c) What does the scale read?

Problem 10.



This problem will help you learn required concepts such as:

- Pressure in a Static Fluid

so please review them before you begin.

That it is dangerous to build a drain for a pool or tub consisting of a single narrow pipe that drops down a long ways before encountering air at atmospheric pressure was demonstrated tragically in an accident that occurred (no fooling!) within two miles from where you are sitting (a baby pool was built with just such a drain, and was being drained one day when a little girl sat down on the drain and was severely injured).

In this problem you will analyze why.

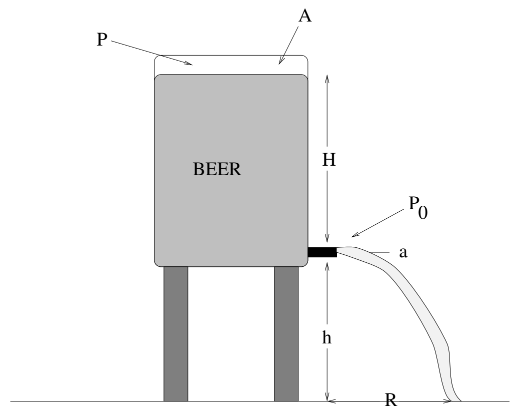
Suppose the mouth of a drain is a circle five centimeters in radius, and the pool has been draining long enough that its drain pipe is filled with water (and no bubbles) to a depth of ten meters below the top of the drain, where it exits in a sewer line open to atmospheric pressure. The pool is 50 cm deep. If a thin steel plate is dropped to suddenly cover the drain with a watertight seal, what is the force one would have to exert to remove it straight up?

Note carefully this force relative to the likely strength of mere flesh and bone (or even thin steel plates!) Ignorance of physics can be actively dangerous.

Problem 11.

This problem will help you learn required concepts such as:

- Bernoulli's Equation
- Toricelli's Law



so please review them before you begin.

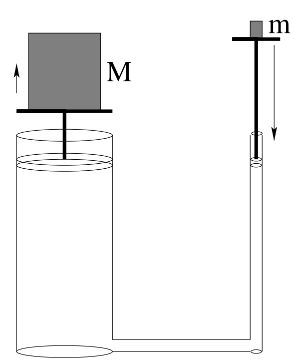
In the figure above, a CO_2 cartridge is used to maintain a pressure P on top of the beer in a beer keg, which is full up to a height H above the tap at the bottom (which is obviously open to normal air pressure) a height h above the ground. The keg has a cross-sectional area A at the top. Somebody has pulled the tube and valve off of the tap (which has a cross sectional area of a) at the bottom.

a) Find the speed with which the beer emerges from the tap. You may use the approximation $A \gg a$, but please do so only at the end. Assume laminar flow and no resistance.

b) Find the value of R at which you should place a pitcher (initially) to catch the beer.

c) Evaluate the answers to a) and b) for $A = 0.25$ m^2, $P = 2$ atmospheres, $a = 0.25$ cm^2, $H = 50$ cm, $h = 1$ meter and $\rho_{\rm beer} = 1000$ kg/m^3 (the same as water).

Problem 12.

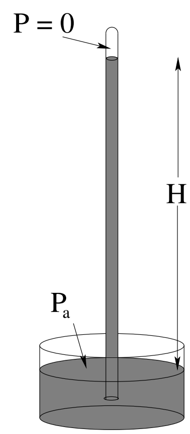


The figure above illustrates the principle of hydraulic lift. A pair of coupled cylinders are filled with an incompressible, very light fluid (assume that the mass of the fluid is zero compared to everything else).

a) If the mass M on the left is 1000 kilograms, the cross-sectional area of the left piston is 100 cm^2, and the cross sectional area of the right piston is 1 cm^2, what mass m should one place on the right for the two objects to be in balance?

b) Suppose one pushes the right piston down a distance of one meter. How much does the mass M rise?

Problem 13.



The idea of a barometer is a simple one. A tube filled with a suitable liquid is inverted into a reservoir. The tube empties (maintaining a seal so air bubbles cannot get into the tube) until the static pressure in the liquid is in balance with the *vacuum* that forms at the top of the tube and the ambient pressure of the surrounding air on the fluid surface of the reservoir at the bottom.

a) Suppose the fluid is water, with $\rho_w = 1000$ kg/m^3. Approximately how high will the water column be? Note that water is not an ideal fluid to make a barometer with because of the height of the column necessary and because of its annoying tendency to boil at room temperature into a vacuum.

b) Suppose the fluid is mercury, with a specific gravity of 13.6. How high will the mercury column be? Mercury, as you can see, *is* nearly ideal for fluids-pr-compare-barometers except for the minor problem with its extreme toxicity and high vapor pressure.

Fortunately, there are many other ways of making good fluids-pr-compare-barometers.

Optional Problems: Start Review for Final!

At this point we are roughly four "weeks" out from our final exam[171]. I thus **strongly suggest** that you devote any extra time you have not to further reinforcement of fluid flow, but to a gradual slow review of all of the basic physics from the first half of the course. Make sure that you still remember and understand all of the basic principles of Newton's Laws, work and energy, momentum, rotation, torque and angular momentum. Look over your old homework and quiz and hour exam problems, review problems out of your notes, and look for help with any ideas that still aren't clear and easy.

[171] ...which might be only *one and a half* weeks out in a summer session!

Week 9: Oscillations

Oscillation Summary

- Springs obey Hooke's Law: $\vec{F} = -k\vec{x}$ (where k is called the *spring constant*. A perfect spring (with no damping or drag force) produces perfect harmonic oscillation, so this will be our archetype.

- A pendulum (as we shall see) has a restoring force or torque proportional to displacement for *small* displacements but is much too complicated to treat in this course for large displacements. It is a simple example of a problem that oscillates harmonically for small displacements but *not* harmonically for large ones.

- An oscillator can be **damped** by dissipative forces such as friction and viscous drag. A damped oscillator can have exhibit a variety of behaviors depending on the relative strength and form of the damping force, but for one special form it can be easily described.

- An oscillator can be ***driven*** by e.g. an external harmonic driving force that may or may not be at the same frequency (in ***resonance*** with the natural frequency of the oscillator.

- The equation of motion for any (undamped) harmonic oscillator is the same, although it may have different dynamical variables. For example, for a spring it is:

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = \frac{d^2x}{dt^2} + \omega^2 x = 0 \tag{9.1}$$

  where for a simple pendulum (for small oscillations) it is:

$$\frac{d^2\theta}{dt^2} + \frac{g}{\ell}x = \frac{d^2\theta}{dt^2} + \omega^2 \theta = 0 \tag{9.2}$$

  (In this latter case ω is the ***angular frequency*** of the oscillator, not the angular velocity of the mass $d\theta/dt$.)

- The general solution to the equation of motion is:

$$x(t) = A\cos(\omega t + \phi) \tag{9.3}$$

  where $\omega = \sqrt{k/m}$ and the amplitude A (units: length) and phase ϕ (units: dimensionless/radians) are the constants of integration (set from e.g. the initial conditions).

Note that we alter the variable to fit the specific problem – for a pendulum it would be:

$$\theta(t) = \Theta \cos(\omega t + \phi) \tag{9.4}$$

with $\omega = \sqrt{g/\ell}$, where the angular amplitude Θ now has units of radians.

- The velocity of the mass attached to an oscillator is found from:

$$v(t) = \frac{dx}{dt} = -A\omega \sin(\omega t + \phi) = -V \sin(\omega t + \phi) \tag{9.5}$$

(with $V = v_{\max} = A\omega$).

- From the velocity equation above, we can easily find the kinetic energy as a function of time:

$$K(t) = \frac{1}{2}mv^2 = \frac{1}{2}mA^2\omega^2 \sin^2(\omega t + \phi) = \frac{1}{2}kA^2 \sin^2(\omega t + \phi) \tag{9.6}$$

- The potential energy of an oscillator is found by integrating:

$$U(x) = -\int_0^x -kx'dx' = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \cos^2(\omega t + \phi) \tag{9.7}$$

if we use the (usual but not necessary) convention that $U(0) = 0$ when the mass is at the equilibrium displacement, $x = 0$.

- The total mechanical energy is therefore obviously a constant:

$$E(t) = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \sin^2(\omega t + \phi) + \frac{1}{2}kA^2 \cos^2(\omega t + \phi) = \frac{1}{2}kA^2 \tag{9.8}$$

- As usual, the relation between the angular frequency, the regular frequency, and the period of the oscillator is given by:

$$\omega = \frac{2\pi}{T} = 2\pi f \tag{9.9}$$

(where $f = 1/T$). SI units of frequency are ***Hertz*** – cycles per second. Angular frequency has units of radians per second. Since both cycles and radians are dimensionless, the units themselves are dimensionally inverse seconds but they are (obviously) related by 2π radians per cycle.

- A (non-ideal) harmonic oscillator in nature is almost always ***damped*** by friction and drag forces. If we assume damping by viscous drag in (low Reynolds number) laminar flow – not unreasonable for smooth objects moving in a damping fluid, if somewhat itself idealized – the equation of motion becomes:

$$\frac{d^2x}{dt^2} + + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x = 0 \tag{9.10}$$

- The solution to this equation of motion is:

$$x_{\pm}(t) = A_{\pm}e^{\frac{-b}{2m}t} \cos(\omega' t + \phi) \tag{9.11}$$

where

$$\omega' = \omega_0\sqrt{1 - \frac{b^2}{4km}} \tag{9.12}$$

9.1: The Simple Harmonic Oscillator

Oscillations occur whenever a force exists that pushes an object back towards a stable equilibrium position whenever it is displaced from it. Such forces abound in nature – things are held together in structured form *because* they are in stable equilibrium positions and when they are disturbed in certain ways, they oscillate.

When the displacement from equilibrium is *small*, the restoring force is often *linearly* related to the displacement, at least to a good approximation. In that case the oscillations take on a special character – they are called *harmonic* oscillations as they are described by harmonic functions (sines and cosines) known from trigonometery.

In this course we will study **simple harmonic oscillators**, both with and without damping (and harmonic driving) forces. The principle examples we will study will be masses on springs and various penduli.

9.1.1: The Archetypical Simple Harmonic Oscillator: A Mass on a Spring

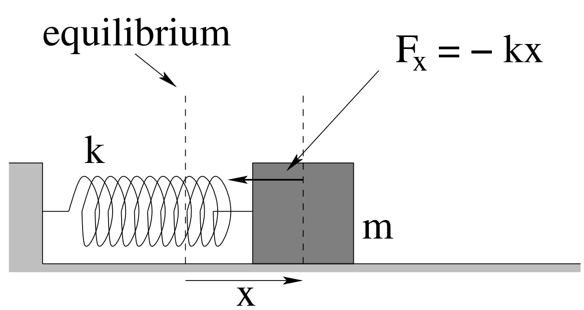


Figure 122: A mass on a spring is displaced by a distance x from its equilibrium position. The spring exerts a force $F_x = -kx$ on the mass (Hooke's Law).

Consider a mass m attached to a perfect spring (which in turn is affixed to a wall as shown in figure 122. The mass rests on a surface so that gravitational forces are cancelled by the normal force and are hence irrelevant. The mass will be displaced only in one direction (call it x) and otherwise constrained so that motion in or out of the plane is impossible and no drag or frictionless forces are (yet) considered to be relevant.

We know that the force exerted by a perfect spring on a mass stretched a distance x from its equilibrium position is given by ***Hooke's Law:***

$$F_x = -kx \tag{9.13}$$

where k is the ***spring constant*** of the spring. This is a ***linear restoring force***, and (as we shall see) is highly characteristic of the restoring forces exerted by *any* system around a point of stable equilibrium.

Although thus far we have avoided trying to determine the most general motion of the mass, it is time for us to tackle that chore. As we shall see, the motion of an undamped simple harmonic oscillator is *very easy to understand* in the ideal case, and easy enough to understand qualitatively or semi-quantitatively that it serves as an excellent springboard to understanding many of the properties of bulk materials, such as ***compressibility, stress, and strain***.

We begin, as one might expect, with Newton's Second Law, used to obtain the (second order, linear, homogeneous, differential) equation of motion for the system. Note well that although this sounds all complicated and everything – like "real math" – we've been solving second order differential equations from day one, so they *shouldn't* be intimidating. Solving the equation of motion for the simple harmonic oscillator isn't *quite* as simple as just integrating twice, but as we will see neither is it all that difficult.

Hooke's Law combined with Newton's Second Law can thus be written and massaged algebraically as follows:

$$\begin{aligned}
ma_x = m\frac{d^2x}{dt^2} &= F_x = -kx \\
m\frac{d^2x}{dt^2} + kx &= 0 \\
\frac{d^2x}{dt^2} + \frac{k}{m}x &= 0 \\
\frac{d^2x}{dt^2} + \omega^2 x &= 0 \qquad (9.14)
\end{aligned}$$

where we have defined the ***angular frequency*** of the oscillator,

$$\omega^2 = k/m \qquad (9.15)$$

This must have units of ***inverse time squared*** (why?). We will momentarily justify this identification, but it won't hurt to start learning it now.

Equation 9.14 (with the ω^2) is the ***standard harmonic oscillator differential equation of motion*** (SHO ODE). As we'll soon see with quite a few examples and an algebraic argument, we can put the equation of motion for ***many*** systems into this form for at least small displacements from a stable equilibrium point. If we can properly solve it *once and for all* now, whenever we can put an equation of motion into this form in the future we can just ***read off the solution*** by ***identifying similar quantities*** in the equation.

To solve it[172], we note that it basically tells us that $x(t)$ must be a function that has a ***second derivative proportional to the function itself***.

We know at least three functions whose second derivatives are proportional to the functions themselves: cosine, sine and exponential. In this particular case, we *could* guess cosine or sine and we would get a perfectly reasonable solution. The bad thing about doing this is that the solution methodology would not generalize at all – it wouldn't work for first order, third order, or even *general* second order ODEs. It would give us a solution to the SHO problem (for example) but *not* allow us to solve the *damped* SHO problem or *damped, driven* SHO problems we investigate later this week. For this reason, although it is a bit more work now, we'll search for a solution ***assuming that it has an exponential form***.

---

[172]Not only it, but *any* homogeneous linear Nth order ordinary differential equation – the method can be applied to first, third, fourth, fifth... order linear ODEs as well.

Note Well!

If you are *completely panicked* by the following solution, if thinking about trying to understand it makes you feel sick to your stomach, you can probably *skip ahead to the next section* (or rather, begin reading again at the end of this chapter after the ***real*** solultion is obtained).

There is a price you will pay if you do. You will never understand where the solution comes from or how to solve the slightly more difficult damped SHO problem, and will therefore have to ***memorize the solutions***, unable to rederive them if you forget (say) the formula for the damped oscillator frequency or the criterion for critical damping.

As has been our general rule above, I think that it is better to *try* to make it through the derivation to where you understand it, even if only a single time and for a moment of understanding, even if you do then move on and just learn and work to retain the result. I think it helps you remember the result with less effort and for longer once the course is over, and to bring it back into mind and understand it more easily if you should ever need to in the future. But I also realize that mastering a chunk of math like this doesn't come easily to some of you and that investing the time to do *given* a limited amount of time to invest might actually reduce your eventual understanding of the general content of this chapter. You'll have to decide for yourself if this is true, ideally after at least giving the math below a look and a try. It's not really as difficult as it looks at first.

The exponential assumption:

$$x(t) = Ae^{\alpha t} \tag{9.16}$$

makes solutions to general linear homogeneous ODEs *simple*.

Let's look at the pattern when we take repeated derivatives of this equation:

$$\begin{aligned}
x(t) &= Ae^{\alpha t} \\
\frac{dx}{dt} &= \alpha Ae^{\alpha t} \\
\frac{d^2x}{dt^2} &= \alpha^2 Ae^{\alpha t} \\
\frac{d^3x}{dt^3} &= \alpha^3 Ae^{\alpha t} \\
&\ldots
\end{aligned} \tag{9.17}$$

where α is an unknown parameter and A is an arbitrary (possibly complex) constant (with the units of $x(t)$, in this case, length) called the ***amplitude***. Indeed, this is a general rule:

$$\frac{d^nx}{dt^n} = \alpha^n Ae^{\alpha t} \tag{9.18}$$

for any $n = 0, 1, 2....$

Substituting this assumed solution and its rule for the second derivative into the SHO

ODE, we get:

$$\begin{aligned} \frac{d^2x}{dt^2} + \omega^2 x &= 0 \\ \frac{d^2Ae^{\alpha t}}{dt^2} + \omega^2 Ae^{\alpha t} &= 0 \\ \alpha^2 Ae^{\alpha t} + \omega^2 Ae^{\alpha t} &= 0 \\ \left(\alpha^2 + \omega^2\right) Ae^{\alpha t} &= 0 \end{aligned} \tag{9.19}$$

There are two ways this equation could be true. First, we could have $A = 0$, in which case $x(t) = 0$ for any value of α. This indeed does solve the ODE, but the solution is *boring* – nothing moves! Mathematicians call this the ***trivial solution*** to a homogeneous linear ODE, and we will reject it out of hand by insisting that we have a *non*trivial solution with $A \neq 0$.

In that case it is necessary for

$$\left(\alpha^2 + \omega^2\right) = 0 \tag{9.20}$$

This is called the ***characteristic equation*** for the linear homogeneous ordinary differential equation. If we can find an α such that this equation is satisfied, then our assumed answer will indeed solve the ODE for nontrivial (nonzero) $x(t)$.

Clearly:

$$\alpha = \pm i\omega \tag{9.21}$$

where

$$i = +\sqrt{-1} \tag{9.22}$$

We now have a solution to our second order ODE – indeed, we have two solutions – but those solutions are ***complex exponentials***[173] and contain the ***imaginary unit***[174] , i.

In principle, if you have satisfied the prerequisites for this course you have almost certainly studied imaginary numbers and complex numbers[175] in a high school algebra class and perhaps again in college level calculus. Unfortunately, because high school math is often indifferently well taught, you may have thought they would never be *good* for anything and hence didn't pay much attention to them, or (however well they were or were not covered) at this point you've now forgotten them.

In any of these cases, now might be a really ***good time to click on over to my online Mathematics for Introductory Physics book***[176] and review at least some of the properties of i and complex numbers and how they relate to trig functions. This book is still (as of this moment) less detailed here than I would like, but it does review all of their most important properties that are used below. Don't hesitate to follow the wikipedia links as well.

If you are a life science student (perhaps a bio major or premed) then *perhaps* (as noted above) you won't ever need to know even this much and can get away with just

[173] Wikipedia: http://www.wikipedia.org/wiki/Euler Formula.

[174] Wikipedia: http://www.wikipedia.org/wiki/Imaginary unit.

[175] Wikipedia: http://www.wikipedia.org/wiki/Complex numbers.

[176] http://www.phy.duke.edu/~rgb/Class/math_for_intro_physics.php There is an entire chapter on this: *Complex Numbers and Harmonic Trigonometric Functions*, well worth a look.

memorizing the real solutions below. If you are a physics or math major or an engineering student, the mathematics of this solution is ***just a starting point*** to an entire, amazing world of complex numbers, quaternions, Clifford (geometric division) algebras, that are not only *useful*, but seem to be *essential* in the development of electromagnetic and other field theories, theories of oscillations and waves, and above all in quantum theory (bearing in mind that everything we are learning this year is *technically* incorrect, because the Universe turns out not to be microscopically classical). Complex numbers also form the basis for one of the most powerful methods of doing certain classes of otherwise enormously difficult integrals in mathematics. So you'll have to decide for yourself just how far you want to pursue the discovery of this beautiful mathematics at this time – we will be presenting only the bare minimum necessary to obtain the desired, general, *real* solutions to the SHO ODE below.

Here are the two ***linearly independent solutions***:

$$\begin{aligned} x_+(t) &= A_+e^{+i\omega t} && (9.23)\\ x_-(t) &= A_-e^{-i\omega t} && (9.24) \end{aligned}$$

that follow, one for each possible value of α. Note that we will always get n independent solutions for an nth order linear ODE, because we will always have solve for the roots of an nth order characteristic equation, and there are n of them! $A_\pm$ are the *complex* constants of integration – since the solution is complex already we might as well construct a general complex solution instead of a less general one where the $A_\pm$ are real.

Given these two independent solutions, an ***arbitrary, completely general*** solution can be made up of a sum of these two independent solutions:

$$x(t) = A_+e^{+i\omega t} + A_-e^{-i\omega t} \tag{9.25}$$

We now use a pair of True Facts (that you can read about and see proven in the wikipedia articles linked above or in the online math review). First, let us note the *Euler Equation*:

$$e^{i\theta} = \cos(\theta) + i\sin(\theta) \tag{9.26}$$

This can be proven a number of ways – probably the easiest way to verify it is by noting the equality of the taylor series expansions of both sides – but we can just take it as "given" from here on in this class. Next let us note that a completely general complex number z can always be written as:

$$\begin{aligned} z &= x + iy \\ &= |z|\cos(\theta) + i|z|\sin(\theta) \\ &= |z|(\cos(\theta) + i\sin(\theta)) \\ &= |z|e^{i\theta} && (9.27) \end{aligned}$$

(where we used the Euler equation in the last step so that (for example) we can quite generally write:

$$\begin{aligned} A_+ &= |A_+|e^{+i\phi_+} && (9.28)\\ A_- &= |A_-|e^{-i\phi_-} && (9.29) \end{aligned}$$

for some ***real*** amplitude $|A_\pm|$ and ***real*** phase angles $\pm\phi_\pm$[177].

If we substitute this into the two independent solutions above, we note that they can be written as:

$$\begin{aligned} x_+(t) &= A_+e^{+i\omega t} = |A_+|e^{i\phi_+}e^{+i\omega t} = |A_+|e^{+i(\omega t+\phi_+)} && (9.30)\\ x_-(t) &= A_-e^{-i\omega t} = |A_-|e^{-i\phi_-}e^{-i\omega t} = |A_-|e^{-i(\omega t+\phi_-)} && (9.31)\end{aligned}$$

Finally, we wake up from our mathematical daze, hypnotized by the strange beauty of all of these equations, smack ourselves on the forehead and say "But what am I *thinking!* I need $x(t)$ to be ***real***, because the physical mass m ***cannot possibly be found at an imaginary (or general complex) position***!". So we *take the real part of either of these solutions:*

$$\begin{aligned} \Re x_+(t) &= \Re|A_+|e^{+i(\omega t+\phi_+)}\\ &= |A_+|\Re\left(\cos(\omega t+\phi_+)+i\sin(\omega t+\phi_+)\right)\\ &= |A_+|\cos(\omega t+\phi_+) && (9.32)\end{aligned}$$

and

$$\begin{aligned} \Re x_-(t) &= \Re|A_-|e^{-i(\omega t+\phi_-)}\\ &= |A_-|\Re\left(\cos(\omega t+\phi_-)-i\sin(\omega t+\phi_-)\right)\\ &= |A_-|\cos(\omega t+\phi_-) && (9.33)\end{aligned}$$

These two solutions are the *same*. They differ in the (sign of the) *imaginary* part, but have exactly the same form for the real part. We have to figure out the amplitude and phase of the solution in any event (see below) and we won't get a *different* solution if we use $x_+(t)$, $x_-(t)$, or any linear combination of the two! We can finally get rid of the $\pm$ notation and with it, the last vestige of the complex solutions we used as an intermediary to get this lovely real solution to the position of (e.g.) the mass m as it oscillates connected to the perfect spring.

If you skipped ahead above, resume reading/studying here!

Thus:

$$x(t) = A\cos(\omega t+\phi) \qquad (9.34)$$

is the ***completely general, real*** solution to the SHO ODE of motion, equation 9.14 above, valid in *any* context, including ones with a different context (and even a different variable) leading to a different algebraic form for ω^2.

A few final notes before we go on to try to *understand* this solution. There are two unknown real numbers in this solution, A and ϕ. These are the ***constants of integration!*** Although we didn't exactly "integrate" in the normal sense, we are still picking out a particular solution from an infinity of two-parameter solutions with different ***initial conditions***,

[177] It doesn't matter if we define A_- with a negative phase angle, since ϕ_- might be a positive *or* negative number anyway. It can also always be reduced via modulus 2π into the interval $[0, 2\pi)$, because $e^{i\phi}$ is periodic.

just as we did for constant acceleration problems eight or nine weeks ago! If you like, this solution has to be able to describe the answer for *any permissible value of the initial position and velocity* of the mass at time $t = 0$. Since we can *independently* vary $x(0)$ and $v(0)$, we must have at least a two parameter *family* of solutions to be able to describe a general solution[178].

9.1.2: The Simple Harmonic Oscillator Solution

As we *formally derived above*, the solution to the SHO equation of motion is;

$$x(t) = A\cos(\omega t + \phi) \tag{9.35}$$

where A is called the ***amplitude*** of the oscillation and ϕ is called the ***phase*** of the oscillation. The amplitude tells you how big the oscillation is at peak (maximum displacement from equilibrium); the phase tells you *when* the oscillator was started relative to your clock (the one that reads t). The amplitude has to have the same units as the variable, as $\sin, \cos, \tan, \exp$ functions (and their arguments) are all necessarily ***dimensionless*** in physics[179]. Note that we could have used $\sin(\omega t+\phi)$ as well, or any of several other forms, since $\cos(\theta) = \sin(\theta + \pi/2)$. But you knew that[180].

A and ϕ are two unknowns and have to be determined from the initial conditions, the givens of the problem, as noted above. They are basically constants of integration *just like* x_0 and v_0 for the one-dimensional constant acceleration problem. From this we can easily see that:

$$v(t) = \frac{dx}{dt} = -\omega A\sin(\omega t + \phi) \tag{9.36}$$

and

$$a(t) = \frac{d^2x}{dt^2} = -\omega^2 A\cos(\omega t + \phi) = -\frac{k}{m}x(t) \tag{9.37}$$

This last result ***proves that $x(t)$ solves the original differential equation*** and is where we would have gotten *directly* if we'd assumed a general cosine or sine solution instead of an exponential solution at the beginning of the previous section.

Note Well!

An ***unfortunately commonly made mistake*** for SHO problems is for students to take $F_x = ma = -kx$, write it as:

$$a = -\frac{k}{m}x \tag{9.38}$$

[178] In future courses, math or physics majors might have to cope with situations where you are given two pieces of *data* about the solution, not necessarily initial conditions. For example, you might be given $x(t_1)$ and $x(t_2)$ for two specified times t_1 and t_2 and be required to find the particular solution that satisfied these as a constraint. However, this problem is much more difficult and can easily be insufficient data to fully specify the solution to the problem. We will avoid it here and stick with initial value problems.

[179] All function in physics that have a power series expansion have to be dimensionless because we do not know how to add a liter to a meter, so to speak, or more generally how to add powers of any dimensioned unit.

[180] I hope. If not, time to review the unit circle and all those trig identities from 11th grade...

and then try to substitute this into the *kinematic* solutions for constant acceleration problems that we tried very hard not to blindly memorize back in weeks 1 and 2. That is, they try to write (for example):

$$x(t) = \frac{1}{2}at^2 + v_0 t + x_0 = -\frac{1}{2}\frac{k}{m}xt^2 + v_0 t + x_0 \tag{9.39}$$

This solution is ***so very, very wrong***, so wrong that it is ***deeply disturbing*** when students write it, as it means that they have *completely failed* to understand either the SHO or how to solve even the constant acceleration problem. Obviously it ***bears no resemblance*** to either the correct answer or the observed behavior of a mass on a spring, which is to *oscillate*, not speed up quadratically in time. The appearance of x on both sides of the equation means that it isn't even a solution.

What it reveals is a student who has tried to learn physics by memorization, not by understanding, and hasn't even succeeded in that. Very sad.

Please do not make this mistake!

9.1.3: Plotting the Solution: Relations Involving ω

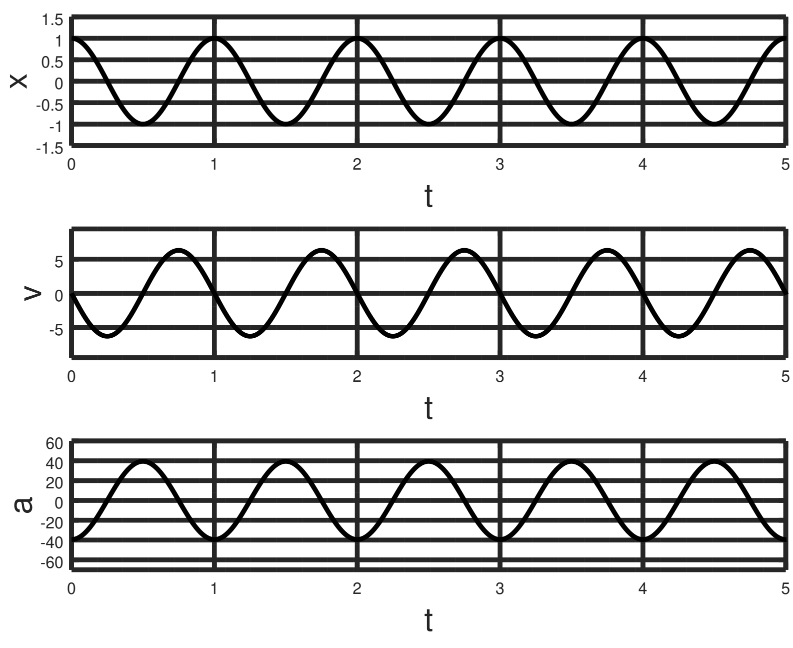


Figure 123: Solutions for a mass on a spring started at $x(0) = A = 1$, $v(0) = 0$ at time $t = 0$ (so that $\phi = 0$). Note well the location of the ***period of oscillation,*** $\boldsymbol{T = 1}$ on the time axis, one full cycle from the beginning.

Since we are going to quite a bit with harmonic oscillators from now on, we should take a few moments to ***plot*** $x(t)$, $v(t)$, and $a(t)$.

We remarked above that $omega$ had to have units of t^{-1}. The following are some True Facts involving ω that You Should Know:

$$\omega = \frac{2\pi}{T} \quad (9.40)$$
$$= 2\pi f \quad (9.41)$$

where T is the *period* of the oscillator the time required for it to return to an identical position *and* velocity) and f is called the *frequency* of the oscillator. Know these relations *instantly*. They are easy to figure out but will cost you valuable time on a quiz or exam if you don't just take the time to completely embrace them now.

Note a very interesting thing. If we build a perfect simple harmonic oscillator, it oscillates at the same frequency *independent of its amplitude*. If we know the period and can count, we have just invented the *clock*. In fact, clocks are nearly *always* made out of various oscillators (why?); some of the earliest clocks were made using a pendulum as an oscillator and mechanical gears to count the oscillations, although now we use the much more precise oscillations of a bit of stressed crystalline quartz (for example) and electronic counters. The idea, however, remains the same.

9.1.4: The Energy of a Mass on a Spring

As we evaluated and discussed in week 3, the spring exerts a ***conservative force*** on the mass m. Thus:

$$U = -W(0 \to x) = -\int_0^x (-kx)dx = \frac{1}{2}kx^2$$
$$= \frac{1}{2}kA^2\cos^2(\omega t + \phi) \quad (9.42)$$

where we have arbitrarily set the zero of potential energy and the zero of the coordinate system to be the equilibrium position[181].

The kinetic energy is:

$$K = \frac{1}{2}mv^2$$
$$= \frac{1}{2}m(\omega^2)A^2\sin^2(\omega t + \phi)$$
$$= \frac{1}{2}m(\frac{k}{m})A^2\sin^2(\omega t + \phi)$$
$$= \frac{1}{2}kA^2\sin^2(\omega t + \phi) \quad (9.43)$$

The total energy is thus:

$$E = \frac{1}{2}kA^2\sin^2(\omega t + \phi) + \frac{1}{2}kA^2\cos^2(\omega t + \phi)$$
$$= \frac{1}{2}kA^2 \quad (9.44)$$

[181] What would it look like if the zero of the energy were at an arbitrary $x = x_0$? What would the force and energy look like if the zero of the coordinates where at the point where the spring is attached to the wall?

and is *constant* in time! Kinda spooky how that works out...

Note that the energy oscillates between being all potential at the extreme ends of the swing (where the object comes to rest) and all kinetic at the equilibrium position (where the object experiences no force).

This more or less concludes our *general* discussion of simple harmonic oscillators in the specific context of a mass on a spring. However, there are many more systems that oscillate harmonically, or nearly harmonically. Let's study another very important one next.

9.2: The Pendulum

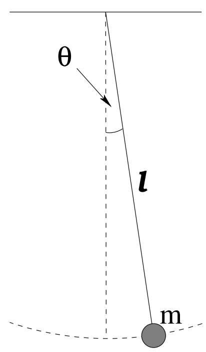


Figure 124: A simple pendulum is just a point like mass suspended on a long string and displaced sideways by a small angle. We will assume no damping forces and that there is no initial velocity into or out of the page, so that the motion is stricly in the plane of the page.

The pendulum is another example of a simple harmonic oscillator, at least for small oscillations. Suppose we have a mass m attached to a string of length ℓ. We swing it up so that the stretched string makes a (small) angle θ_0 with the vertical and release it at some time (not necessarily $t = 0$). What happens?

We write Newton's Second Law for the force component *tangent* to the arc of the circle of the swing as:

$$F_t = -mg\sin(\theta) = ma_t = m\ell\frac{d^2\theta}{dt^2} \tag{9.45}$$

where the latter follows from $a_t = \ell\alpha$ (the angular acceleration). Then we rearrange to get:

$$\frac{d^2\theta}{dt^2} + \frac{g}{\ell}\sin(\theta) = 0 \tag{9.46}$$

This is *almost* a simple harmonic equation. To make it one, we have to use the small angle approximation:

$$\sin(\theta) \approx \theta \tag{9.47}$$

Then:

$$\frac{d^2\theta}{dt^2} + \frac{g}{\ell}\theta = \frac{d^2\theta}{dt^2} + \omega^2\theta = 0 \tag{9.48}$$

where we have defined

$$\omega^2 = \frac{g}{\ell} \tag{9.49}$$

and we can just ***write down the solution***:

$$\theta(t) = \Theta\cos(\omega t + \phi) \tag{9.50}$$

with $\omega = \sqrt{\frac{g}{\ell}}$, Θ the amplitude of the oscillation, and phase ϕ just as before.

Now you see the advantage of all of our hard work in the last section. To solve ***any SHO problem*** one simply puts the differential equation of motion (approximating as necessary) into the form of the SHO ODE ***which we have solved once and for all above!*** We can then just write down the solution and be quite confident that all of its features will be "just like" the features of the solution for a mass on a spring.

For example, if you compute the gravitational potential energy for the pendulum for *arbitrary* angle θ, you get:

$$U(\theta) = mg\ell\,(1 - \cos(\theta)) \tag{9.51}$$

This doesn't initially *look* like the form we might expect from blindly substituting similar terms into the potential energy for mass on the spring, $U(t) = \frac{1}{2}kx(t)^2$. "$k$" for the gravity problem is $m\omega^2$, "$x(t)$" is $\theta(t)$, so:

$$U(t) = \frac{1}{2}mg\ell\Theta^2\sin^2(\omega t + \phi) \tag{9.52}$$

is what we expect.

As an interesting and fun exercise (that really isn't too difficult) see if you can prove that these two forms are really the same, *if* you make the small angle approximation $\theta \ll 1$ in the first form! This shows you pretty much where the approximation will *break down* as Θ is gradually increased. For large enough θ, the period of a pendulum clock *does* depend on the amplitude of the swing. This (and the next section) explains grandfather clocks – clocks with very long penduli that can swing very slowly through very small angles – and why they were so accurate for their day.

9.2.1: The Physical Pendulum

In the treatment of the ordinary pendulum above, we just used Newton's Second Law directly to get the equation of motion. This was possible only because we could neglect the mass of the string and because we could treat the mass like a point mass at its end, so that its moment of inertia was (if you like) just $m\ell^2$.

That is, we *could* have solved it using Newton's Second Law for *rotation* instead. If θ in figure 124 is positive (out of the page), then the torque due to gravity is:

$$\tau = -mg\ell\sin(\theta) \tag{9.53}$$

and we can get to the *same equation of motion* via:

$$\begin{aligned}
I\alpha \;=\; m\ell^2\frac{d^2\theta}{dt^2} &= -mg\ell\sin(\theta) \;=\; \tau \\
\frac{d^2\theta}{dt^2} &= -\frac{g}{\ell}\sin(\theta) \\
\frac{d^2\theta}{dt^2} + \frac{g}{\ell}\sin(\theta) &= 0 \\
\frac{d^2\theta}{dt^2} + \omega^2\theta \;=\; \frac{d^2\theta}{dt^2} + \frac{g}{\ell}\theta &= 0
\end{aligned} \tag{9.54}$$

(where we use the small angle approximation in the last step as before).

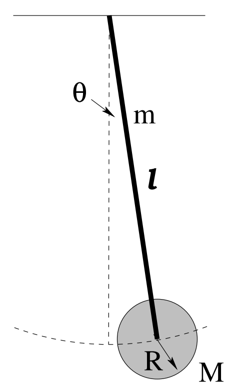


Figure 125: A physical pendulum takes into account the fact that the masses that make up the pendulum have a total ***moment of inertia*** about the pivot of the pendulum.

However, *real* grandfather clocks often have a large, massive pendulum like the one above pictured in figure 125 – a long massive rod (of length ℓ and uniform mass m) with a large round disk (of radius R and mass M) at the end. Both the rod and disk rotate about the pivot with each oscillation; they have *angular momemtum*. Newton's Law for forces alone no longer suffices. We must use torque and the moment of inertia (found using the parallel axis theorem) to obtain the frequency of the oscillator[182].

To do this we go through the *same steps* that I just did for the simple pendulum. The only real difference is that now the weight of both masses contribute to the torque (and the force exerted by the pivot can be ignored), and as noted we have to work harder to compute the moment of inertia.

So let's start by computing the net gravitational torque on the system at an arbitrary (small) angle θ. We get a contribution from the rod (where the weight acts "at the center of mass" of the rod) and from the pendulum disk:

$$\tau = -\left(mg\frac{\ell}{2} + Mg\ell\right)\sin(\theta) \tag{9.55}$$

The negative sign is there because the torque *opposes* the angular displacement from equilibrium and points *into* the page as drawn.

[182] I know, I know, you had hoped that you could finally forget all of that stressful stuff we learned back in the weeks we covered torque. Sorry. Not happening.

Next we set this equal to $I\alpha$, where I is the total moment of inertia for the *system* about the pivot of the pendulum and simplify:

$$\begin{aligned}
I\alpha \;=\; I\frac{d^2\theta}{dt^2} &= -\left(mg\frac{\ell}{2} + Mg\ell\right)\sin(\theta) \;=\; \tau \\
\frac{d^2\theta}{dt^2} &= -\frac{\left(mg\frac{\ell}{2} + Mg\ell\right)}{I}\sin(\theta) \\
\frac{d^2\theta}{dt^2} + \frac{\left(mg\frac{\ell}{2} + Mg\ell\right)}{I}\sin(\theta) &= 0 \\
\frac{d^2\theta}{dt^2} + \frac{\left(mg\frac{\ell}{2} + Mg\ell\right)}{I}\theta &= 0
\end{aligned} \tag{9.56}$$

where we finish off with the small angle approximation as usual for pendulums. We can now recognize that this ODE has the standard form of the SHO ODE:

$$\frac{d^2\theta}{dt^2} + \omega^2\theta = 0 \tag{9.57}$$

with

$$\omega^2 = \frac{\left(mg\frac{\ell}{2} + Mg\ell\right)}{I} \tag{9.58}$$

I left the result in terms of I because it is simpler that way, but of course we have to evaluate I in order to evaluate ω^2. Using the parallel axis theorem (and/or the moment of inertia of a rod about a pivot through one end) we get:

$$I = \frac{1}{3}m\ell^2 + \frac{1}{2}MR^2 + M\ell^2 \tag{9.59}$$

This is "the moment of inertia of the rod plus the moment of inertia of the disk rotating about a parallel axis a distance ℓ away from its center of mass". From this we can read off the angular frequency:

$$\omega^2 = \frac{4\pi^2}{T^2} = \frac{\left(mg\frac{\ell}{2} + Mg\ell\right)}{\frac{1}{3}m\ell^2 + \frac{1}{2}MR^2 + M\ell^2} \tag{9.60}$$

With ω in hand, we know everything. For example:

$$\theta(t) = \Theta\cos(\omega t + \phi) \tag{9.61}$$

gives us the angular trajectory. We can easily solve for the period T, the frequency $f = 1/T$, the spatial or angular velocity, or whatever we like.

Note that the energy of this sort of pendulum can be tricky, not because it is conceptually any different from before but because there are so many symbols in the answer. For example, its potential energy is easy enough – it depends on the elevation of the center of masses of the rod and the disk. The

$$U(t) = (mgh(t) + MgH(t)) = \left(mg\frac{\ell}{2} + Mg\ell\right)(1 - \cos(\theta(t))) \tag{9.62}$$

where hopefully it is obvious that $h(t) = \ell/2\;(1 - \cos(\theta(t)))$ and $H(t) = \ell\;(1 - \cos(\theta(t))) = 2h(t)$. Note that the time dependence is entirely inherited from the fact that $\theta(t)$ is a function of time.

The kinetic energy is given by:

$$K(t) = \frac{1}{2} I \Omega^2 \tag{9.63}$$

where $\Omega = d\theta/dt$ as usual.

We can easily evaluate:

$$\Omega = \frac{d\theta}{dt} = -\omega\Theta\sin(\omega t + \phi) \tag{9.64}$$

so that

$$K(t) = \frac{1}{2} I\Omega^2 = \frac{1}{2} I\omega^2\Theta^2\sin^2(\omega t + \phi) \tag{9.65}$$

Recalling the definition of ω^2 above, this simplifies to:

$$K(t) = \frac{1}{2}\left(mg\frac{\ell}{2} + Mg\ell\right)\Theta^2\sin^2(\omega t + \phi) \tag{9.66}$$

so that:

$$\begin{aligned} E_{\rm tot} &= U + K \\ &= \left(mg\frac{\ell}{2} + Mg\ell\right)(1 - \cos(\theta(t))) + \frac{1}{2}\left(mg\frac{\ell}{2} + Mg\ell\right)\Theta^2\sin^2(\omega t + \phi) \end{aligned} \tag{9.67}$$

which is not, in fact, a constant.

However, for small angles (the only situation where our solution is valid, actually) it is *approximately* a constant as we will now show. The trick is to use the Taylor series for the cosine function:

$$\cos(\theta) = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + ... \tag{9.68}$$

and keep only the first term:

$$1 - \cos(\theta) = \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + ... \approx \frac{\theta^2}{2} = \frac{1}{2}\Theta^2\cos^2(\omega t + \phi) \tag{9.69}$$

You should now be able to see that in fact, the total energy of the oscillator *is* "constant" in the small angle approximation.

Of course, it is actually constant even for *large* oscillations, but proving this requires solving the exact ODE with the $\sin(\theta)$ in it. This ODE is a version of the Sine-Gordon equation and has an elliptic integral for a solution that is way, way beyond the level of this course and indeed is right up there at the edge of some serious (but as always, way cool) math. We will stick with small angles!

9.3: Damped Oscillation

So far, all the oscillators we've treated are *ideal*. There is no friction or damping. In the real world, of course, things *always* damp down. You have to keep pushing the kid on the swing or they slowly come to rest. Your car doesn't *keep* bouncing after going through a pothole in the road. Buildings and bridges, clocks and kids, real oscillators all have damping.

Damping forces can be very complicated. There is kinetic friction, which tends to be independent of speed. There are various fluid drag forces, which tend to depend on speed, but in a sometimes complicated way depending on the shape of the object and e.g. the Reynolds number, as flow around it converts from laminar to turbulent. There may be other forces that we haven't studied yet that contribute at least weakly to damping[183]. So in order to get beyond a very qualitative description of damping, we're going to have to specify a *form* for the damping force (ideally one we can work with, i.e. integrate).

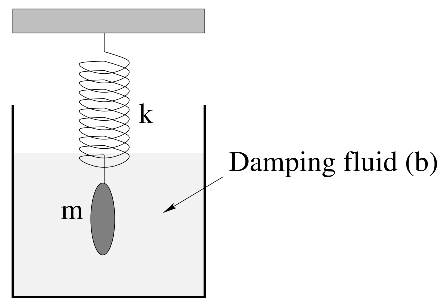


Figure 126: A smooth convex mass on a spring that is immersed in a suitable damping liquid experiences a *linear damping force* due to viscous interaction with the fluid in laminar flow. This idealizes the forces to where we can solve them and understand semi-quantitatively how to describe damped oscillation.

We'll pick the simplest possible one, illustrated in figure 126 above – a ***linear damping force*** such as we would expect to observe in ***laminar flow*** around the oscillating object as long as it moves at speeds too low to excite turbulence in the surrounding fluid.

$$F_d = -bv \tag{9.70}$$

As before (see e.g. week 2) b is called the ***damping constant*** or ***damping coefficient***. With this form we can get an exact solution to the differential equation easily (good), get a preview of a solution we'll need next semester to study LRC circuits (better), and get a very nice qualitative picture of damping even when the damping force is *not* precisely linear (best).

As before, we write Newton's Second Law for a mass m on a spring with spring constant k and a damping force $-bv$:

$$F_x = -kx - bv = ma = m\frac{d^2x}{dt^2} \tag{9.71}$$

Again, simple manipulation leads to:

$$\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x = 0 \tag{9.72}$$

which is the "standard form" for a damped mass on a spring and (within fairly obvious substitutions) for the general linearly damped SHO.

[183]Such as gravitational damping – an oscillating mass interacts with its massive environment to very, very slowly convert its organized energy into heat. We're talking slowly, mind you. Still, fast enough that the moon is gravitationally locked with the earth over geological times, and e.g. tidal/gravitational forces heat the moon Europa (as it orbits Jupiter) to the point where it is speculated that there is liquid water under the ice on its surface...

This is still a linear, second order, homogeneous, ordinary differential equation, but now we ***cannot just guess*** $\boldsymbol{x(t) = A\cos(\omega t)}$ because the first derivative of a cosine is a sine! This time we really ***must*** guess that $x(t)$ is a function that is proportional to its own first derivative!

We therefore guess $x(t) = Ae^{\alpha t}$ ***as before***, substitute for $x(t)$ and its derivatives, and get:

$$\left(\alpha^2 + \frac{b}{m}\alpha + \frac{k}{m}\right) Ae^{\alpha t} = 0 \tag{9.73}$$

As before, we exclude the trivial solution $x(t) = 0$ as being too boring to solve for (requiring that $A \neq 0$, that is) and are left with the characteristic equation for α:

$$\alpha^2 + \frac{b}{m}\alpha + \frac{k}{m} = 0 \tag{9.74}$$

This quadratic equation must be satisfied in order for our guess to be a nontrivial solution to the damped SHO ODE.

To solve for α we have to use the ***dread quadratic formula***:

$$\alpha = \frac{\frac{-b}{m} \pm \sqrt{\frac{b^2}{m^2} - \frac{4k}{m}}}{2} \tag{9.75}$$

This isn't quite where we want it. We expect from experience and intuition that for ***weak damping*** we should get an oscillating solution, indeed one that (in the limit that $b \to 0$) turns back into our familiar solution to the undamped SHO above. In order to get an oscillating solution, the argument of the square root must be ***negative*** so that our solution becomes a complex exponential solution as before!

This motivates us to factor a $-4k/m$ out from under the radical (where it becomes $i\omega_0$, where $\omega_0 = \sqrt{k/m}$ is the frequency of the *undamped* oscillator with the same mass and spring constant). In addition, we simplify the first term and get:

$$\alpha = \frac{-b}{2m} \pm i\omega_0\sqrt{1 - \frac{b^2}{4km}} \tag{9.76}$$

As was the case for the undamped SHO, there are two solutions:

$$x_\pm(t) = A_\pm e^{\frac{-b}{2m}t} e^{\pm i\omega' t} \tag{9.77}$$

where

$$\omega' = \omega_0\sqrt{1 - \frac{b^2}{4km}} \tag{9.78}$$

This, you will note, is ***not terribly simple or easy to remember!*** Yet you are responsible for knowing it. You have the usual choice – work very hard to memorize it, or learn to do the derivation(s).

I personally ***do not remember it at all*** save for a week or two around the time I teach it each semester. Too big of a pain, too easy to derive if I need it. But here you must suit yourself – either memorize it the same way that you'd memorize the digits of π, by lots and lots of mindless practice, or learn how to solve the equation, as you prefer.

Without recapitulating the entire argument, it should be fairly obvious that can take the real part of their sum, get formally identical terms, and combine them to get the general real solution:

$$x_\pm(t) = Ae^{\frac{-bt}{2m}}\cos(\omega' t + \phi) \tag{9.79}$$

where A is the real initial amplitude and ϕ determines the relative phase of the oscillator. The only two differences, then, are that the frequency of the oscillator is *shifted* to ω' and the whole solution is ***exponentially damped in time***.

9.3.1: Properties of the Damped Oscillator

There are several properties of the damped oscillator that are important to know.

- The amplitude damps *exponentially* as time advances. After a certain amount of time, the amplitude is halved. After the *same* amount of time, it is halved again.
- The frequency ω' is shifted so that it is ***smaller than*** ω_0, the frequency of the identical but undamped oscillator with the same mass and spring constant.
- The oscillator can be ***(under)damped***, ***critically damped***, or ***overdamped***. These terms are defined below.
- For exponential decay problems, recall that it is often convenient to define the ***exponential decay time***, in this case:

  $$\tau = \frac{2m}{b} \tag{9.80}$$

  This is the time required for the amplitude to go down to $1/e$ of its value from *any* starting time. For the purpose of drawing plots, you can imagine $e = 2.718281828 \approx 3$ so that $1/e \approx 1/3$. Pay attention to how the damping time ***scales*** with m and b. This will help you develop a conceptual understanding of damping.

Several of these properties are illustrated in figure 127. In this figure the ***exponential envelope*** of the damping is illustrated – this envelope determines the maximum amplitude of the oscillation as the total energy of the oscillating mass decays, turned into heat in the damping fluid. The period T' is indeed longer, but even for this relatively rapid damping, it is still nearly identical to T_0! See if you can determine what ω' is in terms of ω_0 numerically given that $\omega_0 = 2\pi$ and $b/m = 0.3$. Pretty close, right?

This oscillator is ***underdamped***. An oscillator is underdamped if ω' is ***real***, which will be true if:

$$\frac{b^2}{4m^2} = \left(\frac{b}{2m}\right)^2 < \frac{k}{m} = \omega_0^2 \tag{9.81}$$

An underdamped oscillator will exhibit true oscillations, eventually (exponentially) approaching zero amplitude due to damping.

The oscillator is ***critically damped*** if ω' is ***zero***. This occurs when:

$$\frac{b^2}{4m^2} = \left(\frac{b}{2m}\right)^2 = \frac{k}{m} = \omega_0^2 \tag{9.82}$$

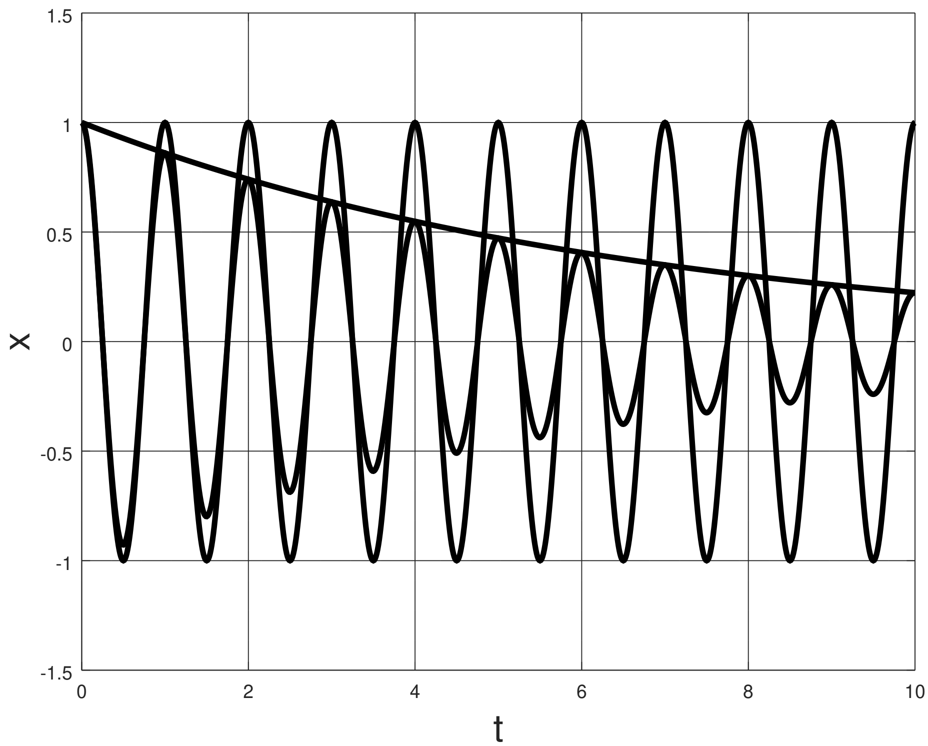


Figure 127: Two identical oscillators, one undamped (with $\omega_0 = 2\pi$, or if you prefer with an undamped period $T_0 = 1$) and one weakly damped ($b/m = 0.3$).

The oscillator will then *not oscillate* – it will go to zero *exponentially* in the shortest possible time. This (and barely underdamped and overdamped oscillators) is illustrated in figure 128.

The oscillator is *overdamped* if ω' is imaginary, which will be true if

$$\frac{b^2}{4m^2} = \left(\frac{b}{2m}\right)^2 > \frac{k}{m} = \omega_0^2 \tag{9.83}$$

In this case α ***is entirely real*** and has a component that damps very slowly. The amplitude goes to zero exponentially as before, but over a longer (possibly much longer) time and does not oscillate through zero at all.

Note that these inequalities and equalities that establish the ***critical boundary*** between oscillating and non-oscillating solutions involve the relative size of the inverse time constant associated with *damping* compared to the inverse time constant (times 2π) associated with *oscillation*. When the former (damping) is larger than the latter (oscillation), damping *wins* and the solution is *overdamped*. When it is smaller, oscillation wins and the solution is underdamped. When they are precisely equal, oscillation precisely disappears, k isn't *quite* strong enough compared to b to give the mass enough momentum to make it across equilibrium to the other side. Keep this in mind for the next semester, where *exactly the same relationship* exists for LRC circuits, which exhibit damped simple harmonic oscillation that is *precisely the same* as that seen here for a linearly damped mass on a perfect spring.

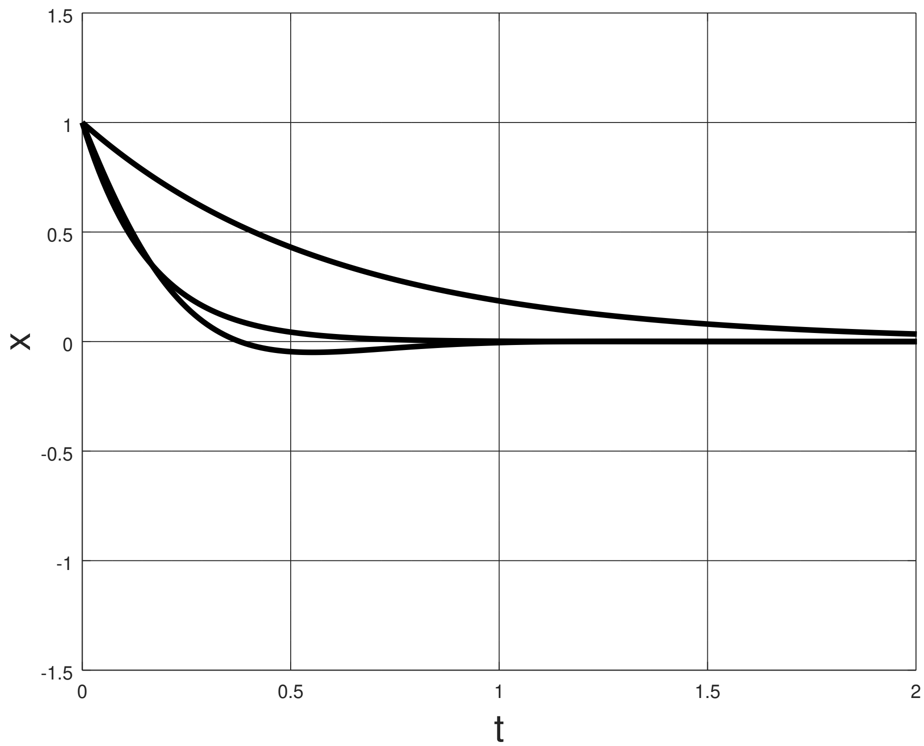


Figure 128: Three curves: Underdamped ($b/m = 2\pi$) barely oscillates. T' is now clearly longer than T_0. Critically damped ($b/m = 4\pi$) goes exponentially to zero in minimum time. Overdamped ($b/m = 8\pi$) goes to zero exponentiall, but much more slowly.

Example 9.3.1: Car Shock Absorbers

This example isn't something one can compute, it is something you experience nearly every day, at least if you drive or ride around in a car.

A car's shock absorbers are there to reduce the "bumpiness" of a bumpy road. Shock absorbers are basically big powerful springs that carry your car suspended in equilibrium between the weight of the car and the spring force.

If your wheels bounce up over a ridge in the road, the shock absorber spring compresses, storing the energy from the "collision" briefly and then giving it back without the car *itself* reacting. However, if the spring is not damped, the subsequent motion of the car would be to bounce up and down for many tens or hundreds of meters down the road, with your control over the car seriously compromised. For that reason, shock absorbers are strongly damped with a suitable fluid (basically a thick oil).

If the oil is *too* thick, however, the shock absorbers become overdamped. The car takes so long to come back to equilibrium after a bump that compresses them that one rides with one's shocks constantly somewhat compressed. This reduces their effectiveness and one feels "every bump in the road", which is *also* not great for safe control.

Ideally, then, your car's shocks should be ***barely underdamped***. This will let the car bounce *through* equilibrium to where it is "almost" in equilibrium even faster than a perfectly critically damped shock and yet still rapidly damp to equilibrium right away, ready for the next shock.

So here's how to test the shocks on your car. Push down (with all of your weight) on each of the corner fenders of your car, testing the shock on that corner. Release your weight suddenly so that it springs back up towards equilibrium.

If the car "bounces" *once* and then returns to equilibrium when you push down on a fender and suddenly release it, the shocks are good. If it bounces three or four time the shocks are too underdamped and dangerous as you could lose control after a big bump. If it doesn't bounce up and back down at all at all and instead slowly oozes back up to level from below, it is overdamped and dangerous, as a succession of sharp bumps could leave your shocks still compressed and unable to absorb the impact of the last one and keep your tires still on the ground.

Damped oscillation is ubiquitous. Pendulums, once started, oscillate for only a while before coming to rest. Guitar strings, once plucked, damp down to quiet again quite rapidly. Charges in atoms can oscillate and give off *light* until the self-force exerted by their very radiation they emit damps the excitation. Cars need barely underdamped shock absorbers. Very tall buildings ("skyscrapers") in a city usually have specially designed dampers in them as well to keep them from swaying too much in a strong wind. Houses are build with lots of damping forces in them to keep them quiet. Fully understanding damped (and eventually driven) oscillation is essential to many sciences as well as both mechanical and electrical engineering.

9.3.2: Energy Damping: Q-value

Suppose that one has a "weakly" damped oscillator. In the terminology above, that means that it is underdamped and, specifically, that $\omega' \approx \omega_0$ (because $1 \gg \frac{b^2}{4km}$). Every full oscillation of the system results in a loss of energy to the nonconservative work of the linear damping force. We can characterize this loss of energy as the ***dimensionless fraction of energy lost per cycle***:

$$\frac{\Delta E}{E} \tag{9.84}$$

We can easily compute this in the specific limit of the weakly damped oscillator. Let's assume that we start the oscillator at any point where it has a maximum amplitude (and hence with zero phase) so that (for $\omega' \approx \omega_0$, recall):

$$x(t) = Ae^{-bt/2m}\cos(\omega_0 t) \tag{9.85}$$

Then at $t = 0$, there is no kinetic energy and:

$$E_0 = U_0 = \frac{1}{2}kA^2 \tag{9.86}$$

After one cycle (at time $t = T_0 = 2\pi/\omega_0$):

$$E_f = U_f = \frac{1}{2}kA^2e^{-bT_0/m} \tag{9.87}$$

or:

$$\frac{\Delta E}{E_0}\frac{E_f - E_0}{E_0} = \frac{\frac{1}{2}kA^2\left(e^{-bT_0/m} - 1\right)}{\frac{1}{2}kA^2} = \left(e^{-bT_0/m} - 1\right) \tag{9.88}$$

This result is actually independent of any particular cycle or phase angle – it is the dimensionless fractional energy loss per cycle.

Weak damping specifically implies that $bT_0/m \ll 1$, so we can expand the exponential, cancel the leading 1, and keep the first surviving term:

$$\frac{\Delta E}{E_0} = ((1 - bT_0/m + ...) - 1) \approx \frac{-bT_0}{m} \tag{9.89}$$

This dimensionless relation looks pretty useful in and of itself, but (as we shall see) a slightly different form is more useful for describing damped *driven* harmonic oscillators. To obtain this form, we express $T_0 = 2\pi/\omega_0$ and rearrange, defining the result to be the ***Q-value*** of the damped oscillator:

$$Q = 2\pi\frac{E}{|\Delta E|} = \frac{m\omega_0}{b} \tag{9.90}$$

(Note that we lose the minus sign in the process – we just remember that this is related to the rate at which the oscillator ***loses*** energy.)

Note that the Q-value is basically $1/2\zeta$ as defined above, where the factor of 2 arises because the energy goes as position *squared*. It has the opposite meaning of ζ as well. ***Weak*** damping is ***large*** Q or *small* ζ. ***Strong*** damping is ***small*** Q or *large* ζ.

In English, the "Q" stands for "quality", and the Q factor is also called the *quality* factor. This now makes verbal sense – a "high quality oscillator" is one that oscillates a long time with slow damping and has large Q-for-quality compared to one that rapidly damps.

9.4: Damped, Driven Oscillation: Resonance

By and large, most of you who are reading this textbook directly experienced ***damped, driven oscillation*** long before you were five years old, in some cases as early as a few months old! This is the physics of ***the swing***, among other things. Babies love swings – one of our sons was colicky when he was very young and would sometimes only be able to get some peace (so we could get some too!) when he was tucked into a wind-up swing. Humans of all ages seem to like a rhythmic swaying motion; children play on swings, adults rock in rocking chairs.

Damped, driven, oscillation is also key in another nearly ubiquitous aspect of human life – the *clock*. Nature provides us with a few "natural" clocks, the most prominent one being the diurnal clock associated with the rotation of the Earth, read from observing the orientation of the sun, moon, and night sky and translating it into a time.

The human body itself contains a number of clocks including the most accessible one, the heartbeat. Although the historical evidence suggests that the size of the second is derived from systematic divisions of the day according to numerological rules in the extremely distant past, surely it is no accident that the smallest common unit of everyday time almost precisely matches the human heartbeat. Unfortunately, the "normal" human heartbeat varies by a factor of around three as one moves from resting to heavy exercise, a range that is further increased by the *ab*normal heartbeat of people with cardiac insufficiency, cardioelectric abnormalities, or taking various drugs. It isn't a very *precise* clock, in other words, although as it turns out it played a key role in the development of precise clocks, which in turn played a *crucial* role in the invention of physics.

Here, there is an interesting story. Galileo Galilei used his own heartbeat to time the oscillations of a large chandelier in the cathedral in Pisa around 1582 and discovered one of the key properties of the oscillations of a pendulum[184] (discussed above), *isochronism*: the fact that the period of a pendulum is independent of both the (small angle) amplitude of oscillation and the mass that is oscillating. This led Galileo and a physician friend to invent both the *metronome* (for musicians) and a simple pendulum device called the *pulsilogium* to use to time the pulse of patients! These were the world's first really accurate clocks, and variations of them eventually became the *pendulum clock*. Carefully engineered pendulum clocks that were compensated for thermal expansion of their rods, the temperature-dependent weather dependent buoyant force exerted by the air on their pendulum, friction and damping were the best clocks in the world and used as international time standards through the late 1920s and early 1930s, when they were superceded by *another*, still more accurate, damped driven oscillator – the quartz crystal oscillator.

Swings, springs, clocks and more – driven, resonant harmonic oscillation has been a part of everyday experience for at least two or three thousand years. Although it isn't obvious at first, *ordinary walking* at a comfortable pace is an example of damped, driven oscillation and resonance.

Military marching – the precise timing of the pace of soldiers in formation – was apparently invented by the Romans. Indeed, this invention gives us one of our most common measures of distance, the ***mile***. The word mile is derived from *milia passuum* – a thousand paces, where a "pace" is a complete cycle of two steps with a length slightly more than five feet – and Roman armies, by marching at a fixed "standard" pace, would consistently cover twenty miles in five summer hours, or by increasing the length of their pace slightly, twenty four miles in the same number of hours. Note well that the mile was originally a *decimal quantity* – a multiple of ten units! Alas, the "pace" did not become the unit of length in England – the "yard" was instead, defined (believe it or not) in terms of the width of a grain of barleycorn[185] . Yes folks, you heard it first here – there is an intimate connection between the *making of beer* (barley is one of the oldest cultivated grains and was used primarily for making beer and as an animal fodder dating back to *neolithic* times) and *the English units of length*.

[184]Wikipedia: http://www.wikipedia.org/wiki/pendulum. I'd strongly recommend that students read through this article, as it is absolutely fascinating. At this point you should already understand that the development of physics required ***good clocks!*** It quite literally *could not have happened* without them, and good clocks, sufficiently accurate to measure e.g. the variations in the apparent gravitational field with height and position around the world, did not exist before the pendulum clock was conceived of and partially invented by Galileo in the late 1500s and invented in fact by Christiaan Huygens in 1656. Properties of the motion of the pendulum were key elements in Newton's invention of both the law of gravitation and his physics.

[185]Wikipedia: http://www.wikipedia.org/wiki/Yard. Yet another fascinating article – three barley grains to the inch, twelve inches to the foot, three feet to the yard, and $22 \times 220 = 4840$ square yards make an acre.

The proper definition of a mile in the English system is thus the length of 190080 ***grains of barleycorn!***! That's almost exactly four cubic feet of barley, which is enough to make roughly 20 gallons of beer. Coincidence? I don't think so. And people wonder why the rest of the world considers Americans and the British to be mad...

Roman soldiers also discovered another important aspect of resonance – it can *destroy human-engineered structures!* The "standard marching pace" of the Roman soldiers was 4000 paces per hour, just over two steps per second. This pace could easily *match* the natural frequency of oscillation of the *bridges* over which the soldiers marched, and driving a bridge oscillation, at resonance, with the weight of a hundred or so men was more than enough to ***destroy the bridge***. Since Roman times, then, although soldiers march with discipline whenever they are on the road, they *break cadence* and cross bridges with an irregular, random step lest they find themselves and the remnants of the bridge in the water, trying to swim in full armor.

This sort of resonance also affects the stability of buildings and bridges today – earthquakes can drive resonances of either one, the wind can drive resonances of either one. Building a sound bridge or tall building requires a careful consideration and damping of the natural frequencies of the structure. The ***Tacoma Bridge Disaster***[186] serves as a modern-times example of the consequences of failing to design for resonance. In more recent times part of the devastation caused by the Haiti Earthquake[187] was caused by the lack of earthquake-proofing – protection against earthquake-driven resonances – in the cheap construction methods used in buildings of all sorts.

From all of this, it seems like establishing at the very least a semi-quantitative understanding of resonance is in order, and as usual math, physics or engineering students will need to go the extra 190080 barleycorn grain lengths and work through the math properly. To manage this, we need to begin with a model.

9.4.1: Harmonic Driving Forces

Let's start by thinking about what you all know from learning to swing on a swing. If you just sit on a swing, nothing happens. You have to "pump" the swing. Pumping the swing is accomplished by pulling the ropes and shifting your weight at the highest point of each oscillation so that the force exerted by the rope no longer passes through your center of mass and hence can *exert a torque in the current direction of rotation*. You know from experience that this torque must be applied *periodically* at a frequency that *matches* the natural frequency of the swing and *in phase with the motion* of the swing in order to increase the amplitude of the swinging oscillation. If you simply jerk backwards and forwards with the same motions as those used to pump "right" but at the wrong (non-resonant) frequency or randomly, you don't ever build up much amplitude. If you apply the torques with the wrong *phase* you will not manage to get the same amplitude that you'd get pumping in phase with the motion.

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[186]Wikipedia: http://www.wikipedia.org/wiki/Tacoma Narrows Bridge (1940). Again, a marvelous article that contains a short clip of the bridge oscillating in resonance to collapse. Note that the resonance in question was due more to the driving of several ***wave*** resonances rather than a simple harmonic oscillator resonance, but the principle is exactly the same.

[187]Wikipedia: http://www.wikipedia.org/wiki/2010 Haiti earthquake.

That's really it, qualitatively. Resonance consists of driving an oscillator at its natural frequency, in phase with the motion, to achieve the greatest possible oscillation amplitude (ultimately limited by things like practical physical constraints and damping).

A pendulum, however, isn't a good system for us to use as a *quantitative* model. For one thing, it isn't really a harmonic oscillator – we automatically adjust our pumping to remain resonant as we swing closer and closer to the angle of $\pi/2$ where the swing chains become limp and we can no longer cheat some torque out of the combination of the pivot force and gravity, but the frequency itself starts to significantly change as the small angle approximation breaks down, and breaking it down is the *point* of swinging on a swing! Who wants to swing only through small angles!

The kind of force we exert in swinging a swing isn't too great, either. It is hardly smooth – we really only pump at all very close to the top of our swinging motion (on both sides) – in between we just coast. We'd prefer instead to assume a periodic driving force that is mathematically relatively easy to treat.

These two things taken together more or less uniquely determine the best model for us to use to understand resonance. This is an ***underdamped mass on a spring*** being driven by an external ***harmonic driving force***, all in one dimension:

$$F_x^{\rm ext}(t) = F_0 \cos(\omega t) \tag{9.91}$$

In this expression, both the angular frequency ω and the amplitude of the applied force are ***free parameters***. Note especially that ω is *not* necessarily the natural or shifted frequency of the mass on the spring – it can be anything, just as you can push your little brother or sister on a swing at the *right* frequency to build up their swing amplitude or the *wrong* one to jerk them back and forth and rattle their teeth without building up much amplitude at all. We'd like to be able to derive the solution for both of these general cases and everything in between!

Although there are a number of ways one can exert such a force in the real world, one relatively simple one is to drive the "fixed" end of the spring harmonically through some amplitude e.g. $A(t) = -A_0\cos(\omega t)$; this will modulate the total force exerted by the spring on the mass in just the right way:

$$\begin{aligned} F_x(t) &= -k(x + A(t)) - bv \\ &= -kx - bv + kA_0\cos(\omega t) \\ &= -kx - bv + F_0\cos(\omega t) \end{aligned} \tag{9.92}$$

where we've included the usual linear damping force $-bv$. The minus sign is chosen deliberately to make the driving force positive on the right in the *in*homogeneous ODE obtained below. The apparatus we might use to observe this under controlled circumstances in the lab is drawn in figure 129.

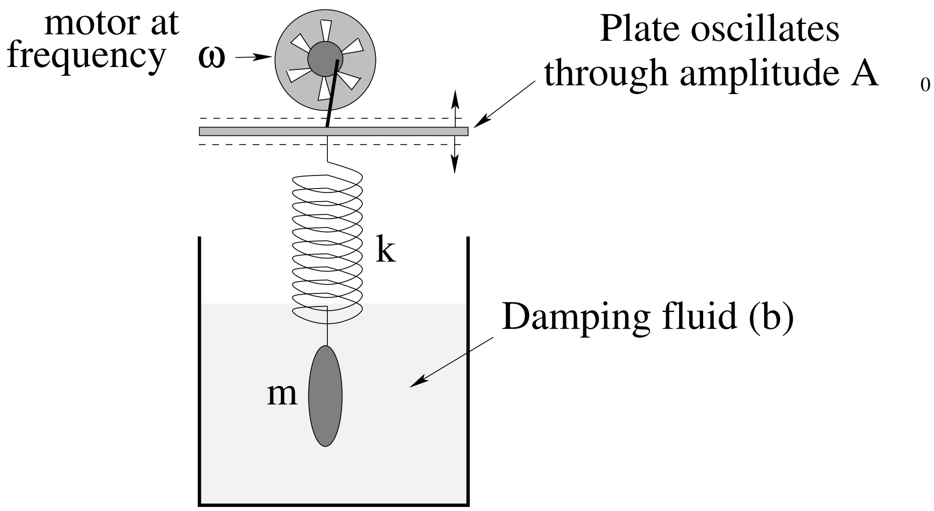


Figure 129: A small frequency-controlled motor drives the "fixed" end of a spring attached to a damped mass up and down through a (variable) amplitude A_0 at (independently adjustable) angular frequency ω, thereby exerting an additional harmonic driving force on the mass.

Putting this all together, our equation of motion can be written:

$$\begin{aligned}
ma &= -kx - bv + F_0\cos(\omega t) \\
m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx &= F_0\cos(\omega t) \\
\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x &= \frac{F_0}{m}\cos(\omega t) \\
\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \omega_0^2 x &= \frac{F_0}{m}\cos(\omega t) \\
\frac{d^2x}{dt^2} + 2\left(\frac{b}{2m\omega_0}\right)\omega_0\frac{dx}{dt} + \omega_0^2 x &= \frac{F_0}{m}\cos(\omega t) \\
\frac{d^2x}{dt^2} + 2\zeta\omega_0\frac{dx}{dt} + \omega_0^2 x &= \frac{F_0}{m}\cos(\omega t) \qquad (9.93)
\end{aligned}$$

where as before $\omega_0 = \sqrt{k/m}$ is the "natural frequency" of the undamped oscillator and where we have written it in terms of the ***damping ratio***:

$$\zeta = \frac{b}{2m\omega_0} \qquad (9.94)$$

that was the *key parameter* that determined whether or not a damped oscillator was underdamped, critically damped or overdamped (for $\zeta < 1, \zeta = 1, \zeta > 1$ respectively). This form makes it easier to write the resonance solutions "generically" so that they can be applied to different problems.

This is a ***second order, linear, inhomogeneous ordinary differential equation*** – the wordk "inhomogeneous" basically means that there is a function of time instead of 0 on the right.

Although we will not treat it in this course, it is possible to write *arbitrary* functions of time as a superposition of harmonic functions of time using either fourier series or more generally, fourier analysis[188] . This means that the solution we develop here (however crudely) can be transformed into part of a *general solution, valid for an arbitrary driving force $F(t)$!* One day, maybe, you[189] might learn how, but obviously that would cause our brains to explode to attempt it *today*, unless you are a math super-genius or at least, have learned a lot more math than undergrads *usually* have learned by the time they take this course.

9.4.2: Solution to Damped, Driven, Simple Harmonic Oscillator

We will not, actually, fully develop this solution this semester. It is a bit easier to do so in the context of LRC circuits next semester, and you'll have had the chance to "sleep on" all that you've learned about driven oscillation *this* semester and will discover that, magically, it is somehow easier and less intimidating the second time around. Math like this is best learned in several passes with a bit of time for our brains to "digest" in between.

First, let us really learn an important property of inhomogeneous ordinary differential equations. Suppose we have both:

$$\frac{d^2x_H}{dt^2} + 2\zeta\omega_0\frac{dx_H}{dt} + \omega_0^2 x_H = 0 \tag{9.95}$$

(the homogeneous ODE for a damped SHO) with solution $x_H(t)$ given in the previous section, and *also* have:

$$\frac{d^2x_I}{dt^2} + 2\zeta\omega_0\frac{dx_I}{dt} + \omega_0^2 x_I = \frac{F_0}{m}\cos(\omega t) \tag{9.96}$$

(the inhomogeneous ODE for a damped *harmonically driven* SHO) with solution $x_I(t)$ yet to be determined. Suppose also that we have found at least one solution to the inhomogeneous ODE $x_p(t)$, usually referred to as a ***particular solution***. Then it is easy to show from the *linearity* of the ODE that:

$$x_I(t) = x_H(t) + x_p(t) \tag{9.97}$$

also solves the inhomogeneous equation:

$$\begin{aligned}
\frac{d^2x_I}{dt^2} + 2\zeta\omega_0\frac{dx_I}{dt} + \omega_0^2 x_I &= \frac{F_0}{m}\cos(\omega t) \\
\frac{d^2(x_H+x_p)}{dt^2} + 2\zeta\omega_0\frac{d(x_H+x_p)}{dt} + \omega_0^2(x_H+x_p) &= \frac{F_0}{m}\cos(\omega t) \\
\left(\frac{d^2x_H}{dt^2} + 2\zeta\omega_0\frac{dx_H}{dt} + \omega_0^2 x_H\right) + \left(\frac{d^2x_p}{dt^2} + 2\zeta\omega_0\frac{dx_p}{dt} + \omega_0^2 x_p\right) &= \frac{F_0}{m}\cos(\omega t) \\
0 + \frac{F_0}{m}\cos(\omega t) &= \frac{F_0}{m}\cos(\omega t)
\end{aligned} \tag{9.98}$$

which is true as an identity by hypothesis (that x_p solves the inhomogeneous ODE).

That is, given one particular solution to the inhomogeneous linear ODE, we can generate an *infinite family* of solutions by adding to it any solution we like to the *homogeneous*

[188] Wikipedia: http://www.wikipedia.org/wiki/Fourier Analysis.

[189] For a very small statistical value of "you", one that pretty much looks like *not* you unless you are a math or physics major or perhaps an electrical engineering student.

linear ODE. This (homogeneous) solution has two free parameters (constants of integration) that can be adjusted to satisfy arbitrary initial conditions, the inhomogeneous solution inherits these constants of integration and freedom, and the mathematical gods are thus pleased.

The solution to the homogeneous equation, however, is *exponentially damped*. If we wait long enough, no matter how we start the system off initially, the homogeneous solution will get as small as we like – basically, it will disappear and we'll be left *only* with the particular solution. For that reason, in the context of driven harmonic oscillation the homogeneous solution is called the ***transient*** solution and depends on initial conditions, the particular solution is called the ***steady state*** solution and does *not* depend on the initial conditions!

We can then search for a steady state solution that does *not* contain any free parameters but is completely determined by the givens of the problem independent of the initial conditions!

Although it is not immediately obvious, given a harmonic driving force at a particular frequency ω, the steady state solution must *also* be a harmonic function at the *same* frequency ω. This is part of that deep math mojo that comes out of fourier transforming not only the driving force, but the entire ODE. The latter effectively reduces the entire ODE to an *algebraic* problem. It is by far easiest to see how this goes by using a complex exponential driving force and solution and then taking the real part at the end, but we will skip doing this for now. Instead I will point out some ***true facts*** about the solution one obtains when one does it all right and completely. Interested parties can always look ahead into the companion volume for this course, Intro Physics 2, in the chapter on AC circuits to see how it really should be done.

a) The steady state solution to the driven SHO is:

$$x_p(t) = A\cos(\omega t - \delta) \tag{9.99}$$

where:

$$A = \frac{F_0/m}{\sqrt{(\omega^2-\omega_0^2)^2 + (2\zeta\omega_0\omega)^2}} \tag{9.100}$$

and the phase angle δ is:

$$\delta = \tan^{-1}\left(\frac{2\zeta\omega_0\omega}{\omega^2-\omega_0^2}\right) \tag{9.101}$$

b) The ***average power*** delivered to the mass by the driving force is a quantity of great interest. This is the rate that work is being done on the mass by the driving force, and (because it is in steady state) is equal to the rate that the damping force *removes* the energy that is being added. Evaluating the instantaneous power is straightforward:

$$P(t) = \vec{\boldsymbol{F}}\cdot\vec{\boldsymbol{v}} = Fv = -F_0A\omega\cos(\omega t)\sin(\omega t - \delta) \tag{9.102}$$

We use a trig identity to rewrite this as:

$$\begin{aligned} P(t) &= -F_0A\omega\cos(\omega t)(\sin(\omega t)\sin(\delta) - \cos(\omega t)\cos(\delta)) \\ &= F_0A\omega\left(\cos^2(\omega t)\sin(\delta) - \cos(\omega t)\sin(\omega t)\cos(\delta)\right) \end{aligned} \tag{9.103}$$

The time average of $\cos^2(\omega t)$ is $1/2$. The time average of $\cos(\omega t)\sin(\omega t)$ is 0. Hence:

$$P_{\rm avg} = \frac{F_0 A\omega}{2}\sin(\delta) \tag{9.104}$$

c) If one sketches out a right triangle corresponding to:

$$\tan(\delta) = \frac{2\zeta\omega_0\omega}{\omega^2 - \omega_0^2} \tag{9.105}$$

one can see that:

$$\sin(\delta) = \frac{2\zeta\omega_0\omega}{\sqrt{(\omega^2 - \omega_0^2)^2 + (2\zeta\omega_0\omega)^2}} \tag{9.106}$$

Finally, substituting this equation and the equation for A above into the expression for the average power, we get:

$$P_{\rm avg}(\omega) = \frac{F_0^2\omega^2\zeta\omega_0}{m\left((\omega^2 - \omega_0^2)^2 + (2\zeta\omega_0\omega)^2\right)} \tag{9.107}$$

This equation is maximum when $\omega = \omega_0$, at ***resonance***. At that point the peak average power can be written:

$$P_{\rm avg}(\omega_0) = P_{\rm max} = \frac{F_0^2}{4m\zeta\omega_0} = \frac{F_0^2}{2b} \tag{9.108}$$

d) This ***shape*** of $P_{\rm avg}(\omega)$ is very important to *semi-quantitatively* understand. It is (for weak damping) a sharply peaked curve with the peak centered on ω_0, the natural frequency of the undamped oscillator. You can see from the expression given for $P_{\rm avg}$ why this would be so – at $\omega = \omega_0$ the quantity in the denominator is minimum. This function is plotted in figure 130 below.

e) The sharpness of the resonance is controlled by the dimensionless ***Q-factor***, or "quality factor", of the resonance. Q for the driven SHO is defined to be:

$$Q = \frac{\omega_0}{\Delta\omega} \tag{9.109}$$

where $\boldsymbol{\Delta\omega}$ ***is the full width of*** $\boldsymbol{P_{\rm avg}(\omega)}$ ***at half-maximum!*** To find $\Delta\omega$, one must use:

$$P_{\rm avg}(\omega) = \frac{P_{\rm max}}{2} \tag{9.110}$$

and solve for the two roots where the equality is satisfied. The difference between them is the full width.

f) In this way one can (for weak damping) determine that:

$$Q = \frac{1}{2\zeta} = \frac{m\omega_0}{b} = \omega_0\tau \tag{9.111}$$

where $\tau = m/b$ is the exponential damping time of the *energy* of the associated damped oscillator. This is one of the reasons defining dimensionless ζ is convenient. ζ is basically the *inverse* of Q, so that the *larger* Q (smaller ζ) is, the weaker the damping and the better the resonance will be!

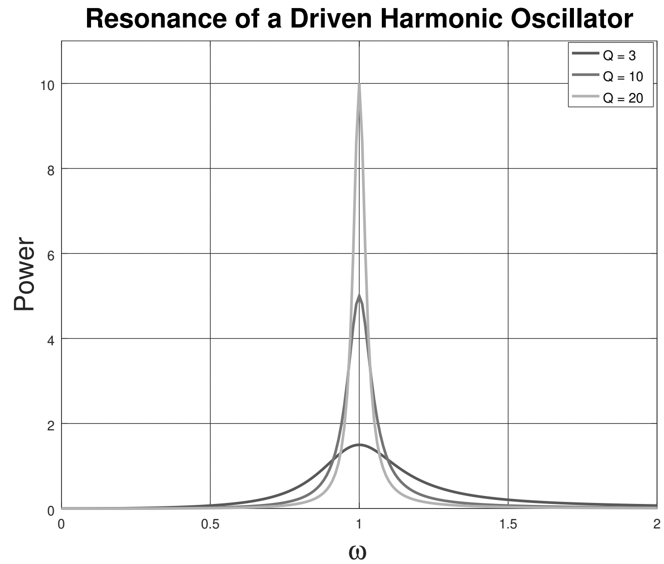


Figure 130: This plots the ***average power*** $P(\omega)$ for three resonances. In these figures, $F_0 = 1$, $k = 1$, $m = 1$ (so $\omega_0 = 1$) and hence b is the only variable. Since $Q = \omega_0/\Delta = m\omega_0/b$ we plot $Q = 3, 10, 20$ by selecting $b = 0.333, 0.1, 0.05$. Thus $P_{\rm max} = F_0^2/2b = 3/2, 5, 10$ respectively (all in suitable units for the quantities involved). Note that the full width of e.g. the $Q = 20$ curve (with the sharpest/highest peak) is $\sim 1/20$ at $P = P_{\rm max}/2 = 5$. P is the average power *added to* the oscillator by the driving force, which in turn in steady state motion must equal the average power *lost to* to dissipative drag forces.

It is worth mentioning in passing that one can easily reformulate the instantaneous or average power in terms of A, the driven amplitude, instead of F_0, the driving force. The point then is that the power will be related to the *amplitude of oscillation squared*. This is intuitively reasonable, as the work removed per cycle is the damping force (proportional to A) integrated over the distance travelled (proportional to A). This is consistent with our developing rule of thumb that oscillator or wave energies or powers are (almost) always proportional to the oscillation or wave amplitude squared.

So what of this are you responsible for knowing? That depends on your interest and the level of your class. If you are a physics or math major or an engineer, you should probably work through the math in some detail. If you are a major in some other science or are premed, you probably don't need to know all of the details, but you should still work to understand the general idea of resonance, as it is actually relevant to various aspects of biology and medicine and can occur in many other disciplines as well. At the very least, *all* students should be able to semi-quantitatively draw, and understand, $P_{\rm avg}(\omega)$ for any given Q in the range from 3 to maybe 20 ***visibly to scale***, specifically so that $P_{\rm max} = F_0^2/2b$ and $Q \approx \omega_0/\Delta\omega$ in your graph. Practice this on your homework! ***I nearly always ask this on one exam or quiz in addition to the homework.***

The point of understanding Q pretty thoroughly is that oscillators with low Q quickly damp and don't build up much amplitude even from a perfectly resonant $\omega = \omega_0$ driving force. This is "good" when you are building bridges and skyscrapers. High Q means you get a *large* amplitude, eventually, even from a *small* but perfectly resonant driving force.

This is "good" for jackhammers, musical instruments, understanding a good walking pace, and many other things.

9.5: Adding Springs in Series and in Parallel

At this point, you should have a pretty good understanding of how an ideal spring will make a mass oscillate (more or less) harmonically, with or without weak linear damping. You should also have the glimmerings of an idea why they are so important – almost *any* system in the near neighborhood of a stable equilibrium point will a very good chance of experiencing linear restoring forces in that neighborhood and hence is likely to make the mass oscillate harmonically around the equilibrium point.

However, we still have a mystery or two to address. One of the most important ones is: ***Why do springs behave like springs?*** Isn't there a bit of a chicken and egg problem here? Granted that linear springs make things oscillate harmonically, why are springs – things made up (we profoundly believe) of many, many atoms bound together by complicated interatomic forces – linear in the first place?

To understand that, as well as to address various issues with engineering devices that might contain more than one spring or use multiple springs at the same time to accomplish some design goal, we first need to learn how to *add up* springs in various arrangements and see how multiple springs in those arrangements will (perhaps surprisingly) behave like a single spring. Along the way, we will learn the spring-equivalents of "a chain is only as strong as its weakest link" and "E pluribus unum" – things that you might already understand just using common sense.

The two arrangements we will concentrate on are springs in ***series*** – one spring connected to another in a linear chain, where the springs are allowed to all have different spring constants – and in ***parallel*** where the springs are all side by side.

Our goal will be to determine algebraically what the spring constant of a *single* spring would have to be if we were to replace the entire series or parallel arrangement with that spring and observe the exact same restoring force for the exact same total stretch (or compression).

The algebra for this isn't terribly difficult, but for series springs in particular, it may not be very *intuitive* – you may need to practice to master the "just right" first steps to keep the algebra simple. It is important to master this algebra because as it turns out, almost *identical* reasoning and algebra is used to determine the total resistance of series and parallel arrangements of resistors or capacitors in electronic circuits covered in the next semester of this course. If you get it right now, and remember it then, it will make understanding these rules of electronics very simple when you get to them.

In the meantime, once we understand these rules for adding ideal springs, well be in a perfect position to understand why ***nearly all materials behave like springs*** when they are stretched, compressed, squeezed down in volume, or deformed. We will come to realize that even the good old normal force that we idealized way back at the beginning of the course is really just an example of this same principle – when we press our finger down on something, it *very slightly deforms* it, compressing tiny "intermolecular" springs that are then what exerts the observed normal force. When you stand on a ladder, the ladder gets *very slightly shorter* by an amount proportional to your weight and the length of the ladder, all due to these rules.

Let's get started.

9.5.1: Adding Springs in Series

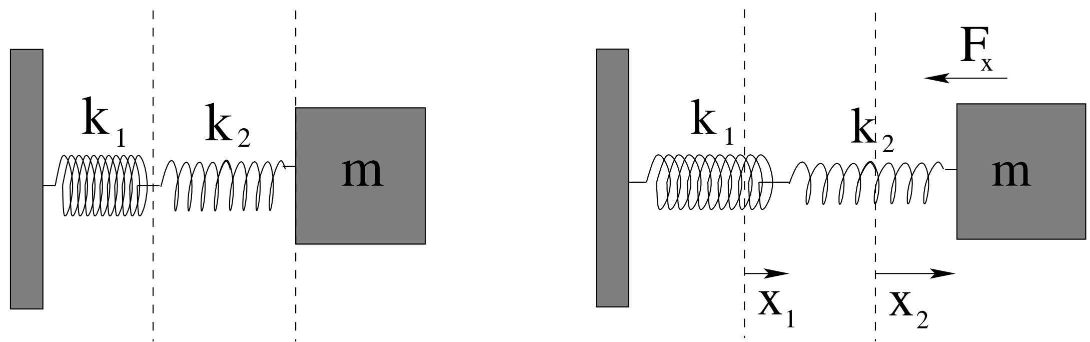


Figure 131: Two springs with spring constants k_1 and k_2 in *series*. The left-hand picture is before either spring is stretched, so that the mass is at equilibrium (no force due to springs). The right-hand picture shows the mass pulled out to the right so that the first spring has stretched a distance x_1 and the second (weaker) spring has stretched x_2.

The figure above illustrates what happens when only *two* springs are connected in series. When the attached mass is pulled out to the right and then held there, it stretches *both* of the two springs. Here's an important argument.

The total force acting on the mass due to the two springs is F_x, pulling back to its equilibrium position at the end of the two unstretched springs. From N3, that mass is pulling spring 2 to the right with an equal and opposite force. But spring 2 itself is in equilibrium! So spring 1 ***must*** be pulling it to the ***left*** with force F_x! This in turn means spring 2 is pulling spring 1 to the right with force F_x (N3 again) and spring 1 isn't moving, so spring 1 must be pulled to the left by the wall with F_x, and the wall is pulled to the right by F_x. Newton's Third Law is *useful!*

Clever students will recognize this is being very similar to our original arguments for why the tension in a stationary *string* is the same throughout the string, pulling any little differential chunk equally to the right and to the left and ultimately just transferring force from one end to the other. We've simply made a "string" out of our two springs!

This gives us a simple insight. We now know (from Hooke's Law and this reasoning) that the force exerted by either spring on its ends is:

$$F_x = -k_1 x_1 = -k_2 x_2$$

Now comes the tricky part. We'd like to find a *single* spring (with spring constant k_{eff}) that produces this *same* force with the ***same total stretch***. That is, we'd like to find:

$$F_x = -k_{\text{eff}} x$$

where

$$x_1 + x_2 = x$$

as shown in figure 132.

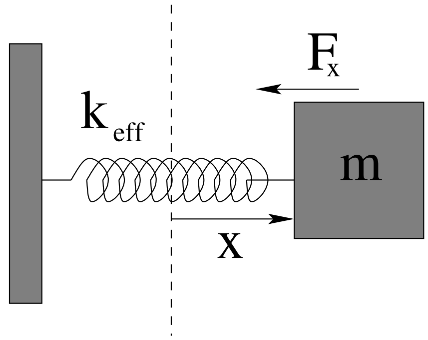


Figure 132: A single spring equivalent to the two in series (or the two in parallel, next).

This isn't all *that* tricky, but the first time you try this on your own, trust me, you are likely to try to ***incorrectly*** add k_1x_1 to k_2x_2 and get yourself into all kinds of algebraic difficulty. I certainly did, the first time I tried it (lo, those many years ago, and an embarrassing number of times afterwards). So don't feel bad if this is you; just learn the trick! *Series* springs add based on the *interesting* fact that ***the length every spring stretches adds up to a total length stretched***, while *the force is the same everywhere throughout the line of springs* and hence is a bit boring and not at all anything you can directly add up.

Boring or not, it is *useful*, so let's use it! We use it to express x_1, x_2, and x all in terms of the k's and the single F_x that is the *same* for both pictures and all three equations:

$$x_1 = F_x/k_1 \qquad x_2 = F_x/k_2 \qquad x = F_x/k_{\text{eff}}$$

Now the remaining algebra is *simple!* We just substitute these three equations into the equation that represents the *interesting* fact regarding the sum of the stretches and cancel the common factor of F_x:

$$x_1 + x_2 = x \quad \Rightarrow \quad \frac{F_x}{k_1} + \frac{F_x}{k_2} = \frac{F_x}{k_{\text{eff}}} \quad \Rightarrow \quad \frac{\cancel{F_x}}{k_1} + \frac{\cancel{F_x}}{k_2} = \frac{\cancel{F_x}}{k_{\text{eff}}} \quad \Rightarrow \quad \boxed{\frac{1}{k_{\text{eff}}} = \frac{1}{k_1} + \frac{1}{k_2}(+...)}$$

I took the liberty of adding a $(+...)$ to this last equation because it is hopefully pretty obvious that if I'd done *three*, or *four*, or N springs in series, they would all have the same force F_x but the sum of the lengths each spring stretched would now be $x_1 + x_2 + ...x_N = x$ (for any value of N, the total number of springs). This lets us write the sum rule for series in a nice, compact form:

$$\boxed{\frac{1}{k_{\text{eff}}} = \sum_{i=1}^{N} \frac{1}{k_i}}$$

or in English, "the reciprocal of the total effective spring constant is the sum of the reciprocals of the spring constants in series".

Here's a single example of adding up three springs in series. The key is that to find k_{eff}, you have to *put the sum over a common denominator and add them, and **only then** invert the result!* Watch out for this!

Example 9.5.1: Three Springs in Series

Suppose $k_1 = 1$, $k_2 = 2$, and $k_3 = 3$ (all in N/m). Then:

$$\frac{1}{k_{\text{eff}}} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} = \frac{2 \times 3}{1 \times 2 \times 3} + \frac{1 \times 3}{1 \times 2 \times 3} + \frac{1 \times 2}{1 \times 2 \times 3} = \frac{6+3+2}{6} = \frac{11}{6}$$

or:

$$k_{\text{eff}} = \frac{6}{11} \text{ N/m}$$

Note Well! A common mistake here would be to do:

$$\frac{1}{k_{\text{eff}}} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} \Longrightarrow k_{\text{eff}} = 1 + 2 + 3 = 6 \text{ N/m}$$

This is very, very wrong! The reciprocal of a sum is not the same thing as the sum of the reciprocals! This is the *whole point* of putting things over a common denominator before adding them!

9.5.2: Adding Springs in Parallel

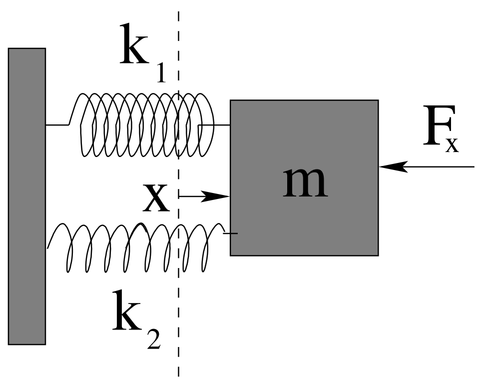


Figure 133: Two springs with spring constants k_1 and k_2 in *parallel*. This is much easier as it is obvious that *the forces of the springs just add for the common stretch x!*

In parallel, the forces of the two springs just add. So:

$$F_x = -k_1 x - k_2 x = -(k_1 + k_2)x = -k_{\text{eff}} x \Longrightarrow \boxed{k_{\text{eff}} = k_1 + k_2 (+...)}$$

and we can write this more generally as:

$$\boxed{k_{\text{eff}} = \sum_{i=1}^{N} k_i}$$

for any number of springs in parallel.

Example 9.5.2: Adding Springs in Parallel

Suppose we are adding the same three springs – $k_1 = 1$, $k_2 = 2$, and $k_3 = 3$ (all in N/m) – but now we'e adding them in parallel. Now we really do get:

$$k_{\text{eff}} = 1 + 2 + 3 = 6 \text{ N/m}$$

Note that this is exactly what you would get if you ***misadded*** the three springs in series! How, you might wonder, can you keep this straight? On an exam, under stress, you have to recognize which is which, and you might need to reason conceptually about this in simple scaling problems as usual. This motivates the following short summary.

9.5.3: Rules of Thumb

If you look at the results above, when adding springs in series, the effective spring constant is always *smaller than the smallest spring constant in the series!* We have an adage for this, already quoted above: "a chain is only as strong as its weakest link". If one spring happens to be much weaker than all of the rest, most of the stretch will happen for that one weak spring, and it is likely to be the one that breaks if overstressed! However, even the stronger springs will stretch *a little*, so the total is *weaker* than this weakest spring.

A second rule for series is that if you have N *identical* springs in series:

$$\frac{1}{k_{\text{eff}}} = \frac{1}{k} + \frac{1}{k} + ... = \frac{N}{k}$$

then:

$$\boxed{k_{\text{eff}} = \frac{k}{N}}$$

This rule is a key component of the following section, so make sure that you understand it! N identical springs in series have a spring constant that is $1/N$th of the constant of any of the springs! A long line of springs gets *weaker* the longer it gets, at a rate *more or less proportional ot the total unstretched length!*

For parallel springs, it is obvious that the effective spring constant is ***larger than the largest*** spring constant in the parallel set. If one of the springs is super strong compared to the others (a spring used in the suspension of a car versus the springs you find on screen doors, for example) you have to stretch that car spring *and* you still have to exert enough additional force to stretch all of the weaker springs as well. This is just common sense. Adages include *E pluribus unum* – from many, one – used to suggest that together we are stronger than any of us is alone. However, the wisdom in this goes all the way back to elementary school, where everybody tried to pick the biggest, strongest kid to be on *their* team in a tug of war, where everybody's strength on a team gets *added*.

In a way, this is one of the motivations for *team based learning*, which is the way I always teach this course. A team of students working together has the double advantage of leveraging the *varied* strengths of its members *and* allowing them to share and pool that knowledge so that every member emerges from problem solving practice stronger than they went in!

Again we have a rule for adding M *identical* springs:

$$\boxed{k_{\rm eff} = k + k + ... = M \times k}$$

You just multiply the spring constant by the number of springs.

These two rules, combined, are the entire basis of the next section. Make sure you understand them before proceeding.

9.6: Elastic Properties of Materials

It is now time to close a very important bootstrapping loop in your understanding of physics. From the beginning, we have used a number of force rules like "the normal force", and "Hooke's Law" because they were simple rules that we could directly observe and use to help us both understand ubiquitous phenomena and learn to use Newton's Laws to quantitatively describe many of them.

We had to do this *first*, because until you understand force, work, energy, equations of motion and conservation principles – basic mechanics – you cannot *start* to understand the microscopic basis for the macroscopic "rules" that govern both everyday Newtonian physics and things like thermodynamics and chemistry and biology (all of which have rules at the macroscopic scale that follow from physics at the microscopic scale).

We've done a bit of this along the way – thought about microscopic causes of friction and drag forces, derived the ways in which a macroscopic object can be thought of as a pointlike microscopic object located at its center of mass (and ways it cannot, e.g. when describing its rotation). We've even thought a *bit* about things like compressibility of fluids, but we haven't really thought about this *enough*.

Let's fix that.

To do so, we need a *microscopic model for a solid*, in particular for the molecular bonds within a solid.

9.6.1: Simple Models for Molecular Bonds

Consider, then a very crude microscopic model for a solid. We know that this solid is made up of *many* elementary particles, and that those elementary particles interact to form nucleons, which bind together to form nuclei, which bind elementary electrons to form atoms and that the atoms in turn are bound together by short range (nearest neighbor) interatomic forces – "chemical bond" if you like – to actually form the solid.

From both the text and some of the homework problems you should have learned that a "generic" potential energy associated with the interaction of a pair of atoms with a chemical bond between them is given by a ***short range repulsion*** followed by a ***long range attraction***. We saw a potential energy form *like* this as the "effective potential energy" in gravitation (the form that contained an angular momentum barrier with L^2 in it), but the physical origin of the terms is very different.

The repulsion in the molecular potential energy comes from first the ***Pauli exclusion principle***[190] in quantum mechanics (that makes the interpenetration of the electron clouds surrounding atoms energetically "expensive" as the underlying quantum states rearrange to satisfy it) plus the penetration of a ***screened Coulomb interaction***. The former is truly beyond the scope of this course – it is quantum magic associated with electrons[191] as fermions[192] , where I'm inserting wikipedia links to lots of these terms so that *interested* students can use them as the starting points of wikipedia romps, as a lot of this is all *absolutely fascinating* and is one of the reasons physicists love physics, it is all just so very amazing.

Next semester you *will* learn about ***Coulomb's Law***[193] , that describes the forces between two charged particles, and (using Gauss's Law[194]) you will be able to understand how the electron cloud normally "screens" the nuclei as eventually the rearrangement brings increasingly "bare" nuclei close enough so that the atoms have a *very* strong net repulsion.

One thing that we won't cover then, however, is how this simple/naive model, which leads to two atoms *not interacting at all* as soon as they are not "touching" (electron clouds interpenetrating) is replaced by one where two neutral atoms have a residual *long range interaction* due to dipole-induced dipole forces, leading to what is called a ***London dispersion force***[195] . This force has the generic form of an *attractive* $-C/r^6$ for a rather complicated C that parameterizes various details of the interatomic interaction.

Physicists and quantum chemists or engineers often idealize the exact/quantum theory with an approximate (semi)classical potential energy function that models these important generic features. For example, two very common models (one of which we already briefly explored in week 4, homework problem 4) is the ***Lennard-Jones potential***[196] (energy):

$$U_{LJ}(r) = U_{\min}\left\{2\left(\frac{r_b}{r}\right)^6 - \left(\frac{r_b}{r}\right)^{12}\right\} \tag{9.112}$$

In this function, $U_{\min}$ is the ***minimum*** of the potential energy curve (evaluated with the usual convention that if $r \to \infty$ then $U_{LJ}(\infty) = 0$), r_b is the radius where the minimum occurs (and hence is the equilibrium ***bond length***), and r is along the bond axis. This function is portrayed as the solid line in figure 134.

An alternative is the ***Morse potential***[197] (energy):

$$U_M(r) = U_{\min}\left(1 - e^{-a(r-r_b)}\right)^2 \tag{9.113}$$

[190] Wikipedia: http://www.wikipedia.org/wiki/Pauli Exclusion Principle.
[191] Wikipedia: http://www.wikipedia.org/wiki/Electron.
[192] Wikipedia: http://www.wikipedia.org/wiki/Fermion.
[193] Wikipedia: http://www.wikipedia.org/wiki/Coulomb's Law.
[194] Wikipedia: http://www.wikipedia.org/wiki/Gauss' Law.
[195] Wikipedia: http://www.wikipedia.org/wiki/London dispersion force. This force is due to ***Fritz London*** who was a Duke physicist of great reknown, although he derived this force from second-order perturbation theory long before he fled the rise of the Nazi party in Germany and eventually moved to the United States and took a position at Duke. London is honored with an special invited "London Lecture" at Duke every year. Just an interesting True Fact for my Duke students.
[196] Wikipedia: http://www.wikipedia.org/wiki/Lennard-Jones potential.
[197] Wikipedia: http://www.wikipedia.org/wiki/Morse potential.

In this expression $U_{\rm min}$ and r_b have the same meaning, but they are joined by a, a parameter than sets the *width* of the well. The only problem with this form of the potential is that it is difficult to compare it to the Lennard-Jones potential energy above because the LJ potential is zero at infinity and the Morse potential energy is $U_{\rm min}$ at in infinity. Of course, potential energy is *always* only defined within an additive constant, so I will actually subtract a constant $U_{\rm min}$ and display:

$$U_M(r) = -U_{\rm min} + U_{\rm min}\left(1 - e^{-a(r-r_b)}\right)^2) \tag{9.114}$$

as the dashed line in figure **??** below. This potential energy now "conventionally" vanishes at ∞.

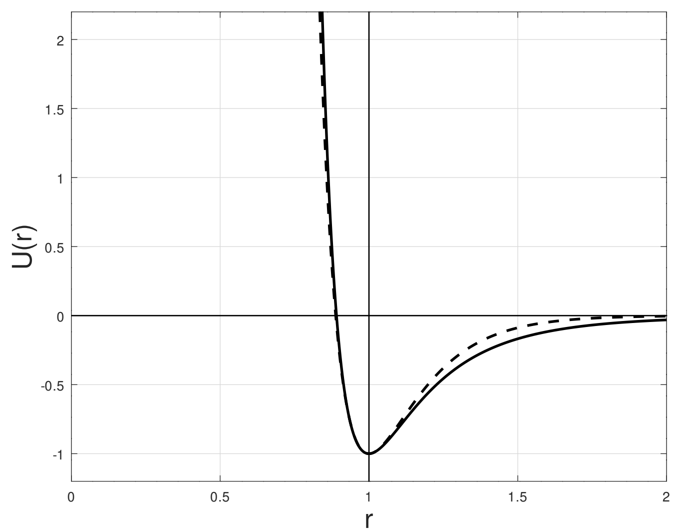


Figure 134: Two "generic" classical potential energy functions associated with atomic bonds on a common scale $U_{\rm min} = -1$, $r_b = 1.0$, and $a = 6.2$, the latter a value that makes the two potential have roughly the same ***force constant*** for small displacements.The solid line is the ***Lennard-Jones*** potential energy $U_L(r)$ and the dashed line is the ***Morse*** potential $U_M(r)$. Note that the two are very closely matched for short range repulsion, but the Morse potential dies off faster than the $1/r^6$ London form expected at longer range.

I should emphasize that neither of these potential energies is in any sense a law of nature. They are effective potential energies, idealized and parameterized model potential energies that have close to the right shape and that can be used to study and understand molecular bonding in a semi-quantitative way – good enough to be compared to experimental results in an understandable if approximate way, but hardly exact.

The exact (relevant) laws of nature are those of electromagnetism and quantum mechanics, where the many electron problem must be solved, which is very difficult. The problem rapidly becomes too complex to really be solvable/computable as the number of electrons grows and more atoms are involved. So even in quantum mechanics people not infrequently work with Lennard-Jones or Morse potential energies (or any of a number of other related forms with more or less virtue for any particular problem) and sacrifice precision in the result for computability.

In the case we are interested in, however, even these relatively *simple* effective potential energies are too complex. We therefore take advantage of something I have written about extensively up above. Nearly *all* potential energy functions that have a true minimum (so that there is a force equilibrium there, recalling that the force is the negative slope (gradient) of the potential energy function) vary ***quadratically*** for small displacements from equilibrium. A quadratic potential energy corresponds to a harmonic oscillator and a linear restoring force. Therefore all solids made up of atoms bound by interactions like the effective molecular potential energies above will inherit certain properties from the tiny "springs" of the bonds between them!

9.6.2: The "Spring Constant" of a Molecular Bond

In both of the cases above, we can derive a quantity that acts as a sort of ***effective "spring constant"*** of the interatomic bonds that hold atoms in their place relative to a neighboring atom. The easiest way to do so is to do a ***Taylor Series Expansion***[198] of the potential energy function around r_b. This is:

$$U(x_0 + \Delta x) = U(x_0) + \left(\frac{dU}{dx}\right)\bigg|_{x=x_0} \Delta x + \frac{1}{2!}\left(\frac{d^2U}{dx^2}\right)\bigg|_{x=x_0} \Delta x^2 + ... \tag{9.115}$$

At a *stable equilibrium* point x_0 the force *vanishes*:

$$F_x(x_0) = \left(\frac{dU}{dx}\right)\bigg|_{x=x_0} = 0 \tag{9.116}$$

so that:

$$U(x_0 + \Delta x) = U(x_0) + \frac{1}{2!}\left(\frac{d^2U}{dx^2}\right)\bigg|_{x=x_0} \Delta x^2 + ... \tag{9.117}$$

We can now identify this with the ***form*** of a potential energy equation for a mass on a spring in a coordinate system where the equilibrium position of the spring is at x_0[199]:

$$U(x_0 + \Delta x) = U(x_0) + \frac{1}{2!}\left(\frac{d^2U}{dx^2}\right)\bigg|_{x=x_0} \Delta x^2 + ... = U_0 + \frac{1}{2}k_{\text{eff}}\Delta x^2 \tag{9.118}$$

where

$$k_{\text{eff}} = \left(\frac{d^2U}{dx^2}\right)\bigg|_{x=x_0} \tag{9.119}$$

What this means is that ***any mass at a stable equilibrium point will usually behave like a mass on a spring for motion "close to" the equilibrium point!*** If pulled a short distance away and released, it will oscillate nearly harmonically around the equilibrium. The terms "usually" and "nearly" can be made more precise by considering the neglected third and higher order derivatives in the Taylor series – as long as they are negligible compared to the second order derivative with its quadratic Δx^2 dependence, the approximation will be a good one.

[198] Wikipedia: http://www.wikipedia.org/wiki/Taylor series expansion.

[199] Remember, any potential energy function is defined only to within an additive constant, so the constant term U_0 simply sets the scale for the potential energy without affecting the actual force derived from the potential energy. The force is what makes things happen – it is, in a sense, all that matters.

We have already seen this to be true and used it for the simple and physical pendulum problem, where we used a Taylor series expansion for the force or torque and for the energy and kept only the leading order term – the small angle approximation for $\sin(\theta)$ and $\cos(\theta)$. You will see it in homework problems later this semester and next semester as well (at least if you continue using this textbook) where you will from time to time use the binomial expansion (the Taylor series for a particular form of polynomial) to transform a force or potential energy associated with a particle near equilibrium into the form that reveals the simple harmonic oscillator within, so to speak. Any time you see a *linear* restoring force or torque, you can expect harmonic oscillation!

At *this* point, however, we wish to put this idea to a different use – to help us bridge the gap between microscopic forces that hold a "rigid" object together and that object's response to forces applied to it. After all, we *know* that there is no such thing as a truly rigid object. Steel is pretty hard, but with enough force we can bend "solid" steel, we can stretch or compress it, we can turn it into a spring, we can fracture it. Bone is also pretty hard and *it does exactly the same thing:* bend, stretch, compress, fracture. The physics of bending, stretching, compressing, and fracturing a solid object is associated with the application of a quantity called *stress* (with units of pressure, note well, but not *precisely* a pressure) to the object. Let's see how.

9.6.3: A Microscopic Picture of a Solid

Let us consider a solid piece of some simple material. If we look at pure elemental solids – for example metals – we often find that when they form a solid the atoms arrange themselves into a *regular lattice* that is "close packed" – arranged so that the atoms more or less touch each other with a minimum of wasted volume. There are a number of kinds of lattices that appear (determined by the subtleties of the quantum mechanical interactions between the atoms and hence beyond the scope of this course) but in many cases the lattice is a variant of the cubic lattice where there are atoms on the corners of a regular cartesian grid in three dimensions (and sometimes additional atoms in the center of the cubes or on the faces of the cubes).

Not all materials are so regular. Materials can be made out of a mixture of atoms, out of molecules made of a mixture of atoms, out of a mixture of molecules, or even out of living cells made out of a mixture of molecules. The resulting materials can be ordered, *structured* (not exactly the same thing as ordered, especially in the case of solids formed by life processes such as bone or coral), or disordered (amorphous).

As usual, we will deal with all of this complexity by ignoring most of it for now and considering an "ideal" case where a single kind of atoms lined up in a regular ***simple cubic lattice*** is sufficient to help us understand properties that will hold, with different values of course, even for amorphous or structured solids. This is illustrated in figure 135, which is basically a mental cartoon model for a generic solid – lots of atoms in a regular cubic lattice with a cube side a, where the interactomic forces that hold each atom in position is represented by a ***spring***, a concept that is valid as long as we don't compress or stretch these interatomic "bonds" by too much.

We need to quantify the numbers of atoms and bonds in a way that helps us understand

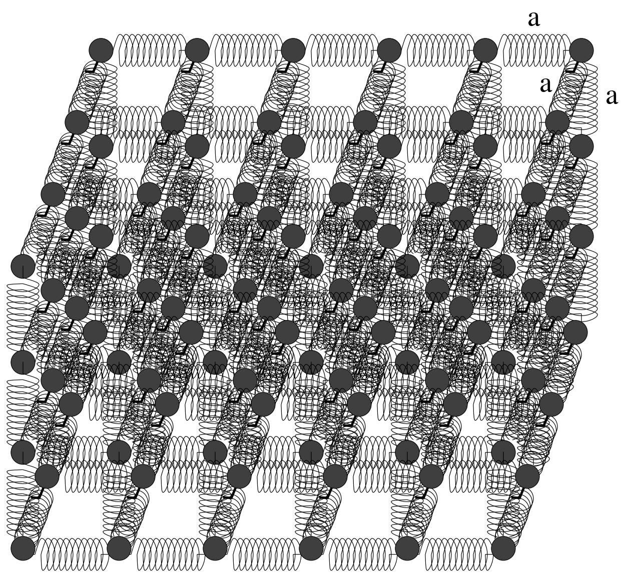


Figure 135: An idealized simple cubic lattice of atoms separated by the “springs” of interatomic forces that hold them in equilibrium positions. The equilibrium separation of the atoms as a.

how stretching or compressing forces are distributed among all of the bonds. Suppose we have N_x atoms in the x-direction, so that the length of the solid is $L = N_x a$. Suppose also that we have N_y and N_z atoms in the y- and z-directions respectively, so that the cross-sectional area $A = N_y N_z a^2$ (at least approximately, as we are not treating atoms on the boundary particularly accurately as they contribute to the problem at “lower order”, meaning we can ignore them for large systems).

Now let us imagine applying a force (magnitude F) *uniformly* to all of the atoms on the left and right ends that *stretches* all of these bonds by a small amount Δx, presumed to be “small” in precisely the sense that leaves the interatomic bonds still behaving like springs. The force F has to be distributed equally among all of the atoms on the end areas on both sides, so that the force applied to each *chain* of atoms in the x-direction end to end is:

$$F_{\rm chain} = \frac{F}{N_y N_z} = \frac{Fa^2}{A} \tag{9.120}$$

This situation is portrayed in figure 136.

Each spring in the chain is stretched by *this force* between the atoms on the ends, so that:

$$F_{\rm chain} = -\frac{Fa^2}{A} = -k_{\rm eff}\Delta x \tag{9.121}$$

The negative sign just means that the springs are trying to go back to their equilibrium length and hence oppose the applied force.

We multiply this by one in the form $\frac{N_x}{N_x}$ and divide the a^2 over to the right-hand side of the equation and split things up in the following clever way:

$$\frac{F}{A} = -k_{\rm eff}\frac{N_x}{a^2 N_x}\Delta x = -\frac{k_{\rm eff}}{a}\frac{N_x \Delta x}{N_x a} \tag{9.122}$$

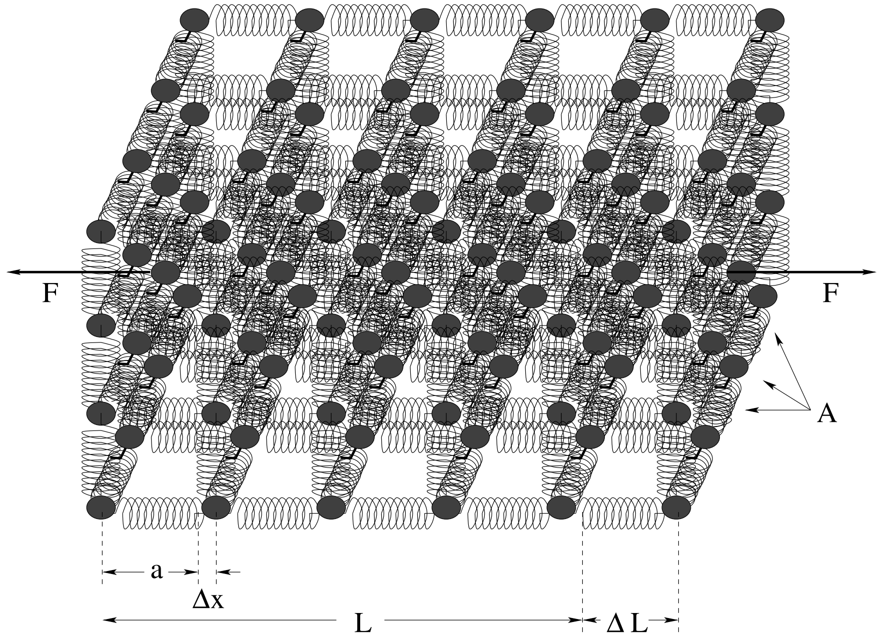


Figure 136: The same lattice stretched by an amount ΔL as a force F is applied to both ends, spread out uniformly across the cross-sectional area of the faces A.

Note that $aN_x = L$ (again, within one depending on how we count the atoms at the ends of the chain, an error that is negligible if $N_x \gg 1$) and that $N_x \Delta x = \Delta L$. Making these substitutions we get:

$$\frac{F}{A} = -\frac{k_{\text{eff}}}{a}\frac{\Delta L}{L} = -Y\frac{\Delta L}{L} \tag{9.123}$$

where we define:

$$Y = \frac{k_{\text{eff}}}{a} \tag{9.124}$$

This (and the further variants we discss below) is one of the most important equations in the mechanics of solids.

We now give English names to the algebraic forms in the last two equations: F/A is called the ***Stress***, $\Delta L/L$ is called the ***Strain***, and Y is defined to be ***Young's Modulus***. With these definitions, the equation above, stated in English, reads thus:

> ***Compressive or extensive Stress applied to a solid equals Young's Modulus times the Strain.***

(where we don't mention the minus sign because it is understood that, like a spring, the solid *oppposes* the stress to restore its unstressed length).

Here is a short list of things you should know about this equation.

- Stress (F/A) has units of ***pressure***. However, it is ***not*** generally a pressure! Note that in the derivations above, I explicitly *stretched* the material with an *extensive* force rather than *compressed* that material with a *compressive* force pointing towards the material on both ends instead of away. The latter one might be forgiven for calling it a pressure, but the former certainly is not! In both cases, though, the SI units would be pascals, although one could certainly express stress in e.g. bar or even torr.

- Strain ($\Delta L/L$) is ***dimensionless***.
- (Therefore) Young's Modulus (Y) *also* must have (matching) units of pressure: pascals, bar, torr. Again, in no sense should Y be confused with being an *actual* pressure – it is a property of the material based on its linearized microscopic physics!
- In most cases one does not actually compute Y using the definition above (in no small part because computing or measuring the intermolecular potential energy function is extremely difficult). Rather, one reads it out of tables that are worked out empirically! Indeed, one is more likely to work the other way – estimate some of the properties of the intermolecular potential energy from a measurement of Y!

Material	Y (Gpa)
Plastics (various)	1-3
Compact Bone	18
Aluminum	69
Spongy Bone	76
Steel (alloys)	180-200
Tungsten (alloys)	400-600
Diamond	1220

Table 5: A short table of Young's Moduli (in *giga*pascals) for a few interesting materials, to give you an idea of their range. Remember that 10 bar is one megapascal, so these all in the general range of millions of atmospheres

Note well that we can rearrange this equation into "Hooke's Law" as follows:

$$F_x = -\frac{YA}{L}\Delta x = -K\ \Delta x \tag{9.125}$$

which basically states that any given chunk of material *behaves like a spring* when forces are applied to stretch or compress it, with a spring constant K that is ***proportional to the cross sectional area A and inversely proportional to the length L!*** Young's modulus is the material-dependent contribution from the actual geometry and interaction potential energy that holds the material together, but the rest of the dependence is *generic*, and applies to all materials.

This ***scaling*** behavior of the response of solid (or liquid, or gaseous) matter to applied forces is the important take-home conceptual lesson of this section. Substances like bone or steel or wood or nearly anything are *stronger* – respond less to an applied force – as they get *thicker*, and are *weaker* – respond more to an applied force – as they get *longer*. Note well that strength per se is not necessarily directly proportional to Y (or M_s); as we will discuss shortly, the stress at which a material fractures is a separate empirical number and depends on other properties of the material, such as "brittleness". Diamond is extremely hard, for example (Young's modulus in the terapascal range) but it is nevertheless easy to chip because it is brittle.

However, we expect strength to *scale* very much like this as well – longer weaker, thicker stronger. This intuitive understanding can be made quantitatively precise to be sure (and e.g. engineers or orthopedic surgeons will *have* to be quantitatively precise as they craft buildings that won't fall down or artificial hips or bones that mimic natural ones) but is sufficient in and of itself for people to see the world through new eyes, to understand why there are building codes for houses and decks governing the lengths and cross-sectional areas of support beams and joists, why you can't just blow an mouse up to the size of an elephant and still have it able to stand without breaking its own bones, and so on.

We'll explore a few of these applications later, but first let us relatively *quickly* extend the linear extension/compression result to *sideways* forces and understand *shear*.

9.6.4: Shear Forces and the Shear Modulus

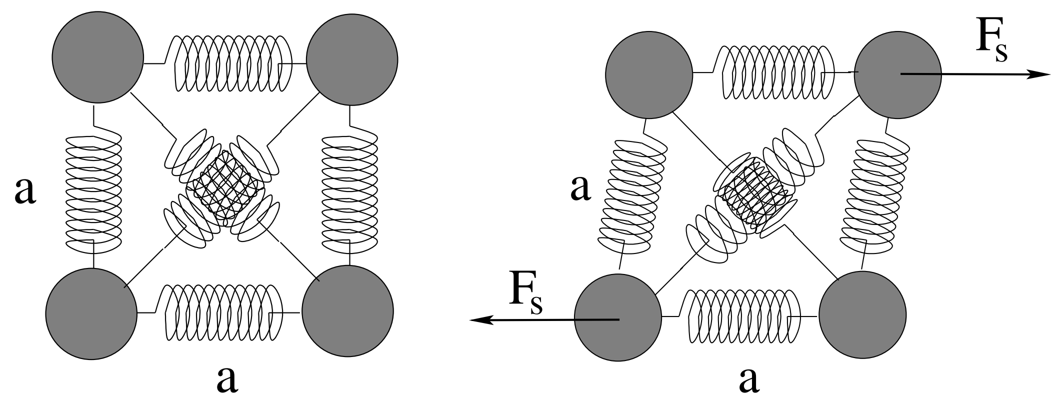


Figure 137: A simple model for shear. This model is not particularly correct – in most materials, molecular bonds resist the bending of the stable bond angles at the level of *quantum mechanics* or are more complex (and shear-resistant) lattices.

Let's look at a single 2D "square" of atoms in an imaginary simple cubic lattice like the one pictured in figure 135 above. One of the sad things about simple 2D rectangular structures is that they are generally *not* resistant to shear unless angled braces are added – they can flop over from rectangles into parallelograms without altering the lengths of the sides, only the angles at junctions. To allow for that and still have a simple model, in figure 137 we portray such a square, but his time we've included some (possibly weaker) molecular bonds that connect opposite corners of the square[200].

In the left-hand figure, no stress is applied. In the right-hand figure, a pair of ***shear stresses*** is applied that form a ***force couple*** (and hence exert a torque!) on the square. We imagine that the bottom of the square is somehow "glued down" so that it cannot move, so instead of rotating, the top of the square is displaced in the direction of the sideways force there, twisting the square into a parallelogram and stretching and compressing the crosswise bonds in the middle as shown.

Now we must imagine an entire block of solid material, "glued" to something along its top and bottom surfaces, with two opposed shear forces applied across the entire area of

[200] In the most common regular atomic lattices found in nature (things like "face centered cubic" or "body centered cubic") the nearest neighbor bonds alone are sufficient to resist shear. Also, as students of chemistry know, molecular bond angles are typically "fixed" by quantum mechanics (one of the factors that determines the specific geometry of the solid lattice favored by any particular material) and resist shear (bending of these angles) even without "crossbraces".

those surfaces. Internally, all of the molecular bonds will be tipped over as portrayed for our single "atomic" square above, resulting in a net displacement of the top surface (in the direction of the force applied there) to the bottom surface.

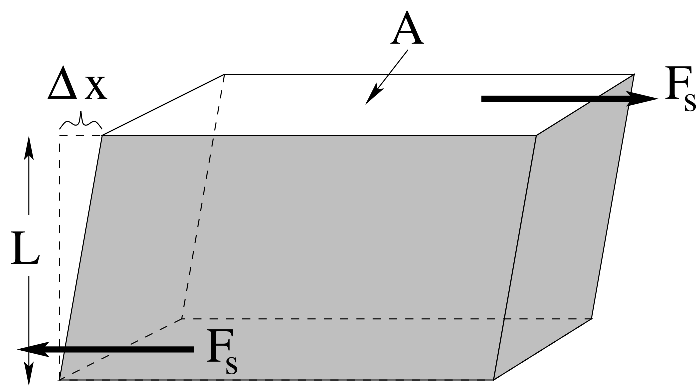


Figure 138: A rectangular block of material with *unstressed* dimensions L in height and cross sectional area A. When a shear stress is applied to this block (within the linear response regime) we expect it to deform as shown.

This situation is portrayed in figure 138, which also shows the relevant dimensions. The block length (height) we will continue to call L, as this rule will usually be applied to e.g. bones or long support beams, where A is the bone *cross-sectional area* and L its actaal length. Δx is the displacement of the top surface in the direction of the shear force F_s relative to the bottom (which we imagine to be "fixed" and hence is being pulled the other way by an equal and opposite force). The result is that the entire material tips over by a (small) angle θ relative to its unstressed orientation.

A fully microscopic derivation of the expected response is still quite possible, but (obviously) will not be quite as simple as that for Young's modulus above. For that reason we will content ourselves with a simple heuristic extension of the *scaling* ideas we learned in the previous section.

The displacement layer to layer required to oppose a given shear force is going to be inversely proportional to the thickness of the material L, because the longer the material is, the more layers there are to split the restoring force among. So we expect the shear force F_s required to produce some given displacement Δx to be proportional to $1/L$. The work we do as the material bends comes primarily from the stretching of the cross-bracing bonds across a single atomic square. Every atom on the top surface has a set of these bonds cross-connecting to the atoms in the direction of the force F_s, and the number of these atoms is proportional to the cross-sectional area A of the material at the shear surface on the top and bottom. The more bonds, the stronger force the force F_s will need to be for the displacement Δx.

We wrap up all of our ignorance of the *specific* microscopic structures that lead to this expected scaling in the linear regime into a modulus that we can more easily *measure in the lab* than *compute from first principles* even as crudely as we did for Young's Modulus, and assemble the result into:

$$F_s = -\frac{M_s A}{L}\Delta x = -K_s \Delta x \tag{9.126}$$

where M_s is the ***shear modulus*** – the equivalent of Young's modulus for shear forces.

Once again we can put this in a form involving *shear stress* (defined to be: F_s/A) and *shear strain* (defined to be $\Delta x/L$):

$$\frac{F_s}{A} = -M_s \frac{\Delta x}{L} \tag{9.127}$$

In words:

Shear Stress applied to a solid equals the Shear Modulus times the Strain.

where note well the minus sign that as always means that the reaction force *opposes* the applied shear force, trying to make Δx *smaller in magnitude* no matter what direction the material is being bent.

As before, $\Delta x/L$ is dimensionless, the units of F_s/A are pressure units (pascals, bar, torr) and hence the units of M_s must be pressure as well. However the shear stress is ***in absolutely no sense an actual pressure!*** The force is being applied *parallel to the surface* and not perpendicular to it!

Shear stress is how springs really work! A "spring" is just a very long piece of material wrapped around into a coil so that when the ends are pulled, the coils *tip* through some small angle in a continuous way, producing shear stress spread out across the entire coil and lengthening the coil itself! So it is not an *accident* that we discover Hooke's Law buried within the discussion in this section – this is a (slightly handwaving) ***microscopic derivation*** of Hooke's Law:

$$F_x = -\frac{M_s A}{L}\Delta x = -k\ \Delta x \tag{9.128}$$

where:

$$k = \frac{M_s A}{L} \tag{9.129}$$

and A is the cross-sectional area of the spring material and L is the entire length of the coiled wire. From this, we can predict how the strength of a spring made of some specific material will vary according to the length of the wire in the coil L and its thickness A. We can also see our addition rules buried in this expression – two identical springs in series are equivalent to one spring twice as long and $k \Rightarrow k/2$; and two identical springs in parallel are equivalent to one spring made with wire with twice the cross sectional area and $k \Rightarrow 2k$! Pretty cool!

9.6.5: Deformation and Fracture

What happens when one stretches or compresses or shears a material by an amount (per bond) that is *not* small? Here is a verbal description of what we might expect based on the microscopic model above and our own experience.

For a range of (relatively small) stresses, the strain remains a ***linear response*** – proportional to the stress, acting in a direction opposed to the stress. As one increases the stress, then, the first thing that happens is that the response becomes *non*-linear. Each additional increment of stress produces *more* than a linear increment of strain when stretching, and quite possibly *less* than a linear increment of strain when compressing, as one can see from the graph of a typical interatomic or intermolecular force in figure 134 above. Materials tend to resist compression better than extension because one can always pull bonded atoms *apart*. However, one cannot cause two bonded atoms to *interpenetrate* with *any* reasonable force – they are *very strongly repulsive* when their electron clouds start to overlap but only weakly attractive as the electron clouds are puleed apart.

A second interesting point is that since the total amount of material is conserved, stretching a system at some point starts to make it *thinner* while compressing it makes it *thicker* – this sort of deformation in one direction in response to stresses in another direction is best described by *tensors* and is generally beyond the scope of this course, although the shear modulus is a simple example of one small part of a tensor response.

At some point, one adds enough energy to the bonds (doing *work* as one e.g. stretches the material) that atoms or molecules start to *dislocate* – leave their normal place in the lattice or structure and migrate someplace else. This leaves behind a ***defect*** – a missing atom or molecule in the structure – and correspondingly weakens the entire structure. This migration tends to be *permanent* – even if one releases the stress on the material, it does not return to its original state but remains stretched, compressed, or bent out of place. This region is one of ***permanent deformation*** of the material, as when one bends a paper clip.

Finally, if one continues to ratchet up the stress, at some point the number of defects introduced reaches a critical point and each new defect produced weakens the material enough that another defect is produced without further increase in stress, as the number of bonds over which the stress is distributed decreases with each new defect. Also, the thinning of material on stretching actually decreases the cross-sectional area, which makes it stretch even more in a positive feedback loop. The number of defects "explodes", creating a *fracture* and the material *breaks* (or tears, or crumbles – it comes irreversibly apart).

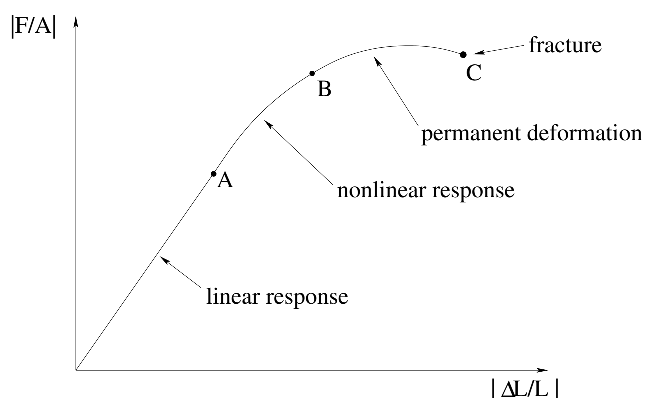


Figure 139: The slope of the linear response part of the curve is Young's Modulus (note well the absolute magnitude signs). A marks the point where nonlinear response is first apparent. B marks the boundary where dislocations occur and permanent deformation of the material begins. C marks the point where there is a chain reaction – each defect produced produces on average at least one more defect at the same stress, so that the material "instantly" fractures .

These behaviors – and the critical points where the behavior changes – are summarized on the graph in figure 139.

A material's strength is not (as noted earlier!) just characterized by its various elastic moduli. Elastic moduli describe only the simple linear response regime. Furthermore, the moduli themselves depend on how far the atoms or molecules of the material are away from equilibrium *without* stress of any sort, because the material has a ***temperature*** (basically, a mechanical energy per atom that is *larger* than the minimum associated with sitting at the equilibrium point, at rest). This causes materials to (usually, but not quite always!) *expand* when they are heated by effectively "trapping" a thermal stress inside the material itself.

In thermal equilibrium, each atom is *already* oscillating back and forth around its equilibrium position and temperature increases alone are sufficient to drive hhe system through the same series of states – linear and nonlinear response states where the heated material it can cool back to the original structure, nonlinear response that introduces defects so that the material "starts to melt" and doesn't precisely come back to its original state, followed by *melting* instead of *fracture* when the energy per bond no longer keeps atoms localized to a lattice or structure at all.

When one mixes heating and stress, one can get many different ranges of physical behavior – stressing a system heats it (try bending a paper clip back and forth until it breaks, and it can get hot enough to burn your fingers where it bends). Heating or cooling a system can cause a stress – water is one of the important exceptions to the "expands when heated rule in the vicinity of its freezing point – it actually *[*expands from its greatest density at 4°C to the extent that ice has a density of around 920 kg/m^3 compared to water at 1000 kg/m^2. Water frozen in a confined space exerts and *enormous stress* on the walls confining it. On the other side of four degrees Centigrade, water *heated* in a confined space will exert a large stress on it (even before the water boils). Many, many design decisions in engineering rely on being aware of this coupling between temperature, stress, and strain. Bridges, roadways, siding boards on houses, and much more have to be installed with "expansion joints" or gaps lest the high stresses associated with thermal expansion in a confined space cause structural deformation or failure or the design!

One last qualitative measure of interest when discussing elasticity and strength is *brittleness* or *toughness* – opposing measures of how likely a material is to *bend* (or elastically deform) versus *fracture* (or inelastically deform) when stresses are applied. Some materials, such as steel, are very tough (or not very brittle) – they do not easily deform or fracture and have very large elastic moduli. Others, like diamond, can be very hard indeed (have very large elastic moduli) but are *easy to fracture* – they are *brittle*.

Human bone as a material that has evolved for the specific purpose of structural support varies tremendously in the range of its brittleness with the lifetime of the human involved, with genetic factors, whhh dietary factors, and with the history of the person involved. Yonng people (on average) have bones that bend easily (lower elastic moduli) but aren't very brittle so they don't always break when stressed. Old people have bones that are inflexible but become progressively more brittle as they decalcify with age or disease. Normal adults tend to fall in between – bones that are not as flexible but that are also not particularly brittle.

In all cases *all things being equal* thick bones are stronger than thin bones, long bones are weaker than short bones. This section should have given you a good chance of understanding at least semi-quantitatively how bone strength varies and can be described with a few empirical parameters that can be connected (with a fair bit of work) all the way back to the intermolecular bonds within the bone itself and its physical structure.

9.7: Human Bone

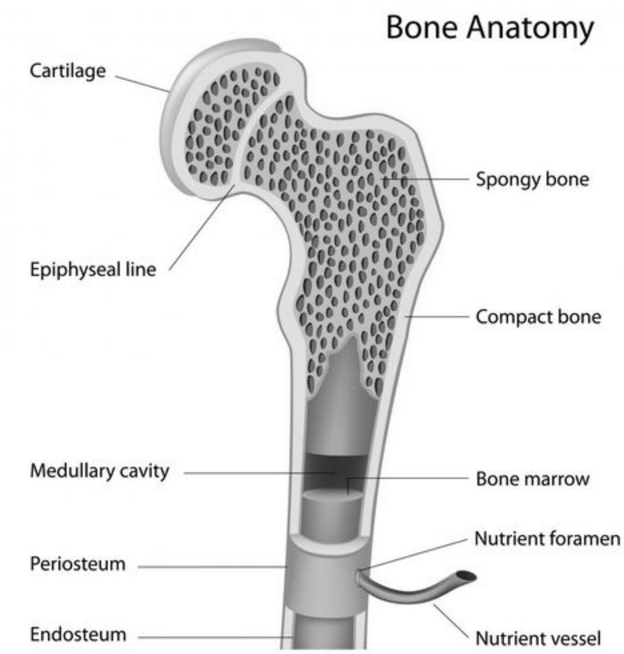


Figure 140: This figure illustrates the principle anatomical features of bone.

The bone itself is a composite material made up of a mix of living and dead cells embedded in a mineralized organic matrix. It has significant tensor structure – looking somewhat like a random honeycomb structure in cross section but with a laminated microstructure along the length of the bone. Its anatomy is illustrated in figure 140.

Bone is layered from the outside in. The very outer hard layer of a bone is called is ***periosteum***[201]. In between is compact bone, or ***osteon*** that gives bone much of its strength. Nutrients flow into living bone tissue through holes in the bone called ***foramen***, and are distributed up and down through the osteon through ***haversian canals*** (not shown) that are basically tubes through the bone for blood vessels that run *along* the bone's length to perfuse it. The periosteum and osteon make up roughly 80% of the mass of a typical long bone.

Inside the osteon is softer inner bone called ***endosteum***[202]. The inner bone is made up of a mix of different kinds of bone and other tissue that include spongy bone called ***trabeculae*** and bone marrow (where blood cells are stored and formed). It has only 20% of the bone's mass, but 90% of the bone's surface area. Much of the spongy bone material is filled with blood, to the point where a good way to characterize the difference between

[201] Latin for "outer bone". But it sounds much cooler in Latin, doesn't it?
[202] Latin for – wait for it – "inner bone".

the osteon and the trabeculae is that in the former, bone surrounds blood but in the latter, blood surrounds bone.

The bone matrix itself is made up of a mix of inorganic and organic parts. The inorganic part is formed mostly of calcium hydroxylapatite (a kind of calcium phosphate that is quite rigid). The organic part is collagen, a protein that gives bone its toughness and elasticity in much the same way that tough steels are often a mix of soft iron and hard cementite particles, with the latter contributing hardness and compressive/extensive strength, the latter reducing the brittleness that often accompanies hardness and giving it a broader range of linear response elasticity.

There are two types of microscopically distinct bone. ***Woven bone*** has collagen fibers mixed haphazardly with the inorganic matrix, and is mechanically weak. ***Lamellar bone*** has a regular parallel alignment of collagen fibers into *sheets* (lamellae) that is mechanically strong. The latter give the osteon a laminar/layered structure aligned with the bone axis. Woven bone is an early developmental state of lamellar bone, seen in fetuses developing bones and in adults as the initial soft bone that forms in a healing fracture. It serves as a sort of template for the replacement/formation of lamellar bone.

Bones are typically connected together with surface layers of cartilage at the joints, augmented by tough connective tissue and tendons smoothly integrated into muscles that permit mobile bones to be articulated at the joints. Together, they make an impressive mechanical structure capable of an extraordinary range of motions and activities while still supporting and protecting softer tissue of our organs and circulatory system. Pretty cool!

Bone is quite strong. It fractures under compression at a stress of around 170 MPa in typical human bones, but has a smaller fracture stress under tension at 120 MPa and is relatively ¡I¿easy¡/i¿ to fracture with shear stresses of around 52 MPa. This is why it is "easy" (and common!) to break bones with shear stresses, less common to break them from naked compression or tension – typically other parts of the skeletal structure – tendons or cartilage in the joints – fail before the actual bone does in these situations.

Bone is basically brittle and easy to chip, but does have a significant degree of compressive, tensile, or shear elasticity (represented by e.g. Young's modulus in the linear regime) contributed primarily by collagen in the bone tissue. Younger humans still have relatively elastic bones; as one ages one's bones become first harder and tougher, and then (as repair mechanisms break down with age) weaker and more brittle.

Figure 141 above shows the changes in bone associated with ***osteoporosis***, the gradual thinning of the bone matrix as the skeleton starts to decalcify. This process is associated with age, especially in post-menopausal women, but it can also occur in association with e.g. corticosteroid therapy, cancer, or other diseases or conditions such as Paget's disease in younger adults.

This process significantly weakens the bones of those afflicted to the point where the static shear stresses associated with muscular articulation (for example, standing up) can break bones. A young person might fall and break their hip where a person with really significant osteoporosis can actually break their hip (from the stress of standing) and then fall. This is not a medical textbook and should not be treated as an authoritative guide to the practice of medicine (but rather, as a basis for understanding what one might by trying

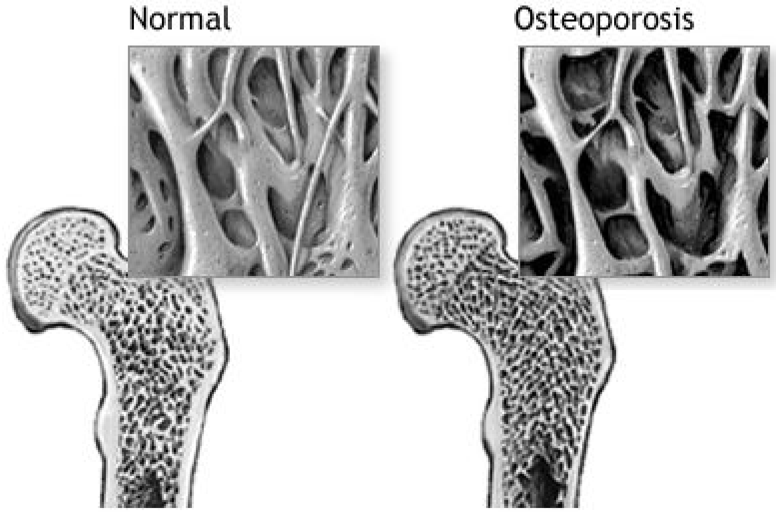


Figure 141: Illustration of the alteration of bone tissue accompanying osteoporosis.

to learn *in* a directed study of medicine), but with that said, osteoporosis can be treated to some extent by things like hormone replacement therapy in women (it seems to go with the reduction of estrogen that occurs in menopause), calcium supplementation to help slow the loss of calcium, and certain drugs such as *Fosamax* (Alendronate) that reduce calcium loss and increase bone density (but that have risks and side effects).

Example 9.7.1: Scaling of Bones with Animal Size

An interesting biological example of *scaling laws* in physics – and the reason I emphasize the dependence of many physically or physiologically interesting quantities on length and/or area – can be seen in the scaling of animal bones with the size of the animal[203] . Let us consider this.

We have seen above that the scaling of the "spring constant" of a given material that governs its change in length or its transverse displacement under compression, tension, or shear is:

$$k_{\rm eff} = \frac{XA}{L} \tag{9.130}$$

where X is the relevant (compression, tension, shear) modulus. Bone strength, including the point where the bone fractures under stresses of these sorts, might very reasonably be expected to be proportional to this constant and to scale similarly.

The leg bones of a four-legged animal have to be able to support the weight of that animal under compressive stress. This enables us to make the following scaling argument:

[203] Wikipedia: http://www.wikipedia.org/wiki/On Being the Right Size. This and many other related arguments were collected by J. B. S. Haldane in an article titled *On Being the Right Size*, published in 1926. Collectively they are referred to as the *Haldane Principle*. However, the original idea (and 3/2 scaling law discussed below) is due to none other than Galileo Galilei!

- In general, the weight of any animal is roughly proportional to its volume. Most animals are mostly made of water, and have a density close to that of water, so the volume of the animal times the density of water is a decent approximate guess of what its weight should be.

- In general, the volume of an animal (and hence its weight) is proportional to any characteristic length scale that describes the animal *cubed*. Obviously this won't work well if one compares a snake, with one very long length and two very short lengths, to a comparatively round hippopotamus, but it won't be crazy comparing mice to dogs to horses to elephants that all have reasonably similar body proportions. We'll choose the animal height.

- The leg bones of the animal have a strength proportional to the cross-sectional area.

We would like to be able to estimate the thickness of an animal's bone it's known height and from a knowledge of the thickness of *one kind* of a "reference" animal's bone and its height.

Our argument then is: The volume, and hence the weight, of an animal increases like the cube of its characteristic length (e.g. its height H). The strength of its bones that must support this weight goes like the square of the diameter D of those bones. Therefore:

$$H^3 = CD^2 \tag{9.131}$$

where C is some constant of proportionality. Solving for $D(H)$ we get:

$$D = \frac{1}{\sqrt{C}} H^{3/2} \tag{9.132}$$

This simple equation is approximately satisfied, although not *exactly* as given because our model for bones breaking does not reflect shear-driven "buckling" and a related need for muscle to scale, for mammals ranging from small rodents through the mighty elephant. Bone thickness does indeed increase nonlinearly with respect to body size.

Homework for Week 9

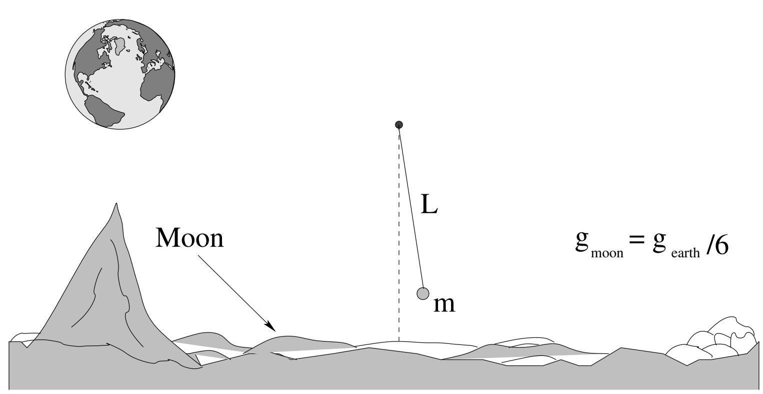
Problem 1.

Physics Concepts: Make this week's physics concepts summary as you work all of the problems in this week's assignment. Be sure to cross-reference each concept in the summary to the problem(s) they were key to, and include concepts from previous weeks as necessary. Do the work carefully enough that you can (after it has been handed in and graded) punch it and add it to a three ring binder for review and study come finals!

Problem 2.

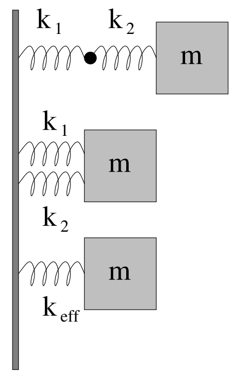
Harmonic oscillation is conceptually very important because (as has been remarked in class) many things that are *stable* will oscillate more or less harmonically if perturbed a small distance away from their stable equilibrium point. Draw an ***energy diagram***, a ***graph*** of a "generic" interaction potential energy with a ***stable equilibrium point*** and explain in words and equations where, and why, this should be.

Problem 3.



A pendulum with a string of length L supporting a mass m on the earth has a certain period T. A physicist on the moon, where the acceleration near the surface is around $g/6$, wants to make a pendulum with the same period. What mass m_m and length L_m of string could be used to accomplish this?

Problem 4.



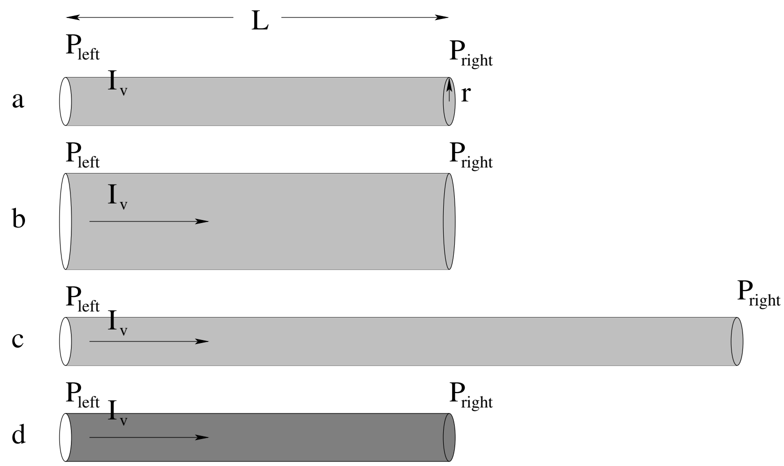
In the figure above identical masses m are attached to two springs k_1 and k_2 in "series" – one connected to the other end-to-end – and in "parallel" – the two springs side by side. In the third picture another identical mass m is shown attached to a single spring with constant $k_{\rm eff}$.

a) What should $k_{\rm eff}$ be in terms of k_1 and k_2 for the series combination of springs so that the force on the mass is the same for any given displacement Δx from equilibrium?

b) What should $k_{\rm eff}$ be in terms of k_1 and k_2 for the parallel combination of springs so that the force on the mass is the same for any given displacement Δx from equilibrium?

Note that this problem illustrates the scaling of Young's modulus with length (series) or area (parallel). It also gives you a head start on how series and parallel addition of resistors and capacitors works next semester! It's therefore well worth puzzling over.

Problem 5.



In this class we usually idealize fluid flow by neglecting resistance (drag) and the viscosity of the fluid as it passes through cylindrical pipes so we can use the Bernoulli equation. As discussed in class, however, it is often necessary to have at sound conceptual understanding of at least the ***qualitative effects*** of including the resistance and viscosity. Use ***Poiseuille's Law*** and the concepts of ***resistance, pressure, and flow*** from the online textbook to answer and discuss the following simple questions. Make sure your conceptual understanding of these concepts is ***qualitatively sound!***

a) Is $\Delta P_a = P_{\rm left} - P_{\rm right}$ greater than, less than, or equal to zero in figure a) above, where blood flows at a rate I_v horizontally through a blood vessel with constant radius r and some length L against the **resistance** of that vessel?

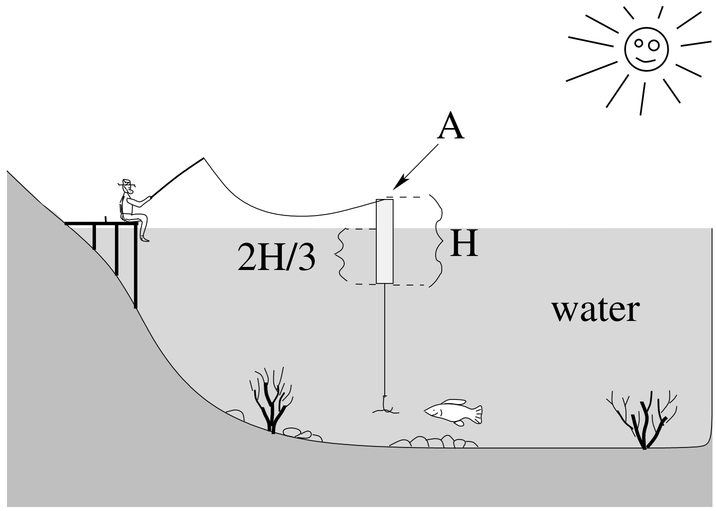
b) If the radius r **increases** (while flow I_v and length L remain the same as in a), does the pressure difference ΔP_b increase, decrease, or remain the same compared to ΔP_a?

c) If the length **increases** (while flow I_v and radius r remains the same as in a), does the pressure difference ΔP_c increase, decrease, or remain the same compared to ΔP_a?

d) If the viscosity μ of the blood **increases** (where flow I_v, radius r, and length L are all unchanged compared to a) do you expect the pressure difference ΔP_d difference across a blood vessel to increase, decrease, or remain the same compared to ΔP_a?

Blood viscosity is chemically related to things like the "stickiness" of platelets and their abundance in the blood. Viscosity and the radius r of blood vessels can be regulated or altered (within limits) by drugs, disease, diet and exercise – all of which have a ***completely understandable*** effect upon blood pressure based on the simple ideas above.

Problem 6.



This problem will help you learn required concepts such as:

- Static equilibrium
- Archimedes Principle
- Simple Harmonic Oscillation

so please review them before you begin.

rgb is fishing in still water off of the old dock. He is using a ***cylindrical*** bobber as shown. The bobber has a cross sectional area of A and a length of H, and is balanced so that it remains vertical in the water. When it is floating ***at equilibrium*** (supporting the weight of the hook and worm dangling underneath) $2H/3$ of its length is submerged in the water. You can neglect the volume of the line, hook and worm (that is, their ***buoyancy*** is negligible), and neglect all drag/damping forces. Answer the following questions in terms of A, H, ρ_w (the density of water), g:

a) What is the combined mass M of the bobber, hook and worm from the data?

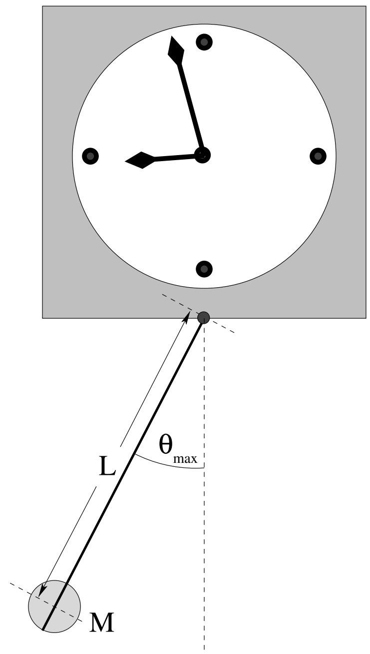
b) A fish gives a tug on the worm and pulls the bobber straight down an ***additional distance*** (from equilibrium) y. What is the ***net*** restoring force on the bobber as a function of y?

c) Use this force and the calculated mass M from a) to write Newton's 2nd Law for the motion of the bobber up and down (in y) and turn it into an equation of motion in standard form for a harmonic oscillator.

d) Assume that the bobber is pulled down the specific distance $y = H/3$ and ***released from rest*** at time $t = 0$. Neglecting the damping effects of the water, write an equation for the displacement of the bobber from its equilibrium depth as a function of time, $y(t)$ (the solution to the equation of motion).

e) With what frequency does the bobber bob? Evaluate your answer for $H = 10$ cm, $A = 1$ cm^2 (reasonable values for a fishing bobber).

Problem 7.



This problem will help you learn required concepts such as:

- Simple Harmonic Oscillation
- Torque
- Newton's Second Law for Rotation
- Moments of Inertia
- Small Angle Approximation

so please review them before you begin.

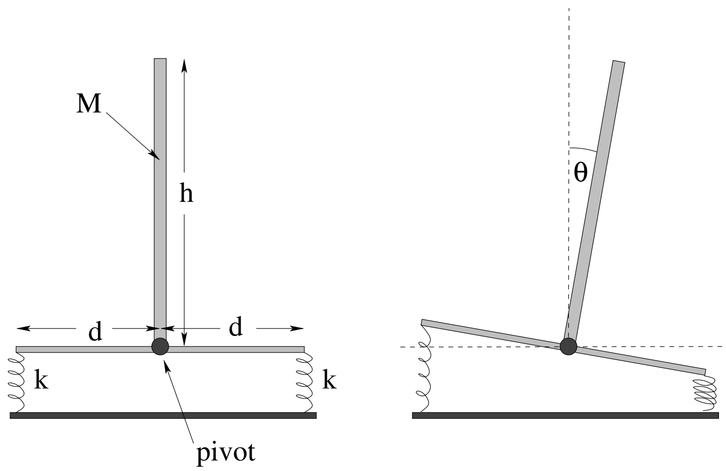
A grandfather clock is constructed with a pendulum that consists of a long, light (assume massless) rod and a small, heavy (assume point-like) mass M that can be slid up and down the rod changing L to "tune" the clock. The clock keeps perfect time when the period of oscillation of the pendulum is $T = 2$ seconds. When the clock is running, the maximum angle the rod makes with the vertical is $\theta_{\rm max} = 0.05$ radians (a "small angle").

a) *Derive* the equation of motion for the rod when it freely swings and solve for $\theta(t)$ assuming it starts at $\theta_{\rm max}$ at $t = 0$. You may use either force or torque.

b) At what distance L from the pivot should the mass be set so that the clock keeps correct time? As always, solve the problem algebraically *first* and only then worry about numbers.

c) In your answer above, you idealized by assuming the rod to be massless and the ball to be pointlike. In the real world, of course, the rod ***has a small mass*** m and the ball is ***a ball of radius*** r, not a point mass. Will the clock run slow or fast if you set the mass *exactly* where you computed it in part b)? Should you reduce L or increase L a bit to compensate and get better time? Explain.

Problem 8.



This problem will help you learn required concepts such as:

- Simple Harmonic Oscillation
- Springs (Hooke's Law)
- Small Angle Approximation

so please review them before you begin.

The pole in the figure has mass M and length h is supported by two identical springs with spring constant k that connect its platform to a table as shown. The springs attach to the platform a distance d from the base of the pole, which is pivoted on a frictionless hinge so it can rotate only in the plane of the page as shown. Gravity acts down. Neglect the mass of the platform.

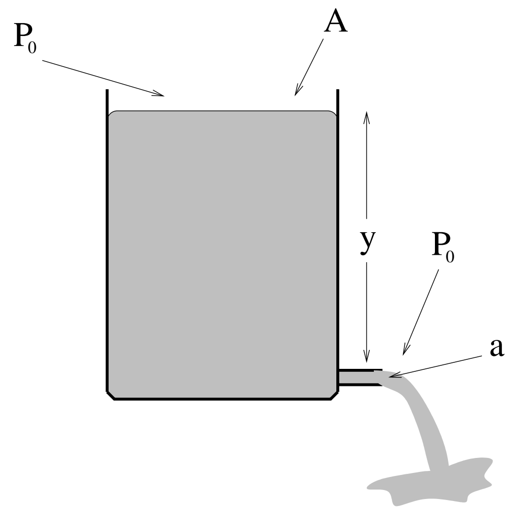
a) On a copy of the right-hand figure, indicate the forces acting on the pole and platform when the pole tips over as shown, deviating from the vertical by a (small) angle θ.

b) Using your answer to a), find the ***total torque*** acting on the system (pole and platform) about the hinged bottom of the pole, and note its direction. Use the small angle approximations $\sin(\theta) \approx \theta$; $\cos(\theta) \approx 1$.

c) Find the ***minimal value*** for the spring constant k of the two springs such that the pole is stable in the vertical position. This means that a small deviation as above produces a torque that restores the pole to the vertical.

d) For $M = 50$ kg, $h = 1.0$ meter, $d = 0.5$ meter, and springs with spring constant $k = 9600$ N/m, find the angular frequency with which the pole oscillates about the vertical.

Advanced Problem 9.



This problem will help you learn required concepts such as:

- Bernoulli's Equation
- Torricelli's Law

so please review them before you begin.

In the figure above, a large drum of water is open at the top and filled up to a height y above a tap at the bottom (which is also open to normal air pressure). The drum has a cross-sectional area A at the top and the tap has a cross sectional area of a at the bottom.

The questions below help you use ***calculus*** to determine how long it will take for the drum to empty. The first two questions and the last question anybody can answer. The one where you actually have to integrate may be difficult for non-math/physics/engineering students, but feel free to give it a try!

a) Find the speed with which the water emerges from the tap. Assume laminar flow without resistance. Compare your answer to the speed a mass has after falling a height y in a uniform gravitational field (after using $A \gg a$ to simplify your final answer, Torricelli's Law).

b) ***Start by guessing*** how long it will take water to flow out of the tank by ***using dimensional analysis*** and the insight gained from a). That is, think about how you *expect* the time to vary with each quantity that describes the problem and form a simple expression with the relevant parameters that has the right units and goes up where it should go up and down where it should go down.

c) Next, find an ***algebraic expression*** for the velocity of the top. This is dy/dt.

d) Manupulate the expression and integrate dy on one side, an expression with dt on the other side, to find the time it takes for the top of the water to reach the bottom of the tank. Compare your answer to b) to your answer to d) and the time it takes a mass to fall a height y in a uniform gravitational field. Does the correct answer make dimensional and physical sense? You might want to think a bit about Toricelli's Law here as well...

e) Evaluate the answers to a) and d) for $A = 0.50$ m^2, $a = 0.5$ cm^2, $y(0) = 100$ cm.

Optional Problems

Continue studying for the final exam! Only three “weeks” (chapters) to go in this textbook, finals are coming apace!

Week 10: The Wave Equation

Wave Summary

- **Wave Equation**
$$\frac{d^2y}{dt^2} - v^2\frac{d^2y}{dx^2} = 0 \tag{10.1}$$
where for waves on a string:
$$v = \pm\sqrt{\frac{T}{\mu}} \tag{10.2}$$

- **Superposition Principle**
$$y(x,t) = Ay_1(x,t) + By_2(x,t) \tag{10.3}$$
(sum of solutions is solution). Leads to interference, standing waves.

- **Travelling Wave Pulse**
$$y(x,t) = f(x \pm vt) \tag{10.4}$$
where $f(u)$ is an arbitrary functional shape or pulse

- **Harmonic Travelling Waves**
$$y(x,t) = y_0 \sin(kx \pm \omega t). \tag{10.5}$$
where frequency f, wavelength λ, wave number $k = 2\pi/\lambda$ and angular frequency ω are related to v by:
$$v = f\lambda = \frac{\omega}{k} \tag{10.6}$$

- **Stationary Harmonic Waves**
$$y(x,t) = y_0 \cos(kx + \delta)\cos(\omega t + \phi) \tag{10.7}$$
where one can select k and ω so that waves on a string of length L satisfy fixed or free boundary conditions.

- **Energy (of wave on string)**
$$E_{\rm tot} = \frac{1}{2}\mu\omega^2A^2\lambda \tag{10.8}$$
is the total energy in a wavelength of a travelling harmonic wave. The wave transports the power
$$P = \frac{E}{T} = \frac{1}{2}\mu\omega^2A^2\lambda f = \frac{1}{2}\mu\omega^2A^2 v \tag{10.9}$$
past any point on the string.

- **Reflection/Transmission of Waves**

  a) Light string (medium) to heavy string (medium): Transmitted pulse right side up, reflected pulse inverted. (A fixed string boundary is the limit of attaching to an “infinitely heavy string”).

  b) Heavy string to light string: Transmitted pulse right side up, reflected pulse right side up. (a free string boundary is the limit of attaching to a “massless string”).

10.1: Waves

We have seen how a particle on a spring that creates a restoring force proportional to its displacement from an equilibrium position oscillates harmonically in time about that equilibrium. What happens if there are *many* particles, *all* connected by tiny “springs” to one another in an extended way? This is a good metaphor for many, many physical systems. Particles in a solid, a liquid, or a gas both attract and repel one another with forces that maintain an average particle spacing. Extended objects under tension or pressure such as strings have components that can exert forces on one another. Even fields (as we shall learn next semester) can interact so that changes in one tiny element of space create changes in a neighboring element of space.

What we observe in all of these cases is that changes in any part of the medium “propagate” to other parts of the medium in a very systematic way. The motion observed in this propagation is called a *wave*. We have all observed waves in our daily lives in many contexts. We have watched water waves propagate away from boats and raindrops. We listen to sound waves (music) generated by waves created on stretched strings or from tubes driven by air and transmitted invisibly through space by means of radio waves. We read these words by means of light, an electromagnetic wave. In advanced physics classes one learns that all matter is a sort of quantum wave, that indeed *everything* is really a manifestation of waves.

It therefore seems sensible to make a first pass at understanding waves and how they work in general, so that we can learn and understand more in future classes that go into detail.

The concept of a wave is simple – it is an *extended structure* that oscillates in both *space* and in *time*. We will study two kinds of waves at this point in the course:

- **Transverse Waves** (e.g. waves on a string). The displacement of particles in a transverse wave is *perpendicular* to the direction of the wave itself.

- **Longitudinal Waves** (e.g. sound waves). The displacement of particles in a longitudinal wave is in the same direction that the wave propagates in.

Some waves, for example water waves, are simultaneously longitudinal and transverse. Transverse waves are probably the most important waves to understand for the future; light is a transverse wave. We will therefore start by studying transverse waves in a simple context: waves on a stretched string.

10.2: Waves on a String

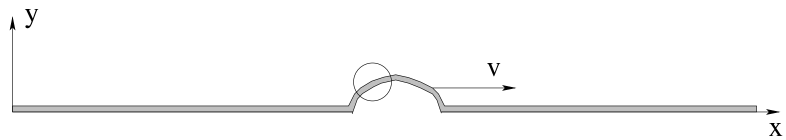


Figure 142: A uniform string is plucked or shaken so as to produce a *wave pulse* that travels at a speed v to the right. The circled region is examined in more detail in the next figure.

Suppose we have a uniform *stretchable* string (such as a guitar string) that is pulled at the ends so that it is under tension T. The string is characterized by its mass per unit length μ – thick guitar strings have more mass per unit length than thin ones but little else. It is fairly harmless at this point to imagine that the string is fixed to pegs at the ends that maintain the tension. Our experience of such things leads us to expect that the stretched string will form an almost perfectly straight line unless we pluck it or otherwise bend some other shape upon it. We will impose coordinates upon this string such that x runs along its (undisplaced) equilibrium position and y describes the vertical displacement of any given bit of the string.

Now imagine that we have plucked the string somewhere between the end points so that it is displaced in the y-direction from its equilibrium (straight) stretched position and has some curved shape, as portrayed in figure 142. The tension T, recall, acts all along the string, but because the string is *curved* the force exerted on any small bit of string does not balance. This inspires us to try to write Newton's Second Law not for the entire string itself, but for just a tiny bit of string, indeed a *differential* bit of string.

We thus zoom in on just a small chunk of the string in the region where we have stretched or shaken a *wave pulse* as illustrated by the figure above. If we blow this small segment up, perhaps we can find a way to write the unbalanced forces out in a way we can deal with algebraically.

This is shown in figure 143, where I've indicated a short segment/chunk of the string of length Δx by cross-hatching it. We would like to write Newton's 2nd law for that chunk. As you can see, the same *magnitude* of tension T pulls on both ends of the chunk, but the tension pulls in slightly different directions, tangent to the string at the end points.

If θ is small, the components of the tension in the x-direction:

$$\begin{aligned} F_{1x} &= -T\cos(\theta_1) \approx -T \\ F_{2x} &= T\cos(\theta_1) \approx T \end{aligned} \tag{10.10}$$

are nearly equal and in opposite directions and hence nearly perfectly cancel (where we have used the small angle approximation to the Taylor series for cosine:

$$\cos(\theta) = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - ... \approx 1 \tag{10.11}$$

for small $\theta \ll 1$).

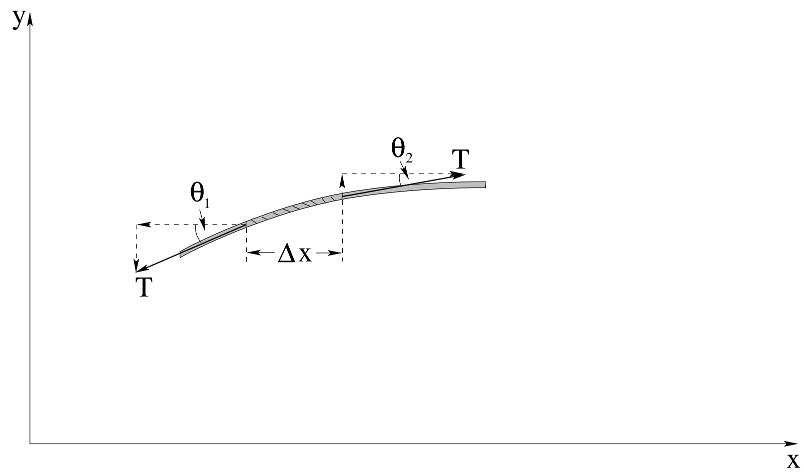


Figure 143: The forces exerted on a small chunk of the string by the tension T in the string. Note that we neglect gravity in this treatment, assuming that it is much too small to bend the stretched string significantly (as is indeed the case, for a taut guitar string).

Each bit of string therefore moves more or less *straight up and down*, and a useful solution is described by $y(x, t)$, the y **displacement** of the string at position x and time t. The permitted solutions must be *continuous* if the string does not break. It is worth noting that not *all* waves involving moving up and down only – this initial example is called a *transverse* wave, but other kinds of waves exist. Sound waves, for example, are *longitudinal* compression waves, with the motion back and forth along the direction of motion and not at right angles to it. Surface waves on water are a mixture, they involve the water *both* moving up and down *and* back and forth. But for the moment we'll stick with these very simple transverse waves on a string because they already exhibit most of the properties of far more general waves you might learn more about later on.

In the y-direction, we find write the force law by considering the total y-components of the sum of the force exerted by the tension on the ends of the chunk:

$$\Delta F_{y,tot} = F_{1y} + F_{2y} = -T\sin(\theta_1) + T\sin(\theta_2) = \Delta m a_y = \mu \Delta x a_y \tag{10.12}$$

Here ΔF_y is not the "change in the force" but rather the total force acting on the chunk of length Δx. These two quantities will obviously scale together.

We make the small angle approximation: $\sin(\theta) \approx \tan(\theta) \approx \theta$:

$$\Delta F_y = T\tan(\theta_2) - T\tan(\theta_1) = T(\tan(\theta_2) - \tan(\theta_1)) = \mu \Delta x a_y, \tag{10.13}$$

Next, we note that $\tan(\theta) = \frac{dy}{dx}$ (the slope of the string is the tangent of the angle the string makes with the horizon) and divide out the $\mu\Delta x$ to isolate $a_y = d^2y/dt^2$. We end up with the following expression for what might be called *Newton's Second Law per unit length* of the string:

$$\frac{\Delta F_y}{\Delta x} = T\frac{\Delta(\frac{dy}{dx})}{\Delta x} = \mu\frac{d^2y}{dt^2} \tag{10.14}$$

In the limit that $\Delta x \to 0$, this becomes:

$$f_y = \frac{dF_y}{dx} = T\frac{d^2y}{dx^2} = \mu\frac{d^2y}{dt^2} \tag{10.15}$$

where we might consider $f_y = dF_y/dx$ the *force density* (force per unit length) acting at a point on the string. Finally, we rearrange to get:

$$\frac{d^2y}{dt^2} - \frac{T}{\mu}\frac{d^2y}{dx^2} = 0 \tag{10.16}$$

The quantity $\frac{T}{\mu}$ has to have units of $\frac{L^2}{t^2}$ which is a velocity squared.

We therefore formulate this as the one dimensional *wave equation*:

$$\frac{d^2y}{dt^2} - v^2\frac{d^2y}{dx^2} = 0 \tag{10.17}$$

where

$$v = \pm\sqrt{\frac{T}{\mu}} \tag{10.18}$$

are the velocity(s) of the wave on the string. This is a second order linear homogeneous differential equation and has (as one might imagine) well known and well understood solutions.

Note well: At the tension in the string increases, so does the wave velocity. As the mass density of the string increases, the wave velocity decreases. This makes *physical sense*. As tension goes up the restoring force is greater. As mass density goes up one accelerates less for a given tension.

10.3: Solutions to the Wave Equation

In one dimension there are at least three distinct solutions to the wave equation that we are interested in. Two of these solutions *propagate* along the string – energy is *transported* from one place to another by the wave. The third is a *stationary* solution, in the sense that the wave doesn't propagate in one direction or the other (not in the sense that the string doesn't move). But first:

10.3.1: An Important Property of Waves: Superposition

The wave equation is *linear*, and hence it is easy to show that if $y_1(x,t)$ solves the wave equation and $y_2(x,t)$ (independent of y_1) *also* solves the wave equation, then:

$$y(x,t) = Ay_1(x,t) + By_2(x,t) \tag{10.19}$$

solves the wave equation for arbitrary (complex) A and B.

This property of waves is most powerful and sublime.

10.3.2: Arbitrary Waveforms Propagating to the Left or Right

The first solution we can discern by noting that the wave equation equates a second derivative in time to a second derivative in space. Suppose we write the solution as $f(u)$ where

u is an unknown function of x and t and substitute it into the differential equation and use the chain rule:

$$\frac{d^2 f}{du^2}(\frac{du}{dt})^2 - v^2 \frac{d^2 f}{du^2}(\frac{du}{dx})^2 = 0 \tag{10.20}$$

or

$$\frac{d^2 f}{du^2}\left\{(\frac{du}{dt})^2 - v^2(\frac{du}{dx})^2\right\} = 0 \tag{10.21}$$

$$\frac{du}{dt} = \pm v \frac{du}{dx} \tag{10.22}$$

with a simple solution:

$$u = x \pm vt \tag{10.23}$$

What this tells us is that *any* function

$$y(x,t) = f(x \pm vt) \tag{10.24}$$

satisfies the wave equation. *Any* shape of wave created on the string and propagating to the right or left is a solution to the wave equation, although not all of these waves will vanish at the ends of a string.

10.3.3: Harmonic Waveforms Propagating to the Left or Right

An interesting special case of this solution is the case of *harmonic* waves propagating to the left or right. Harmonic waves are simply waves that oscillate with a given harmonic frequency. For example:

$$y(x,t) = y_0 \sin(x - vt) \tag{10.25}$$

is one such wave. y_0 is called the *amplitude* of the harmonic wave. But what sorts of parameters describe the wave itself? Are there more than one harmonic waves?

This particular wave looks like a sinusoidal wave propagating to the right (positive x direction). But this is not a very convenient parameterization. To better describe a general harmonic wave, we need to introduce the following quantities:

- The **frequency** f. This is the number of cycles per second that pass a point or that a point on the string moves up and down.
- The **wavelength** λ. This the distance one has to travel down the string to return to the same point in the wave cycle at any given instant in time.

To convert x (a distance) into an angle in radians, we need to multiply it by 2π radians per wavelength. We therefore define the *wave number*:

$$k = \frac{2\pi}{\lambda} \tag{10.26}$$

and write our harmonic solution as:

$$\begin{aligned} y(x,t) &= y_0 \sin(k(x - vt)) && (10.27) \\ &= y_0 \sin(kx - kvt) && (10.28) \\ &= y_0 \sin(kx - \omega t) && (10.29) \end{aligned}$$

where we have used the following train of algebra in the last step:

$$kv = \frac{2\pi}{\lambda} v = 2\pi f = \frac{2\pi}{T} = \omega \tag{10.30}$$

and where we see that we have two ways to write v:

$$v = f\lambda = \frac{\omega}{k} \tag{10.31}$$

As before, you should simply *know* every relation in this set of algebraic relations between λ, k, f, ω, v to save time on tests and quizzes. Of course there is also the harmonic wave travelling to the left as well:

$$y(x,t) = y_0 \sin(kx + \omega t). \tag{10.32}$$

A final observation about these harmonic waves is that because arbitrary functions can be *expanded* in terms of harmonic functions (e.g. Fourier Series, Fourier Transforms) and because the wave equation is linear and its solutions are superposable, the two solution forms above are not really distinct. One can expand the "arbitrary" $f(x - vt)$ in a sum of $\sin(kx - \omega t)$'s for special frequencies and wavelengths. In one dimension this doesn't give you much, but in two or more dimensions this process helps one compute the *dispersion* of the wave caused by the wave "spreading out" in multiple dimensions and reducing its amplitude.

10.3.4: Stationary Waves

The third special case of solutions to the wave equation is that of *standing waves*. They are especially apropos to waves on a string fixed at one or both ends. There are two ways to find these solutions from the solutions above. A harmonic wave travelling to the right and hitting the end of the string (which is fixed), it has no choice but to reflect. This is because the *energy* in the string cannot just disappear, and if the end point is fixed no work can be done by it on the peg to which it is attached. The reflected wave has to have the same amplitude and frequency as the incoming wave. What does the *sum* of the incoming and reflected wave look like in this special case?

Suppose one adds two harmonic waves with equal amplitudes, the same wavelengths and frequencies, but that are travelling in *opposite* directions:

$$\begin{aligned} y(x,t) &= y_0 \left(\sin(kx - \omega t) + \sin(kx + \omega t)\right) && (10.33)\\ &= 2y_0 \sin(kx)\cos(\omega t) && (10.34)\\ &= A\sin(kx)\cos(\omega t) && (10.35) \end{aligned}$$

(where we give the standing wave the arbitrary amplitude A). Since all the solutions above are independent of the phase, a second useful way to write stationary waves is:

$$y(x,t) = A\cos(kx)\cos(\omega t) \tag{10.36}$$

Which of these one uses depends on the details of the boundary conditions on the string.

In this solution a sinusoidal form oscillates harmonically up and down, but the solution has some very important new properties. For one, it is always *zero* when $x = 0$ for all possible $lambda$:

$$y(0, t) = 0 \tag{10.37}$$

For a given λ there are certain other x positions where the wave vanishes at all times. These positions are called *nodes* of the wave. We see that there are nodes for any L such that:

$$y(L, t) = A \sin(kL) \cos(\omega t) = 0 \tag{10.38}$$

which implies that:

$$kL = \frac{2\pi L}{\lambda} = \pi, 2\pi, 3\pi, \ldots \tag{10.39}$$

or

$$\lambda = \frac{2L}{n} \tag{10.40}$$

for $n = 1, 2, 3, ...$

Only waves with these wavelengths and their associated frequencies can persist on a string of length L fixed at both ends so that

$$y(0, t) = y(L, t) = 0 \tag{10.41}$$

(such as a guitar string or harp string). Superpositions of these waves are what give guitar strings their particular tone.

It is also possible to stretch a string so that it is fixed at one end but so that the *other* end is *free to move* – to slide up and down without friction on a greased rod, for example. In this case, instead of having a node at the free end (where the wave itself vanishes), it is pretty easy to see that the *slope* of the wave at the end has to vanish. This is because if the slope were not zero, the terminating rod would be pulling up or down on the string, contradicting our requirement that the rod be frictionless and not *able* to pull the string up or down, only directly to the left or right due to tension.

The slope of a sine wave is zero only when the sine wave itself is a maximum or minimum, so that the wave on a string free at an end must have an *antinode* (maximum magnitude of its amplitude) at the free end. Using the same standing wave form we derived above, we see that:

$$kL = \frac{2\pi L}{\lambda} = \pi/2, 3\pi/2, 5\pi/2 \ldots \tag{10.42}$$

for a string fixed at $x = 0$ and free at $x = L$, or:

$$\lambda = \frac{4L}{2n - 1} \tag{10.43}$$

for $n = 1, 2, 3, ...$

There is a second way to obtain the standing wave solutions that particularly exhibits the relationship between waves and harmonic oscillators. One assumes that the solution $y(x, t)$ can be written as the *product* of a fuction in x alone and a second function in t alone:

$$y(x, t) = X(x)T(t) \tag{10.44}$$

If we substitute this into the differential equation and divide by $y(x,t)$ we get:

$$\frac{d^2y}{dt^2} = X(x)\frac{d^2T}{dt^2} = v^2\frac{d^2y}{dx^2} = v^2T(t)\frac{d^2X}{dx^2} \tag{10.45}$$

$$\frac{1}{T(t)}\frac{d^2T}{dt^2} = v^2\frac{1}{X(x)}\frac{d^2X}{dx^2} \tag{10.46}$$

$$= -\omega^2 \tag{10.47}$$

where the last line follows because the second line equations a function of t (only) to a function of x (only) so that both terms must equal a constant. This is then the two equations:

$$\frac{d^2T}{dt^2} + \omega^2 T = 0 \tag{10.48}$$

and

$$\frac{d^2X}{dt^2} + k^2 X = 0 \tag{10.49}$$

(where we use $k = \omega/v$).

From this we see that:

$$T(t) = T_0\cos(\omega t + \phi) \tag{10.50}$$

and

$$X(x) = X_0\cos(kx + \delta) \tag{10.51}$$

so that the most general stationary solution can be written:

$$y(x,t) = y_0\cos(kx+\delta)\cos(\omega t + \phi) \tag{10.52}$$

10.4: Reflection of Waves

We argued above that waves have to reflect of of the ends of stretched strings because of energy conservation. This is true independent of whether the end is fixed or free – in neither case can the string do work on the wall or rod to which it is affixed. However, the behavior of the reflected wave is different in the two cases.

Suppose a wave *pulse* is incident on the fixed end of a string. One way to “discover” a wave solution that apparently conserves energy is to imagine that the string *continues* through the barrier. At the same time the pulse hits the barrier, an *identical* pulse hits the barrier from the other, “imaginary” side.

Since the two pulses are identical, energy will clearly be conserved. The one going from left to right will transmit its energy onto the imaginary string beyond at the same rate energy appears going from right to left from the imaginary string.

However, we still have two choices to consider. The wave from the imaginary string could be *right side up* the same as the incident wave or *upside down*. Energy is conserved either way!

If the right side up wave (left to right) encounters an upside down wave (right to left) they will always be *opposite* at the barrier, and when superposed they will *cancel* at the barrier. This corresponds to a *fixed string*. On the other hand, if a right side up wave encounters a right side up wave, they will *add* at the barrier with opposite *slope*. There will be a maximum at the barrier with zero slope – just what is needed for a free string.

From this we deduce the general rule that wave pulses *invert* when reflected from a fixed boundary (string fixed at one end) and reflect right side up from a free boundary.

When two strings of different weight (mass density) are connected, wave pulses on one string are both transmitted onto the other and are generally partially reflected from the boundary. Computing the transmitted and reflected waves is straightforward but beyond the scope of this class (it starts to involve real math and studies of boundary conditions). However, the following qualitative properties of the transmitted and reflected waves should be learned:

- Light string (medium) to heavy string (medium): Transmitted pulse right side up, reflected pulse inverted. (A fixed string boundary is the limit of attaching to an "infinitely heavy string").

- Heavy string to light string: Transmitted pulse right side up, reflected pulse right side up. (a free string boundary is the limit of attaching to a "massless string").

10.5: Energy

Clearly a wave can carry energy from one place to another. A cable we are coiling is hung up on a piece of wood. We flip a pulse onto the wire, it runs down to the piece of wood and knocks the wire free. Our lungs and larnyx create sound waves, and those waves trigger neurons in ears far away. The sun releases nuclear energy, and a few minutes later that energy, propagated to earth as a light wave, creates sugar energy stored inside a plant that is still later released while we play basketball[204]. Since moving energy around seems to be important, perhaps we should figure out how a wave manages it.

10.5.1: Energy in Travelling Waves

Let us restrict our attention to a travelling harmonic wave of known angular frequency ω, although many of our results will be more general, because arbitrary wave pulses can be fourier decomposed as noted above.

Consider a small piece of the string of length dx and mass $dm = \mu dx$. This piece of string, displaced to its position $y(x,t)$, will have kinetic energy:

$$\begin{aligned} dK &= \frac{1}{2} dm \left(\frac{dy}{dt}\right)^2 && (10.53) \\ &= \frac{1}{2} A^2 \mu \omega^2 \cos^2(kx - \omega t) dx && (10.54) \end{aligned}$$

[204] Painfully and badly, in my case. As I'm typing this, my ribs and ankles hurt from participating in the Great Beaufort 3 on 3 Physics Basketball Tournament, Summer Session 1, 2011! But what the heck, we won and are in the finals. And it was energy propagated by a wave, stored (in my case) as fat, that helped get us there...

We can easily integrate this over any specific interval. Let us pick a particular time $t = 0$ and integrate it over a single wavelength:

$$\begin{aligned}
\Delta K &= \int_0^{\Delta} K dK = \int_0^{\lambda} \frac{1}{2} A^2 \mu \omega^2 \cos^2(kx) dx && (10.55) \\
&= \frac{1}{2k} A^2 \mu \omega^2 \int_0^{\lambda} \cos^2(kx) k dx && (10.56) \\
&= \frac{1}{2k} A^2 \mu \omega^2 \int_0^{2\pi} \cos^2(\theta) d\theta && (10.57) \\
&= \frac{1}{4} A^2 \mu \omega^2 \lambda && (10.58)
\end{aligned}$$

(where we have used the easily proven relation:

$$\int_0^{2\pi} \sin^2(\theta) d\theta = \int_0^{2\pi} \cos^2(\theta) d\theta = \pi \tag{10.59}$$

to do the final form of the integral). If we assume that the string has length $L \gg \lambda$, we can average this over many wavelengths and get the ***average*** kinetic energy per unit length

$$\kappa = K/L \approx \Delta K/\lambda = \frac{1}{4} A^2 \mu \omega^2 \tag{10.60}$$

In the limit of an infinitely long string, or a length of string that contains an integer number of wavelengths, this expression is exact. Note also that this answer does not depend on time, because ωt only corresponds to a different phase and the integral of $\sin^2(\theta) + \delta)$ does not change.

The average potential energy of the string is more difficult. There is a temptation to treat the potential energy of a chunk of the string of length Δx as the work done against that force bending the string into its instantaneous shape. This approach will lead to a constant energy per unit length for the string, as it basically treats every string element as an "independent" mass oscillating as if it is attached to a spring. Unfortunately, this picture doesn't explain how energy can *propagate* in the case of a travelling wave, nor does it give us any insight into just where the energy is being stored, as there *are* no springs.

It is better to treat it as the work done *stretching* the local pieces of the string, while maintaining the assumption above that the pieces of string engage in transverse motion *only*, that is, they move straight up or down and not back and forth in the x-direction.

This approach basically treats the string as a very long chain of mass elements, each of which *is* a little "spring". Each mass element has kinetic energy as it moves straight up and down, and has potential energy to the extent that it is stretched. Note that the string *must* stretch if it is not straight, because a straight line is the shortest distance between two points and $y(x, t)$ cannot be zero everywhere or it is the boring trivial solution, no actual wave at all.

In this approach, the potential energy does not depend on value of $y(x, t)$ at all, but rather on the *slope* of the string $\frac{dy}{dx}$ at a point. This is illustrated in figure 144. A small (eventually differentially small) chunk of the string of unstretched length Δx is seen to be stretched to a new length $\Delta \ell = \sqrt{(\Delta x)^2 + (\Delta y)^2}$ when it is moved straigth up to the height $y(x, t)$ so that it has some slope there. If we assume that this slope is very small ($\Delta y \ll \Delta x$) then:

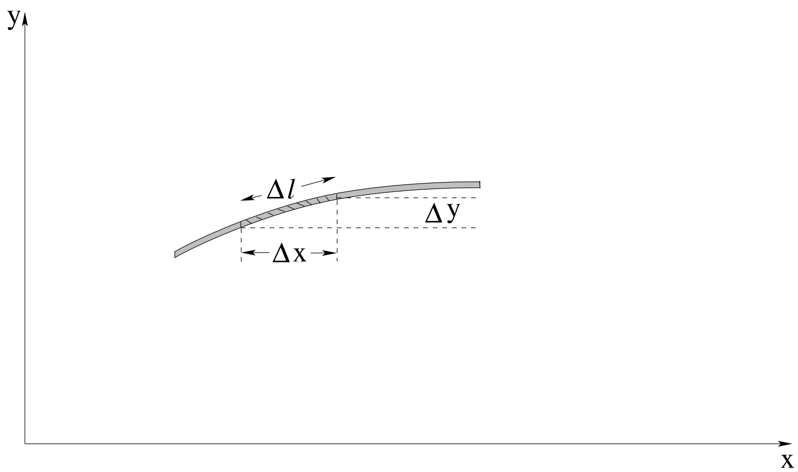


Figure 144: A small chunk of the string. The chunk that (unstretched at $y(x,t) = 0$ has length Δx is stretched by an amount $\Delta\ell - \Delta x$.

$$\begin{aligned}
\Delta\ell - \Delta x &= \left((\Delta x)^2 + (\Delta y)^2\right)^{1/2} - \Delta x \\
&= \Delta x \left(1 + \frac{(\Delta y)^2}{(\Delta x)^2}\right)^{1/2} - \Delta x \\
&= \Delta x \left(1 + \frac{1}{2}\left(\frac{\Delta y}{\Delta x}\right)^2 + ...\right) - \Delta x \\
&\approx \frac{\Delta x}{2}\left(\frac{\Delta y}{\Delta x}\right)^2
\end{aligned} \tag{10.61}$$

In the limit that $\Delta x \to 0$, this becomes:

$$d\ell - dx = \frac{1}{2}\left(\frac{dy}{dx}\right)^2 dx \tag{10.62}$$

(where really, the expression should use partial derivatives e.g. $d\ell - dx = \frac{1}{2}\left(\frac{\partial y}{\partial x}\right)^2 dx$ as usual but either way it is the instantaneous slope of $y(x,t)$). The work *we* do to stretch this small chunk of string to its new length is considered to be the potential energy of the chunk:

$$dU = T(d\ell - dx) = \frac{T}{2}\left(\frac{dy}{dx}\right)^2 dx \tag{10.63}$$

or the potential energy per unit length of the string is just:

$$\frac{dU}{dx} = \frac{T}{2}\left(\frac{dy}{dx}\right)^2 \tag{10.64}$$

For the *travelling harmonic wave* $y(x,t) = A\sin(kx - \omega t)$ we considered before:

$$\frac{dy}{dx} = kA\cos(kx - \omega t) \tag{10.65}$$

and

$$dU = \frac{1}{2}Tk^2A^2\cos^2(kx - \omega t)dx \tag{10.66}$$

so that we can form the *total energy density* of the string dE/dx in this formulation:

$$\begin{aligned}\frac{dE}{dx} &= \frac{1}{2}\mu\omega^2 A^2\cos^2(kx-\omega t) + \frac{1}{2}Tk^2A^2\cos^2(kx-\omega t) \\ &= \frac{1}{2}\mu\omega^2 A^2\left(\cos^2(kx-\omega t)+\cos^2(kx-\omega t)\right) \\ &= \mu\omega^2A^2\cos^2(kx-\omega t) \end{aligned} \tag{10.67}$$

because $\mu\omega^2 = Tk^2$ from $\frac{T}{\mu} = \frac{\omega^2}{k^2} = v^2$.

There is apparently exactly as much kinetic energy as there is potential energy in the differentially small chunk of string, and that energy in the chunk is *not a constant*. Not only that, but we note that the energy density is itself a wave propagating from left to right as it has the form $f(x - vt)$. Our interpretation of this is that the string is ***carrying energy from left to right*** as the wave propagates, with the peak energy in the string concentrated where $y(x,t) = 0$, where both the speed of the string and its local stretch (magnitude of its slope) are maximal.

It is useful to find the average energy per unit length of this wave. Since the wave is lots of copies of a single wavelength, we can get the average over a very long string by just evaluating the energy per wavelength from a single wavelength:

$$\frac{E}{\lambda} = \frac{1}{\lambda}\mu\omega^2A^2\int_0^\lambda \cos^2(kx-\omega t)dx = \frac{1}{2}\mu\omega^2A^2 \tag{10.68}$$

(where the integral is considered below, if it isn't familiar to you).

10.5.2: Power Transmission by a Travelling Wave

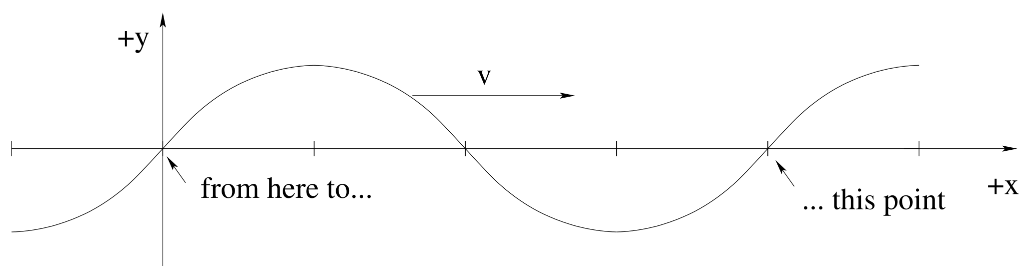


Figure 145: *All* of the energy in a single wavelength of the wave is transmitted past an (arbitrary) point on the string in the direction of motion in *one* period of oscillation.

We haven't discussed how to put a travelling wave *onto* a string or how to take it *off* at the other end, but in a nutshell, we put it on by doing *work* on (say) the left hand end of the string and letting the string do work on something attached to the right hand end of the string. In that way, energy added at the left can be removed and used at the right. Waves transmit energy[205]!

[205] In case you never did this as a child, if you take two plastic cups and stretch a long piece of fishing line attached to the ends in between them, you can speak into one cup while a friend holds the other end up to their ear and *hear* the result of energy transmission through the string. In the old days when tin cans were more prevalent than plastic cups, this was called a "tin can telephone" and most kids would build them at least once because they are pretty cool/nifty/radical/fresh (the terminology changes but the idea remains the same).

The figure above gives us a simple model of energy transmission that is nearly independent of just *how* the energy is distributed. A travelling wave going from left to right on a string that has a formula like:

$$y(x,t) = A\sin(kx - \omega t)$$

has (as shown above) a total energy in one wavelength given by:

$$\Delta E = \frac{1}{2}\mu\omega^2 A^2 \lambda \tag{10.69}$$

All of this energy moves from left to right along with the wave. In one period, all of the energy in one wavelength passes any given point on the string, so the (average) energy per unit time passing that point is:

$$P = \frac{\Delta E}{\Delta t} = \frac{1}{2}\mu\omega^2 A^2 \frac{\lambda}{T} = \frac{1}{2}\mu\omega^2 A^2 v \tag{10.70}$$

In words, the average power transmitted by the string is the energy density times the speed of the travelling wave on the string.

10.5.3: Energy in Standing Waves

Next, Consider a generic *standing wave* $y(x,t) = A\sin(kx)\cos(\omega t)$ for a string that (perhaps) is fixed at ends located at $x = 0$ and $x = L$. We can easily evaluate the kinetic energy of a chunk of length dx:

$$dK = \frac{1}{2}\mu\omega^2 A^2 \sin^2(kx)\sin^2(\omega t) dx \tag{10.71}$$

and the potential energy of the same chunk:

$$\begin{aligned} dU &= \frac{1}{2}Tk^2 A^2 \cos^2(kx)\cos^2(\omega t) dx \\ &= \frac{1}{2}\mu\omega^2 A^2 \cos^2(kx)\cos^2(\omega t) dx \end{aligned} \tag{10.72}$$

If we now form the total energy density of this chunk of string as before:

$$\frac{dE}{dx} = \frac{1}{2}\mu\omega^2 A^2 \left(\sin^2(kx)\sin^2(\omega t) + \cos^2(kx)\cos^2(\omega t)\right) \tag{10.73}$$

we note that the energy density of the string is *not a constant*.

At first this appears to be a problem, but really it is not. All this tells us is that the nodes and antinodes of the string (where $\sin(kx)$ or $\cos(kx)$ are zero, respectively) carry the energy of the string *out of phase* with one another. At times $\omega t = 0, \pi, 2\pi ...$, the potential energy is maximal, the kinetic energy is zero, and the potential energy is concentrated at the points where $\cos(kx) = \pm 1$. These are the *nodes*, which have maximum magnitude of slope and stretch as the string reaches its maximum positive or negative amplitude. At times $\omega t = \pi/2, 3\pi/2 ...$ the kinetic energy is maximal, the potential energy is zero (indeed, the string is now flat and unstretched at all) and the kinetic energy is concentrated at the *antinodes*, where the velocity of the string is the greatest.

This makes perfect sense! The energy in the string oscillates from being all potential (for the entire string) as it momentarily comes to rest to all kinetic as the string is momentarily straight and moving at maximum speed (for the location) everywhere. The one question we need to answer, though, is: Is the *total* energy of the string constant? We would be sad if it were not, because we know that the endpoints of the standing wave are (for example) fixed, and hence no work is done there. Whatever the energy in the wave on the string, it has nowhere to go.

We note that no matter what the boundary conditions, the string contains an integer number of quarter wavelengths. All we have to do is show that the total energy in any quarter wavelength of the string is constant in time and we're good to go. Consider the following integral:

$$\begin{aligned}\int_0^{\lambda/4} \cos^2(kx)dx = \int_0^{\lambda/4} \sin^2(kx)dx &= \frac{1}{k}\int_0^{\lambda/4} \sin^2(kx)kdx \\ &= \frac{1}{k}\int_0^{\pi/2} sin^2(\theta)d\theta \\ &= \frac{\lambda}{2\pi}\frac{\pi}{4} = \frac{\lambda}{8} \end{aligned} \tag{10.74}$$

(where $\sin^2$ and $\cos^2$ have the same integral from symmetry, one equals the other with the addition or subtraction of the constant angle $\pi/2$ which doesn't change the integral). Note also that it doesn't matter if we are doing quarter wavelengths or quarter periods of a time-dependent harmonic wave:

$$\int_0^{T/4} \cos^2(\omega t)dt = \int_0^{T/4} \sin^2(\omega t)dt = \frac{T}{8} \tag{10.75}$$

The time average of these quantities (over any integer number of quarter periods or quarter wavelengths) are thus always $\frac{1}{2}$:

$$\frac{4}{T}\int_0^{T/4} \cos^2(\omega t)dt = \frac{4}{T}\int_0^{T/4} \sin^2(\omega t)dt = \frac{4}{T}]\frac{T}{8} = \frac{1}{2} \tag{10.76}$$

Finally, if we let the time or length go to ∞ to do our time or space average, we always asymptotically approach $\frac{1}{2}$ because even if there is a non-quarter wavelength fraction of a wave leftover at one end (the remainder after summing all the full quarter waves) its contribution vanishes as time (or distance) goes to infinity.

We summarize as follows:

The time or space average of any function that goes
like $\sin^2$ ***or*** $\cos^2$ ***(of*** ωt ***or*** kx ***respectively) is one half!***

This is worth remembering, as we use it quite often.

From this we can easily find the energy per unit length of any standing wave on a string of length L (which as noted has an integer number of quarter waves on it). We'll express this as the energy per wavelength of string to facilitate comparison with a travelling wave and just remember that it works just as well for a quarter wave, half wave, etc.

$$\begin{aligned}
\frac{E}{\lambda} &= \frac{1}{\lambda}\frac{1}{2}\mu\omega^2 A^2 \left(\sin^2(\omega t)\int_0^\lambda \sin^2(kx)dx + \cos^2(\omega t)\int_0^\lambda \cos^2(kx)\right) \\
&= \frac{1}{\lambda}\frac{1}{2}\mu\omega^2 A^2 \left(\sin^2(\omega t)\frac{\lambda}{2} + \cos^2(\omega t)\frac{\lambda}{2}\right) \\
&= \frac{1}{4}\mu\omega^2 A^2 \left(\sin^2(\omega t) + \cos^2(\omega t)\right) \\
&= \frac{1}{4}\mu\omega^2 A^2
\end{aligned} \tag{10.77}$$

If we compare this to the expression for the energy per wavelength of a traveling wave with the same amplitude, we get:

$$\frac{E_s}{\lambda} = \frac{1}{2}\frac{E_t}{\lambda} \tag{10.78}$$

We can understand this easily enough. The standing wave consists of two travelling waves of half of the amplitude going in opposite directions. Their total energy per wavelength are independent – we just add them. Energy per wavelength is proportional to amplitude squared, so each wave with amplitude $A/2$ has $1/4$ of the energy of a travelling wave with amplitude A. But there are two of them, hence a factor of $1/2$.

Homework for Week 10

Problem 1.

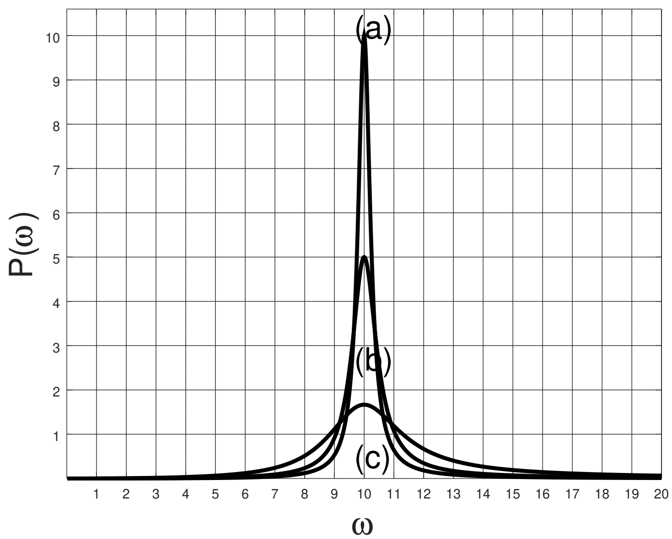
Physics Concepts: Make this week's physics concepts summary as you work all of the problems in this week's assignment. Be sure to cross-reference each concept in the summary to the problem(s) they were key to, and include concepts from previous weeks as necessary. Do the work carefully enough that you can (after it has been handed in and graded) punch it and add it to a three ring binder for review and study come finals!

Problem 2.

This (and several of the following) problems are from ***resonance and damped and/or driven oscillation***, not from waves! Be sure to use the correct concepts on your concept summary!

Roman soldiers (like soldiers the world over even today) marched in step at a constant frequency – except when crossing wooden bridges, when they broke their march and walked over with random pacing. Why? What might have happened (and originally did sometimes happen) if they marched across with a collective periodic step?

Problem 3.



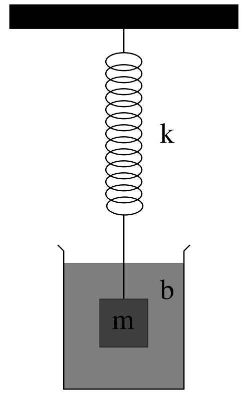
In the figure above, three ***resonance curves*** are drawn showing the power delivered to a steady-state driven oscillator, $P(\omega)$. In all three cases the resonance frequency ω_0 is the same. Put down an *estimate* of the Q-value of each oscillator by looking at the graph. It may help for you to put down the definition of Q most relevant to the process of estimation on the page.

a)

b)

c)

Problem 4.



A mass m is attached to a spring with spring constant k and immersed in a medium with damping coefficient b. (Gravity, if present at all, is irrelevant as shown in class). The net force on the mass when displaced by x from equilibrium and moving with velocity v_x is thus:

$$F_x = ma_x = -kx - bv_x$$

(in one dimension).

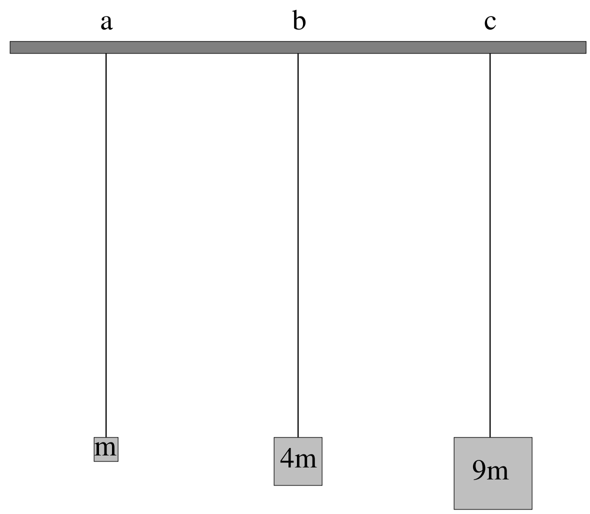
a) Convert this equation (Newton's second law for the mass/spring/damping fluid arrangement) into the equation of motion for the system, a "second order linear homogeneous differential equation" as done in class.

b) Optionally *solve* this equation, finding in particular the exponential damping rate of the solution (the real part of the exponential time constant) and the shifted frequency ω', assuming that the motion is underdamped. You can put down any form you like for the answer; the easiest is probably a sum of exponential forms. However, you may also simply **put down the solution** derived in class if you plan to *just* memorize this solution instead of learn to derive and understand it.

c) Using your answer for ω' from part b), write down the criteria for damped, underdamped, and critically damped oscillation.

d) Draw *three* qualitatively correct graphs of $x(t)$ if the oscillator is pulled to a position x_0 and released at rest at time $t = 0$, one for each damping. Note that you should be able to do this part even if you cannot derive the curves that you draw or ω'. Clearly label each curve.

Problem 5.



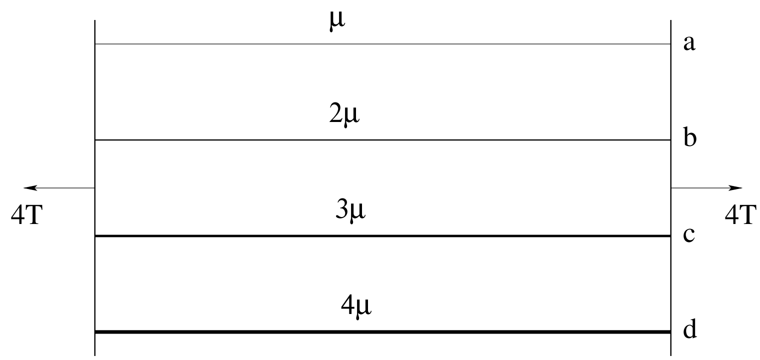
Three strings of length L (not shown) with the same mass per unit length μ are suspended vertically and blocks of mass m, $4m$ and $9m$ are hung from them. The total mass of each string $\mu L \ll m$ (the strings are *much* lighter than the masses hanging from them). If the speed of a wave pulse on the first string is v_0, fill in the following table with entries for b) and c):

a) v_0

b)

c)

Problem 6.



In the figure above, the neck of a stringed instrument is schematized. Four strings of different *thickness* (and hence different μ as shown) and the same length are stretched in such a way that the tension in each is about the same (T) in each string. This produces a total of $4T$ between the end bridges – if this were not so, the neck of the guitar or ukelele or violin would tend to bow towards the side with the greater tension.

If the speed of a wave pulse on the first (lightest) string is v_0, fill in the following table for the speed of a wave pulse for the other three:

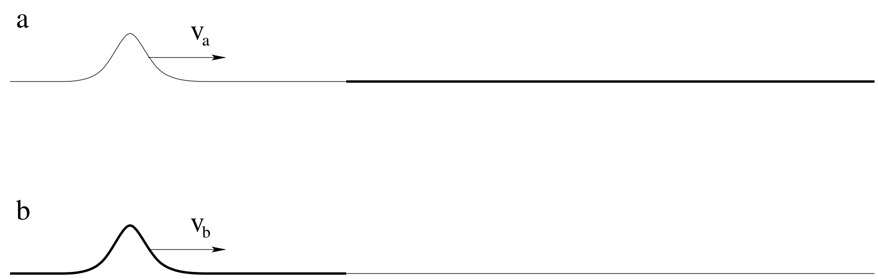
a) v_0

b)

c)

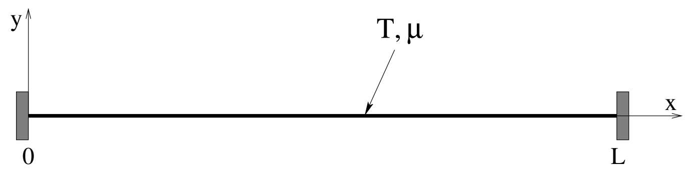
d)

Problem 7.



Two combinations of two strings with different mass densities are drawn above that are connected in the middle. In both cases the string with the greatest mass density is drawn darker and thicker than the lighter one, and the strings have the same tension T in both a and b. A wave pulse is generated on the string pairs that is travelling from left to right as shown. The wave pulse will arrive at the junction between the strings at time t_a (for a) and t_b (for b). Sketch reasonable *estimates* for the ***transmitted and reflected wave pulses*** onto (copies of) the a and b figures at time $2t_a$ and $2t_b$ respectively. Your sketch should correctly represent things like the ***relative speed of the reflected and transmitted wave*** and any changes you might reasonably expect for the ***amplitude and appearance*** of the pulses.

Problem 8.

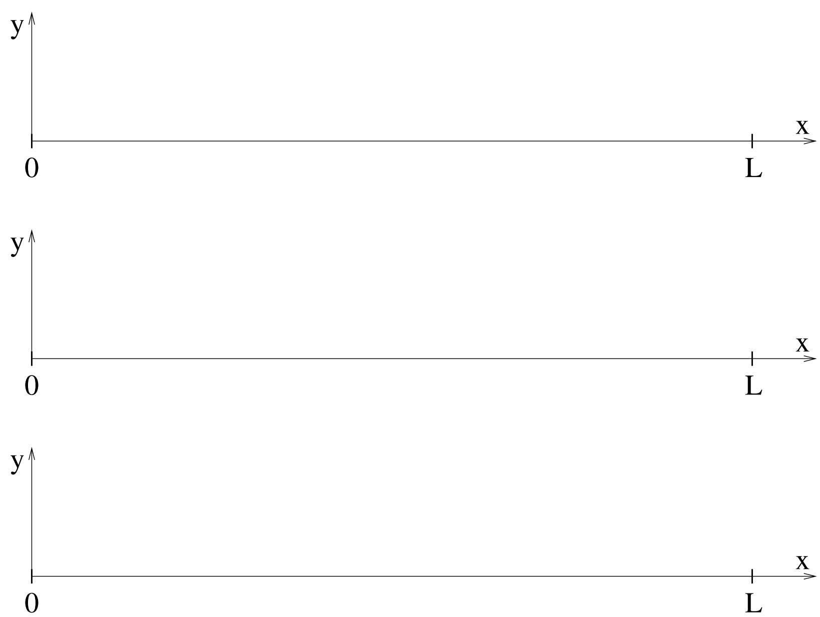


A string of mass density μ is stretched to a tension T and is ***fixed at both $x = 0$ and $x = L$.*** The transverse string displacement is measured in the y direction. All answers should be given in terms of these quantities or new quantities you define in terms of these quantities.

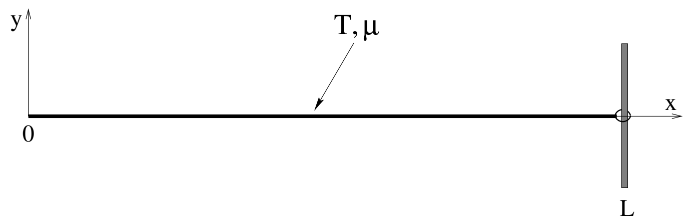
a) Following the text, derive the wave equation (the equation of motion) for waves on a string and identify the wave velocity squared in terms of T and μ. This one derivation suffices for this and the next problem.

b) Write down the equation $y_n(x, t)$ for a generic ***standing wave*** on this string with mode index n, assuming that the string is maximally displaced at $t = 0$. Verify that it is a solution to the ODE in a). Remember that the string is fixed at both ends!

c) Find $k_n, \omega_n, f_n, \lambda_n$ for the ***first three modes supported by the string***. Sketch them in on the axes below, labelling nodes and antinodes. Note that you should be able to draw the modes and find at least the wavelengths from the pictures alone.



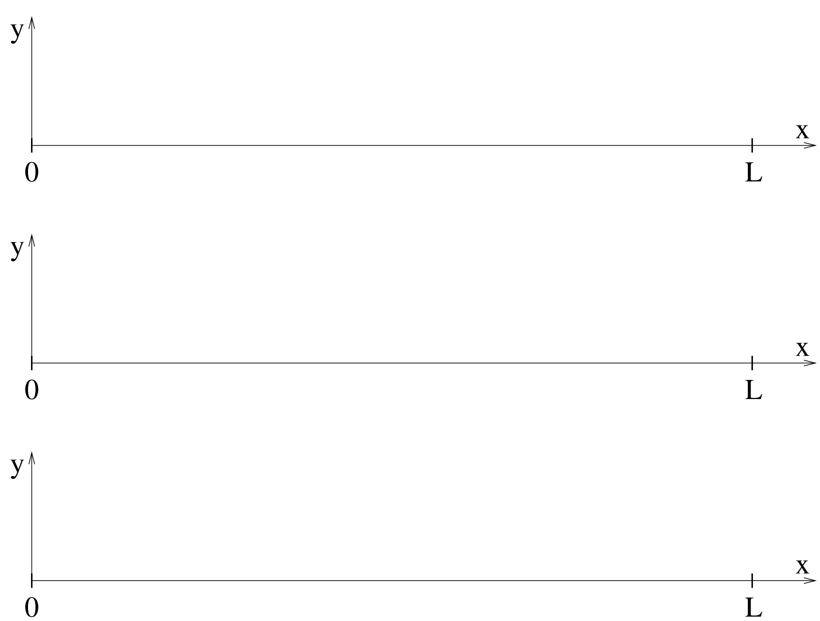
Problem 9.



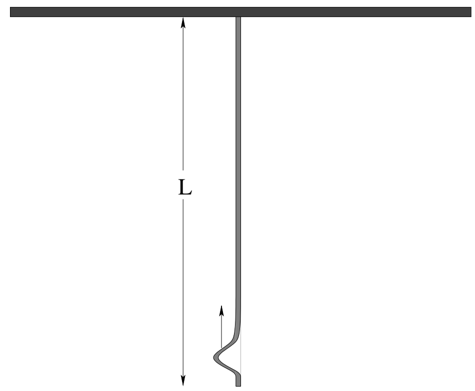
A string of mass density μ is stretched to a tension T and is ***fixed at $x = 0$ and free (frictionless loop) at $x = L$***. The transverse string displacement is measured in the y direction. All answers should be given in terms of these quantities or new quantities you define in terms of these quantities.

a) Write down the equation $y_n(x, t)$ for a generic ***standing wave*** on this string with mode index n, assuming that the string is maximally displaced at $t = 0$. Verify that it is a solution to the ODE in a). Remember that the string is free at one end!

b) Find $k_n, \omega_n, f_n, \lambda_n$ for the ***first three modes supported by the string***. Sketch them in on the axes below, labelling nodes and antinodes. Note that you should be able to draw the modes and find at least the wavelengths from the pictures alone.



Problem 10.



This problem will help you learn required concepts such as:

- Speed of Wave on String
- Static Equilibrium
- Relationship betweeen Distance, Velocity, and Time

so please review them before you begin.

A string of total length L with a mass density μ is shown hanging from the ceiling above.

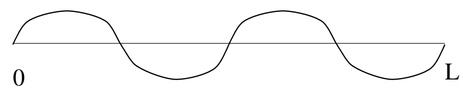
a) Find the tension $T(y)$ in the string as a function of y, the distance *up* from its bottom end. Note that the string is not massless, so each small bit of string must be in ***static equilibrium***.

b) Find the velocity $v(y)$ of a small wave pulse cast into the string at the bottom that is travelling upward.

c) Find the amount of time it will take this pulse to reach the top of the string, reflect, and return to the bottom. Neglect the size (width in y) of the pulse relative to the length of the string.

Hint for last part. Set $v = dy/dt$, rearrange to get all the y-dependent parts on one side and dt and some constants on the other side, then integrate both sides, the y part from 0 to L, the t part from 0 to t_0. Solve for t_0, double it. This is what calculus is *for!*

Problem 11.



A string of total mass M and total length L is fixed at both ends, stretched so that the speed of waves on the string is v. It is plucked so that it harmonically vibrates in its $n = 4$ mode:

$$y(x, t) = Asin(k_4 x)cos(\omega_4 t).$$

Find the instantaneous total energy in the string in terms of M, L, $n = 4$, v and A (although it will simplify matters to use λ_4 and ω_4 ***once you define them in terms of the givens***).

If you are a ***physics or math major***, the word "find" should be interpreted as "derive", following the derivation presented in the book. All others can get by from remembering the general way the total energy ***scales with the givens*** in understandable ways to lead to the formula derived in the book above.

For those who attempt the derivation, remember (or FYI):

$$\int_0^{n\pi} \sin^2(u)du = \int_0^{n\pi} \cos^2(u)du = \frac{n\pi}{2}$$

Optional Problems

Continue studying for the final exam! Only ***one more week of class*** (and one chapter) to go in this textbook! Don't wait until the last moment to start!

Week 11: Sound

Sound Summary

- **Bulk Modulus**

  The ***Bulk Modulus*** of a fluid (or a solid) is defined by:

  $$\Delta P = -B\frac{\Delta V}{V} \tag{11.1}$$

  where B is the bulk modulus. The bulk modulus of a solid is clearly related to Young's modulus, where compressive forces are applied in all three dimensions at once. In a gas, it tells us how much the gas compresses (at a given temperature) when the pressure in the gas is changed.

  For an ideal gas at fixed temperature, we can easily derive B using calculus:

  $$P = \frac{NkT}{V} \tag{11.2}$$

  so

  $$\frac{dP}{dV} = -\frac{NkT}{V^2} \tag{11.3}$$

  and

  $$dP = -\frac{NkT}{V}\frac{dV}{V} = -P\frac{dV}{V} \tag{11.4}$$

  and we see that the bulk modulus of an ideal gas *at constant temperature* equals its pressure!

- **Speed of Sound** in a fluid

  $$v = \sqrt{\frac{B}{\rho}} \tag{11.5}$$

  where B is the bulk modulus of the fluid and ρ is the density.

  In an ideal *monoatomic* gas

  $$v = \sqrt{\frac{P}{\rho}} \tag{11.6}$$

  We usually assume that sound waves are ***adiabatic*** (energy conserving), not ***isothermal***. Also, liquids are nothing like ideal gases!

  For near-ideal-gases (such as air) we will use:

  $$B = \gamma P \tag{11.7}$$

where $\gamma = 1.4$ for diatomic gases (see a chapter on thermodynamics for the derivation and explanation of γ). For liquids and non-ideal gases we will ignore further discussions of the thermodynamics and use B, or v itself, as a given.

- **Speed of Sound in Air**

  Air is to a decent approximation an ideal *diatomic* gas). $\gamma = 1.4$ for an ideal diatomic gas, so:

$$v = \sqrt{\frac{B}{\rho}} = \sqrt{\gamma\frac{P}{\rho}} \approx= \sqrt{1.4\frac{10^5}{1.225}} \approx 343 \quad \text{m/sec} \tag{11.8}$$

  where we've used $P \approx 10^5$ pascals and $\rho = 1.225$ kg/m^3 for dry air at one atmosphere. This is in *very good agreement* with observation.

  We will often use approximations for this in class to facilitate arithmetic "in your head", such as $v_a \approx 340$ m/sec or $v_a \approx 333$ m/sec (so that $v_a \approx 1/3$ km/sec). This is very close to $v_a = 1000$ ft/sec in the English system, or $v_a = 1/5$ mi/sec, which can be quite useful for estimating distances to approaching thunderstorms.

- **Travelling Sound waves:**

  Plane (displacement) waves (in the x-direction):

$$s(x,t) = s_0 \sin(kx - \omega t)$$

  Spherical waves:

$$s(r,t) = s_0 \frac{R}{(\sqrt{4\pi})\, r} \sin(kr - \omega t)$$

  where R is a reference length needed to make the units right corresponding physically to the "size of the source" (e.g. the smallest ball that can be drawn that completely contains the source). The $\sqrt{4\pi}$ is needed so that the intensity has the right functional form for a spherical wave (see below).

- **Pressure Waves:** The pressure waves that correspond to these two displacement waves are:

$$P(x,t) = P_0 \cos(kx - \omega t)$$

  and

$$P(r,t) = P_0 \frac{R}{(\sqrt{4\pi})\, r} \cos(kr - \omega t)$$

  where $P_0 = v_a \rho \omega s_0 = Z \omega s_0$ with $Z = v_a \rho = \sqrt{B\rho}$ (a conversion factor that scales microscopic displacement to pressure). Note that:

$$P(x,t) = Z \frac{d}{dt} s(x,t)$$

  or, the displacement wave is a scaled derivative of the pressure wave.

  The pressure waves represent the oscillation of the pressure *around* the baseline ambient pressure P_a, e.g. 1 atmosphere. The total pressure is really $P_a + P(x,t)$ or $P_a + P(r,t)$ (and we could easily put a "Δ" in front of e.g. $P(r,t)$ to emphasize this point but don't so as to not confuse variation around a baseline with a derivative).

- **Sound Intensity:** The intensity of sound waves can be written:
$$I = \frac{1}{2} v_a \rho \omega^2 s_0^2$$
or
$$I = \frac{1}{2} \frac{1}{v_a \rho} P_0^2 = \frac{1}{2Z} P_0^2$$

- **Spherical Waves:** The intensity of spherical sound waves drops off like $1/r^2$ (as can be seen from the previous two points). It is usually convenient to express it in terms of the ***total power emitted by the source*** $\boldsymbol{P_{\mathrm{tot}}}$ as:
$$I = \frac{\boldsymbol{P_{\mathrm{tot}}}}{4\pi r^2}$$
This is "the total power emitted divided by the area of the sphere of radius r through which all the power must symmetrically pass" and hence it *makes sense!* One can, with some effort, take the intensity at some reference radius r and relate it to P_0 and to s_0, and one can easily relate it to the intensity at other radii.

- **Decibels:** Audible sound waves span some 20 orders of magnitude in intensity. Indeed, the ear is barely sensitive to a *doubling* of intensity – this is the smallest change that registers as a change in audible intensity. This motivate the use of ***sound intensity level*** measured in ***decibels***:
$$\beta = 10 \log_{10} \left(\frac{I}{I_0} \right) \ (\mathrm{dB})$$
where the reference intensity $I_0 = 10^{-12}$ watts/meter^2 is the ***threshold of hearing***, the weakest sound that is audible to a "normal" human ear.

  Important reference intensities to keep in mind are:

  - 60 dB: Normal conversation at 1 m.
  - 85 dB: Intensity where long term continuous exposure *may* cause gradual hearing loss.
  - 120 dB: Hearing loss is ***likely*** for anything more than brief and highly intermittent exposures at this level.
  - 130 dB: Threshold of ***pain***. Pain is bad.
  - 140 dB: Hearing loss is ***immediate and certain*** – you are actively losing your hearing during any sort of prolonged exposure at this level and above.
  - 194.094 dB: The upper limit of undistorted sound (overpressure equal to one atmosphere). This loud a sound will instantly rupture human eardrums 50% of the time.

- **Doppler Shift: Moving Source**
$$f' = \frac{f_0}{(1 \mp \frac{v_s}{v_a})} \tag{11.9}$$
where f_0 is the unshifted frequency of the sound wave for receding (+) and approaching (-) source, where v_s is the speed of the source and v_a is the speed of sound in the medium (air).

- **Doppler Shift: Moving Receiver**

$$f' = f_0(1 \pm \frac{v_r}{v_a}) \tag{11.10}$$

where f_0 is the unshifted frequency of the sound wave for receding (-) and approaching (+) receiver, where v_r is the speed of the source and v_a is the speed of sound in the medium (air).

- **Stationary Harmonic Waves**

$$y(x,t) = y_0 \sin(kx)\cos(\omega t) \tag{11.11}$$

for displacement waves in a pipe of length L closed at one or both ends. This solution has a node at $x = 0$ (the closed end). The permitted resonant frequencies are determined by:

$$kL = n\pi \tag{11.12}$$

for $n = 1, 2...$ (both ends closed, nodes at both ends) or:

$$kL = \frac{2n-1}{2}\pi \tag{11.13}$$

for $n = 1, 2, ...$ (one end closed, nodes at the closed end).

- **Beats** If two sound waves of equal amplitude and slightly different frequency are added:

$$\begin{aligned} s(x,t) &= s_0 \sin(k_0 x - \omega_0 t) + s_0 \sin(k_1 x - \omega_1 t) && (11.14) \\ &= 2 s_0 \sin(\frac{k_0 + k_1}{2} x - \frac{\omega_0 + \omega_1}{2} t)\cos(\frac{k_0 - k_1}{2} x - \frac{\omega_0 - \omega_1}{2} t) && (11.15) \end{aligned}$$

which describes a wave with the average frequency and twice the amplitude modulated so that it "beats" (goes to zero) at the difference of the frequencies $\delta f = |f_1 - f_0|$.

11.1: Sound Waves in a Fluid

Waves propagate in a fluid much in the same way that a disturbance propagates down a closed hall crowded with people. If one shoves a person so that they knock into their neighbor, the neighbor falls against *their* neighbor (and shoves back), and their neighbor shoves against their still further neighbor and so on.

Such a wave differs from the transverse waves we studied on a string in that the displacement of the medium (the air molecules) is *in the same direction* as the direction of propagation of the wave. This kind of wave is called a *longitudinal* wave.

Although different, sound waves can be related to waves on a string in many ways. Most of the similarities and differences can be traced to one thing: a string is a one dimensional medium and is characterized only by length; a fluid is typically a three dimensional medium and is characterized by a volume.

Air (a typical fluid that supports sound waves) does not support "tension", it is under pressure. When air is compressed its molecules are shoved closer together, altering its density and occupied volume. For small changes in volume the pressure alters approximately *linearly* with a coefficient called the "bulk modulus" B describing the way the pressure increases as the fractional volume decreases. Air does not have a mass per unit length μ, rather it has a mass per unit volume, ρ.

The velocity of waves in air (treated as an ideal diatomic gas at 1 atmosphere) is given by:

$$v = \sqrt{\frac{B}{\rho}} = \sqrt{\gamma \frac{P}{\rho}} \approx = \sqrt{1.4 \frac{10^5}{1.225}} \approx 343 \quad \mathrm{m/sec} \tag{11.16}$$

The "approximately" here is fairly serious. The speed obviously varies like the square root of the air pressure over the density, which both vary significantly with altitude and with the weather at any given altitude as low and high pressure areas move around on the earth's surface. Both also vary with the temperature (hotter molecules push each other apart more strongly at any given density). Consequently the speed of sound can vary by a few percent from the approximate value given above over the course of as little as a day at any given location, in addition to varying by more than that as one travels from sea level to the tops of mountains.

11.2: The Wave Equation for Sound

The derivation of the wave equation for sound in a gas is, to put it bluntly, "difficult". It involves synthesizing three separate ideas – the "state equation" for the gas (and the concept of an "adiabatic process" expressed in the calculus of a bulk medium), the law of conservation of mass (the continuity equation), and a force law – Newton's second law for a chunk of the fluid. If you are a non-physics major student (reading this) for whom things like this are moderately terrifying, be at peace – as was the case for the derivation of the solutions to the harmonic oscillator equation above, you will not be held responsible in any way for deriving the wave equation for sound, and you can without any penalty skip from the break just below to the similar line ending the break below. Even though you won't be held responsible for this, though, you may find it interesting. Physics majors absolutely should go through the derivation and try to understand it, and it is up to their instructor as to whether or not it is required.

The Following is an Advanced Topic and May be Skipped!

We start with the physicist version of the Ideal Gas Law, as the equation of state of an ideal gas. Air at standard temperature and pressure is a very good approximation of an ideal gas, as are most gases far away from the liquid-gas phase transition

$$PV = NkT \tag{11.17}$$

We assume that this equation applies to any small chunk ΔV of the gas large enough to contain many molecules but small enough to be (eventually) treatable as a differential volume. You will recognize this as the coarse-graining hypothesis that we've used from the beginning to describe microscopically discrete matter as "mass density".

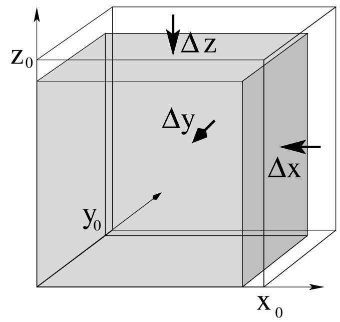


Figure 146: A volume $V = xyz$ containing N molecules is compressed across all of its faces so that it shrinks by Δx in the x-direction, Δy in the y-direction, and Δz in the z-direction. This reduces its volume by $\Delta V = xy\Delta z + yz\Delta x + zx\Delta y$.

Next, we consider what happens when we compress this volume of gas. Suppose that a fixed number N of gas molecules occupy a box with sides x, y, and z so that its volume $V_0 = x_0y_0z_0$. Such a box is pictured in figure 146. Now imagine that the box is compressed (by pressure in the surrounding fluid) so that it shrinks by a (small!) Δx in the x-direction, Δy in the y-direction, and Δz in the z-direction while the number of molecules in the box does not change. The change in volume of the box is $\Delta V = x_0y_0z_0 - (x_0 - \Delta x)(y_0 - \Delta y)(z_0 - \Delta z) \approx x_0y_0\Delta z + y_0z_0\Delta x + z_0x_0\Delta y$ plus terms with products of small pieces such as $x_0\Delta y\Delta z$ which we will neglect.

Extending our previous discussion of Young's Modulus, stress, and strain to three dimensions, we write (for the three directions independently):

$$\begin{aligned}
\Delta P &= \frac{\Delta F_x}{y_0z_0} = -Y\frac{\Delta x}{x_0} \\
\Delta P &= -Y\frac{\Delta y}{y_0} \\
\Delta P &= -Y\frac{\Delta z}{z_0}
\end{aligned}$$

If we multiply each term by a clever form of 1:

$$\begin{aligned}
\Delta P &= -Y\frac{\Delta x y_0 z_0}{x_0y_0z_0} \\
\Delta P &= -Y\frac{\Delta y x_0 z_0}{x_0y_0z_0} \\
\Delta P &= -Y\frac{\Delta z x_0 y_0}{x_0y_0z_0}
\end{aligned}$$

and then add them and divide both sides by 3, we get:

$$\Delta P = \frac{-Y}{3}\frac{\Delta V}{V_0} = -B\frac{\Delta V}{V_0} \tag{11.18}$$

which relates a new quantity, the ***bulk modulus*** B, to Young's modulus. This equation holds for solids, liquids and gases, but we only need to use it for our ideal gas above. This is one of the key equations needed to derive the wave equation for sound. Note well that the bulk modulus has units of pressure in pascals (or atmospheres or bar or torr), just as do Young's modulus and the shear modulus.

In this equation *for* an ideal gas, $\Delta P = P - P_0$, the increase in pressure relative to a "reference pressure" P_0 associated with the number of molecules N, the "reference volume" volume V_0 (which is really any small chunk of volume in the bulk gas, it needn't be an actual box of gas with solid walls), and temperature T of the gas:

$$P_0 V_0 = NkT \tag{11.19}$$

If we divide the volume over to the other side and multiply both sides by the molecular mass m we obtain the (reference) density of the gas at the reference pressure/volume:

$$\rho_0 = \frac{Nm}{V_0} = P_0 \frac{kT}{m} \tag{11.20}$$

Let's linearize the change in density in terms of the change in volume using the binomial expansion, holding (in our minds) the number of molecules in the volume unchanged. "Linearizing" means that we just keep the first term in ΔV (the linear term) and ignore the terms that are higher order in ΔV^2, etc:

$$\begin{aligned}
\Delta\rho &= \rho - \rho_0 = \frac{mN}{V_0 - \Delta V} - \frac{mn}{V_0} \\
&= \frac{mN}{V_0}(1 - \frac{\Delta V}{V_0})^{-1} - \frac{mn}{V_0} = \frac{mN}{V_0}\left(1 + \frac{\Delta V}{V_0} + \frac{\Delta V^2}{2V_0^2} + \ldots\right) - \frac{mn}{V_0} \\
&= \frac{mN}{V_0}\frac{\Delta V}{V_0} + \ldots \\
&\approx \rho_0 \frac{\Delta V}{V_0}
\end{aligned} \tag{11.21}$$

Note well that ΔV is the *positive* magnitude of the *reduction* in volume according to our use of signs. This means that :

$$\frac{\Delta\rho}{\rho_0} = \frac{\Delta V}{V_0}$$

As one might expect, reducing the volume increases the density in identical ways relative to the initial values. With the same sign convention for ΔV, we also have:

$$\Delta P = B\frac{\Delta V}{V_0} = B\frac{\Delta\rho}{\rho_0} \tag{11.22}$$

where reducing the volume increases the density and increases the pressure of the fluid. This equation *also* holds for bulk compression of ANY substance, solid, gas or fluid.

The definition of B and its extension to ρ is not complete or universal. For example, it does not account for temperature changes, which can also cause pressure changes at a constant volume! Volume changes can also occur at a constant pressure if one varies the temperature. There are literally an infinite number of ways one can compress a chunk of gas (doing work) and distribute the work as a mix of an increase in the average kinetic energy (temperature) of molecules in the gas and "heat", energy that flows in or out through the sides of the volume. The two limiting cases of this are ***isothermal*** compression where *all* of the work done to compress the fluid flows out of the volume as heat (keeping the volume at a constant temperature) and ***adiabatic*** (or "isentropic") compression where *none* of the work goes out of the volume; it remains as an increase in average kinetic energy, hence increases the absolute temperature.

Sound waves are not precisely either of these "pure" processes, but dry air at one atmosphere is a *good thermal insulator* and most sound waves vary on timescales that are short relative to the time required for significant heat to flow. We therefore assume that sound waves in air are composed of ***adiabatic compression*** of small volumes of the air (treated as an ideal gas) by the surrounding air.

We pause for a very brief discussion of adiabatic processes for a volume of ideal gas with a constant number of particles. The justification for the starting equation is beyond the scope of this course but is to be found in any introductory physics textbook that covers the thermodynamics of an ideal gas. The relation between pressure and volume in an adiabatic compression of an ideal gas is:

$$PV^{\gamma} = A \tag{11.23}$$

where $\gamma = 5/3 = 1.67$ for monoatomic ideal gases and $\gamma = 1.4$ for diatomic ideal gases (such as air) and A is a constant with the appropriate units. If we form the differential of both sides, we get:

$$\gamma P V^{\gamma-1} dV - V^{\gamma} dP = 0 \tag{11.24}$$

or (dividing both sides by V^{γ} and rearranging):

$$dP = \gamma P \frac{dV}{V} \tag{11.25}$$

or (comparing with the equations defining the bulk modulus above):

$$B = \gamma P \tag{11.26}$$

This will make it easy to at least estimate the expected speed of sound in air at the end.

Let's return to the expression defining the bulk modulus above, evaluating the (small) changes ***relative to*** the reference pressure and density for air, P_0 and ρ_0 (eventually one atmosphere and 1.225 kg/m^3 at room temperature and sea level).

$$\Delta P = P - P_0 = B \frac{(\rho - \rho_0)}{\rho_0} = B \frac{\Delta \rho}{\rho_0} \tag{11.27}$$

We now make a simplifying change of variables. Sound waves are ultimately going to be described by a change in pressure *relative* to some reference/background pressure and changes in density *relative* to the corresponding reference/background density. Carrying along the reference values and/or constantly writing Δ's will be tedious. So we now define the ***relative pressure change***:

$$p = P - P_0 \tag{11.28}$$

which I will sometimes refer to as the "overpressure", although it can be positive or negative and hence may be an underpressure as well. We will also refer to the (dimensionless!)

relative density change scaled by the reference density, which is also called the "condensation" (hence symbol c):

$$c = \frac{\rho - \rho_0}{\rho_0} \tag{11.29}$$

Thus

$$p = Bc \tag{11.30}$$

or in words, the overpressure is the bulk modulus times the condensation.

Next, we need a fundamental concept from our discussion of fluid flow. Suppose we have a small block of fluid:

$$\Delta V = \Delta x \Delta y \Delta z$$

and imagine holding its *volume constant*. The only way the density of the fluid in the block can change if the volume is held constant is by fluid particles *flowing into or out of the block*. That is, let us think for a moment about what happens when we do *not* insist that the number of molecules in a small block of fluid is constant.

We are only concerned at the moment with a single dimension – the direction of the ***longitudinal*** motion of molecules in the sound wave – so we can consider only the motion of particles in the x direction and assume that at least, $\bar{v}_y = \bar{v}_z = 0$, there is no net motion in the y and z directions associated with the sound wave. Conservation of mass now requires that the change in the number of particles *in* the volume ΔV equals the number flowing in on the left face minus the number flowing out on the right face. Subject to these conditions this works out to become the ***continuity equation***:

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u}{\partial x} = 0 \tag{11.31}$$

where $u = \bar{v}_x$ is the local average x-directed velocity of the fluid. More generally (allowing for nonzero average motion in the other two directions) one obtains the three dimensional form:

$$\frac{\partial \rho}{\partial t} + \vec{\boldsymbol{\nabla}} \cdot \rho \vec{\boldsymbol{u}} = 0 \tag{11.32}$$

but we won't need this in this simplified treatment. This is basically the law of conservation of mass applied to the fluid.

Again we linearize this so that we express the changes in density relative to the reference density, using the condensation c as defined above to substitute $\rho = \rho_0 + \rho_0 c$ into both terms:

$$\frac{\partial}{\partial t}(\rho_0 + \rho_0 c) + \frac{\partial}{\partial x}(\rho_0 u + \rho_0 c u) = 0 \tag{11.33}$$

Note that ρ_0 is a *constant* as far as these derivatives are concerned. We can therefore cancel it out of the entire expression and rewrite it relating only c and u:

$$\frac{\partial c}{\partial t} + \frac{\partial}{\partial x}(u + cu) = 0 \tag{11.34}$$

c is necessarily "small" for the linearization above to work, so the product $cu \ll u$. We neglect this term and end up with:

$$\frac{\partial c}{\partial t} + \frac{\partial u}{\partial x} = 0 \tag{11.35}$$

or, the time derivative of the relative change in density is the negative space derivative of the (x) velocity of the fluid at each point. This makes sense.

Next, we need what amounts to "Newton's Second Law" for a chunk of fluid. Suppose again we imagine a chunk of fluid with sides $\Delta x, \Delta y, \Delta z$. The force on this chunk in the x-direction is basically:

$$F_x = P(x)\Delta y\Delta z - P(x+\Delta x)\Delta y\Delta z = -\frac{\partial P}{\partial x}\Delta x\Delta y\Delta z \tag{11.36}$$

Newton's Law becomes:

$$F_x = -\frac{\partial P}{\partial x}\Delta V = \frac{d(\Delta m u)}{dt} = \rho\Delta V\frac{du}{dt} \tag{11.37}$$

where as before, u is the x-component of the local velocity only. This is called the Euler force equation. We cancel the ΔV and rearrange this into:

$$\rho\frac{du}{dt} + \frac{\partial P}{\partial x} = 0 \tag{11.38}$$

Again, we have to linearize this around ρ_0 and use the relevant part of the *total derivative with respect to* t (this is multivariate calculus you may not have learned yet):

$$(\rho_0 + \rho_0 c)(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x}) + \frac{\partial(P_0 + p)}{\partial x} = 0 \tag{11.39}$$

In this equation (recall) c is very small. P_0 is the background pressure, presumed constant. The variation of u with x is small. Neglecting or cancelling these terms gives us:

$$\rho_0\frac{\partial u}{\partial t} + \frac{\partial p}{\partial x} = 0 \tag{11.40}$$

which we can view as the linearized Euler equation, a.k.a. Newton's second law for a coarse grained chunk of the fluid small enough to be treated as a differential, but large enough that densities and pressures make sense.

If we line up these two equations:

$$\begin{aligned} \frac{\partial c}{\partial t} + \frac{\partial u}{\partial x} &= 0 \\ \rho_0\frac{\partial u}{\partial t} + \frac{\partial p}{\partial x} &= 0 \end{aligned}$$

and take the partial time derivative of the first and the partial space derivative of the second, we get:

$$\begin{aligned} \frac{\partial^2 c}{\partial t^2} + \frac{\partial^2 u}{\partial t\partial x} &= 0 \\ \rho_0\frac{\partial^2 u}{\partial t\partial x} + \frac{\partial^2 p}{\partial x^2} &= 0 \end{aligned}$$

If we divide the second equation by ρ_0 and subtract, we get:

$$\frac{\partial^2 c}{\partial t^2} - \frac{1}{\rho}\frac{\partial^2 p}{\partial x^2} = 0 \tag{11.41}$$

Finally, we multiply both sides by B and use our linearized state equation $p = Bc$ to eliminate c (the condensation) in favor of p:

$$\frac{\partial^2 p}{\partial t^2} - \frac{B}{\rho}\frac{\partial^2 p}{\partial x^2} = \frac{\partial^2 p}{\partial t^2} - v^2\frac{\partial^2 p}{\partial x^2} = 0 \tag{11.42}$$

or (rearranging to a more or less standard form):

$$\frac{\partial^2 p}{\partial x^2} - \frac{1}{v^2}\frac{\partial^2 p}{\partial t^2} = 0 \tag{11.43}$$

with:

$$v = \sqrt{\frac{B}{\rho_0}}$$

Finally, since we can express sound as *either* a pressure wave *or* as a displacement wave, we have to find a relationship between s (displacement of some individual arbitrary (ith) molecule from a linearized "equilibrium position" s_i) and p. We do this using $u = \frac{\partial s}{\partial t}$ in the x-direction and:

$$\begin{aligned}
\rho_0\frac{\partial u}{\partial t} + \frac{\partial p}{\partial x} &= 0 \\
\rho_0\frac{\partial^2 s}{\partial t^2} + \frac{\partial p}{\partial x} &= 0 \\
\rho_0\frac{\partial^3 s}{t^2 x} + \frac{\partial^2 p}{\partial x^2} &= 0 \\
\rho_0\frac{\partial^3 s}{t^2 x} + \frac{1}{v^2}\frac{\partial^2 p}{\partial t^2} &= 0 \\
\frac{\partial^2(\rho_0\frac{\partial s}{\partial x} + \frac{1}{v^2}p)}{\partial t^2} &= 0
\end{aligned}$$

There are infinitely many solutions for s from this, but they boil down to the freedom to make s_i any constant position in the fluid and to add any constant speed u_0 to the entire fluid – the usual freedom for zero-net-force problems. The condition we are interested (which is sufficient, though not unique):

$$\rho_0\frac{\partial s}{\partial x} = -\frac{1}{v^2}p \tag{11.44}$$

Suppose

$$p(x,t) = p_0\cos(kx - \omega t) \tag{11.45}$$

(a simple harmonic travelling wave). Then:

$$\begin{aligned}
s(x,t) &= \frac{1}{v^2\rho_0}p_0\int\cos(kx-\omega t)dx \\
s(x,t) &= \frac{1}{kv^2\rho_0}p_0\sin(kx-\omega t) = \frac{1}{\omega v\rho_0}p_0\sin(kx-\omega t) \\
s(x,t) &= \frac{1}{Z\omega}p_0\sin(kx-\omega t) = s_0\sin(kx-\omega t)
\end{aligned}$$

(plus, as noted above, any $u_0 t + s_i$ term you like) where:

$$Z = \rho_0 v \tag{11.46}$$

and:

$$p_0 = Zs_0 \tag{11.47}$$

Obviously, displacement will satisfy an identical wave equation to pressure, but the pressure and displacement solutions are $\pi/2$ out of phase – where displacement is $s_0 \sin(kx - \omega t)$, pressure is $p_0 \cos(kx - \omega t) = p_0 \sin(kx - \omega t + \pi/2$.

There are a number of ways to write the relationship between p and s as derivatives. We already have:

$$p(x,t) = -v^2 \rho_0 \frac{\partial s}{\partial x} = -Zv \frac{\partial s}{\partial x} \tag{11.48}$$

We can also take a partial with respect to time of the $s(x,t)$ result we obtained to get:

$$\frac{\partial s}{\partial t} = \frac{1}{Z\omega} p_0 \frac{\partial \sin(kx - \omega t)}{\partial t} = -\frac{1}{Z} p_0 \cos(kx - \omega t) \tag{11.49}$$

or:

$$p(x,t) = Z \frac{\partial s}{\partial t} \tag{11.50}$$

Note Well: $s(x,t)$ is the ***relative*** displacement of any given molecule in the fluid from an equilibrium position s_i. s_i itself is a function of x. s_0 is the amplitude of the oscillation of molecules at the equilibrium position of x from this position. All of this makes no sense unless s_0 is small, its mean velocity is small, and hence x can accurately pick out a small block of fluid in the coarse grained linearized limit! The pressure wave is a little bit easier to understand than the displacement wave, as bulk displacement can occur even in a fluid with no waves – e.g. wind.

It is beyond the scope of this course to go any further into this, which is (as you can see) a derivation where we repeatedly had to throw away large terms. A similar derivation (essentially the derivation above repeated in all three dimensions) can be used to obtain a three-dimensional wave equation:

$$\nabla^2 p - \frac{1}{v^2} \frac{\partial^2 p}{\partial t^2} = 0$$

that has as solutions (among others) the symmetric spherical waves indicated below. Beyond this, the next stop in fluid dynamics is the full Navier-Stokes equation, which is so difficult that it cannot be generally solved (yet), where the results above are basically the results of linearizing it and neglecting certain things.

If You Skipped The Section Above, Start Reading Again Here!

Let's summarize the from the (omitted) section above:

- For a longitudinal plane wave, the pressure at a longitudinal point x at time t is given by solutions to the one dimensional wave equation:

$$\frac{\partial^2 p}{\partial x^2} - \frac{1}{v^2} \frac{\partial^2 p}{\partial t^2} = 0 \tag{11.51}$$

  with:

$$v = \sqrt{\frac{B}{\rho_0}}$$

- We can find the *relative* displacement of molecules whose equilibrium position is x at time t from the pressure or vice versa from the relations:

$$p(x,t) = -v^2\rho_0\frac{\partial s}{\partial x} = -Zv\frac{\partial s}{\partial x} \tag{11.52}$$

or

$$\frac{\partial s}{\partial t} = -\frac{1}{Z}p_0\cos(kx - \omega t) \tag{11.53}$$

or:

$$p(x,t) = Z\frac{\partial s}{\partial t} \tag{11.54}$$

where

$$Z = \rho_0 v = \sqrt{\rho_0 B} = \sqrt{\gamma\rho_0 P_0} \tag{11.55}$$

(the latter for an ideal gas in the adiabatic limit only). Pressure waves and displacement waves are thus ***out of phase by*** $\pi/2$ with the pressure wave ***leading*** the displacement wave by this amount.

- If we manipulate the last two equations, it is easy to show that displacement ***also*** satisfies the *same* wave equation with the *same* speed $v = \sqrt{B/\rho_0}$:

$$\frac{\partial^2 s}{\partial x^2} - \frac{1}{v^2}\frac{\partial^2 s}{\partial t^2} = 0 \tag{11.56}$$

11.3: Sound Wave Solutions

From the previous section (whether or not you read it) sound waves can be characterized one of two ways: as organized fluctuations in the *displacement* of the molecules of the fluid as they oscillate around an equilibrium position or as organized fluctuations in the *pressure* (or density/concentration, see above) of the fluid as molecules are crammed closer together or are diven farther apart than they are on average in the quiescent fluid. We can visualize this and understand it from figure 147 below.

One dimensional sound waves propagate in one direction (out of three) at any given point in space. This means that in the direction *perpendicular* to propagation, the wave is spread out to form a "wave front" – a planar region where all of the molecules are moving together.

In three dimensions, sounds waves satisfy a three-dimensional wave equation. In this case the wave front can be nearly arbitrary in shape initially (corresponding to the shape of a speaker surface producing the wave, for example). Thereafter it evolves according to the mathematics of the wave equation in three dimensions, which is similar to but a wee bit more complicated than the wave equation in one dimension. Or possibly even a *lot* more complicated... more complicated than is appropriate for this course. However, we will indicate what one of the *simplest* solutions in three dimensions is like, the one appropriate to a "spherical speaker", or a sound source that radiates sound in all directions uniformly.

We don't need to do any more work to write these solutions down. All the hard part was done in the last chapter. For one-dimensional plane wave solutions we can make wave

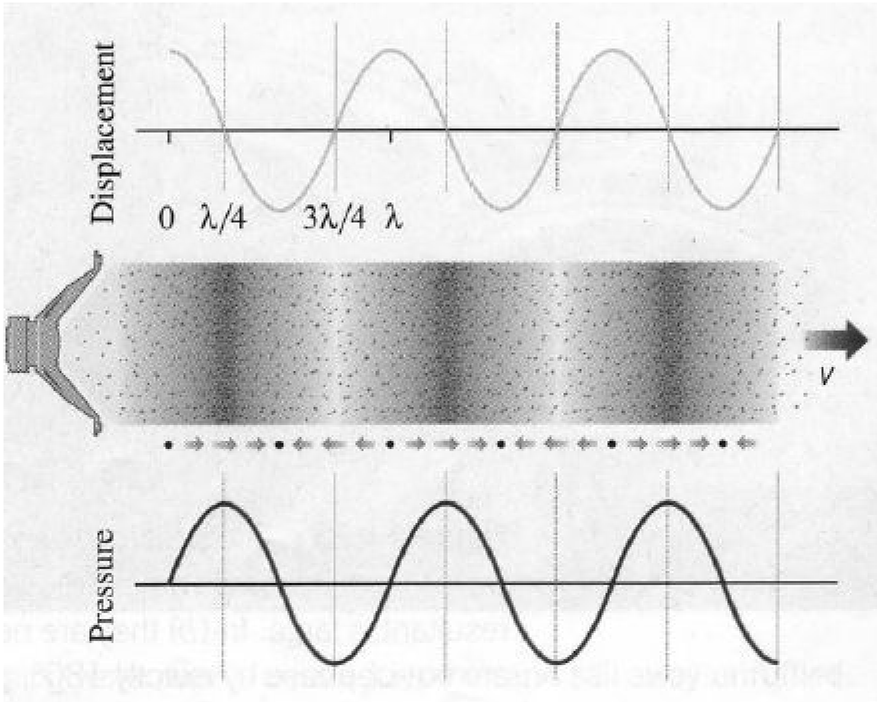


Figure 147: Schematic representation of a sound wave, illustrating how air molecules move ***longitudinally*** back and forth around their equilibrium position at the same time they generate ***over and under pressure waves*** around the background pressure of P_0 ~1 atm.

pulses or harmonic waves exactly like we did before, but now the "wave" is (over)pressure p or displacement s. For three I'll just give the appropriate solution for you to use below.

Let's consider ***only harmonic waves*** with a fixed frequency and wavelength, since wave pulses are not that common. Harmonic waves dominate our utilization of sound for communication and musical entertainment and awareness of our environment.

a) *Plane Wave* solutions. In these solutions, the entire wave moves in one direction (say the x direction) and the wave front is a 2-D plane perpendicular to the direction of propagation. These (displacement) solutions can be written as (e.g.):

$$s(x, t) = s_0 \sin(kx - \omega t) \tag{11.57}$$

where s_0 is the maximum displacement in the travelling wave (which moves in the x direction) and where all molecules in the entire *plane* at position x are displaced by the same amount.

Even spherical waves or waves produced by an arbitrary source behave like and are well described by plane waves near any given point in space. So are waves propagating down a constrained environment such as a tube that permits waves to only travel in "one direction". Most of the properties of sound of interest can be easily understood in terms of plane waves alone.

As we see above, we can *equally* well describe this specific sound plane wave in terms of its *pressure*, instead of *displacement*:

$$p(x, t) = P_{\rm atm} + p_0 \cos(kx - \omega t) \tag{11.58}$$

Now the wave amplitude p_0 describes the maximum ***overpressure*** of a wave oscillating around the equilibrium pressure of e.g. 1 atmosphere. Note well that pressure waves are $\pi/2$ ***ahead of*** the displacement waves as I indicated by using cosine instead of sine. The pressure waves are at maxima and minima where the displacement is zero and vice versa.

b) *Spherical Wave* solutions. Sound is often emitted from a source that is highly localized (such as a hammer hitting a nail, or a loudspeaker). If the sound is emitted equally in all directions from the source, a spherical wavefront is formed. Even if it is not emitted equally in all directions, sound from a localized source will generally form a spherically curved wavefront as it travels away from the point with constant speed. The displacement of a spherical wavefront *decreases* as one moves further away from the source because the *energy* in the wavefront is spread out on larger and larger surfaces. Its form is given by:

$$s(r,t) = s_0 \frac{R}{r} \sin(kr - \omega t) \tag{11.59}$$

where R is a "reference length". In practical terms, R might be the radius of a spherical speaker's surface, and s_0 might be the amplitude of that surface's oscillation to create the wave. Once the wave leaves the surface, its amplitude (as we will see) has to diminish like $1/r$ in order for energy to be conserved.

We can also express this as a pressure wave.

$$p(r,t) = p_0 \frac{R}{r} \cos(kr - \omega t) \tag{11.60}$$

where r is the radial distance away from the point-like source. Note again that the pressure wave is $\pi/2$ out of phase with (ahead of) the displacement wave.

Remember that this pressure wave is oscillating *around* one atmosphere of pressure, so that the actual total air pressure is:

$$p_{\rm tot}(r,t) = P_{\rm atm} + p_0 \frac{R}{r} \cos(kr - \omega t) \tag{11.61}$$

just as the displacement wave is for the displacement of *each* molecule in the air relative to its equilibrium position.

The key thing to remember about both of these spherical wave solutions is that the pressure wave or displacement wave *amplitudes* die down like $\boldsymbol{1/r}$, where r is the distance of the observer from the source. Plane waves, on the other hand, do not change their amplitude as they propagate (or rather, they do so only very gradually due to damping phenomena in the air that removes energy from the wave to slightly heat the air).

Next, let's discuss the *energy* carried by sound waves, as it is very important to anyone who wants to talk or listen or make music.

11.4: Sound Wave Intensity

The energy density of sound waves is given by:

$$\frac{dE}{dV} = \frac{1}{2}\rho\omega^2 s^2 \tag{11.62}$$

(again, very similar in form to the energy density of a wave on a string). However, this energy per unit *volume* is propagated in a single direction. It is therefore spread out so that it crosses an *area*, not a single point. Just how much energy an object receives therefore depends on how much *area* it intersects in the incoming sound wave, not just on the energy density of the sound wave itself.

For this reason the energy carried by sound waves is best measured by *intensity*: the energy per unit time per unit area perpendicular to the direction of wave propagation. Imagine a box with sides given by ΔA (perpendicular to the direction of the wave's propagation) and $v\Delta t$ (in the direction of the wave's propagation. All the energy in this box crosses through ΔA in time Δt. That is:

$$\Delta E = (\frac{1}{2}\rho\omega^2 s^2)\Delta A v \Delta t \tag{11.63}$$

or

$$I = \frac{\Delta E}{\Delta A \Delta t} = \frac{1}{2}\rho\omega^2 s^2 v \tag{11.64}$$

which looks very much like the *power* carried by a wave on a string. In the case of a plane wave propagating down a narrow tube, it is very similar – the power of the wave is the intensity times the tube's cross section.

However, consider a spherical wave. For a spherical wave, the intensity looks something like:

$$I(r,t) = \frac{1}{2}\rho\omega^2 \frac{s_0^2(4\pi)R^2\sin^2(kr-\omega t)}{4\pi r^2} v \tag{11.65}$$

which can be written as:

$$I(r,t) = \frac{\boldsymbol{P}_{\mathbf{tot}}}{4\pi r^2} \tag{11.66}$$

where $\boldsymbol{P}_{\mathbf{tot}}$ is the total power in the wave. Expressing it this way helps a lot with all of those constants (the R and 4π and ρ etc.). All we really need to know is the total power emitted by the source in watts, and we can predict how the sound intensity will drop off with distance!

This makes sense from the point of view of energy conservation and symmetry. If a source emits a power $\boldsymbol{P}_{\mathbf{tot}}$, that energy has to cross each successive spherical surface that surrounds the source. Those surfaces have an area equal to $A = 4\pi r^2$. Thus the surface at $r = 2r_0$ has 4 times the area of one at $r = r_0$, but the *same* total power has to go through both surfaces. Consequently, the intensity at the $r = 2r_0$ surface has to be $1/4$ the intensity at the $r = r_0$ surface.

It is important to remember this argument, simple as it is. Next week we will learn about Newton's law of gravitation. There we will learn that the gravitational field diminishes as $1/r^2$ with the distance from the source. Electrostatic field also diminishes as $1/r^2$. There seems to be a shared connection between symmetric propagation and spherical geometry; this will form the basis for *Gauss's Law* in electrostatics and much beautiful math, as all of these ideas are *connected* by the geometry of the sphere.

11.4.1: Sound Displacement and Intensity In Terms of Pressure

The pressure in a sound wave (as noted) oscillates around the mean/baseline ambient pressure of the air (or water, or whatever). The pressure *wave* in sound is thus the time varying pressure *difference* – the amplitude of the pressure oscillation around the *mean* of normal atmospheric pressure.

As always, we will be interested in writing the pressure *as* a harmonic wave (where we can) and hence will use the peak pressure difference as the amplitude of the wave. We will call this pressure the (peak) "overpressure":

$$P_0 = P_{\max} - P_a \tag{11.67}$$

where P_a is the baseline atmospheric pressure. It is easy enough to express the pressure wave in terms of the displacement wave (and vice versa). The amplitudes are related by:

$$P_0 = v_a \rho \omega s_0 = Z \omega s_0 \tag{11.68}$$

where $Z = v_a \rho$, the product of the speed of sound in air and the air density. The pressure and displacement waves are $\boldsymbol{\pi/2}$ ***out of phase!*** with the pressure wave leading the displacement wave:

$$P(t) = P_0 \cos(kx - \omega t) \tag{11.69}$$

(for a one-dimensional "plane wave", use kr and put it over $1/r$ to make a spherical wave as before). The displacement wave is (Z times) the time derivative of the pressure wave, note well.

The intensity of a sound wave can also be expressed in terms of ***pressure*** (rather than displacement). The expression for the intensity is then very simple (although not so simple to derive):

$$I = \frac{P_0^2}{2Z} = \frac{P_0^2}{2 v_a \rho} \tag{11.70}$$

11.4.2: Sound Pressure and Decibels

Source of Sound	P (Pa)	I	dB
Auditory threshold at 1 kHz	2×10^{-5}	10^{-12}	0
Light leaf rustling, calm breathing	6.32×10^{-5}	10^{-11}	10
Very calm room	3.56×10^{-4}	3.16×10^{-12}	25
A Whisper	2×10^{-3}	10^{-8}	40
Washing machine, dish washer	6.32×10^{-3}	10^{-7}	50
Normal conversation at 1 m	2×10^{-2}	10^{-6}	60
Normal (Ambient) Sound	6.32×10^{-2}	10^{-5}	70
"Loud" Passenger Car at 10 m	2×10^{-1}	10^{-4}	80
Hearing Damage *Possible*	0.356	3.16×10^{-4}	85
Traffic On Busy Roadway at 10 m	0.356	3.16×10^{-4}	85
Jack Hammer at 1 m	2	10^{-2}	100
Normal Stereo at Max Volume	2	10^{-2}	100
Jet Engine at 100 m	6.32-200	10^{-1} to 10^{3}	110-140
Hearing Damage *Likely*	20	1	120
Vuvuzela Horn	20	1	120
Rock Concert ("The Who" 1982) at 32 m	20	1	120
Threshold of *Pain*	63.2	10	130
Marching Band (100-200 members, in front)	63.2	10	130
"Very Loud" Car Stereo	112	31.6	135
Hearing Damage *Immediate, Certain*	200	10^{2}	140
Jet Engine at 30 m	632	10^{3}	150
Rock Concert ("The Who" 1982) at *speakers*	632	10^{3}	150
30-06 Rifle 1 m to side	6,320	10^{5}	170
Stun Grenades	20,000	10^{6}	180
Limit of Undistorted Sound (human eardrums rupture 50% of time)	101,325 (1 atm)	2.51×10^{7}	194.094

Table 6: Table of (approximate) P_0 and sound pressure levels in decibels relative to the threshold of human hearing at 10^{-12} watts/m^2. Note that ordinary sounds only extend to a peak overpressure of $P_0 = 1$ atmosphere, as one cannot oscillate symmetrically to underpressures pressures *less* than a vacuum.

The one real problem with the very simple description of sound intensity given above is one of scale. The human ear is ***routinely exposed to and sensitive to*** sounds that vary by ***twenty orders of magnitude*** – from sounds so faint that they barely can move our eardrums to sounds so very loud that they immediately *rupture them!* Even this isn't the full range of sounds out there – microphones and amplifiers allow us to detect even weaker sounds – as much as eight orders of magnitude weaker – and much stronger "sounds" called *shock waves*, produced by *supersonic* events such as explosions. The strongest supersonic "sounds" are some 32 orders of magnitude "louder" than the weakest sounds the human ear can detect, although even the weakest shock waves are almost strong enough to kill people.

It is very inconvenient to have to describe sound intensity in scientific notion across this wide a range – basically 40 orders of magnitude if not more (the limits of technology not being well defined at the low end of the scale). Also the human *mind* does not respond to sounds linearly. We do not psychologically perceive of sounds twice as intense as being twice as loud – in fact, a doubling of intensity is barely perceptible. Both of these motivate our using a different scale to represent sound intensities a relative *logarithmic* scale, called ***decibels***[206] .

The definition of a sound decibel is:

$$\beta = 10 \log_{10} \left(\frac{I}{I_0} \right) \text{ dB} \tag{11.71}$$

where "dB" is the abbreviation for decibels (tenths of "bels", the same unit without the factor of 10). Note that $\log_{10}$ is log ***base 10***, not the natural log, in this expression. Also in this expression,

$$I_0 = 10^{-12} \text{ watts/meter}^2 \tag{11.72}$$

is the reference intensity, called the ***threshold of hearing***. It is, by definition, the "faintest sound the human ear can hear" although naturally Your Mileage May Vary here – the faintest sound *my* relatively old and deaf ears is very likely much louder than the faintest sound a young child can hear, and there obviously some normal variation (a few dB) from person to person at any given age.

The smallest increments of sound that the human ear can differentiate as being "louder" are typically ***two decibel increases*** – more likely 3. Let's see what a *doubling the intensity* does to the decibel level of the sound.

$$\begin{aligned} \Delta\beta &= 10 \left(\log_{10} \left(\frac{2I}{I_0} \right) - \log_{10} \left(\frac{I}{I_0} \right) \right) \\ &= 10 \log_{10} \left(\frac{2I}{I_0} \frac{I_0}{I} \right) \\ &= 10 \log_{10} (2) \\ &= 3.01 dB \end{aligned} \tag{11.73}$$

In other words, ***doubling the sound intensity from any value corresponds to an increase in sound intensity level of 3 dB!*** This is such a simple rule that it is religiously learned as a rule of thumb by engineers, physicists and others who have reason to need to work with intensities of any sort on a log (decibel) scale. 3 dB per factor of two in intensity can carry you a long, long way! Note well that we used one of the magic properties of logarithms/exponentials in this algebra (in case you are confused):

$$\log(A) - \log(B) = \log \left(\frac{A}{B} \right) \tag{11.74}$$

You should remember this; it will be very useful next year.

Table 6 presents a number of fairly common sounds, sounds you are likely to have directly or indirectly heard (if only from far away). Each sound is cross-referenced with the

[206]Wikipedia: http://www.wikipedia.org/wiki/Decibel. Note well that the term "decibel" is not restricted to sound – it is rather a way of transforming any quantity that varies over a very large (many orders of magnitude) range into a log scale. Other logarithmic scales you are likely to encounter include, for example, the Richter scale (for earthquakes) and the F-scale for tornadoes.

approximate peak ***overpressure*** P_0 in the sound pressure wave (in pascals), the sound intensity (in watts/meter^2), and the sound intensity level relative to the threshold of hearing in decibels.

The overpressure is the pressure *over* the background of (a presumed) 1 atm in the sinusoidal pressure wave. The actual peak pressure would then be $P_{\max} = P_a + P_0$ while the minimum pressure would be $P_{\min} = P_a - P_0$. $P_{\min}$, however, ***cannot be negative*** as the lowest possible pressure is a vacuum, $P = 0$! At overpressures greater than 1 atm, then, it is no longer possible to have a pure sinusoidal sound wave. Waves in this category are a train of highly compressed peak amplitudes that drop off to (near) vacuum troughs in between, and are given their own special name: ***shock waves***. Shock waves are typically generated by very powerful phenomena and often travel faster than the speed of sound in a medium. Examples of shock waves are the sonic boom of a jet that has broken the sound barrier and the compression waves produced by sufficiently powerful explosives (close to the site of the explosion).

Shock waves as table 7 below clearly indicates, are capable of tearing the human body apart and accompany some of the most destructive phenomena in nature and human affairs – exploding volcanoes and colliding asteroids, conventional and nuclear bombs.

Source of Sound	P_0 (Pa)	dB
All sounds beyond this point are nonlinear ***shock waves***		
Human Death from Shock Wave Alone	200,000	200
1 Ton of TNT	632,000	210
Largest Conventional Bombs	2,000,000	220
1000 Tons of TNT	6,320,000	230
20 Kiloton Nuclear Bomb	63,200,000	250
57 Megaton (Largest) Nuclear Bomb	2,000,000,000	280
Krakatoa Volcanic Explosion (1883 C.E.)	63,200,000,000	310
Tambora Volcanic Explosion (1815 C.E.)	200,000,000,000	320

Table 7: Table of (approximate) P_0 and sound pressure levels in decibels relative to the threshold of human hearing at 10^{-12} watts/m^2 of ***shock waves***, events that produce distorted overpressures greater than one atmosphere. These "sounds" can be quite extreme! The Krakatoa explosion cracked a 1 foot thick concrete wall 300 miles away, was heard 3100 miles away, ejected 4 cubic miles of the earth, and created an audible pressure antinode on the opposite side of the earth. Tambora ejected 36 cubic miles of the earth and was equivalent to a *14 gigaton* nuclear explosion (14,000 1 megaton nuclear bombs)!

11.5: Doppler Shift

Everybody has heard the doppler shift in action. It is the rise (or fall) in frequency observed when a source/receiver pair approach (or recede) from one another. In this section we will derive expressions for the doppler shift for moving source and moving receiver.

11.5.1: Moving Source

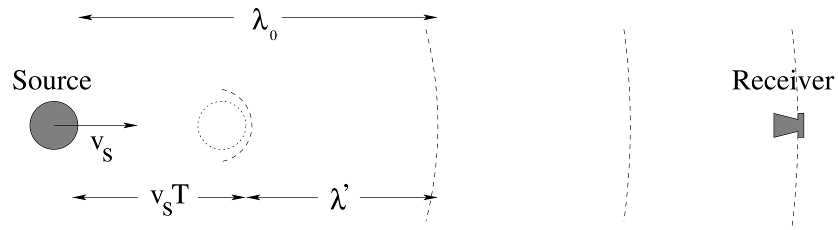


Figure 148: Waves from a source moving towards a stationary receiver have a foreshortened wavelength because the source moves ***in*** to the wave it produces. The *key* to getting the frequency shift is to recognize that the new (shifted) *wavelength* is $\lambda' = \lambda_0 - v_s T$ where T is the unshifted period of the source.

Suppose your receiver (ear) is stationary, while a source of harmonic sound waves at fixed frequency f_0 is approaching you. As the waves are emitted by the source they have a fixed wavelength $\lambda_0 = v_a/f_0 = v_a T$ and expand spherically from the point where the source was at the time the wavefront was emitted.

However, that point moves in the direction of the receiver. In the time between wavefronts (one period T) the source moves a distance $v_s T$. The shifted distance between successive wavefronts in the direction of motion (λ') can easily be determined from an examination of figure 148 above:

$$\lambda' = \lambda_0 - v_s T \tag{11.75}$$

We would really like the *frequency* of the doppler shifted sound. We can easily find this by using $\lambda' = v_a/f'$ and $\lambda = v_a/f$. We substitute and use $f = 1/T$:

$$\frac{v_a}{f'} = \frac{v_a - v_s}{f_0} \tag{11.76}$$

then we factor to get:

$$f' = \frac{f_0}{1 - \frac{v_s}{v_a}} \tag{11.77}$$

If the source is moving away from the receiver, everything is the same except now the wavelength is shifted to be bigger and the frequency smaller (as one would expect from changing the sign on the velocity):

$$f' = \frac{f_0}{1 + \frac{v_s}{v_a}} \tag{11.78}$$

11.5.2: Moving Receiver

Now imagine that the source of waves at frequency f_0 is stationary but the receiver is moving towards the source. The source is thus surrounded by spherical wavefronts a distance $\lambda_0 = v_a T$ apart. At $t = 0$ the receiver crosses one of them. At a time T' later, it has moved a distance $d = v_r T'$ in the direction of the source, and the wave from the source has moved a distance $D = v_a T'$ toward the receiver, and the receiver encounters the next wave front.

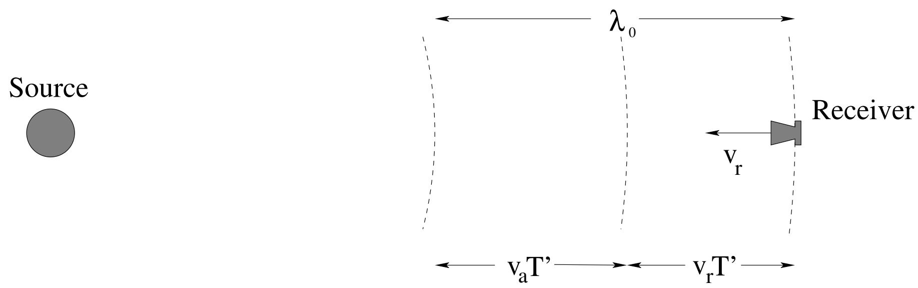


Figure 149: Waves from a stationary source are picked up by a moving receiver. They have a shortened ***period*** because the receiver doesn't wait for the next wavefront to reach it, at receives it when it has only moved part of a wavelength forward. The *key* to getting the frequency shift is to recognize that the sum of the distance travelled by the wave and the receiver in a new period T' must equal the original unshifted wavelength.

This can be visualized in figure 149 above. From it we can easily get:

$$\begin{aligned}
\lambda_0 &= d + D && (11.79)\\
&= v_r T' + v_a T' && (11.80)\\
&= (v_r + v_a)T' && (11.81)\\
v_a T &= (v_r + v_a)T' && (11.82)
\end{aligned}$$

We use $f_0 = 1/T$, $f' = 1/T'$ (where T' is the apparent time between wavefronts to the receiver) and rearrange this into:

$$f' = f_0(1 + \frac{v_r}{v_a}) \tag{11.83}$$

Again, if the receiver is moving away from the source, everything is the same but the sign of v_r, so one gets:

$$f' = f_0(1 - \frac{v_r}{v_a}) \tag{11.84}$$

11.5.3: Moving Source and Moving Receiver

This result is just the product of the two above – moving source causes one shift and moving receiver causes another to get:

$$f' = f_0 \frac{1 \mp \frac{v_r}{v_a}}{1 \pm \frac{v_s}{v_a}} \tag{11.85}$$

where in both cases *relative* approach shifts the frequency up and *relative* recession shifts the frequency down.

I do *not* recommend memorizing these equations – I don't have them memorized myself. It is *very* easy to confuse the forms for source and receiver, and the derivations take a few seconds and are likely worth points in and of themselves. If you're going to memorize anything, memorize the *derivation* (a process I call "learning", as opposed to "memorizing"). In fact, this is excellent advice for 90% of the material you learn in this course!

11.6: Standing Waves in Pipes

Everybody has created a stationary resonant harmonic sound wave by whistling or blowing over a beer bottle or by swinging a garden hose or by playing the organ. In this section we will see how to compute the harmonics of a given (simple) pipe geometry for an imaginary organ pipe that is open or closed at one or both ends.

The way we proceed is straightforward. Air cannot penetrate a closed pipe end. The air molecules at the very end are therefore "fixed" – they cannot displace into the closed end. The *closed* end of the pipe is thus a *displacement node*. In order *not* to displace air the closed pipe end has to exert a force on the molecules by means of pressure, so that the closed end is a pressure antinode.

At an open pipe end the argument is inverted. The pipe is open to the air (at fixed background/equilibrium pressure) so that there must be a pressure node at the open end. Pressure and displacement are $\pi/2$ out of phase, so that the *open* end is also a *displacement antinode*.

Actually, the air pressure in the standing wave doesn't instantly equalize with the background pressure at an open end – it sort of "bulges" out of the pipe a bit. The displacement antinode is therefore just *outside* the pipe end, not *at* the pipe end. You may still draw a displacement antinode (or pressure node) as if they occur at the open pipe end; just remember that the distance from the open end to the first displacement node is *not* a very accurate measure of a quarter wavelength and that open organ pipes are a bit "longer" than they appear from the point of view of computing their resonant harmonics.

Once we understand the boundary conditions at the ends of the pipes, it is pretty easy to write down expressions for the standing waves and to deduce their harmonic frequencies.

11.6.1: Pipe Closed at Both Ends

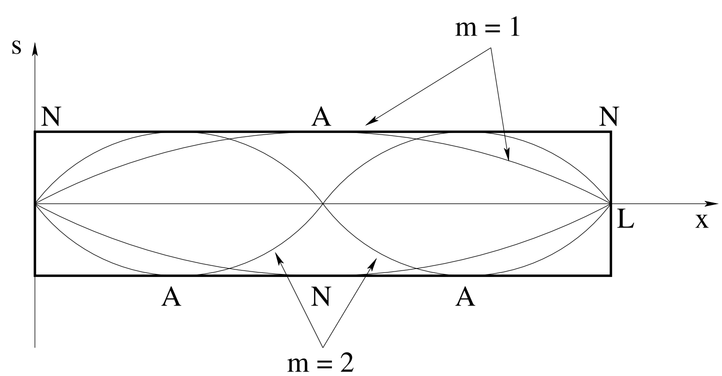


Figure 150: The pipe closed at both ends is ***just like a string fixed at both ends***, as long as one considers the ***displacement*** wave.

As noted above, we expect a ***displacement node*** (and hence ***pressure antinode*** at the closed end of a pipe, as air molecules cannot move through a solid surface. For a pipe closed at both ends, then, there are displacement nodes at both ends, as pictured above in figure 150. This is just like a string fixed at both ends, and the solutions thus have the same functional form:

$$s(x, t) = s_0 \sin(k_m x) \cos(\omega_m t) \tag{11.86}$$

This has a node at $x = 0$ for all k. To get a node at the other end, we require (as we did for the string):

$$\sin(k_m L) = 0 \tag{11.87}$$

or

$$k_m L = m\pi \tag{11.88}$$

for $m = 1, 2, 3....$ This converts to:

$$\lambda_m = \frac{2L}{m} \tag{11.89}$$

and

$$f_m = \frac{v_a}{\lambda_m} = \frac{v_a m}{2L} \tag{11.90}$$

The $m = 1$ solution (first harmonic) is called the *principle harmonic* as it was before. The actual tone of a flute pipe with two closed ends will be a superposition of harmonics, usually dominated by the principle harmonic.

11.6.2: Pipe Closed at One End

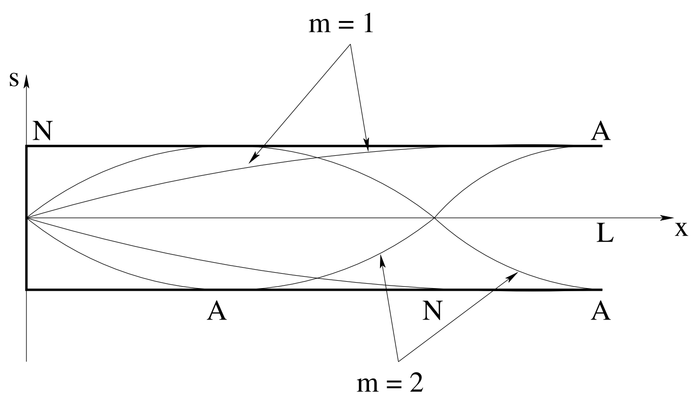


Figure 151: The pipe closed at both ends is ***just like a string fixed at one end***, as long as one considers the ***displacement*** wave.

In the case of a pipe open at only *one* end, there is a ***displacement node*** at the ***closed end***, and a ***displacement antinode*** at the ***open end***. If one considers the pressure wave, the positions of nodes and antinodes are ***reversed***. This is just like a string fixed at one end and free at the other. Let's arbitrarily make $x = 0$ the closed end. Then:

$$s(x, t) = s_0 \sin(k_m x) \cos(\omega_m t) \tag{11.91}$$

has a node at $x = 0$ for all k. To get an antinode at the other end, we require:

$$\sin(k_m L) = \pm 1 \tag{11.92}$$

or

$$k_m L = \frac{2m - 2}{2}\pi \tag{11.93}$$

for $m = 1, 2, 3...$ (odd half-integral multiples of π. As before, you will see different conventions used to *name* the harmonics, with some books asserting that only odd harmonics are supported, but I prefer to make the harmonic index do exactly the same thing for both pipes so it counts the actual number of harmonics that *are* supported by the pipe. This is much more consistent with what one will do next semester considering e.g. interference, where one often encounters similar series for a phase angle in terms of odd-half integer multiples of π, and makes the second harmonic the lowest frequency actually present in the pipe in *all three cases* of pipes closed at neither, one or both ends.

This converts to:

$$\lambda_m = \frac{4L}{2m-1} \tag{11.94}$$

and

$$f_m = \frac{v_a}{\lambda_m} = \frac{v_a(2m-1)}{4L} \tag{11.95}$$

11.6.3: Pipe Open at Both Ends

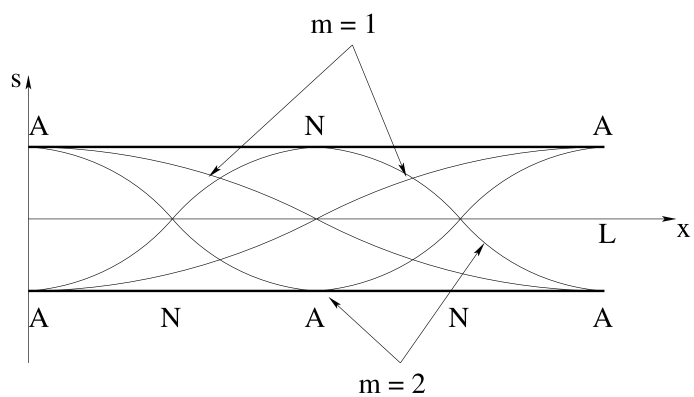


Figure 152: A pipe open at both ends is the exact *opposite* of a pipe (or string) closed (fixed) at both ends: It has *displacement antinodes* at the ends. Note well the principle harmonic with a single node in the center. The *resonant frequency* series for the pipe is the same, however, as for a pipe closed at both ends!

This is a ***panpipe***, one of the most primitive (and beautiful) of musical instruments. A panpipe is nothing more than a tube, such as a piece of hollow bamboo, open at both ends. The modes of this pipe are driven at resonance by blowing gently across one end, where the random fluctuations in the airstream are amplified only for the resonant harmonics.

To understand the frequencies of those harmonics, we note that there are ***displacement antinodes*** at ***both ends***. This is just like a string free at both ends. The displacement solution must thus be a *cosine* in order to have a displacement antinode at $x = 0$:

$$s(x,t) = s_0 \cos(k_m x)\cos(\omega_m t) \tag{11.96}$$

and

$$\cos(k_m L) = \pm 1 \tag{11.97}$$

We can then write $k_m L$ as a series of suitable multiples of π and proceed as before to find the wavelengths and frequencies as a function of the mode index $m = 1, 2, 3....$ This is left as a (simple) exercise for you.

Alternatively, we could *also* note that there are *pressure* nodes at both ends, which makes them like a string fixed at both ends again as far as the *pressure* wave is concerned. This gives us *exactly* the same result (for frequencies and wavelengths) as the pipe closed at both ends above, although the pipe *open* at both ends is probably going to be a bit louder and easier to drive at resonance (how can you "blow" on a closed pipe to get the waves in there in the first place? How can the sound get out?

Either way one will get the same frequencies but the *picture* of the displacement waves is different from the *picture* of the pressure waves – be sure to draw displacement antinodes at the open ends if you are asked to draw a displacement wave, or vice versa for a pressure wave!

You might try drawing the first 2-3 harmonics on a suitable picture like the first two given above, with displacement antinodes at both ends. What does the principle harmonic look like? Show that the supported frequencies and wavelengths match those of the string fixed at both ends, or pipe closed at both ends.

11.7: Beats

If you have ever played around with a guitar, you've probably noticed that if two strings are fingered to be the "same note" but are really slightly out of tune and are struck together, the resulting sound "beats" – it modulates up and down in intensity at a low frequency often in the ballpark of a few cycles per second.

Beats occur because of the superposition principle. We can add any two (or more) solutions to the wave equation and still get a solution to the wave equation, even if the solutions have different frequencies. Recall the identity:

$$\sin(A) + \sin(B) = 2\sin(\frac{A+B}{2})\cos(\frac{A-B}{2}) \tag{11.98}$$

If one adds two waves with different wave numbers/frequencies and uses this rule, one gets

$$\begin{aligned} s(x,t) &= s_0\sin(k_0x - \omega_0 t) + s_0\sin(k_1x - \omega_1 t) && (11.99)\\ &= 2s_0\sin(\frac{k_0+k_1}{2}x - \frac{\omega_0+\omega_1}{2}t)\cos(\frac{k_0-k_1}{2}x - \frac{\omega_0-\omega_1}{2}t) && (11.100)\end{aligned}$$

This describes a wave that has twice the maximum amplitude, the *average* frequency (the first term), and a second term that (at any point x) oscillates like $\cos(\frac{\Delta\omega t}{2})$.

The "frequency" of this second modulating term is $\frac{f_0-f_1}{2}$, but the ear *cannot hear* the inversion of phase that occurs when it is negative and the difference is small. It just hears maximum amplitude in the rapidly oscillating *average* frequency part, which goes to *zero* when the slowing varying cosine does, twice per cycle. The ear then *hears* two beats per cycle, making the "beat frequency":

$$f_{\rm beat} = \Delta f = |f_0 - f_1| \tag{11.101}$$

11.8: Interference and Sound Waves

We will not cover interference and diffraction of harmonic sound waves in this course. Beats are a common experience in sound as is the doppler shift, but sound wave interference is not so common an experience (although it can definitely and annoyingly occur if you hook up speakers in your stereo out of phase). Interference *will* be treated next semester in the context of coherent light waves. *Just* to give you a head start on that, we'll indicate the basic ideas underlying interference here.

Suppose you have two sources that are at the *same* frequency and have the *same* amplitude and phase but are at different locations. One source might be a distance x away from you and the other a distance $x + \Delta x$ away from you. The waves from these two sources add like:

$$\begin{aligned} s(x,t) &= s_0 \sin(kx - \omega t) + s_0 \sin(k(x+\Delta x) - \omega t) &(11.102)\\ &= 2 s_0 \sin(k(x + \frac{\Delta x}{2} - \omega t) \cos(k\frac{\Delta x}{2}) &(11.103)\end{aligned}$$

The sine part describes a wave with twice the amplitude, the same frequency, but shifted slightly in phase by $k\Delta x/2$. The cosine part is *time independent* and *modulates* the first part. For some values of Δx it can *vanish*. For others it can have magnitude one.

The intensity of the wave is what our ears hear – they are insensitive to the phase (although certain echolocating species such as bats may be sensitive to phase information as well as frequency). The average intensity is proportional to the wave amplitude *squared*:

$$I_0 = \frac{1}{2}\rho\omega^2 s_0^2 v \tag{11.104}$$

With two sources (and a maximum amplitude of two) we get:

$$\begin{aligned} I &= \frac{1}{2}\rho\omega^2(2^2 s_0^2 \cos^2(k\frac{\Delta x}{2})v &(11.105)\\ &= 4I_0\cos^2(k\frac{\Delta x}{2}) &(11.106)\end{aligned}$$

There are two cases of particular interest in this expression. When

$$\cos^2(k\frac{\Delta x}{2}) = 1 \tag{11.107}$$

one has *four times* the intensity of one source at peak. This occurs when:

$$k\frac{\Delta x}{2} = n\pi \tag{11.108}$$

(for $n = 0, 1, 2 ...$) or

$$\Delta x = n\lambda \tag{11.109}$$

If the *path difference* contains an *integral number of wavelengths* the waves from the two sources arrive *in phase*, add, and produce sound that has twice the amplitude and four times the intensity. This is called *complete constructive interference*.

On the other hand, when

$$\cos^2(k\frac{\Delta x}{2}) = 0 \tag{11.110}$$

the sound intensity *vanishes*. This is called *destructive* interference. This occurs when

$$k\frac{\Delta x}{2} = \frac{2n+1}{2}\pi \tag{11.111}$$

(for $n = 0, 1, 2...$) or

$$\Delta x = \frac{2n+1}{2}\lambda \tag{11.112}$$

If the path difference contains a *half integral* number of wavelengths, the waves from two sources arrive exactly out of phase, and *cancel*. The sound intensity vanishes.

You can see why this would make hooking your speakers up out of phase a bad idea. If you hook them up out of phase the waves *start* with a phase difference of π – one speaker is pushing out while the other is pulling in. If you sit equidistant from the two speakers and then harmonic waves with the same frequency from a single source coming from the two speakers *cancel* as they reach you (usually not perfectly) and the music sounds very odd indeed, because other parts of the music are not being played equally from the two speakers and don't cancel.

You can also see that there are many other situations where constructive or destructive interference can occur, both for sound waves and for other waves including water waves, light waves, even waves on strings. Our "standing wave solution" can be rederived as the superposition of a left- and right-travelling harmonic wave, for example. You can have interference from more than one source, it doesn't have to be just two.

This leads to some really excellent engineering. Ultrasonic probe arrays, radiotelescope arrays, sonar arrays, diffraction gratings, holograms, are all examples of interference being put to work. So it is worth it to learn the general idea as early as possible, even if it isn't assigned.

11.9: The Ear

Figure 153 shows a cross-section of the human ear, our basic transduction device for sound. This is not a biology course, so we will not dwell upon *all* of the structure visible in this picture, but rather will concentrate on the parts relevant to the physics.

Let's start with the ***outer ear***. This structure collects sound waves from a larger area than the ear canal per se and reflects them down to the ear canal. You can easily experiment with the kind of amplification that results from this by cupping your hands and holding them immediately behind your ears. You should be able to hear both a qualitative change in the frequencies you are hearing and an effective amplification of the sounds from in *front* of you at the expense of sounds originating *behind* you. Many animals have larger outer

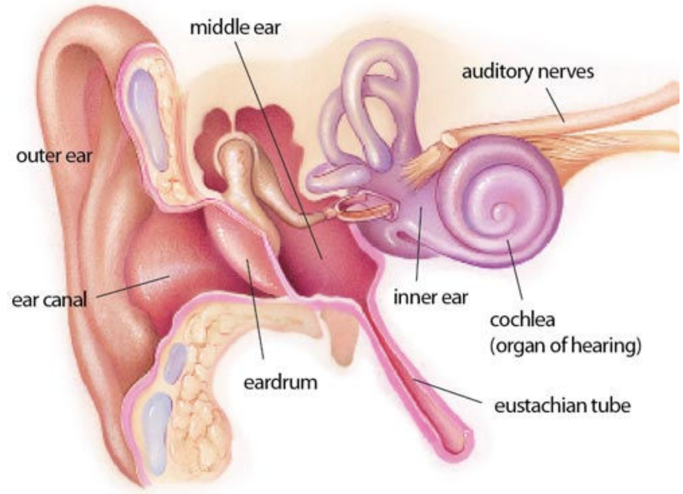


Figure 153: The anatomy of the human ear.

ears oriented primarily towards the front, and have muscles that permit them to further alter the direction of most favorable sound collection without turning their heads. Human ears are more nearly omnidirection.

The ***auditory canal*** (ear canal in the figure above) acts like a resonant cavity to effectively amplify frequencies in the 2.5 kHz range and tune energy deliver to the tympanic membrane or eardrum. This membrane is a strong, resilient, tightly stretched structure that can vibrate in response to driving sound waves. It is connected to a collection of small bones (the ossicles) that conduct sound from the eardrum to the inner ear and that constitute the ***middle ear*** in the figure above. The common name of the ossicles are: hammer, anvil and stirrup, the latter so named because its shape strongly resembles that of the stirrup on a horse saddle. The anvil effectively amplifies oscillations by use of the principle of leverage, as a fulcrum attachment causes the stirrup end to vibrate through a much larger amplitude than the hammer end. The stirrup is directly connected to the ***oval window***, the gateway into the inner ear.

The middle ear is connected to the ***eustachian tube*** to your throat, permitting pressure inside your middle ear to equalize with ambient air pressure outside. If you pinch your nose, close your mouth, and try to breath out hard, you can actually blow air out through your ears although this is unpleasant and can be dangerous. This is one way your ears equilbrate by “popping” when you ride a car up a hillside or fly in an airplane. If/when this does not happen, pressure differences across the tympanic membrane reduce its response to ambient sounds reducing auditory acuity.

Sound, amplifed by focal concentration in the outer ear, resonance in the auditory canal, and mechanical leverage in the ossicles, enters the ***cochlea***, a shell-shaped spiral that is the primary organ of hearing that transduces sound energy into impulses in our nervous system through the oval window. The cochlea contains ***hair cells*** of smoothly varying length lining the narrowing spiral, each of which is ***resonant*** to a particular auditory frequency. The arrangement of the cells in a cross-section of the cochlea is shown in figure 154.

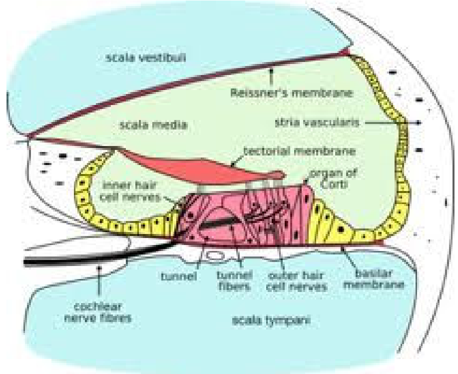


Figure 154: A cross-section of the spiral structure of the cochlea.

The nerves stretching from these cells are collected into the ***auditory nerve bundle*** and from thence carries the impulses they give off when they receive sound at the right frequency in to the ***auditory cortex*** (not shown) where it becomes, eventually and through a process still not fully understood, our perception of sound. Our brains take this frequency resolved information – the biomechanical equivalent of a fourier transform of the sound signal, in a way – and synthesize it back into a detailed perception of sound and music within the general frequency range of 10 Hz to 20,000 Hz.

As you can see, there are many individual parts that can fail in the human auditory system. Individuals can lose or suffer damage to their outer ears through accident or disease. The ear canal can become clogged with cerumen, or ***earwax***, a waxy fluid that normally cleans and lubricates the ear canal and eardrum but that can build up and dry out to both load the tympanic membrane so it becomes less responsive and physically occlude part of the canal so less sound energy can get through. The eardrum itself is vulnerable to sudden changes in sound pressure or physical contact that can puncture it. The middle ear, as a closed, warm, damp cavity connected to the throat, is an ideal breeding ground for certain bacteria that can cause infections and swelling that both interfere with or damage hearing and that can be quite painful. The ossicles are susceptible to physical trauma and infectious damage.

Finally, the hair cells of the cochlea itself, which are safely responsive over at least twelve to fourteen orders of magnitude of transient sound intensity (and safely responsive over eight or nine orders of magnitude of sustained sound intensity) are highly vulnerable to *both* sudden transient sounds of still higher intensity (e.g. sound levels in the vicinity of 120 to 140 decibels and higher *and* to sustained excitation at sound levels from roughly 90 decibels and higher. Both disease and medical conditions such as diabetes (that produces a progressive neuropathy) can further contribute to gradual or acute hearing loss at the neurological level.

When hair cells die, they do not regenerate and hearing loss of this sort is thus cumulative over a lifetime. It is therefore a really good idea to wear ear protectors if, for example, you play an instrument in a marching band or a rock and roll band where your hearing is routinely exposed to 100 dB and up sounds. It is also a good reason not to play music too loudly when you are young, however pleasurable it might seem. One is, after all, very probably trading listening to very loud music at age seventeen against listening to music *at all* at age seventy. Hearing aids do not really fix the problem, although they can help restore enough function for somebody to get by.

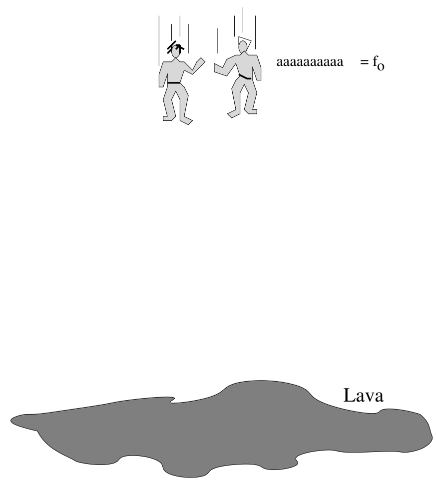
However, it is quite possible that over the next few decades the bright and motivated physics students of today will help create the bioelectronic and/or stem cell replacements of key organs and nervous tissue that will relegate age-related deafness to the past. I would certainly wish, as I sit here typing this with eyes and ears that are gradually failing as I age, that whether or not it come in time for me, it comes in time to help you.

Homework for Week 11

Problem 1.

Physics Concepts: Make this week's physics concepts summary as you work all of the problems in this week's assignment. Be sure to cross-reference each concept in the summary to the problem(s) they were key to, and include concepts from previous weeks as necessary. Do the work carefully enough that you can (after it has been handed in and graded) punch it and add it to a three ring binder for review and study come finals!

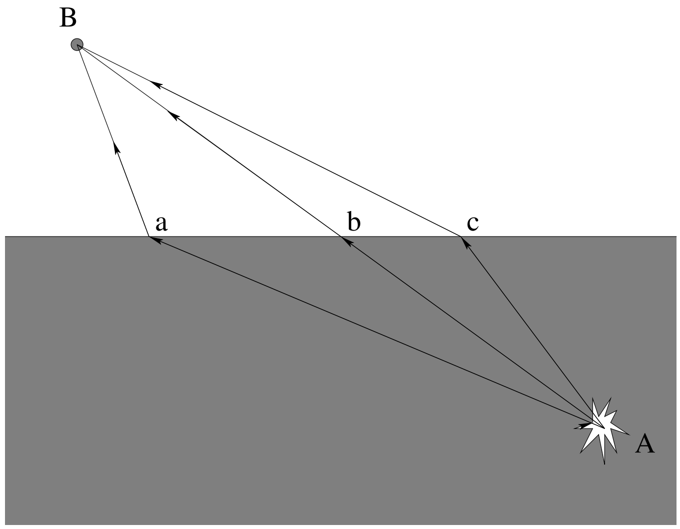
Problem 2.



Bill and Ted are falling into hell at a ***constant speed*** (terminal velocity), and are screaming at the frequency f_0. As they fall, they hear their own voices reflecting back to them from the puddle of molten rock that lies below at a frequency of $2f_0$.

How fast are they falling relative to the speed of sound in warm, dry hellish air?

Problem 3.



Sound waves travel faster in water than they do in air. Light waves travel faster in air than they do in water. Based on this, which of the three paths pictured above are more likely to *minimize* the time required for the

a) Sound:

b) Light:

produced by an underwater explosion to travel from the explosion at A to the pickup at B? Why (explain your answers)?

Problem 4.

Fred is standing on the ground and Jane is blowing past him at a closest distance of approach of a few meters at twice the speed of sound in air. Both Fred and Jane are holding a loudspeaker that has been emitting sound at the *frequency* f_0 for some time.

a) Who hears the sound produced by the *other* person's speaker as *single frequency sound* when they are approaching one another and what frequency do they hear?

b) What does the *other* person hear (when they hear anything at all)?

c) What frequenc(ies) do each of them hear *after* Jane has passed and is receding into the distance?

Problem 5.

Discuss and answer the following questions:

a) Sunlight reaches the surface of the earth with *roughly* 1000 Watts/meter^2 of intensity. What is the "sound intensity level" of a ***sound*** wave that carries as much ***energy per square meter***, in ***decibels***?

b) In table 6, what kind of sound sources produce this sort of intensity? Bear in mind that the Sun is *150 million kilometers away* where sound sources capable of reaching the same intensity are typically only a few meters away. The the Sun produces a *lot* of (electromagnetic) energy compared to terrestrial sources of (sound) energy.

c) The human body produces energy at the rate of roughly 100 Watts. *Estimate* the fraction of this energy that goes into my lecture when I am speaking in a loud voice in front of the class (loud enough to be heard as loudly as normal conversation ten meters away).

d) Again using table 6, how far away from a jack hammer do you need to stand in order for the sound to (marginally) no longer be dangerous to your hearing?

Problem 6.

String one has mass per unit length μ and is at tension T and has a travelling harmonic wave on it:

$$y_1(x,t) = A\sin(kx - \omega t)$$

String one is also very long compared to a wavelength: $L \gg \lambda$.

Identical string two has the superposition of two harmonic travelling waves on it:

$$y_2(x,t) = A\sin(kx - \omega t) + 3A\sin(kx + \omega t)$$

If the average energy density (total mechanical energy per unit length) of the first string is E_1, what is the average energy density of the second string E_2 in *terms* of E_1?

Problem 7.

Two identical strings of length L have mass μ and are fixed at both ends. One string has tension T. The other has tension $1.21T$. When plucked, the first string produces a tone at frequency f_1, the second produces a tone at frequency f_2.

a) What is the *beat* frequency produced if the two strings are is plucked at the same time, in terms of f_1?

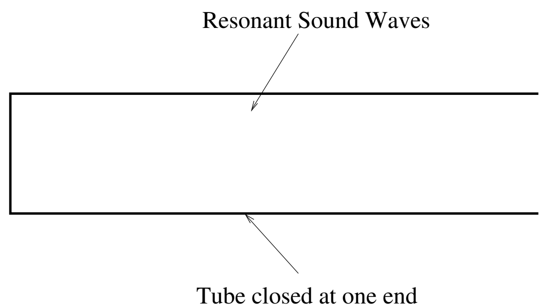
b) Are the beats likely to be audible if f_1 is 500 Hz? How about 50 Hz? Why or why not?

Problem 8.

You measure the sound intensity level of a single frequency sound wave produced by a loudspeaker with a calibrated microphone to be 80 dB. At that intensity, the *peak* pressure in the sound wave at the microphone is P_0. The loudspeaker's amplitude is turned up until the intensity level is 100 dB. What is the peak pressure of the sound wave now (in terms of P_0)?

Note that you could look this up in the table, but don't. The point is for you to know how the peak pressure *scales* with the intensity, as well as how the intensity varies with the sound intensity level in decibels.

Problem 9.



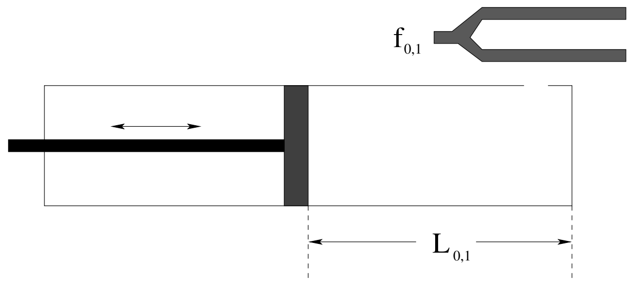
An organ pipe is made from a brass tube closed at one end as shown. The pipe has length L and the speed of sound is v_s. When played, it produces a sound that is a mixture of the first, third and seventh harmonic:

a) What are the frequencies of these harmonics?

b) Qualitatively sketch the wave amplitudes for the first and the third harmonic modes (only) in on the figure, indicating the nodes and antinodes. Be sure to indicate whether the nodes or antinodes drawn are for **pressure/density** waves or **displacement** waves!

c) Evaluate your answers numerically when $L = 3.4$ meters long, and $v_s = 340$ meters/second (as usual).

Problem 10.



You crash land on a strange planet and all your apparatus for determining if the planet's atmosphere is like Earth's is wrecked. In desperation you decide to measure the speed of sound in the atmosphere before taking off your helmet. You do have a barometer handy and can see that the air pressure outside is approximately one atmosphere and the temperature seems to be about 300 °K, so if the speed of sound is the same as on Earth the air *might* be breathable.

You jury rig a piston and cylinder arrangement like the one shown above (where the cylinder is ***closed at both ends*** but has a small hole in the side to let sound energy in to resonate) and take out your two handy tuning forks, one at $f_0 = 3400$ Hz and one at $f_1 = 6800$ Hz.

a) Using the 3400 Hz fork as shown, what do you *expect* (or rather, hope) to hear as you move the piston in and out (varying L_0). In particular, what are the shortest few values of L_0 for which you expect to hear a maximum resonant intensity from the tube if the speed of sound in the unknown atmosphere is indeed the same as in air (which you will cleverly note I'm *not* telling you as you are supposed to know this number)?

b) Using the 6800 Hz fork you hear your first maximum (for the smallest value of L_1) at $L_1 = 5$ cm. Should you sigh with relief and rip off your helmet?

c) What is the *next* value for L_1 for which you should hear a maximum (given the measurement in b) and what should the difference between the two equal in terms of the wavelength of the 6800 Hz wave in the unknown gas? Draw the *displacement* wave for this case only schematically in on the diagram above (assuming that the L shown is this second-smallest value of L_1 for the f_1 tuning fork), and indicate where the nodes and antinodes are.

Optional Problems

Study for the final exam! This is the last week of class, and this wraps up both the chapter and the texbook. Students looking for more problems to work on are directed to the online review guide for introductory physics 1 and the online math review, the latter as needed.

Week 12: Gravity

Gravity Summary

- Early western (Greek) cosmology was both ***geocentric*** – simple earth-centered model with fixed stars "lamps" or "windows" on big solid bowl, moon and stars and planets orbiting the (usually flat) Earth "somehow" in between. The simple geocentric models failed to explain ***retrograde motion*** of the planets, where for a time they seem to go backwards against the fixed stars in their general orbits. There were also early *heliocentric* – sun centered – models, in particular one by Aristarchus of Samos (270 B.C.E.), who used ***parallax*** to measure the size of the earth and the sizes of and distances to the Sun and Moon.

- Ptolemy[207] (140 C.E.) "explained" retrograde motion with a ***geometric geocentric model*** involving complex ***epicycles***. Kudos to Ptolemy for inventing geometric modelling in physics! The model was a genuine scientific hypothesis, in principle ***falsifiable***, and a good starting place for further research.

  Sadly, a few hundred years later the state religion of the western world's largest empire embraced this geocentric model as being ***consistent with The Book of Genesis*** in its theistic scriptural mythology (and with many other passages in the old and new testaments) and for over a thousand years alternative explanations were considered heretical and could only be made at substantial personal risk throughout the Holy Roman Empire.

- Copernicus[208] (1543 C.E.) (re)invented a ***heliocentric*** – sun-centered model, explained retrograde motion with ***simpler*** plain circular geometry, regular orbits. The work of Copernicus, *De Revolutionibus Orbium Coelestium*[209] (On the Revolutions of the Heavenly Spheres) was forthwith banned by the Catholic Church as heretical at the same time that Galileo was both persecuted and prosecuted.

- Wealthy Tycho Brahe accumulated data and his paid assistant, Johannes Kepler, fit that data to specific orbits and deduced ***Kepler's Laws***. All Brahe got for his efforts was a lousy moon crater named after *him*[210] .

- **Kepler's Laws:**

[207]Wikipedia: http://www.wikipedia.org/wiki/Ptolemy.
[208]Wikipedia: http://www.wikipedia.org/wiki/Copernicus.
[209]Wikipedia: http://www.wikipedia.org/wiki/De_revolutionibus_orbium_coelestium.
[210]Wikipedia: http://www.wikipedia.org/wiki/Tycho_(crater).

a) All planets move in elliptical orbits with the sun at one focus.

b) A line joining any planet to the sun sweeps out equal areas in equal times (dA/dt = constant).

c) The square of the period of any planet is proportional to the cube of the planet's mean distance from the sun ($T^2 = CR^3$). Note that the semimajor or semiminor axis of the ellipse will serve as well as the mean, with different contants of proportionality.

- Galileo[211] (1564-1642 C.E.) is known as the Copernican heliocentric model's most famous early defender, not so much because of the quality of his science as for his infamous ***prosecution by the Catholic church***. In truth, Galileo was a contemporary of Kepler and his work was nowhere nearly as carefully done or mathematically convincing (or correct!) as Kepler's, although using a ***telescope*** he made a number of important discoveries that added considerable further weight to the argument in favor of heliocentrism in general.

- Newton[212] (1642-1727 C.E.) was the inheritor of the tremendous advances of Brahe, Descartes[213] (1596-1650 C.E.), Kepler, and Galileo. Applying the analytic geometry invented by Descartes to the empirical laws discovered by Kepler and the kinematics invented by Galileo, he was able to deduce ***Newton's Law of Gravitation***:

$$\vec{F} = -\frac{GMm}{r^2}\hat{r} \tag{12.1}$$

  (a simplified form valid when mass $M \gg m$, $\vec{r}$ are coordinates centered on the larger mass M, and $\vec{F}$ is the force acting on the smaller mass); we will learn a more precisely stated version of this law below. This law fully explained at the limit of observational resoution, and continues to ***mostly*** explain, Kepler's Laws and the motions of the planets, moons, comets, and other visible astronomical objects! Indeed, it allows their orbits to be precisely computed and extrapolated into the distant past or future from a sufficient knowledge of initial state.

- In Newton's Law of Gravitation the constant G is a considered to be a ***constant of nature***, and was measured by Cavendish[214] in a famous experiment, thus (as we shall see) "weighing the planets". The value of G we will use in this class is:

$$G = 6.67 \times 10^{-11} \quad \frac{\text{N-m}^2}{\text{kg}^2} \tag{12.2}$$

  You are responsible for knowing this number! Like g, it is enormously important and useful as a key to the relative strength of the forces of nature and explanation for why it takes an entire planet to produce a force on your body that is easily opposed by (for example) a thin nylon rope.

- The ***gravitational field*** is a simplification of Newton's theory of gravitation that emerged over a considerable period of time with no clear author that attempts to resolve the

[211] Wikipedia: http://www.wikipedia.org/wiki/Galileo.
[212] Wikipedia: http://www.wikipedia.org/wiki/Newton.
[213] Wikipedia: http://www.wikipedia.org/wiki/Descartes.
[214] Wikipedia: http://www.wikipedia.org/wiki/Cavendish.

problem Newton first addressed of ***action at a distance*** – the need for a ***cause*** for the gravitational force that propagates ***from*** one object ***to*** the other. Otherwise it is difficult to understand how one mass "knows" of the mass, direction and distance of its partner in the gravitational force! It is (currently) defined to be the ***gravitational force per unit mass*** or ***gravitational acceleration*** produced at and associated with every ***point in space*** by a ***single*** massive object. This field acts on any mass placed at that point and thereby exerts a force. Thus:

$$\vec{g}(\vec{r}) = -\frac{GM}{r^2}\hat{r} \tag{12.3}$$

$$\vec{F}_m(\vec{r}) = m\vec{g}(\vec{r}) = -\frac{GMm}{r^2}\hat{r} \tag{12.4}$$

- Important true facts about the gravitational field:
  - The gravitational field produced by a (thin) spherically symmetric shell of mass ΔM vanishes inside the shell.
  - The gravitational field produced by this same shell equals the usual

    $$\vec{g}(\vec{r}) = -\frac{G\Delta M}{r^2}\hat{r} \tag{12.5}$$

    outside of the shell. As a consequence the field outside of any spherically symmetric distribution of mass is just

    $$\vec{g}(\vec{r}) = -\frac{G\Delta M}{r^2}\hat{r} \tag{12.6}$$

  These two results can be proven by direct integration or by using Gauss's Law for the gravitational field (using methodology developed next semester for the electrostatic field).

- The gravitational force is conservative. The gravitational potential energy of mass m in the field of mass M is:

  $$U_m(\vec{r}) = -\int_\infty^{\vec{r}} \vec{F}\cdot d\vec{\ell} = -\frac{GMm}{r} \tag{12.7}$$

  By convention, the zero of gravitational potential energy is at $r_0 = \infty$ (in all directions).

- The ***gravitational potential*** is to the potenial energy as the gravitational field is to the force. That is:

  $$V(\vec{r}) = \frac{U_m(\vec{r})}{m} = -\int_\infty^{\vec{r}} \vec{g}\cdot d\vec{\ell} = -\frac{GM}{r} \tag{12.8}$$

  It as a scalar field that depends only on distance, it is the simplest of the ways to describe gravitation. Once the potential is known, one can always find the gravitational potential energy:

  $$U_m(\vec{r}) = mV(\vec{r}) \tag{12.9}$$

  or the gravitational field:

  $$\vec{g}(\vec{r}) = -\vec{\nabla}V(\vec{r}) \tag{12.10}$$

  or the gravitational force:

  $$\vec{F}_m(\vec{r}) = -m\vec{\nabla}V(\vec{r}) = m\vec{g}(\vec{r}) \tag{12.11}$$

- ***Escape velocity*** is the minimum velocity required to escape from the surface of a planet (or other astronomical body) and coast in free-fall all the way to infinity so that the object "arrives at infinity at rest". Since $U(\infty) = 0$ by definition, the ***escape energy*** for a particle is:

$$E_{\rm escape} = K(\infty) + U(\infty) = 0 + 0 = 0 \tag{12.12}$$

  Since mechanical energy is conserved moving through the (presumed) vacuum of space, the total energy must be zero on the surface of the planet as well, or:

$$\frac{1}{2}mv_e^2 - \frac{GMm}{R} = 0 \tag{12.13}$$

  or

$$v_e = \sqrt{\frac{2GM}{R}} \tag{12.14}$$

  On the earth:

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR_e} = 11.2 \times 10^3 \text{ meters/second} \tag{12.15}$$

  (11.2 kilometers per second). This is also the most reasonable starting estimate for the speed with which falling astronomical objects, e.g. meteors or asteroids, will ***strike*** the earth. A large falling mass loses basically all of its kinetic energy on impact, so that even a fairly small asteroid can easily strike with an explosive power greater than that of a ***nuclear bomb***, or ***many nuclear bombs***. It is believed that just such a collision was responsible for at least the final Cretaceous extinction event that brought an end to the age of the dinosaurs some sixty million years ago, and similar collisions may have caused other great extinctions as well.

- A (point-like) object in a plane orbit has a kinetic energy that can be written as:

$$K = K_{\rm rot} + K_r = \frac{L^2}{2mr^2} + \frac{1}{2}mv_r^2 \tag{12.16}$$

  The total mechanical energy of this object is thus:

$$E = K + U = \frac{1}{2}mv_r^2 + \frac{L^2}{2mr^2} - \frac{GMm}{r} \tag{12.17}$$

  $\vec{L}$ for an orbit (in a central force, recall) is ***constant***, hence L^2 is constant in this expression. The total energy and the angular momentum thus become convenient ways to parameterize the orbit.

- The ***effective potential energy*** is of a mass m in an orbit with (magnitude of) angular momentum L is:

$$U'(r) = \frac{L^2}{2mr^2} - \frac{GMm}{r} \tag{12.18}$$

  and the total energy can be written in terms of the ***radial kinetic energy only*** as:

$$E = \frac{1}{2}mv_r^2 + U'(r) \tag{12.19}$$

This is a convenient form to use to make ***energy diagrams*** and determine the ***radial turning points of an orbit***, and permits us to easily classify orbits not only as ellipses but as general ***conic sections***. The term $L^2/2mr^2$ is called the ***angular momentum barrier*** because it's negative derivative with respect to r can be interpreted as a strongly (radially) repulsive pseudoforce for small r.

- The orbit classifications (for a given nonzero L) are:
  - Circular: Minimum energy, only one permitted value of r_c in the energy diagram where $E = U'(r_c)$.
  - Elliptical: Negative energy, always have two turning points.
  - Parabolic: Marginally unbound, $E = 0$, one radial turning point. This is the "escape orbit" described above.
  - Hyperbolic: Unbound, $E > 0$, one radial turning point. This orbit has enough energy to reach infinity while still moving, if you like, although a better way to think of it is that its asymptotic radial kinetic energy is greater than zero.

12.1: Cosmological Models

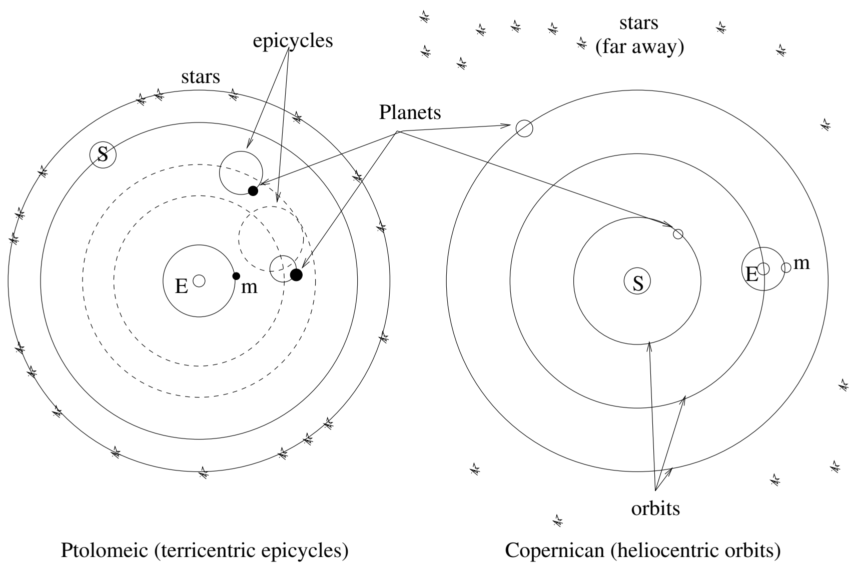


Figure 155: The Ptolemaic ***geocentric*** model with ***epicycles*** that sufficed to explain the observational data of retrograde motion. The Copernican ***geocentric*** model also explained the data and was somewhat simpler. To determine which was correct required the use of ***parallax*** to determine *distances* as well as angles.

Early western (Greek) cosmology was both ***geocentric***, with fixed stars "lamps" or "windows" on a big solid bowl, the moon and sun and planets orbiting a fixed, stationary Earth in the center. Plato represented the Earth (approximately correctly) as a sphere and located it at the center of the Universe. Astronomical objects were located on transparent "spheres" (or circles) that rotated uniformly around the Earth at differential rates. Euxodus and then Aristotle (both students of Plato) elaborated on Plato's original highly idealized description, adding spheres until the model "worked" to some extent, but left a number of phenomena either unexplained or (in the case of e.g. lunar phases) not particularly *believably* explained.

The principle failure of the Aristotelian geocentric model is that it fails to explain ***retrograde motion*** of the planets, where for a time they seem to go backwards against the fixed stars in their general orbits. However, in the second century Claudius Ptolemaeus constructed a somewhat simpler geocentric model that is currently known as the ***Ptolemaic model*** that still involved Plato's circular orbits with stars embedded on an outer revolving sphere, but added to this the notion of *epicycles* – planets orbiting in circles *around* a point that was itself in a circular orbit around the Earth. The model was very complex, but it actually *explained the observational data including retrograde motion* well enough that – for a variety of political, psychological, and religious reasons – it was adopted as ***the "official" cosmology of Western Civilization, endorsed and turned into canonical dogma by the Catholic Church*** as geocentrism agreed (more or less) with the cosmological assertions of the Bible.

In this original period – during which the Greeks invented things like mathematics and philosophy and the earliest rudiments of physics – geocentrism was not the only model. The Pythogoreans, for example, postulated that the earth orbited a "great circle of fire" that was always beneath one's feet in a flat-earth model, while the sun, stars, moon and so on orbited the whole thing. An "anti-Earth" was supposed to orbit on the far side of the great fire, where we cannot see it. All one can say is gee, they must have had really good recreational/religious hallucinogenic drugs back then...[215]. Another "out there" model – by the standards of the day – was the ***heliocentric*** model.

The first person known to have proposed a heliocentric system, however, was Aristarchus of Samos (c. 270 BC). Like Eratosthenes, Aristarchus calculated the size of the Earth, and measured the size and distance of the Moon and Sun, in a treatise which has survived. From his estimates, he concluded that the Sun was six to seven times wider than the Earth and thus hundreds of times more voluminous. His writings on the heliocentric system are lost, but some information is known from surviving descriptions and critical commentary by his contemporaries, such as Archimedes. Some have suggested that his calculation of the relative size of the Earth and Sun led Aristarchus to conclude that it made more sense for the Earth to be moving than for the huge Sun to be moving around it.

Archimedes was familiar with, and apparently endorsed, this model. This model explained the lack of motion of the stars by putting them ***very far away*** so that distances to them could not easily be detected using parallax! This was the first hint that the Universe was ***much larger than geocentric models assumed***, which eliminated any *need* for parallax by approximately fixing the earth itself relative to the stars.

The heliocentric model explained may things, but it wasn't clear how it would explain (in particular) retrograde motion. For a variety of reason (mostly political and religious) the platonic geocentric model was preserved and the heliocentric model officially forgotten and ignored until the early 1500's, when a catholic priest and polymath[216] resurrected it and showed how it *explained* retrograde motion with far less complexity than the Ptolemaic model. Since the work of Aristarchus was long forgotten, this reborn heliocentric model was called the ***Copernican model***[217] , and was perhaps the spark that lit the early Enlightenment[218] .

Initially, the Copernican model, published in 1543 by a Copernicus who was literally on his deathbed, attracted little attention. Over the next 70 years, however, it gradually caused more and more debate, in no little part because it directly contradicted a number of passages in the Christian holy scriptures and thereby strengthened the position of an increasing number of contemporary philosophers who challenged the divine inspiration and fidelity of those writings. This drew attention from scholars within the established Catholic

[215] Which in fact, they did...

[216] One who is skilled at many philosophical disciplines. Copernicus made contributions to astronomy, mathematics, medicine, economics, spoke four languages, and had a doctorate in law.

[217] Wikipedia: http://www.wikipedia.org/wiki/Copernican heliocentrism.

[218] Wikipedia: http://www.wikipedia.org/wiki/Age of Enlightenment. The Enlightenment was the philosophical revolution that led to the invention of physics and calculus as the core of "natural philosophy" – what we now call science – as well as economics, democracy, the concept of "human rights" (including racial and sexual equality) within a variety of social models, and to the rejection of scriptural theism as a means to knowledge that had its roots in the discoveries of Columbus (that the world was not flat), Descartes (who invented analytic geometry), Copernicus (who proposed that the non-flat Earth was not the center of creation after all), setting the stage in the sixteenth century for radical and rapid change in the seventeenth and eighteenth centuries.

church as well as from the new Protestant churches that were starting to emerge, as well as from other philosophers.

The most important of these philosophers was another polymath by the name of Galileo Galilei[219] . The first refracting telescopes were built by spectacle makers in the Netherlands in 1608; Galileo heard of the invention in 1609 and immediately built one of his own that had a magnification of around 20. With this instrument (and successors also of his own design) he performed an amazing series of astronomical observations that permitted him to *empirically support* the Copernican model in preference over the Ptolemaic model.

It is important to note well that *both* models explain the observations available to the naked eye. Ptolemaeus' model was somewhat more complex than the Copernican model (which weighs against it) but one common early complaint against the latter was that it wasn't provable by observation and all of the sages and holy fathers of the church for nealy 2000 years considered geocentrism to be true on observational grounds.

Galileo's telescope – which was little more powerful than an ordinary pair of hand-held binoculars today – was sufficient to provide that proof. Galileo's instrument clearly revealed that the moon was a *planetoid object*, a truly massive ball of rock that orbited the Earth, so large that it had its own mountains and "seas". It revealed that Jupiter had not one, but four similar moons of its own that orbited *it* in similar manner (moons named "The Galilean Moons" in his honor). He observed the phases of Venus as it orbited the sun, and correctly interpreted this as positive evidence that Venus, too, was a huge world orbiting the sun as the Earth orbits the sun while revolving and being orbited by its own moon. He was one of the first individuals in modern times to observe sunspots (although he incorrectly interpreted them as Mercury in transit across the Sun) and set the stage for centuries of solar astronomical observations and sunspot counts that date from roughly this time. His (independent) observations on gravity even helped inspire Newton to develop gravity as the *universal cause* of the observed orbital motions.

However, the publication of his own observations defending Copernicus corresponded almost exactly with the Church finally taking action to condemn the work of Copernicus and ban his book describing the model. In 1600 the Roman Inquisition had found Catholic priest, freethinker, and philosopher Giordano Bruno[220] guilty of heresy and burned him at the stake, establishing a dangerous precedent that put a damper on the development of science everywhere that the Roman church held sway.

Bruno not only embraced the Copernican theory, he went far beyond it, recognizing that the Sun is a star like other stars, that there were far, far more stars than the human eye could see without help, and he even asserted that many of those stars have planets like the Earth and that those planets were likely to be inhabited by intelligent beings. While Galileo was aided in his assertions by the use of the telescope, Bruno's were all the more remarkable because they *preceded* the invention of the telescope. Note well that the human eye can only make out some ***3500 stars altogether*** unaided on the darkest, clearest

[219]Wikipedia: http://www.wikipedia.org/wiki/Galileo Galilei. It would take too long to recite all of Galileo's discoveries and theories, but Galileo has for good reason been called "The Father of Modern Science".

[220]Wikipedia: http://www.wikipedia.org/wiki/Giordano Bruno. Bruno is, sadly, almost unknown as a philosopher and early scientist for all that he was braver and more honest in his martyrdom that Galileo in his capitulation.

nights. This leap from 3500 to "infinity", and the other inferences he made to accompany them, were quite extraordinary. His guess that the stars are effectively numberless was validated shortly afterwards by means of the very first telescopes, which revealed more and more stars in the gaps between the visible stars as the power of the telescopes was systematically increased.

We only discovered positive evidence of the first confirmed exoplanet[221] in *1988* and are *still* in the process of searching for evidence that might yet validate his further hypothesis of life spread throughout the Universe, some of it (other than our own) intelligent. Galileo had written a letter to Kepler in 1597, a mere three years before Bruno's ritualized murder, stating his belief in the Copernican system (which was not, however, the direct cause of Bruno's conviction for heresy). The stakes were indeed high, and piled higher still with wood.

Against this background, Galileo developed a careful and observationally supported argument in favor of the Copernican model and began cautiously to publish it within the limited circles of philosophical discourse available at the time, proposing it as a "theory" only, but arguing that it did not contradict the Bible. This finally attracted the attention of the church. Cardinal and Saint Robert Bellarmine wrote a famous letter to Galileo in 1615[222] explaining the Church's position on the matter. This letter should be required reading for all students, and since if you are reading this textbook you are, in a manner of speaking, *my* student, please indulge me by taking a moment and following the link to read the letter and some of the commentary following.

In it Bellarmine makes the following points:

- If Copernicus (and Galileo, defending Copernicus and advancing the theory in his own right) are correct, the heliocentric model "is a very dangerous thing, not only by irritating all the philosophers and scholastic theologians, but also by injuring our holy faith and rendering the Holy Scriptures false."

  In other words, if Galileo is correct, the holy scriptures are incorrect. Bellarmine correctly infers that this would reduce the degree of belief in the infallibility of the holy scriptures and hence the entire basis of belief in the religion they describe.

- Furthermore, Bellarmine continues, Galileo is *disagreeing with established authorities* with his hypothesis, who "...all agree in explaining literally (ad litteram) that the sun is in the heavens and moves swiftly around the earth, and that the earth is far from the heavens and stands immobile in the center of the universe. Now consider whether in all prudence the Church could encourage giving to Scripture a sense contrary to the holy Fathers and all the Latin and Greek commentators. Nor may it be answered that this is not a matter of faith, for if it is not a matter of faith from the point of view of the subject matter, it is on the part of the ones who have spoken."

- Finally, Bellarmine concludes that "if there were a true demonstration that the sun was in the center of the universe and the earth in the third sphere, and that the sun did

[221] Wikipedia: http://www.wikipedia.org/wiki/Extrasolar planet. As of today, some 851 planets in 670 systems have been discovered, with more being discovered almost every day using a dazzling array of sophisticated techniques.

[222] http://www.fordham.edu/halsall/mod/1615bellarmine-letter.asp

> not travel around the earth but the earth circled the sun, then it would be necessary to proceed with great caution in explaining the passages of Scripture which seemed contrary, and we would rather have to say that we did not understand them than to say that something was false which has been demonstrated." He goes on to assert that "the words 'the sun also riseth and the sun goeth down, and hasteneth to the place where he ariseth, etc.' were those of Solomon, who not only spoke by divine inspiration but was a man wise above all others and most learned in human sciences and in the knowledge of all created things, and his wisdom was from God."

Interested students are invited to play *Logical Fallacy Bingo*[223] with the text of the entire document. Opinion as fact, appeal to consequences, wishful thinking, appeal to tradition, historian's fallacy, argumentum ad populum, thought-terminating cliche, and more. The argument of Bellarmine boils down to the following:

- If the heliocentric model is true, the Bible is false where that model contradicts it.
- If the Bible is false *anywhere*, it cannot be trusted *everywhere* and Christianity itself can legitimately be doubted.
- The Bible and Christianity are true. Even if they appear to be false they are *still* true, but don't worry, they don't even appear to be false.
- Therefore, while it is all very well to show how a heliocentric model could *mathematically, or hypothetically* explain the observational data, it ***must be false***.

In 1633, this same Bellarmine (later made into a saint of the church) prosecuted Galileo in the Inquisition. Galileo was found "vehemently suspect of heresy" for precisely the reasons laid out in Bellarmine's original letter to Galileo. He was forced to publicly recant, his book laying out the reasons for believing the Copernican model was added along with the book of Copernicus to the list of banned books, and he was sentenced to live out his life under house arrest, praying all day for forgiveness. He died in 1642 a broken man, his prodigious and productive mind silenced by the active defenses of the locally dominant religious mythology for almost ten years.

I was fortunate enough to be teaching gravitation in the classroom on October 31, 1992, when Pope John Paul II (finally) *publicly apologized* for how the entire Galileo affair was handled. On Galileo's behalf, I accepted the apology, but of course I must also point out that *Bellarmine's argument is essentially correct*. The conclusions of modern science have, almost without exception, contradicted the assertions made in the holy scriptures not just of Christianity but of all faiths. They therefore stand as direct evidence that those scriptures are *not*, in fact, divinely inspired or perfect truth, at least where we can check them. While this does not *prove* that they are incorrect in other claims made elsewhere, it certainly and legitimately makes them less plausible.

[223] http://lifesnow.com/bingo/ http://lifesnow.com/bingo/

12.2: Kepler's Laws

Galileo was not, in fact, the person who made the greatest contributions to the rejection of the Ptolemaic model as the first step towards first the (better) heliocentric Copernican model, then to the invention of physics and science as a systematic methodology for successively improving our beliefs about the Universe that does not depend on authority or scripture. He wasn't even one of the top two. Let's put him in the third position and count up to number one.

The person in the second position (in my opinion, anyway) was *Tycho Brahe*[224] , a wealthy Danish nobleman who in 1571, upon the death of his father, established an observatory and laboratory equipped with the most modern of contemporary instrumentation in an abbey near his ancestral castle. He then proceeded to spend a substantial fraction of his life, including countless long Danish winter nights, *making and recording systematic observations of the night sky!*

His observations bore almost immediate fruit. In 1572 he observed a supernova in the constellation Cassiopeia. This one observation refuted a major tenet of Aristotelian and Church philosophy – that the Universe beyond the Moon's orbit was immutable. A new star had appeared where none was observed before. However, his most important contributions were immense tables of very precise measurements of the locations of objects visible in the night sky, over time. This was in no small part because his *own* hybrid model for a mixture of Copernican and Ptolemaic motion proved utterly incorrect.

If you are a wealthy nobleman with a hobby who is generating a huge pile of data but who also has no particular mathematical skill, what are you going to do? You hire a lab rat, a flunky, an *assistant* who can do the annoying and tedious work of analyzing your data while you continue to have the pleasure of accumulating still more. And as has been the case many a time, the servant exceeds the master. The number one philosopher who contributed to the Copernican revolution, more important than Brahe, Bruno, Galileo, or indeed any natural philosopher before Newton was Brahe's assistant, *Johannes Kepler*[225]
.

Kepler was a brilliant young man who sought geometric order in the motions of the stars and planets. He was also a protestant living surrounded by Catholics in predominantly Catholic central Europe and was persecuted for his religious beliefs, which had a distinctly negative impact on his professional career. In 1600 he came to the attention of Tycho Brahe, who was building a new observatory near Prague. Brahe was impressed with the young man, and gave him access to his closely guarded data on the orbit of Mars and attempted to recruit him to work for him. Although he was was trying hard to be appointed as the mathematician of Archiduke Ferdinand, his religious and political affilations worked against him and he was forced to flee from Graz to Prague in 1601, where Brahe supported him for a full year until Brahe's untimely death (either from possibly deliberate mercury poisoning or a bladder that ruptured from enforced continence at a state banquet – it isn't clear which even today). With Brahe's support, Kepler was appointed an Imperial mathematician and "inherited" at least the use of Brahe's voluminous data. For the next

[224]Wikipedia: http://www.wikipedia.org/wiki/Tycho Brahe.
[225]Wikipedia: http://www.wikipedia.org/wiki/Johannes Kepler.

eleven years he put it to very good use.

Although he was largely ignored by contemporaries Galileo and Descartes, Kepler's work laid, as we shall see, the foundation upon which one Isaac Newton built his physics. That foundation can be summarized in ***Kepler's Laws*** describing the motion of the orbiting objects of the solar system. They were *observational* laws, propounded on the basis of careful analysis of the Brahe data and further observations to verify them. Newton was able to *derive* trajectories that rather precisely agreed with Kepler's Laws on the basis of his physics and law of gravitation.

The laws themselves are surprisingly simple and geometric:

a) Planets move around the Sun in ***elliptical orbits*** with the Sun at one focus (see next section for a review of ellipses).

b) Planets sweep out ***equal areas in equal times*** as they orbit the Sun.

c) The ***mean radius*** of a planetary orbit (in particular, the semimajor axis of the ellipse) ***cubed*** is directly proportional to the ***period*** of the planetary orbit ***squared***, with the same constant of proportionality for all of the planets.

The first law can be proven directly from Newton's Law of Gravitation (although we will not prove it in this course, as the proof is mathematically involved). Instead we will content ourselves with the observation that a circular orbit is certainly consistent, and by using energy diagrams we will see that elliptical orbits are at least rather plausible. The second law will turn out to be equivalent to the ***conservation of angular momentum*** of the orbits, because gravitation is a ***central force*** and exerts no torque. The third, again, is difficult to formally prove for elliptical orbits but straightforward to verify for circular orbits.

Since most planets have *nearly* circular orbits, we will not go far astray by idealizing and restricting our analysis of orbits to the circular case. After all, not even elliptical orbits are precisely correct, because Kepler's results and Newton's demonstration *ignore the influence of the planets on each other* as they orbit the Sun, which constantly perturb even elliptical orbits so that they are at best a not-quite-constant approximation. The best one can do is directly and numerically integrate the equations of motion for the entire solar system (which can now be done to quite high precision) but even that eventually fails as small errors from ignored factors accumulate in time.

Nevertheless, the path from Ptolemy to Copernicus, Galileo and Kepler to Newton stands out as a great triumph in the intellectual and philosophical development of the human species. It is for that reason that we study it.

12.2.1: Ellipses and Conic Sections

The following is a short review of the properties of ellipses (and, to a lesser extent, the other conic sections). Recall that a conic section is the intersection of a plane with a right circular cone aligned with (say) the z-axis, where the intersecting plane can intercept at any value of z and parallel, perpendicular, or at an angle to the x-y plane.

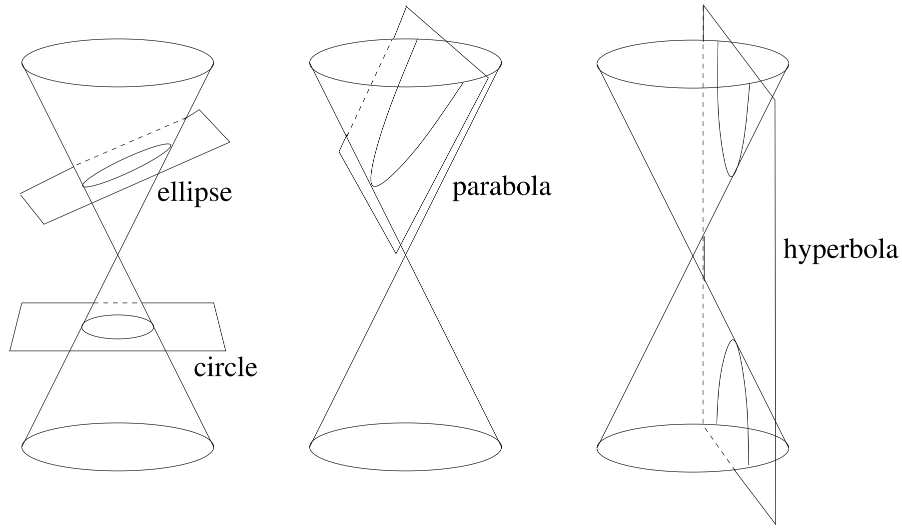


Figure 156: The various conic sections. Note that a circle is really just a special case of the ellipse.

A circle is the intersection of the cone with a plane parallel to the x-y plane. An ellipse is the intersection of the cone with a plane tipped at an angle *less* than the angle of the cone with the cone. A parabola is the intersection of the cone with a plane *at the same angle* as that of the cone. A hyperbola is the intersection of the cone with a plane tipped at a *greater* angle than that of the cone, so that it produces two disjoint curves and has *asymptotes*. An example of each is drawn in figure 156, the hyperbola for the special case where the intersecting plane is parallel to the z-axis.

Properly speaking, gravitational two-body orbits are conic sections: hyperbolas, parabolas, ellipses, or circles, not just ellipses per se. However, *bound planetary orbits* are elliptical, so we will concentrate on that.

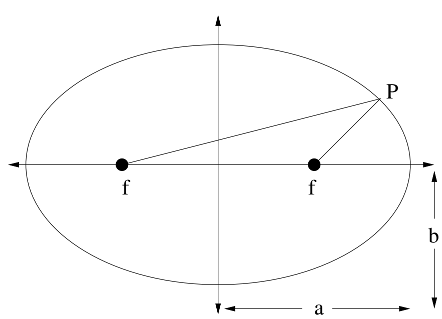


Figure 157:

Figure 157 illustrates the general geometry of the ellipse in the x-y plane drawn such that its major axis is aligned with the x axis. In this simple case the equation of the ellipse can be written:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \tag{12.20}$$

There are certain terms you should recall that describe the ellipse. The major axis is the longest "diameter", the one that contains both foci and the center of the ellipse. The minor axis is the shortest diameter and is at right angles to the major axis. The semimajor axis is the long-direction "radius" (half the major axis); the semiminor axis is the short-direction "radius" (half the minor axis).

In the equation and figure above, a is the semimajor axis and b is the semiminor axis.

Not all ellipses have major/minor axes that can be easily chosen to be x and y coordinates. Another general parameterization of an ellipse that is useful to us is a parametric cartesian representation:

$$x(t) = x_0 + a\cos(\omega t + \phi_x) \tag{12.21}$$

$$y(t) = y_0 + b\cos(\omega t + \phi_y) \tag{12.22}$$

This equation will describe *any* ellipse centered on (x_0, y_0) by varying ωt from 0 to 2π. Adjusting the phase angles ϕ_x and ϕ_y and amplitudes a and b vary the orientation and eccentricity of the ellipse from a straight line at arbitrary angle to a circle.

The ***foci*** of an ellipse are defined by the property that the sum of the distances from the foci to every point on an ellipse is a constant (so an ellipse can be drawn with a loop of string and two thumbtacks at the foci). If f is the distance of the foci from the origin, then the sum of the distances must be $2d = (f + a) + (a - f) = 2a$ (from the point $x = a$, $y = 0$. Also, $a^2 = f^2 + b^2$ (from the point $x = 0$, $y = b$). So $f = \sqrt{a^2 - b^2}$ where by convention $a \geq b$.

This is all you need to know (really more than you need to know) about ellipses in order to understand Kepler's First and Third Laws. The key things to understand are the meanings of the terms "focus of an ellipse" (because the Sun is located at one of the foci of an elliptical orbit) and "semimajor axis" as a measure of the "average radius" of a periodic elliptical orbit. As noted above, we will concentrate in *this* course on circular orbits because they are easy to solve for and understand, but in future, more advanced physics courses students will actually solve the equations of motion in 2 dimensions (the third being irrelevant) for planetary motion using Newton's Law of Gravitation as the force and prove that the solutions are parametrically described ellipses. In some versions of even *this* course, students might use a tool such as octave, mathematica, or matlab to solve the equations of motion numerically and graph the resulting orbits for a variety of initial conditions.

12.3: Newton's Law of Gravitation

In spite of the church's opposition, the early seventeenth century saw the formal development of the heliocentric hypothesis, supported by Kepler's empirical laws. Instrumentation improved, and the geometric methods involving *parallax* to determine distance produced a systematically improving picture of the solar system that was not only heliocentric but verified Kepler's Laws in detail for additional planetary bodies. The debate with the geocentric/ptolemaic model supporters continued, but in countries far away from Rome where its influence waned, a consensus was gradually forming that the geocentric hypothesis was incorrect. The observations of Brahe and Galileo and analysis of Kepler was compelling.

However, the *cause* of heliocentric motion was a mystery. There was clearly substantial geometry and order in the motion of the planets, although it was not precisely the geometry proposed by Plato and advanced by Aristotle and Ptolemaeus and others. This geometry was subtle, and best described within the confines of the new Analytic Geometry invented by Descartes[226] where ellipses (as we can see above) were not "just" conic sections or objects visualized in a solid geometry: They could be represented by *equations*.

Descartes was another advocate of the heliocentric theory, but when, in 1633, he heard that Galileo had been condemned for his advocacy of Copernicus and arguments against the Ptolemaic geocentric model, he abruptly changed his mind about publishing a work to that effect! As noted above, these were dangerous times for freethinking philosophers who were literally forbidden by the rulers of the predominant religion under threat of torture and murder from speculating in ways that contradicted the scriptures of that religion. A powerful voice was thus silenced and the geocentric model persisted without any *open* challenge for fifty more years.

So things remained until one of the most brilliant and revered men of all time came along: Isaac Newton. Born on December 25, 1642, Newton was only 8 in 1650 when Descartes died, but he was taught Descartes' geometry at Cambridge (before it closed in the midst of a bout of the plague so that he was sent home for a while) and by the age of 24 had transformed it into a theory of "fluxions" – the first rudimentary description of calculus. Calculus, or the mathematics of related rates of change established on top of a coordinatized geometry, was the missing ingredient, the key piece needed to transform the *strictly geometric* observations of philosophers from Plato through Kepler into an *analytic description* of both the causes and effects of motion.

Even so, Newton worked *thirteen more years* producing and presenting advances in mathematics, optics, and alchemy before (in 1679), having recently completed a speculative theory of optics, he turned his attention wholly towards the problem of celestial mechanics and Kepler's Laws. In this he was reportedly inspired by the intuition that the force of *gravity* – the same force that makes the proverbial apple fall from the tree – was responsible for holding the moon in its orbit around the Earth.

Initially he corresponded heavily with *Robert Hooke*[227] , known to us through *Hooke's Law* in the text above, who had been appointed secretary of the brand new *Royal Society*[228] , the world's first "official" scientific organization, devoted to an eclectic mix of mathematics, philosophy, and the brand new "natural philosophy" (the correct and common termin for "science" almost to the end of the nineteenth century). Hooke later claimed (quite possibly correctly) that he suggested the inverse-square force law to Newton, but what Hooke did not do that Newton did is to take the postulated inverse square force law, add to it a set of axioms (Newton's Laws) that *defined* force in a particular mathematical

[226]Wikipedia: http://www.wikipedia.org/wiki/Rene Descartes. Descartes was another of the "renaissance man" polymaths of the age. He was brilliant and led a most interesting life, making contributions to mathematics (where "Cartesian Coordinates" are named in his honor), physics, and philosophy. He reportedly liked to sleep late, never rising before 11 a.m., and when an opportunity to become a court mathematician and tutor arose that forced him to change his habits and arise at 5 a.m. every day, he sickened and died (in 1650) a short while thereafter!

[227]Wikipedia: http://www.wikipedia.org/wiki/Robert Hooke.

[228]Wikipedia: http://www.wikipedia.org/wiki/Royal Society.

way, and then show that the equations of motion that followed from an inverse square force law, evaluated through the use of calculus, *completely predicted and explained Kepler's Laws and more* by means of explicit functional solutions built on top of Descartes' analytic geometry, where the "more" was the apparent non-elliptical orbits of other celestial bodies, notably comets.

It is difficult to properly explain how revolutionary, how world-shattering this combination of invention and discovery was. Initially it was communicated privately to the Royal Society itself in 1684; three years later it was formally published as the *Philosophiae Naturalis Principia Mathematica*[229] , or "The Mathematical Principles of Natural Philosophy". This book changed everything. It utterly destroyed, forever, any possibility that the geocentric hypothesis was correct. The reader must determine for themselves if it initiated the very process anticipated and feared by Robert Bellarmine – as the consequences of Newton's work unfolded, they have proven the Bible and all of the other religious mythologies and scriptures of the world *literally* false time and again.

As we have seen from a full semester of work with its core principles, Newton's Laws and a small set of actual force laws permit the nearly full description and prediction of virtually all everyday mechanical phenomena, and its *ideas* (in some cases extended far beyond what Newton originally anticipated) survive to some extent even in its eventual replacement, quantum mechanics. *Principia Mathematica* laid down a template for the *process* of scientific endeavor – a mix of accumulation and analysis of experimental data, formal axiomatic mathematics, and analytic reasoning leading to a detailed description of the visible Universe of ever-improving consistency. It was truly a *system of the world*, the basis of the ***scientific worldview***. It was a radically different worldview than the one based on faith, authority, and the threat of violence divine or mundane to any that dared challenge it that preceded it.

Let us take a look at the force law invented or discovered (as you please) by Newton and see how it works to explain Kepler's Laws, at least for simple cases we can readily solve without much calculus.

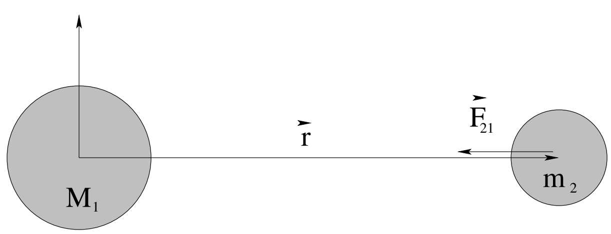


Figure 158:

Here are Newton's axioms, the essential individual assumptions that are assembled compactly into the law of gravitation. Note that these assumptions were initially applied to objects like the Sun and the planets and moons that are spherically symmetric to a close approximation; the also apply to "particles" of mass or chunks of mass small enough to be treated as particles. Following along with figure 158 above:

a) The force of gravity is a ***two body force*** and does not change if three or more bodies

[229]Wikipedia: http://www.wikipedia.org/wiki/Philosophiae Naturalis Principia Mathematica.

are present.

b) The force of gravity is ***action at a distance*** and does not require the two objects to "touch" in order to act.

c) The force of gravity acts along (in the direction of) a line ***joining centers of spherically symmetric masses***, in this case along $\vec{r}$.

d) The force of gravity is ***attractive***.

e) The force of gravity is ***proportional to each mass***.

f) The force of gravity is ***inversely proportional to the distance between the centers of the masses***.

We will add to this list the assumption that one of the two masses is *much larger than the other* so that the center of mass and the center of coordinates can both be placed at the center of the larger mass. This is *not at all necessary* and proper treatments dating all the way back to Newton account for motion around a more general center of mass, but for us it will greatly simplify our pictures and treatments if we idealize in this way and in the case of systems like the Earth and the moon, or the Sun and the Earth, it isn't a terrible idealization. The Sun's mass is a thousand times larger than even that of Jupiter!

These axioms are rather prolix in words, but in the form of an *algebraic equation* they are rather *beautiful*:

$$\vec{\boldsymbol{F}}_{21} = -\frac{G\, M_1\, m_2}{r^2}\hat{\boldsymbol{r}} \tag{12.23}$$

where $G = 6.67 \times 10^{-11}$ N-m^2/kg^2 is the textbfuniversal gravitational constant, added as the constant of proportionality that establishes the connections between all of the different *units* in question. Note that we continue to use the convention that $\vec{\boldsymbol{F}}_{21}$ stands for the force acting on mass 2 due to mass 1; the force $\vec{\boldsymbol{F}}_{12} = -\vec{\boldsymbol{F}}_{21}$ both from Newton's third law and because the force is *attractive* for both masses.

Kepler's first law follow from solving Newton's laws and the equations of motion in three dimensions for this particular force law. Even though one dimension turns out to be irrelevant (the motion is strictly in a plane), even though the motion turns out to have two constants of the motion that permits it to be further simplified (the energy and the angular momentum) the actual solution of the resulting differential equations is a bit difficult and beyond the scope of this course. We will instead show that circular orbits are one special solution that easily satisfy Kepler's First and Third Laws, while Kepler's Second Law is a trivial consequence of conservation of angular momentum.

Let us begin with Kepler's Second Law, as it stands alone (the other two proofs are related). It is proven by observing that the force is ***radial***, and hence exerts ***no torque***. Thus the angular momentum of a planetary orbit is constant!

We start by noting that the area enclosed by an parallelogram formed out of two vectors is the magnitude of the the cross product of those vectors. Hence the area in the shaded

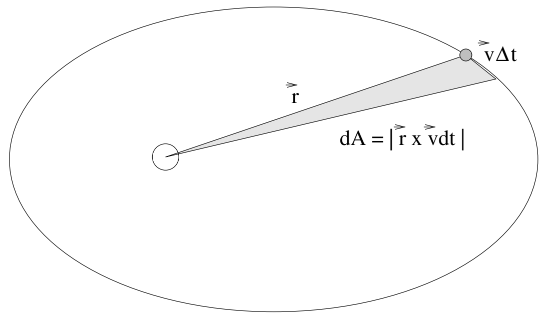


Figure 159: The area swept out in an elliptical orbit in time Δt is shaded in the ellipse above.

triangle in figure 159 is half of that:

$$dA = \frac{1}{2}|\vec{\boldsymbol{r}} \times \vec{\boldsymbol{v}}\,dt| = \frac{1}{2}|\vec{\boldsymbol{r}}||\vec{\boldsymbol{v}}\,dt|\sin\theta \quad (12.24)$$

$$= \frac{1}{2m}|\vec{\boldsymbol{r}} \times m\vec{\boldsymbol{v}}\,dt| \quad (12.25)$$

If we divide the Δt over to the other side we get the area per unit time being swept out by the orbit:

$$\frac{dA}{dt} = \frac{1}{2m}|\vec{\boldsymbol{r}} \times \vec{\boldsymbol{p}}| = \frac{1}{2m}|\vec{\boldsymbol{L}}| = \text{a constant} \quad (12.26)$$

because angular momentum is conserved for a central force (see the chapter/week on torque and angular momentum if you have forgotten this argument) and Kepler's second law is proved for this force.

That was pretty easy! Let's reiterate the point of this demonstration:

Kepler's Second Law is equivalent to the Law of Conservation of Angular Momentum and is true for any central force (not just gravitation)!

The proofs of Kepler's First and the Third laws *for circular orbits* rely on a common algebraic argument, so we group them together. They key formula is, as one might expect the fact that *if* an orbiting mass moves in a circular orbit, *then* the gravitational force has to be equal to the mass times the centripetal acceleration:

$$\frac{G\,M_s m_p}{r^2} = m_p a_r = m_p \frac{v^2}{r} \quad (12.27)$$

where M_s is the mass of the central attracting body (which we implicitly assume is much larger than the mass of the orbiting body so that its center of mass is more or less at the center of mass of the system), m_p is the mass of the planet, v is its speed in its circular orbit of radius r. This situation is illustrated in figure 160.

This equation in and of itself "proves" that Newton's Laws plus Newton's Law of Gravitation have a solution consisting of a circular orbit, where a circle is a special case of an ellipse. This proof isn't very exciting, however, as *any* attractive radial force law we might attempt would have a similarly consistent circular solution. The kinematic radial acceleration of a particle moving in uniform circular motion is independent of the particular force law that produces it!

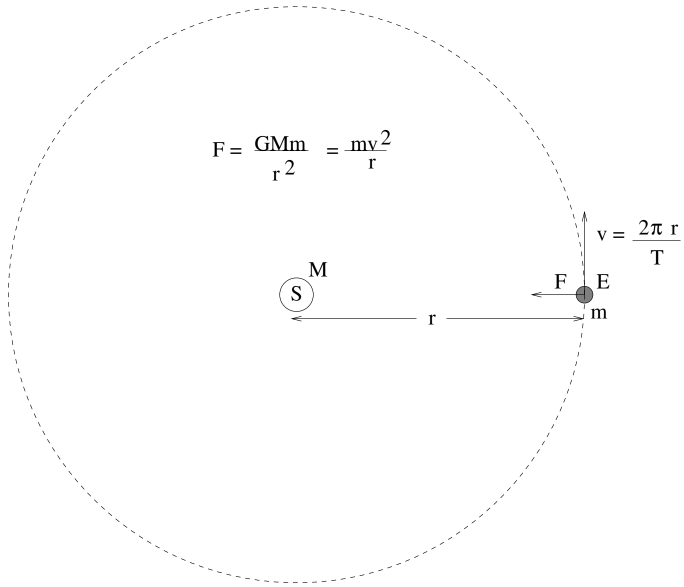


Figure 160: The geometry used to prove Kepler's and Third Laws for a circular (approximately) orbit like that of the Earth around the Sun.

What is a lot more interesting is the demonstration that the circular orbit satisfies Kepler's *Third* Law, as this law quite specifically defines the relationship between the radius of the orbit and its period. We can easily see that only *one* radial force law will lead to consistency with the observational data for circular orbits.

We start by cancelling the mass of the planet and one of the factors of r:

$$v^2 = \frac{G\ M_s}{r} \tag{12.28}$$

But, v is related to r and the period T by:

$$v = \frac{2\pi r}{T} \tag{12.29}$$

so that

$$v^2 = \frac{4\pi^2 r^2}{T^2} = \frac{G\ M_s}{r} \tag{12.30}$$

Finally, we isolate the powers of r:

$$r^3 = \left(\frac{G\ M_s}{4\pi^2}\right) T^2 \tag{12.31}$$

and Kepler's third law is proved for circular orbits.

Since there is nothing unique about circular orbits and *all* closed elliptical orbits around the *same* central attracting body have to have the same constant of proportionality, we have both proven that Newton's Law of Gravitation has circular solutions that satisfy Kepler's Third Law *and* we have evaluated the *universal* constant of proportionality, valid for all of the planets in the solar system! We can then write the law more compactly:

$$R_{sm}^3 = \left(\frac{G\ M_s}{4\pi^2}\right) T^2 \tag{12.32}$$

where now R_{sm} is the semimajor axis of the elliptical orbit, which happens to be r for a circular orbit.

Note well that this constant is easily measured! In fact we can evaluate it from our knowledge of the semimajor axis of Earth's nearly circular orbit – $R_E \approx 1.5 \times 10^{11}$ meters (150 million kilometers) plus our knowledge of its period – $T = 3.153 \times 10^7$ seconds (1 year, in seconds). These two numbers are well worth remembering – the first is called an ***astronomical unit*** and is one of the fundamental lengths upon which our knowledge of the distances to the nearer stars is based; the second physicists tend to remember as "ten million times π seconds per year" because that is accurate to well within one percent and easier to remember than 3.153.

Combining the two we get:

$$\left(\frac{G\ M_s}{4\pi^2}\right) = \frac{T^2}{R_{sm}^3} = \frac{\pi^2 \times 10^{14}}{3.375 \times 10^{33}} \approx 3 \times 10^{-19} \tag{12.33}$$

where we used *another* physics geek cheat: $\pi^2 \approx 10$, and then approximated $10/3.375 \approx 3$ as well. That way we can get an answer, good to within a couple of percent, without using a calculator or looking anything up!

Note well! If only we knew G, we'd know the mass of the Sun! If we use the same logic to determine the *same* constant for objects orbiting the *Earth* (where we might use the semimajor axis of the moon's orbit, 384,000 kilometers, and the period of the moon's orbit, 27.3 days, to get $GM_E/4\pi^2$) we would also be able to determine the mass of the Earth!

Of course we *do* know G *now*, but when Newton proposed his theory, it wasn't so easy to figure out! This is because *gravitation is the weakest of the forces of nature, by far!* It is so weak that it is remarkably difficult to measure the direct gravitational force between two objects of known masses separated by a known distance in the laboratory, so that all of the quantities in Newton's Law of Gravitation were measured *but* G.

In fact, it took over a century for Henry Cavendish[230] to build a clever apparatus that was sufficiently sensitive that it could measure G from these three known quantities. This experiment was said to "weigh the Earth" not because it actually did so – far from it – but because once G was known experiments that had long since been done instantly gave us the mass of the Sun, the mass of the Earth, the mass of Jupiter and Saturn and Mars (any planet where we can remotely observe the semimajor axis and period of a moon) and much more.

These in turn gave us some serious conundrums! The Sun turns out to be 1.4 *million kilometers* in diameter, and to have a mass of 2×10^{30} kilograms! With a surface temperature of some 6000 K, what mechanism keeps it so hot? Any sort of chemical fire would soon burn out!

Laboratory experiments plus astronomical observations based on the use of parallax with the entire diameter of the Earth's orbit used as a triangle base and with exquisitely sensitive measurements of the angles between the lines of sight to the nearer stars (which allowed us to determine the distance to these stars) all analyzed by means of Newton's Laws (including gravitation), allowed astronomers to rapidly infer a startling series of facts

[230] Wikipedia: http://www.wikipedia.org/wiki/Cavendish Experiment.

about the Solar system, our local galaxy (the Milky Way), and the Earth.

Not only was the *geocentric* hypothesis wrong, so was the *heliocentric* hypothesis. The Earth turned out to be a mostly unremarkable planet, a relatively small one of a rather large number orbiting an entirely unremarkable star that itself was orbiting in a huge collection of stars, that was only one of a truly staggering number of similar collections of stars, where every new generation of telescopes revealed still more of everything, still further away. At the moment, there appear to be on the order of a hundred *billion* galaxies, containing somewhere in the ballpark of 10^{23} stars, in the visible Universe, which is (allowing for its original inflation, 13.7 billion years ago) around 46 billion light years in radius. At least one method of estimation has claimed to establish a radius around twice this large as a *lower bound* for its size (so that all of these estimates are probably low by an order of magnitude) – and there is no upper bound.

Exoplanets are being discovered at a rate that suggests that planetary systems around those stars are *common*, not rare (especially so given that we can only "see" or infer the existence of extremely large planets so far – we would find it almost impossible to detect a planet as small as the Earth). Bruno's original assertion that the Universe is infinite, contains and infinite number of stars, with an infinite number of planets, an infinite number of which have some sort of intelligent or otherwise *life* may be impossible to verify or refute, but infinite or not the Universe is *enormous* compared to the scale of the Solar system, which is *huge* compared to the scale of the Earth, and contains many, many stars with many, many planetary systems.

In fact, the only thing about the Earth that is remarkable may turn out to be – us!

12.4: The Gravitational Field

As noted above, Newton proposed the gravitational force as the *cause* of the observed orbital motions of the celestial objects. However, this force was *action at a distance* – it exists between two objects that are not touching and that indeed are separated by *nothing*: a vacuum! What then, causes the gravitational force itself? Let us suggest that there must be *something* that is produced by one planet acting as a *source* that is present at the location of the other planet that is the proximate cause of the force that planet experiences. We define the ***gravitational field*** to be this ***cause*** of the gravitational force, the thing that is present at all points in space surrounding a mass *whether or not some other mass is present there to be acted on!*

We define the gravitational field conveniently to be the force per unit mass, a quantity that has the units of *acceleration*:

$$\vec{\boldsymbol{g}}(\vec{\boldsymbol{r}}) = -\frac{G\,M}{r^2}\hat{\boldsymbol{r}} = \frac{\vec{\boldsymbol{F}}}{m} \tag{12.34}$$

The magnitude of the gravitational field at the surface of the earth is thus:

$$g = g(R_E) = \frac{F}{m} = \frac{G\,M_E}{R_E^2} \tag{12.35}$$

and we see that the quantity that we have been calling the gravitational *acceleration* is in fact more properly called the near-Earth gravitational *field*.

This is a very useful equation. It can be used to find any one of g, R_E, M_E, or G, from a knowledge of any of the other three, depending on which ones you think you know best. g is easy; students typically measure g in physics labs at some point or another several different ways! R_E is actually also easy to measure independently and some classical methods were used to do so *long* before Columbus.

M_E, however is hard! This is because it always appears in the company of G, so that knowing g and R_E only gives you their product. This turns out to be the case nearly everywhere – any ordinary measurement you might make turns out to tell you GM_E together, not either one separately.

What about G?

To measure G in the laboratory, one needs a very sensitive apparatus for measuring forces. Since we know already that G is on the order of 10^{-10} N-m^2/kg^2, we can see that gravitational forces between kilogram-scale masses separated by ten centimeters or so are on the order of a few *billionths* of a Newton.

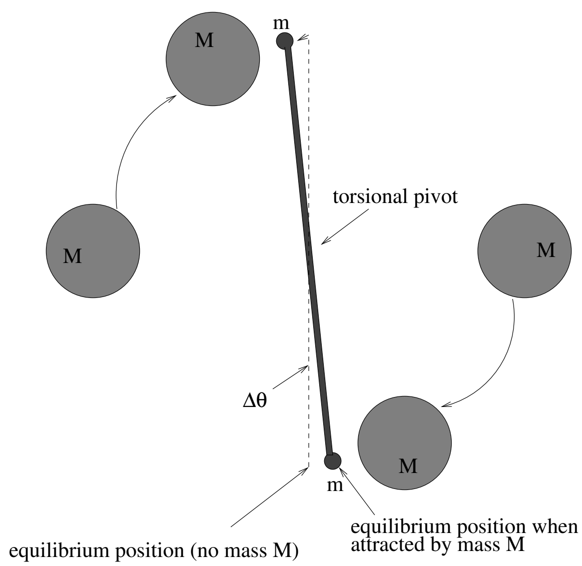


Figure 161: The apparatus associated with the ***Cavendish experiment***, which established the first accurate estimates for G and thereby "weighed the Earth", the Sun, and many of the other objects we could see in the sky.

Henry Cavendish made the first direct measurement of G using a torsional pendulum – basically a barbell suspended by a very thin, strong thread – and some really massive balls whose relative position could be smoothly adjusted to bring them closer to and farter from the barbell balls. As you can imagine, it takes very little torque to twist a long thread from its equilibrium angle to a new one, so this apparatus has – when utilized by someone with a great deal of patience, using a light source and a mirror to further amplify the resolution of the twist angle – proven to be sufficiently sensitive to measure the tiny forces required to determine G, even to some reasonable precision.

Using this apparatus, he was able to find G and hence to "weigh the earth" (find M_E).

By measuring $\Delta\theta$ as a function of the distance r measured between the centers of the balls, and calibrating the torsional response of the string using known forces, he managed to get 6.754 (vs 6.673 currently accepted) $\times 10^{-11}$ N-m^2/kg^2. This is within just about one percent. Not bad!

12.4.1: Spheres, Shells, General Mass Distributions

So far, our empirically founded expression for gravitational force (and by inheritance, field) applies only to ***spherically symmetric mass distributions*** – planets and stars, which are generally almost perfectly round *because* of the gravitational field – or particles small enough that they can be treated like spheres. Our pathway towards the gravitational field of more general distributions of mass starts by formulating the field of a single point-like chunk of mass in such a distribution:

$$d\vec{\boldsymbol{g}} = -\frac{G\ dm_0}{|\vec{\boldsymbol{r}} - \vec{\boldsymbol{r}}_0|^3}(\vec{\boldsymbol{r}} - \vec{\boldsymbol{r}}_0) \tag{12.36}$$

This equation can be integrated as usual over an arbitrary mass distribution using the usual connection: The mass of each chunk is the mass per unit volume times the volume of the chunk, or $dm = \rho dV_0$.

$$\vec{\boldsymbol{g}} = -\int \frac{G\ \rho\ dV_0}{|\vec{\boldsymbol{r}} - \vec{\boldsymbol{r}}_0|^3}(\vec{\boldsymbol{r}} - \vec{\boldsymbol{r}}_0) \tag{12.37}$$

where for example $dV_0 = dx_0 dy_0 dz_0$ (Cartesian) or $dV_0 = r_0^2 \sin(\theta_0) d\theta_0 d\phi_0 dr_0$ (Sphereical Polar) etc. This integral is not always easy, but it can generally be done very accurately, if necessary numerically. In simple cases we can actually do the calculus and evaluate the integral.

In *this* part of *this* course, we will avoid doing the integral, although we will tackle many examples of doing it in simple cases next semester. We will content ourselves with learning the following **True Facts** about the gravitational field:

- The gravitational field produced by a (thin) spherically symmetric shell of mass ΔM vanishes inside the shell.
- The gravitational field produced by this same shell equals the usual

  $$\vec{\boldsymbol{g}}(\vec{\boldsymbol{r}}) = -\frac{G\Delta M}{r^2}\hat{\boldsymbol{r}} \tag{12.38}$$

  outside of the shell. As a consequence the field outside of any spherically symmetric distribution of mass is just

  $$\vec{\boldsymbol{g}}(\vec{\boldsymbol{r}}) = -\frac{G\Delta M}{r^2}\hat{\boldsymbol{r}} \tag{12.39}$$

These two results can be proven by direct integration or by using Gauss's Law for the gravitational field (using methodology developed next semester for the electrostatic field). The latter is so easy that it is hardly worth the time to learn the former for this special case.

Note well the most important consequence for our purposes in the homework of this rule is that when we descend a tunnel into a uniformly dense planet, the gravity will *diminish* as we are only pulled down by the mass *inside* our radius. This means that the gravitational field we experience is:

$$\vec{g}(\vec{r}) = -\frac{G\Delta M(r)}{r^2}\hat{r} \tag{12.40}$$

where $M(r) = \rho 4\pi r^3/3$ for a uniform density, something more complicated in cases where the density itself changes with r. You will use this expression in several homework problems.

12.5: Gravitational Potential Energy

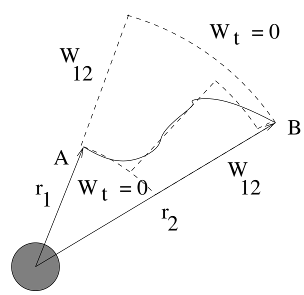


Figure 162: A crude illustration of how one can show the gravitational force to be conservative (so that the work done by the force is independent of the path taken between two points), permitting the evaluation of a potential energy function.

If you examine figure 162 above, and note that the force is always "down" along $\vec{r}$, it is easy to conclude that gravity must be a conservative force. Gravity produced by some (spherically symmetric or point-like) mass does work on another mass only when that mass is moved in or out along $\vec{r}$ connecting them; moving at right angles to this along a surface of constant radius r involves no gravitational work. Any path between two points near the source can be broken up into approximating segments parallel to $\vec{r}$ and perpendicular to $\vec{r}$ at each point, and one can make the approximation as good as you like by choosing small enough segments.

This permits us to easily compute the gravitational potential energy as the negative work done moving a mass m from a reference position $\vec{r}_0$ to a final position $\vec{r}$:

$$\begin{aligned} U(r) &= -\int_{r_0}^{r} \vec{F}\cdot d\vec{r} && (12.41)\\ &= -\int_{r_0}^{r} -\frac{GMm}{r^2}dr && (12.42)\\ &= -(\frac{GMm}{r} - \frac{GMm}{r_0}) && (12.43)\\ &= -\frac{GMm}{r} + \frac{GMm}{r_0} && (12.44) \end{aligned}$$

Note that the potential energy function depends only on the *scalar magnitude* of $\vec{r}_0$ and

$\vec{r}$, and that r_0 is in the end the radius of an arbitrary point where we define the potential energy to be zero.

By convention, unless there is a good reason to choose otherwise, we require the zero of the gravitational potential energy function to be at $r_0 = \infty$. Thus:

$$U(r) = -\frac{GMm}{r} \tag{12.45}$$

Note that since energy in some sense is more fundamental than force (the latter is the negative derivative of the former) we could just as easily have learned Newton's Law of Gravitation directly as this *scalar* potential energy function and then evaluated the force by taking its negative gradient (multidimensional derivative).

The most important thing to note about this function is that it is *always negative*. Recall that the force points in the direction that the potential energy *decreases most strongly* in. Since $U(r)$ is negative and gets larger in magnitude for smaller r, gravitation (correctly) points *down* to smaller r where the potential energy is "smaller" (more negative).

The potential energy function will be very useful to us when we wish to consider things like escape velocity/energy, killer asteroids, energy diagrams, and orbits. Let's start with energy diagrams and orbits.

12.6: Energy Diagrams and Orbits

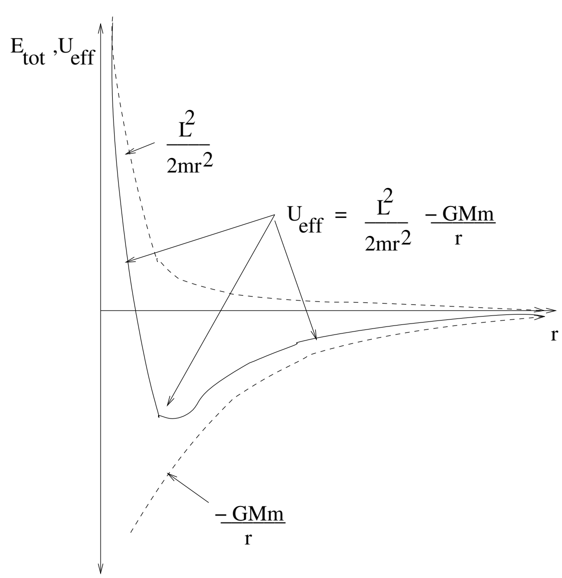


Figure 163: A typical energy diagram illustrating the *effective potential energy*, which is basically the sum of the radial potential energy and the angular kinetic energy of the orbiting object.

Let's write the total energy of a particle moving in a gravitational field in a clever way that isolates the ***radial kinetic energy***:

$$\begin{aligned}
E_{\rm tot} &= \frac{1}{2}mv^2 - \frac{GMm}{r} &(12.46)\\
&= \frac{1}{2}mv_r^2 + \frac{1}{2}mv_t^2 - \frac{GMm}{r} &(12.47)\\
&= \frac{1}{2}mv_r^2 + \frac{1}{2mr^2}(mv_t r)^2 - \frac{GMm}{r} &(12.48)\\
&= \frac{1}{2}mv_r^2 + \frac{L^2}{2mr^2} - \frac{GMm}{r} &(12.49)\\
&= \frac{1}{2}mv_r^2 + U_{\rm eff}(r) &(12.50)
\end{aligned}$$

In this equation, $\frac{1}{2}mv_r^2$ is the radial kinetic energy, and

$$U_{\rm eff}(r) = \frac{L^2}{2mr^2} - \frac{GMm}{r} \tag{12.51}$$

is the radial potential energy *plus* the rotational kinetic energy of the orbiting particle, formed out of the transverse velocity v_t as $K_{\rm rot} = \frac{1}{2}mv_t^2 = L^2/2mr^2$. If we plot the effective potential (and its pieces) we get a one-dimensional radial energy plot as illustrated in figure 163.

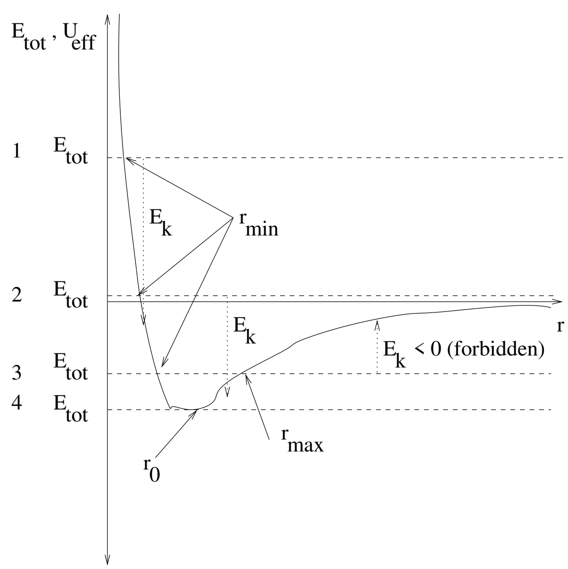


Figure 164: A radial total energy diagram illustrating the four distinct named orbits in terms of their total energy: 1) is a ***hyperbolic*** orbit. 2) is a ***parabolic*** orbit. 3) is an ***elliptical*** orbit. 4) is a ***circular*** orbit. Note that all of these orbits are conic sections, and that the classical elliptic orbits have two *radial* turning points at the apogee and perigee along the major axis of the ellipse.

By drawing a constant total energy on this plot, the difference between $E_{\rm tot}$ and $U_{\rm eff}(r)$ is the radial kinetic energy, which must be positive. We can determine lots of interesting things from this diagram.

In figure 164, we show orbits with a given fixed angular momentum $\vec{L} \neq 0$ and four generic total energies $E_{\rm tot}$. These orbits have the following characteristics and names:

a) $E_{\rm tot} > 0$. This is a **hyperbolic** orbit.

b) $E_{\rm tot} = 0$. This is a **parabolic** orbit. This orbit defines **escape velocity** as we shall see later.

c) $E_{\rm tot} < 0$. This is generally an **elliptical** orbit (consistent with Kepler's First Law).

d) $E_{\rm tot} = U_{\rm eff,min}$. This is a circular orbit. This is a special case of an elliptical orbit, but deserves special mention.

Note well that all of the orbits are **conic sections**. This interesting geometric connection between $1/r^2$ forces and conic section orbits was a tremendous motivation for important mathematical work two or three hundred years ago.

12.7: Escape Velocity, Escape Energy

As we noted in the previous section, a particle has "escape energy" if and only if its total energy is greater than or equal to zero, provided that we set the zero of potential energy at infinity in the first place. We define the ***escape velocity*** (a misnomer!) of the particle as the minimum ***speed*** (!) that it must have to escape from its current gravitational field – typically that of a moon, or planet, or star. Thus:

$$E_{\rm tot} = 0 = \frac{1}{2}mv_{\rm escape}^2 - \frac{GMm}{r} \tag{12.52}$$

so that

$$v_{\rm escape} = \sqrt{\frac{2GM}{r}} = \sqrt{2gr} \tag{12.53}$$

where in the last form $g = \frac{GM}{r^2}$ (the magnitude of the gravitational field – see next item).

To escape from the ***Earth's surface***, one needs to start with a speed of:

$$v_{\rm escape} = \sqrt{\frac{2GM_E}{R_E}} = \sqrt{2gR_E} = 11.2 \quad \text{km/sec} \tag{12.54}$$

Note: Recall the form derived by equating Newton's Law of Gravitation and mv^2/r in an earlier section for the velocity of a mass m in a circular orbit around a larger mass M:

$$v_{\rm circ}^2 = \frac{GM}{r} \tag{12.55}$$

from which we see that $v_{\rm escape} = \sqrt{2}v_{\rm circ}$.)

It is often interesting to contemplate this reasoning in reverse. If we drop a rock onto the earth from a state of rest "far away" (much farther than the radius of the earth, far enough away to be considered "infinity"), it will REACH the earth with escape (kinetic) energy and a total energy close to zero. Since the earth is likely to be much larger than the rock, it will undergo an *inelastic* collision and release nearly ***all its kinetic energy as heat***. If the rock is small, this is not necessarily a problem. If it is large – say, 1 km and up – it releases a *lot* of energy.

Example 12.7.1: How to Cause an Extinction Event

How much energy? Time to do an estimate, and in the process become just a tiny bit scared of a very, very unlikely event that could conceivably cause the extinction of *us*.

Let's take a "typical" rocky asteroid that might at any time decide to "drop in" for a one-way visit. While the asteroid might well have any shape – that of a potato, or *pikachu*[231] , we'll follow the usual lazy physicist route and assume that it is a simple spherical ball of rock with a radius r. In this case we can estimate its total mass as a function of its size as:

$$M = \frac{4\pi\rho}{3} r^3 \tag{12.56}$$

Of course, now we need to estimate its density, ρ. Here it helps to know two numbers: The density of water, or ice, is around 10^3 kg/m^3 (a metric ton per cubic meter), and the ***specific gravity or rock*** is highly variable, but in the ballpark of 2 to 10 (depending on how much of what kinds of metals the rock might contain, for example), say around 5.

If we then let $r \approx 1000$ meters (a bit over a mile in diameter), this works out to $M \approx 1.67 \times 10^{12}$ kg, or around 2 billion metric tons of rock, about the mass of a small mountain.

This mass will land on earth with *escape velocity*, 11.2 km/sec, if it falls in "from rest" from far away. Or more, of course – it may have started with velocity and energy from some other source – this is pretty much a minimum. As an exercise, compute the number of Joules this collision would release to toast the dinosaurs – or us! As a further exercise, convert the answer to "tons of TNT" (a unit often used to describe nuclear-grade explosions – the original nuclear fission bombs had an explosive power of around 20,000 tons of TNT, and the largest nuclear fusion bombs built during the height of the cold war had an explosive power on the order of 1 to 15 million tons of TNT.

The conversion factor is 4.184 gigajoules per ton of TNT. You can easily do this by hand, although the internet now boasts of calculators that will do the entire conversion for you. I get ballpark of ***ten to the twentieth*** joules or 25 gigatons – that is ***billions of tons*** – of TNT. In contrast, wikipedia currently lists the combined explosive power of all of the world's 30,000 or so extant nuclear weapons to be around 5 gigatons. The explosion of Tambora (see last chapter) was estimated to be around 1 gigaton. The asteroid that might have caused the K-T extinction event that ended the Cretaceous and wiped out the dinosaurs and created the 180 kilometer in diameter ***Chicxulub crater***[232] had a diameter estimated at around 10 km and would have released around 1000 times as much energy, between 25 and 100 ***teratons*** of TNT, the equivalent of some 25,000 Tambora's happening all at once.

Such impacts are geologically rare, but obviously can have enormous effects on the climate and environment. On a smaller scale, they are one very good reason to oppose the military exploitation of space – it is all too easy to attack any point on Earth by dropping rocks on it, where the asteroid belt could provide a virtually unlimited supply of rocks.

[231] Wikipedia: http://www.wikipedia.org/wiki/Pikachu. If you don't already know, don't ask...
[232] Wikipedia: http://www.wikipedia.org/wiki/Chicxulum Crater.

12.8: Bridging the Gap: Coulomb's Law and Electrostatics

This concludes our treatment of basic mechanics. Gravitation is our first actual *law of nature* – a force or energy law that describes the way we think the Universe actually works at a fundamental level.

Gravity is, as we have seen, important in the sense that we live gravitationally bound to the outer surface of a planet that is itself gravitationally bound to a star that is gravitationally compressed at its core to the extent that thermonuclear fusion keeps the entire star white hot over billions of years, providing us with our primary source of usable energy. It is *unimportant* in the sense that it is *very weak*, the weakest of all of the known forces.

Next, in the second volume of this book, you will study one of the *strongest* of the forces, the one that dominates almost every aspect of your daily life. It is the force that binds atoms and molecules together, mediates chemistry, permits the exchange of energy we call light, and indeed is the fundamental source of *nearly every* of the "forces" we treated in this semester in collective form: The electromagnetic interaction.

Just to whet your interest (and explain why we have spent so long on gravity when it is weak and mostly irrelevant outside of its near-Earth form in everyday affairs) let is take note of ***Coulomb's Law***, the force that governs the all-important electrostatic interaction that binds electrons to atomic nuclei to make atoms, and binds atoms together to make molecules. It is the force that exists between two *charges*, and can be written as:

$$\vec{F}_{12} = \frac{k_e\ q_1\ q_2}{r_{12}^2}\hat{r}_{12} \tag{12.57}$$

Hmmm, this equation looks rather familiar! It is *almost identical* to Newton's Law of Gravitation, only it seems to involve the *charge* (q) of the particles involved, not their mass, and an *electrostatic constant* k_e instead of the gravitational constant G.

In fact, it is *so* similar that you instantly "know" lots of things about electrostatics from this one equation, plus your knowledge of gravitation. You will, for example, learn about the electrostatic field, the electrostatic potential energy and potential, you will analyze circular orbits, you will analyze trajectories of charged particles in uniform fields – all pretty much the same idea (and algebra, and calculus) as their gravitational counterparts.

The one really *interesting* thing you will learn in the first couple of weeks is how to *properly* describe the geometry of $1/r^2$ force laws and their underlying fields – a result called ***Gauss's Law***. This law and the other Maxwell Equations will turn out to govern nearly everything you experience. In some very fundamental sense, you *are* electromagnetism.

Good luck!

Homework for Week 12

Problem 1.

Physics Concepts: Make this week's physics concepts summary as you work all of the problems in this week's assignment. Be sure to cross-reference each concept in the summary to the problem(s) they were key to, and include concepts from previous weeks as necessary. Do the work carefully enough that you can (after it has been handed in and graded) punch it and add it to a three ring binder for review and study come finals!

Problem 2.

It is a horrible ***misconception*** that astronauts in orbit around the Earth are ***weightless***, where weight (recall) is a measure of the actual gravitational force exerted on an object. Suppose you are in a space shuttle orbiting the Earth at a distance of two times the Earth's radius ($R_e = 6.4 \times 10^6$ meters) from its center.

a) What is your weight *relative* to your weight on the Earth's surface?

b) Does your weight depend on whether or not you are moving at a constant speed? Does it depend on whether or not you are accelerating?

c) Why would you *feel* weightless inside an orbiting shuttle?

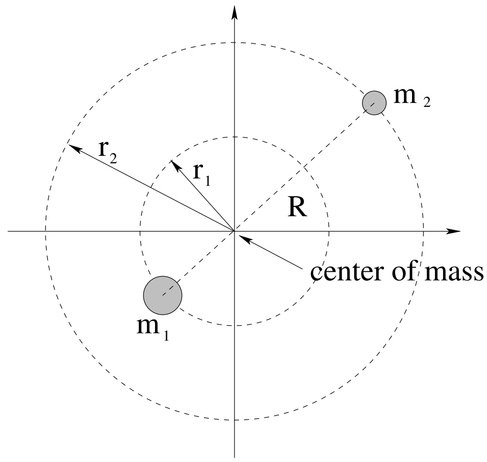
d) Can you ***feel*** as "weightless" as an astronaut on the space shuttle (however briefly) in your own dorm room? How?

Problem 3.

Physicists are working to understand "dark matter", a phenomenological hypothesis invented to explain the fact that things such as the orbital periods around the centers of *galaxies* cannot be explained on the basis of estimates of Newton's Law of Gravitation using the total *visible* matter in the galaxy (which works well for the mass we can see in planetary or stellar context). By adding mass we cannot see until the orbital rates are explained, Newton's Law of Gravitation is preserved (and so are its general relativistic equivalents).

However, there are alternative hypotheses, one of which is that Newton's Law of Gravitation is *wrong*, deviating from a $1/r^2$ force law at very large distances (but remaining a central force). The orbits produced by such a $1/r^n$ force law (with $n \neq 2$) would not be elliptical any more, and $r^3 \neq CT^2$ – but would they still sweep out equal areas in equal times? Explain.

Problem 4.



In the discussion of gravitation and orbits above, we have implicitly assumed that one of the two objects – the Sun in the case of planetary orbits or the planet (e.g. Earth) in the case of lunar orbits – has a much greater mass than the other. In this case, the center of mass of the system is "inside" the larger object and we can pretend that it remains at rest while the lighter one orbits it.

In reality, though, both objects are in opposing orbits around the *center of mass* of the two objects. In this problem, you will try to figure out what happens if the two objects have *similar* masses.

Suppose two stars, a lighter one with mass m_2 and a heavier one with mass $m_1 = 2m_2$ are each orbiting their mutual center of mass in ***circular orbits*** with radii r_1 and r_2 respectively as drawn above. Answer the following questions as you analyze their orbits:

a) Suppose $R = r_1 + r_2$ is the distance between the two stars. What are r_1 and r_2 in terms of R?

b) What is the ***magnitude*** of the gravitational force F_1 acting on mass m_1? Is the magnitude of the force F_2 acting on m_2 the same or different?

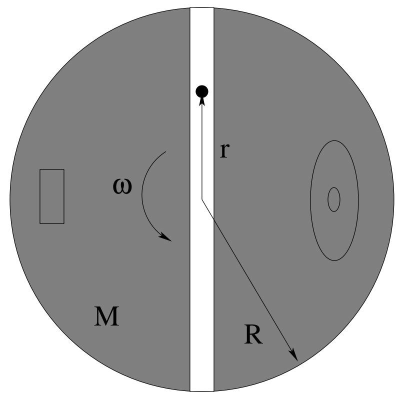
c) The two stars are far away from all other masses so that there is no net external force on them from objects *outside* of this system. The center of mass (in the center of mass reference frame illustrated above) remains at rest. Using this, is ω_1, the angular velocity of mass m_1 the same or different from ω_2, the angular velocity of mass m_2?

d) Using your answers to part b), write ***Newton's Second Law*** for each mass. Express the radial acceleration a_i of each mass in terms of ω_i, the angular velocity of the mass (which may or may not be the same for both masses) and r_i, the radius of its circular orbit.

e) Add the two equations and show that:

$$T^2 = \frac{4\pi^2}{3Gm_2}R^3 \qquad \text{for either/both of the planets.}$$

Problem 5.



This problem will help you learn required concepts such as:

- Newton's Law of Gravitation
- Circular Orbits
- Centripetal Acceleration
- Kepler's Laws

so please review them before you begin.

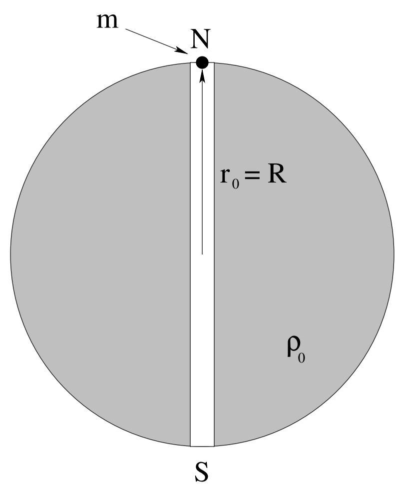
A straight, smooth (frictionless) transit tunnel is dug through a spherical asteroid of radius R and mass M that has been converted into Darth Vader's *death star*. The tunnel is in the equatorial plane and passes through the center of the death star. The death star moves about in a hard vacuum, of course, and the tunnel is open so there are no drag forces acting on masses moving through it.

a) Find the force acting on a car of mass m a distance $r < R$ from the center of the death star.

b) You are commanded to find the precise rotational frequency of the death star ω such that objects in the tunnel will orbit *at* that frequency and hence will appear to *remain at rest* relative to the tunnel at any point along it. That way Darth can Use the Dark Side to move himself along it almost without straining his midichlorians. In the meantime, he is reaching his crooked fingers towards you and you feel a choking sensation, so better start to work.

c) Which of Kepler's laws does your orbit satisfy, and why?

Problem 6.



This problem will help you learn required concepts such as:

- Newton's Second Law.
- Newton's Law of Gravitation
- Gravitational Field/Force Inside a Spherical Shell or Solid Sphere.
- Harmonic Oscillation Given Linear Restoring Forces.
- Definitions and Relations Involving ω and T.

so please review them before you begin.

A straight, smooth (frictionless) transit tunnel is dug through a planet of radius R whose mass density ρ_0 is constant. The tunnel passes through the center of the planet and is lined up with its axis of rotation (so that the planet's rotation is **irrelevant** to this problem). All the air is evacuated from the tunnel to eliminate drag forces.

a) Find the force acting on a car of mass m a distance $r < R$ from the center of the planet.

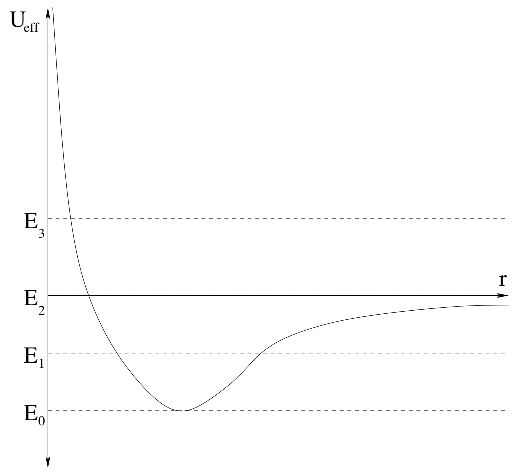
b) Write Newton's second law for the car, and extract the differential equation of motion. From this find $r(t)$ for the car, assuming that it ***starts at rest*** at $r_0 = R$ on the North Pole at time $t = 0$.

c) How long does it take the car to get to the South Pole starting from rest at the North Pole? How long does it take to get back to the North Pole? Compare this (second answer) to the ***period of a circular orbit inside the death star*** you found (disguised as ω) in the previous problem.

d) A final thought question: Suppose it is released at rest from an initial position $r_0 = R/2$ (halfway to the center) instead of from r_0. How long does it take for the mass to get back to this point *now* (compare the periods)?

All answers should be given in terms of G, ρ_0, R and m.

Problem 7.



The *effective radial potential* of a planetary object of mass m in an orbit around a star of mass M is:

$$U_{\text{eff}}(r) = \frac{L^2}{2mr^2} - \frac{GMm}{r}$$

(a form you already explored in a previous homework problem). The total energy of four orbits are drawn as dashed lines on the figure above for some given value of L. Name the kind of orbit (circular, elliptical, parabolic, hyperbolic) each energy represents and mark its turning point(s).

Problem 8.

In a few lines prove Kepler's third law for *circular* orbits around a planet or star of mass M:

$$r^3 = CT^2$$

and determine the constant C and then answer the following questions:

a) Jupiter has a mean radius of orbit around the sun equal to 5.2 times the radius of Earth's orbit. How long does it take Jupiter to go around the sun (what is its orbital period or "year" T_J)?

b) Given the distance to the Moon of 3.84×10^8 meters and its (sidereal) orbital period of 27.3 days, find the mass of the Earth M_e.

c) Using the mass you just evaluated and your knowledge of g on the surface, estimate the radius of the Earth R_e.

Check your answers using google/wikipedia. Think for just one short moment how much of the physics you have learned this semester is verified by the correspondance. Remember, I don't want you to believe anything I am teaching you because of my *authority* as a teacher but because it *works*.

Problem 9.

It is very costly (in energy) to lift a payload from the surface of the earth into a circular orbit, but once you are there, it only costs you that same amount of energy again to get from that circular orbit to anywhere you like – if you are willing to wait a long time to get there. Science Fiction author Robert A. Heinlein succinctly stated this as: "By the time you are in orbit, you're halfway to anywhere."

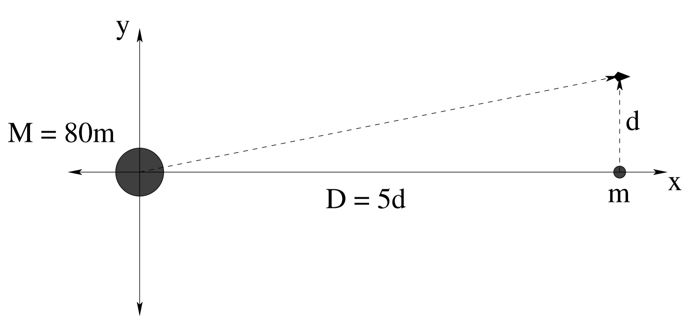
Prove this by comparing the total energy of a mass:

a) On the ground. Neglect its kinetic energy due to the rotation of the Earth.

b) In a (very low) circular orbit with at radius $R \approx R_E$ – assume that it is still more or less the same distance from the center of the Earth as it was when it was on the ground.

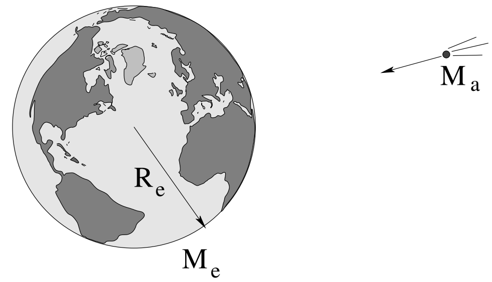
c) The orbit with minimal escape energy (that will arrive, at rest, "at infinity" after an infinite amount of time).

Problem 10.



The large mass above is the Earth, the smaller mass the Moon. Find the **vector gravitational field** acting on the spaceship on its way from Earth to Mars (swinging past the Moon at the instant drawn) in the picture above.

Problem 11.



This problem will help you learn required concepts such as:

- Gravitational Energy
- Fully Inelastic Collisions

so please review them before you begin.

A bitter day comes: a roughly ***spherical*** asteroid of radius R_a and density ρ is discovered that is falling in from ***far away*** so that it will strike the Earth. Ignore the gravity of the Sun in this problem. Determine:

a) If it strikes the Earth (an ***inelastic collision*** if there ever was one) how much energy will be liberated as heat? Express your answer in terms of R_e and either g or G and M_e as you prefer. It is probably very safe to say that $M_a \ll M_e$...

b) Chuck Norris lands on the surface of the asteroid to save the Earth, but instead of screwing around with drills and nuclear bombs Chuck jumps up from the surface of the asteroid at a speed of v_{cn} to deliver a roundhouse kick that would surely break the asteroid in half and cause it to miss the earth – if it knows what's good for it (this *is* Chuck Norris, after all). However, if you jump up too fast on an asteroid, you don't come down again! *Does* Chuck ever fall down onto the asteroid after his jump?

c) Evaluate your answers to a-b above for the following data:

$$\begin{aligned}
\rho &= 6 \times 10^3 \text{ kilograms/meter}^3 \\
R_a &= 10^4 \text{ meters} \\
R_e &= 6.4 \times 10^6 \text{ meters} \\
g &= 10 \text{ meters/second}^2 \\
M_e &= 6 \times 10^{24} \text{ kilograms} \\
v_{cn} &= 5 \text{ meters/second}
\end{aligned} \tag{12.58}$$

Express your answer to a) both in joules and in "tons" (of TNT) where 1 ton-of-TNT = 4.2×10^9 joules. Compare the answer to (say) 30 Gigatons as a safe upper bound for the total combined explosive power of every weapon (including all the *nuclear* weapons) on earth.

d) Find the size of an asteroid that (when it hits) liberates only the energy of a typical thermonuclear bomb, 1 megaton of TNT.

Problem 12.

There is an old physics joke involving cows, and you will need to use its punchline to solve this problem.

A cow is standing in the middle of an open, flat field. A plumb bob with a mass of 1 kg is suspended via an unstretchable string 10 meters long so that it is hanging down roughly 2 meters away from the center of mass of the cow. Making any reasonable assumptions you like or need to, *estimate* the angle of deflection of the plumb bob from vertical due to the gravitational field of the cow.

Optional Problems

The following problems are **not required or to be handed in**, but are provided to give you some extra things to work on or test yourself with *after* mastering the required problems and concepts above and to prepare for quizzes and exams.

Continue Studying for Finals
using problems from the online review!